Order Rodentia

As rodents are generally small bodied and often herbivorous, they are of tremendous ecological importance in nearly all terrestrial habitats. They also comprise about 40% of all mammalian species. Unfortunately, except for domesticated lab species, knowledge of their reproductive biology is not in keeping with either their importance or their diversity.

Details of anatomy (11372), sperm (4826a), developmental time (1449a), reproductive cyclicity (9350), copulation (2513, 6207), and copulatory plugs (6235) are available.

Order Rodentia, Family Aplodontidae, Genus *Aplodontia*

The 800-1800 g mountain beavers (*Aplodontia rufa*) are semi-fossorial herbivores of temperate forests in western North America. Mountain beavers exhibit little sexual dimorphism (9578, 10720). The hemochorial, labyrinthine, discoidal placenta implants in a duplex uterus (8470, 8474). Naked neonates develop eye slits at 10 d but may not open their eyes completely until 45-54 d (3200, 4817, 6609), and are suckled from 6 teats (8470, 8474). Reproductive females have dense patches of black hair around their nipples which can be induced by estrogen (8471). Males have semi-scrotal testes, a forked baculum, and seminal vesicles (1473, 6609, 8470, 8473). Details of birth (2107) are available.

NEONATAL MASS		
Aplodontia rufa	19.8 g; r18.0-21.6; n2, 2 d	6609
	27 g; r25.0-29.5; n3, 1 d	2107
NEONATAL SIZE		
Aplodontia rufa	66 mm; n2	6609
	87.3 mm; r85-90; n3, 1 d	2107
WEANING MASS		
Aplodontia rufa	347 g; n2, 8 wk	6608
	m: 357.44 g; se19.69; r280-440; n9; 1st trapped	6608
	f: 473.13 g; se35.62; r330-590; n8; 1st trapped	6608
LITTER SIZE		
Aplodontia rufa	2-4 yg	5048
	2.18 emb, yg; r2-3; n16 f	9578
	2.4 emb; n12 f	8470, 8474
	2.5 ps; n25 f	8474
	2.8-3.1 cl	8470, 8474
	3 yg; n1 lt	2107
	3 yg; n1 nest	4817
	3 yg; r2-4; n2 f	6608, 6609
	3-4 yg	5146
SEXUAL MATURITY		
Aplodontia rufa	f: 1st ovulate: yearlings, but none preg	8470
	f: 1st mate: 2 yr	1474
	f: 1st birth: 2 yr	3200
	m: hypertrophy of genital tract: some yearlings	8470
	m: 1st breed: 2 yr	5048
GESTATION		
Aplodontia rufa	est 28-30 d	8474
	~30 d	9578
	~1 mo; field data	8470
LACTATION		
Aplodontia rufa	solid food: 7 wk	6609
	wean: 6-8 wk	6608
	den emergence: ~10 wk	3200
LITTERS/YEAR		
Aplodontia rufa	1	1474
	1	3200

SEASONALITY

Aplodontia rufa	large testes: Jan-Mar; WA	8473
	large testes: Jan-Mar; WA	5048
	mate: Feb-Mar; WA	9578
	estrus: mid-Feb; w N America	3200
	ovulate: Jan-Mar; CA	8470
	birth: Feb-Apr; w US	8474
	birth: end Mar-early Apr; few data; captive, OR	6609
	birth: Apr; WA, BC	2088
	birth: July; WA	8470
	lact f: Apr-May; OR	6608

Order Rodentia, Family Sciuridae

Squirrels (*Aeretes* ~305 mm; *Aeromys* 1128-1250 g; *Ammospermophilus* 80-186 g; *Atlantoxerus* 300-350 g; *Belomys* 180-260 mm; *Callosciurus* 45-500 g; *Cynomys* 465-1700 g; *Dremomys* 140-270 g; *Epixerus* 388-604 g; *Eupetaurus* 515-610 mm; *Exilisciurus* 15 g; *Funambulus* 70-168 g; *Funisciurus* 200-350 g; *Glaucomys* 32-185 g; *Glyphotes* ~128 mm; *Heliosciurus* 200-480 g; *Hylopetes* 30-570 g; *Hyosciurus* 200-240 mm; *Iomys* 119-191 g; *Lariscus* 120-230 g; *Marmota* 1.6-13 kg; *Menetes* 184 g; *Microsciurus* 120-160 mm; *Myosciurus* 16 g; *Nannosciurus* 75-115 mm; *Paraxerus* 37-700 g; *Petaurillus* 25 g; *Petaurista* 1-2.5 kg; *Petinomys* 22-110 g; *Prosciurillus* 100-135 mm; *Protoxerus* 388-1000 g; *Pteromys* 140 g; *Pteromyscus* 134-315 g; *Ratufa* 0.9-3 kg; *Rheithrosciurus* 1-2 kg; *Rhinosciurus* 187-255 g; *Rubrisciurus* 150-500 g; *Sciurillus* 90-110 mm; *Sciurotamias* 195-250 mm; *Sciurus* 60-1080 g; *Spermophilopsis* 520-620 g; *Spermophilus* 85-1260 g; *Sundasciurus* 55-420 g; *Syntheosciurus* 150-170 mm; *Tamias* 25-142 g; *Tamiasciurus* 127-312 g; *Tamiops* 27-85 g; *Trogopterus* 270-305 mm; *Xerus* 260-1022 g) are primarily diurnal, arboreal to terrestrial herbivores from the New World, Eurasia, and Africa.

Males are reported larger in *Cynomys* (5916, 8570, 8572), *Funambulus* (8500; cf 8726), and *Spermophilus* (5658, 9890, 10220, 10862; cf 1552, 7287), while females are larger in *Funisciurus* (9945) and *Tamias* (2805, 5658, 10315). Each sex has been reported larger for *Ratufa* (2313, 8501) and *Sciurus* (1361, 4230, 8188).

Ovulation is induced by coitus in *Spermophilus* (2729, 3432, 4850, 10862). *Cynomys*, *Spermophilus*, and *Sciurus* have copulatory plugs (589, 7286, 7633, 10437). The hemochorial, labyrinthine, discoid, deciduate placenta has a superficial, eccentric or mesometrial implantation into a bicornuate or duplex uterus (2015, 3011, 4561, 6353, 6354, 7152, 7563, 7567, 7575, 7944, 8050, 8053, 9707, 10187, 10310, 10532, 11242; cf 231). A postpartum estrus is recorded for *Funisciurus* (2997), *Paraxerus* (11298), and *Tamias* (12014).

Sciurids are usually blind and naked at birth (40, 139, 502, 554, 1474, 1552, 2060, 4083, 4274, 5803, 6504, 6798, 7161, 7235, 8050, 8053, 8573, 8798, 9175, 10445, 10584; cf 2901). Eyes open at 1-3 wk (*Funambulus* 105, 8500, 8501, 9175), 1-5 wk (*Spermophilus* 978, 1286, 1864, 2896, 3238, 3281, 3595, 5020, 5210, 5663, 6992, 7080, 7082, 7550, 7788, 8050, 8348, 9873, 10235, 10862, 10948, 11224, 11322, 11379; cf 9872; *Xerus* 10158, 12134), 20-28 d (*Marmota* 3571, 4270, 8770, 8771, 8798), 22 d (*Callosciurus* 11975; *Ratufa* 33), 20-30 d (*Trogopterus* 11448), 25-33 d (*Tamias* 175, 1295, 3390, 4825, 9416, 11385), 26-36 d (*Tamiasciurus* 3238, 6297, 8745, 10585), 29-36 d (*Ammospermophilus* 7788, 8348), 30 d (*Petaurista* 9175), 4-6 wk (*Glaucomys* 1117, 1672, 3238, 3239, 6497a, 7685, 10206, 10211; *Sciurus* 589, 1361, 2923, 3571, 3749, 5002, 7433, 8053, 8798, 9214, 9943, 10293, 10809, 11024, 12139; cf 1361), or 33-42 d (*Cynomys* 339, 1474, 5342, 8798, 10139). *Paraxerus* is finely furred at birth and opens its eyes within 1-2 wk (2997, 2998, 10158, 11297, 11302). *Funisciurus* also has fine hair at birth, but opens its eyes at 16-20 d (184, 5760, 11302).

Females have 2-4 mammae (*Myosciurus* 2997, 5393), 4 mammae (*Exilisciurus* 7502; *Funambulus* 7502, 7383, 8500, 105; *Nannosciurus* 7502; *Rhinosciurus* 7502; *Rubrisciurus* 7502), 4-6 mammae (*Callosciurus* 182, 7161, 7383, 7502, 10826, with lactation hair 6172; *Funisciurus* 2955, 2997, 5760, 7502, 9945, 10158; *Xerus* 5760, 7502, 9945, 10158), 4-10 mammae (*Tamiasciurus* 1474, 4496, 7502, 8798), 6 mammae (*Belomys* 7383; *Dremomys* 7502; *Heliosciurus* 331, 2955, 2997, 7502; *Lariscus* 7502; *Menetes* 7502; *Microsciurus* 7502; *Paraxerus* 2955, 2997, 7502, 9945, 10158; *Petaurista* 9175; *Prosciurillus* 7502; *Sundasciurus* 2313, 7502; *Tamiops* 7161, 7502), 6-8 mammae (*Atlantoxerus* 7502; *Sciurillus* 7502; *Sciurotamias* 182, 7502; *Syntheosciurus* 7502), 6-10 mammae (*Ratufa* 7161, 7502, 8798; *Sciurus* 1361, 1474, 3022, 3024, 3571, 4494, 4745, 7433, 7502, 8798, 10516), 6-14 (*Spermophilus* 145, 182, 1474, 3571, 4081, 6489, 7433, 7502, 8050, 8798), 8 mammae

(*Epixerus* 2997, 7502; *Glaucomys* 1474; *Protoxerus* 2955, 2997, 7502; *Rheithrosciurus* 7502; *Spermophilopsis* 7502, 8053; *Tamias* 1295, 1474, 4745, 7502, 8053, 9612, 11385), 8-10 mammae (*Cynomys* 1474, 4745, 6561, 7502, 8572), 8-12 mammae (*Marmota* 182, 1474, 3571, 7433, 7502, 8798; cf 8771), 10-12 mammae (*Ammospermophilus* 1474, 4081, 7502). Males have a baculum and seminal vesicles (164, 182, 338, 1473, 2435, 4105, 4782, 7570, 7572, 8053, 8734, 8735, 9964, 10862, 11296; cf 3402). *Glaucomys* sperm form rouleaux in the epididymis (6868).

Details of anatomy (*Cynomys* 3403, 10438, 10439; *Funambulus* 11242; *Marmota* 1828a, 8934; *Sciurus* 2426, 4510, 8777, 8779; *Spermophilus* 3432), endocrinology (*Cynomys* 3401, 3404, 3405; *Funambulus* 5516, 8995, 9798, 10013; *Marmota* 533, 2015, 2016, 2017, 2469, 3337, 8934; *Sciurus* 8779, 10015, 10630, 11513, 11514, 11515, 11516; *Spermophilus* 598, 599, 600, 601, 602, 603, 1480, 1993, 2729, 2983, 2984, 3256, 3502, 4917, 4918, 4919, 5344, 5346, 6429, 7079, 10806, 11580, 11582, 12054, 12055, 12115, 12116), spermatogenesis (6869, 8997, 10629, 11581, 11582, 12056), reproductive energetics (4373a, 4509, 5662, 5663, 7182, 10617), and light (*Ammospermophilus* 5654, 5656, 5659; *Cynomys* 3400; *Funambulus* 4215; *Glaucomys* 7686; *Marmota* 3337; *Spermophilus* 2983, 5655, 6429, 7493, 8349, 10307, 11579, 11583) are available.

NEONATAL MASS

Ammospermophilus harrisii	3.6 g; r3.0-4.1; n12 yg, 2 lt	7788
Ammospermophilus leucurus	2.92 g; sd1.02; r2.0-3.6; n17	6988
	3-4 g; n4 lt	8348
Ammospermophilus nelsoni	4.88 g; n9 yg, 1 lt	4512a
Callosciurus prevosti	16.35 g; r15.9-16.8; n2	11975
Cynomys ludovicianus	15.5 g; n1	5342
	16 g	10139
Funisciurus congicus	9.91 g; ±0.64; n4	11302
Glaucomys sabrinus	5.01 g; r4.8-5.4; n14, 1 d	3239
	5.8 g; n4	7685
	6.0 g; n1	1117
Glaucomys volans	3.0 g; n3, 1 lt	1672
	3.4 g; n1	10584
	3.63 g; r3-5.3; n10	10206
	4.2 g; se0.12; n17	6497a
Marmota camtschatica	33 g	5518
Marmota flaviventris	33.8 g; n3	3481
Marmota marmota	29-30 g	3571
	30 g	8798
Marmota monax	23.7 g; n22	3240
	26.5 g; r22-39; n20	4270
	27.2 g; n11	4083
	32.3 g; r30.2-34.0; n3	10188
Paraxerus cepapi	~10.0 g	10158
	12.75 g; ±2.14; n6	11302
	13 g	11297
Paraxerus palliatus	12.75 g; ±1.06; n2	11302
	17.67 g; ±4.05; n3	11302
Paraxerus poensis	9.91 g; r9.52-10.3; n2	2997, 2998
Petaurista leucogenys	5 g; n1	295
Ratufa bicolor	77 g; n1	33
Sciurus aberti	12 g; n7	5618a
Sciurus carolinensis	13-17 g	9943
	f: 13.9 g; r13.3-14.5; n2	9943
	14.5-16.0 g	10158
	15-18 g; 1 d	11024
Sciurus granatensis	9.5 g; r9-10; n2, 1 lt	7940
Sciurus niger	14-17 g	170
	14.65 g; r14.6-14.7; n2	7943a
Sciurus vulgaris	7-8.5 g	8053
	8-12 g	3571
	8-12 g	8798
Spermophilus beecheyi	8.9 g; n42 yg, 6 lt; r7.2-11.7, r by lt	3281
	9.3 g; r7.7-12.6; n34	10862
	11.8 g; n5	2896

Spermophilus beldingi	6.87 g; sd0.62; r5.24-7.82; n33	7550
Spermophilus columbianus	8.6 g; n23 yg, 5 lt	9873
	~9 g	6798
	10.1 g; n46 yg, 7 lt	5909
	11.3 g; n22	11959
Spermophilus elegans	5.9 g; sd0.45; r5.5-6.6; n6 yg, 2 d	1864
	5.96 g; r5-7; n6 lt	2484
	6.2 g	5909
Spermophilus lateralis	3.6 g; n4	7080
	6-8 g; n14 lt	8348
	6.1 g; 1 d	1871
	6.6 g; n3	7080
Spermophilus mexicanus	4.31 g; r3.85-5.05; n4	2901
Spermophilus mohavensis	4-5 g; n4 lt	8348
Spermophilus richardsoni	6.8 g; r4.4-9.6; n147	7289
Spermophilus saturatus	5.97 g; sd0.39; r5.22-6.55; n18 lt	5658, 5663
Spermophilus tereticaudus	3-5 g; n10 lt	8348
	3.7 g; r2.7-4.7; n26 yg, 4 lt	7788
Spermophilus townsendii	3.26 g; n19 yg, 2 lt	9873
	3.7 g; se0.7; r2.2-4.9; n33	9116
	3.87 g; sd0.62; r2.72-5.15; n21	9115
Spermophilus tridecemlineatus	2.6 g; n10	11379
	3 g; n2	11379
	3.2 g; min 2; n67 yg, 9 lt	5343
Spermophilus variegatus	7.8 g; n4	8019
Spermophilus undulatus	9.7 g; r8.1-11.5; n8	6992
	r9.7-13.2 g; n6 lt	6992
	10.8 g; r9-12.1; n9	6992
	12 g; r11.1-12.8; n5	6992
Tamias amoenus	2.65 g; r2.6-2.8; n10	1295
Tamias minimus	~2.3 g	5210
Tamias quadrivittatus	2-3 g; from graph?	11385
Tamias striatus	2.5-5 g	10182
	3.0 g; n7	175
Tamias townsendii	r3.2-3.9 g; n8	3391
Tamiasciurus douglasii	est 5 g	5922
Tamiasciurus hudsonicus	6 g; n1	10585
	6.7 g; n1	6297
	6.8 g	8745
	7.5 g	4274
	7.5 g; r7.0-8.3; n4	10585
Xerus inauris	20 g	10158
NEONATAL SIZE		
Ammospermophilus nelsoni	58 mm; n2	4512a
Cynomys gunnisoni	CR: 38 mm; near term emb	8572
Cynomys ludovicianus	65-70 mm	10139
Glaucomys sabrinus	HB: 48.7 mm; r47-50; n14, 1 d	3239
	TL: ~71 mm	5210
Glaucomys volans	10 mm; n3 yg, 1 lt	1672
Marmota camtschatica	107 mm	5518
Marmota flaviventris	111.0 mm; n3	3481
Marmota monax	85 mm; n22; from graph	3240
	89 mm; n8	4270
Sciurus aberti	TL: 60 mm	5618a
Sciurus carolinensis	78-80 mm; 1 d	11024
	109-119 mm	589
Sciurus niger	50-60 mm	1871a
Sciurus vulgaris	51-56 mm	8053
Spermophilus beldingi	5.07 cm; sd0.15; r4.91-5.28	7550
Spermophilus citellus	25-30 mm	1552
Spermophilus elegans	CR: 46 mm; n6 lt	2484
Spermophilus fulvus	125-155 mm	8050
Spermophilus lateralis	~61 mm	1871a
Spermophilus mexicanus	TL: 66.5 mm; r63-69; n4	2901

Spermophilus richardsoni	HB: 55.7 mm	7289
Spermophilus townsendii	52.6 mm; se0.4; r42-56; n33	9116
Spermophilus undulatus	55 mm	8050
Tamias amoenus	35-38 mm	1295
Tamias striatus	TL: 66 mm; n1	10182
Tamias townsendii	55-60 mm; n8	3391
WEANING MASS		
Ammospermophilus leucurus	35 g; 6 wk	5661
Ammospermophilus nelsoni	41 g; r40-42; n2, ~8 d after emergence	4512a
Cynomys ludovicianus	f: 145.8 g; se1.6; n287, emergence	3371
	m: 150.9 g; se1.5; n328, emergence	3371
Glaucomys sabrinus	39.74 g; r25.2-53.7; n14, 55 d	3239
Glaucomys volans	46.2 g; se1.11; n26, 8 wk	6497a
Marmota flaviventris	f: 419 g; n13, 1st capture	259
	446 g; 33 d	259
	~454 g; ~30 d	2708
	m: 464 g; n20, 1st capture	259
	f: 500 g	3481
	m: 590 g	3481
Marmota monax	247 g	4270
	249 g; n22, 42 d	3240
Sciurus aberti	355 g; 9 wk	5618a
Sciurus carolinensis	200 g	1361
Spermophilus armatus	130 g; 30 d	5909
Spermophilus beecheyi	50-70 g; emergence	10920
	73 g; ~8 wk	2896
Spermophilus beldingi	36.4 g; r33.5-41.2; emergence	7548
	59.5 g; se2.4	1480
	69.00 g; sd13.27; r47.70-92.13; n33, 25 d	7550
Spermophilus columbianus	70.6-75.7 g; 28 d	5909
	85 g; 30 d	5909
	138.7 g; se10.5; n15 captive; 143.4 g; se7.99; n22 wild; 30 d	11959
	151.2 g; se8.03; n15 captive; 130.6 g; se5.79; n16 wild; 30 d	11959
Spermophilus elegans	80 g; 30 d	5909
Spermophilus lateralis	72.3 g; 35 d partially weaned	1871
	100.8 g; 42 d	1871
Spermophilus major	200 g; 1 mo	8050
Spermophilus richardsoni	105 g; 30 d	5909
Spermophilus saturatus	85 g; 1st capture	5658
Spermophilus spilosoma	40-50 g; emergence	10496
Spermophilus townsendii	42.16 g; sd9.40; r17.88-55.78; n41, 34 d	9115
Spermophilus undulatus	109.9 g; 20 d	6992
Spermophilus washingtoni	22-44 g; emergence	9119
Tamias palmeri	21.22 g; se0.63; n12	4825
Tamias panamintinus	17.68 g; se1.20; n10	4825
Tamias striatus	50.5 g; r27.9-70.9; emergence; regression estimate	8511
Tamias townsendii	35 g; 1st capture	5658
Tamiasciurus hudsonicus	~70-80 g; 7-8 wk	6297
	125 g; r120-142; juveniles	2671
WEANING SIZE		
Glaucomys sabrinus	HB: 10.77 cm; r9.2-12.4; n14, 55 d	3239
Marmota monax	HB: 21 cm; n22, 42 d; from graph	3240
Spermophilus beldingi	13.78 cm; sd1.23; r11.26-15.21; n33	7550
Spermophilus xanthoprymnus	HB: 13-15 cm	8050
LITTER SIZE		
Ammospermophilus harrisii	6 emb; n1 f	7145
	6.4 ps; r4-8; n11 f	7789
	6.5 yg; r6-7; n2 lt	7788, 7789
	6.5; n34 records	7788, 7789
	7.3 emb; r6-10; n7 f	7789
Ammospermophilus interpres	5-14 yg	10110a
Ammospermophilus leucurus	5-11 yg	10110a
	6-8 emb	502
	6 emb; n1 f	6560

	7.40 emb, ps, yg; sd1.73; r5-11; n43 f	5661
	8.0 yg; r7-9	6988
	8-11; n4 lt	8348
	~9 emb; r5-14; n9 f	4081
Ammospermophilus nelsoni	8.9 emb; r6-11	4512a
	12 ps; n1 f	4512a
Callosciurus adamsi	2 emb; n1 f	2313
Callosciurus caniceps	2.2; mode 2; r1-5; n14 lt	4406
Callosciurus erythraeus	2 emb; n1 f	10826, 10827
	2 emb; n1 f	182
Callosciurus nigrovittatus	2.2; mode 2; r1-4; n23 lt	4406
Callosciurus notatus	2.2; mode 2; r1-4; n25 lt	4406
	3 emb ea; n3 f	2313
	3-4 yg	557
Callosciurus prevosti	1 yg; n1 lt	2904
	1.2; mode 1; r1-2; n5 lt	11975
	2-3 yg	557
	2.33 emb; r2-3; n3 f	2313
Cynomys gunnisoni	3-4	9583
	3.5 ps; r3-4; n2 f	145
	mode 4	8572
	4.1 ps; n12 young f	6561
	4.17 emerge; n12 lt	3291
	4.69 emb; mode 4; r1-7; n29 f	145
	4.9 ps; n34 f	6561
	mode 5; r4-6	502
	5.6 emerge; r3-10; n39 lt	8989
Cynomys leucurus	4.90 ps; sd0.77; n20 f	531
	5 yg; r1-8	1871a
	5.40 cl; sd0.72; n48 f	531
	5.48 emb; r2-10; n67 f	10436, 10437
	5.64 emb; sd0.74; r3-8; n25 f	531
	5.65 ps; r2-8; n40 f	10835
Cynomys ludovicianus	2.7 yg; sd1.2; n28	3612
	3.11 emerge; se0.08; r1-6; n198 lt, 124 f	3371
	3.3 emb; yearling f	5916
	3.3 yg; sd1.0; n17	3612
	r4-7 emb; n>12 f	11380
	4 yg; n1 f	5342
	4.00 ps; sd0.30; n10 f yearlings	10441
	4.24 ps; sd0.49; n21 f	10441
	4.3 ps; yearling f	5916
	4.4; sd1.2; n153 lt	5895
	4.56 ps; r2-7; n25 f	10835
	4.6 emb; r1-8; n13 f	339
	4.61 ps; sd0.24; n38 f	10441
	5.0 emb; adult f	5916
	5.02 emb; mode 5; r2-10; n50 f	11380
	5.4 ps; adult f	5916
	5.5 yg; n8 lt	1649a
	5.61 ps; sd0.35; n36	10441
	8 ps, 8 ca; n1 f	6166
Cynomys mexicanus	3 emb; n1 f	522
	6 yg; n1 lt, 12 d	8573
Cynomys parvidens	2-8	10809
	4.8 yg; r3-6; n4 lt	8571
Dremomys lokriah	2-5	7383
	4 emb; n1 f	7383
Epixerus ebii	2; n1	2997, 2998
Exilisciurus exilis	2 emb ea; n2 f	2313
	2.5 yg; r2-3; n2 f	2006
	3 emb ea; n2 f	2006
Funambulus layardi	2.5 emb; r2-3; n2 f	8500, 8501

Funambulus palmarum	2-3	8501
	2.75; r2-3; n4 lt	6163, 8500
Funambulus pennanti	2.9 emb; r1-4; n20 f	6588
	3 yg; 2,4 rare	554
	3 emb ea; n10 f	8720
	3.23 emb; ±0.119; mode 3; r2-5; n30 f	1764
	5 yg ea; n2	8723
Funambulus sublineatus	2 yg; n1 nest	8501
Funisciurus anerythrus	1.08 emb; mode 1; r1-2; n37 f	2446a
	1.2 emb; r1-2; n77 f	8855
	1.5; r1-2; n4	2997, 2998
	2 emb ea; n2 f	8610
Funisciurus congicus	2 yg ea; n3 nests	11300
	2; n8 lt	10158
	2.0 yg; n8 lt	11302
Funisciurus isabella	1 emb ea; n6 f	2997, 2998
Funisciurus lemniscatus	1.8; r1-3; n4	2997, 2998
Funisciurus leucogenys	1 emb; n1 f	2953, 2955
Funisciurus pyrrhopus	1 emb; n1 f	2998
	1 emb; n1 f	187
	mode 1 emb; r1-2	8855
	1.6 yg; n10 lt	2821, 9522
	1.6; r1-3; n31 lt	6755
	2; n1 lt	184
	2 yg	5304
Glaucomys sabrinus	2 emb; n1 f	2085, 2088
	mode 2-3 yg	9385
	3 yg ea; n3 nest	2085, 2088
	4 yg; r2-6	1871a
	4; n1 lt	7685
	5 ps; n1 f	7809
Glaucomys volans	2 emb; n1 f	11896
	r2-4 yg; n9 nests	4716
	2.1 yg; n41 lt	3853
	2.5 emb; r2-3; n2 f	495
	2.54; se0.18; r2-4; n13 lt	3790
	2.6 yg; r2-4; n5 nests	1290
	2.8 old yg; n60 nests	8978a
	3 yg; n1 lt	1672
	3 yg; n1 nest	5210
	3 yg; n1 nest	5747
	3 yg; n1 nest	5803
	3 yg; r2-5; n6 f	10206
	3.0 emb, yg; r2-4; n9 f	4716
	3.1 yg; r1-4; n24 f	10206
	3.17 yg; sd0.36; r2-6; n24 lt	6497a
	3.40 emb, yg; r1-6; n63 f	11025
	3.5 yg; n33 nests	8978a
	3.67; r3-4; n3 lt	12128
	4 yg; n1 f	10584
	4 emb; n1 f	576
	4 emb; n1 f	4716
	4 yg; n1 f	10205
	4.5 yg; n2 nests	495
Heliosciurus gambianus	3.0 emb; n2 f	331
Heliosciurus rufobrachium	1-5	5760
	1-5	10153
	1.5 emb; r1-2; n2 f	185
	mode 2 yg	4354a
	4 emb; n1 f	10158
Heliosciurus ruwenzorii	3 emb; n1 f	8856
Hylopetes alboniger	2 yg; n1 nest	7382
	mode 2-3 yg	7383

Hylopetes fimbriatus	mode 3-4	9175
	4; n1	105
Hylopetes lepidus	2.0 emb; r1-4; n39 f	7687
Iomys horsfieldii	2.5 yg; n2 lt	7161
	3 emb; n1 f	7161
Lariscus hosei	2 emb; n1 f	2313
Marmota baibacina	6; r3-8	8050
Marmota bobak	4-5	5172
	4-5; 6 occasional	8050
	5-6	10516
	max 7	4645
	max 12	8798
Marmota caligata	4-5	1474
	mode 4-5	2088
	5 emb; n1 f	5029
Marmota camtschatica	mode 5; r3-11	5518
Marmota caudata	3; r2-4; 5 occasional	8050
	3-6 forest; 4-9; subalpine	2361
	3.4 emerge; r2-4; n5 lt	9175
Marmota flaviventris	3.64-5.13 yg; n7 locations	11194a
	est 4-5; n1 lt	502
	4.07 weaned; se0.45; n15 lt	5328
	4.14; se0.857; r2-6; n7 lt satellite	2708
	4.15; se0.19; r1-8; n65 lt; colonial	366
	4.2 cl; n4 subadult	7803
	4.46; se0.39; r1-8; n13 lt; satellite	366
	4.50; se0.187; r2-8; n52 lt; colonial	2708
	4.7 cl; n7 adult	7803
	4.8 emb; r3-6	7803
	5 ps; r2-8; n2 f	10954
	mode 5-6 emb; r3-6; n5 f	5029
	5.26; 19 lt	366
	5.3 yg; r5-6; n3 f	7803
	5.5 emb; r5-6; n2 f	10954
	8 emb; n1 f	5029
Marmota marmota	2-4	7433
	2-7	8798
	mode 3-5; r2-8	3571
	7; n1 lt	8770
	7; n1 lt	8771
Marmota menzbieri	1-5	6924
Marmota monax	2.5 cl/ovary; se0.3; n5 postparturient f	2016
	mode 3-4; r1-9	12043
	3.7 ps; se0.4; n8 lt	2015
	3.75 emerge; r2-5; n8 lt	2414
	4 emb; n1 f	9639a
	4 yg; n1	5038
	4-6 yg; max 9	5029
	4.07 emb, ps; n49 f	4270
	4.1 yg; se0.2; r2-6; n17 lt	2015
	4.32 emb; n122 f	10187
	4.58 ps; n169 f	10187
	4.6 emb, ps; max 8; n29 f	4083
	4.7 implantations; se0.6; n6 f	2015
	4.7; r4-5; n3 lt	570
	5 emb; n1 f	10217
	5 yg; n1 f	11956
	5.19 cl; n238 f	10187
	6 ps; 1 f	10839
	9 yg; n1 lt	5210
	13.6 cl/ovary; se1.5; n7 nonparturient f	2016
	15.3 cl/ovary; se2.2; n3 nonparturient yearling f	2016
Marmota sibirica	3-6	8050

Marmota vancouverensis	2.7 emerge; n6 lt	7746
	3.0 emerge; n5 lt	7746
	4.6 emerge; n5 lt	7746
Myosciurus pumilio	2	2997
Paraxerus alexandri	mode 1 emb; r1-2	8855
Paraxerus boehmi	1 emb; n1 f	8856
	1.1 emb; r1-2; n61 f	8855
	1.12 emb; mode 1; r1-2; n34 f	2446a
Paraxerus cepapi	1.93 yg; ±0.59; n29	11302
	mode 2	11295, 11299
	2 emb; n1 f	6272
	2-4	11253
	2-4	9945
	2.1 emb; r1-3; n39 f	10154, 10158
	3 emb; n1 f	334
Paraxerus flavivittis	1; n1 lt	187
	1-2 yg	5760
Paraxerus ochraceus	2 emb ea; n2 f	4925
Paraxerus palliatus	1-2	10158
	1.60 yg; ±0.52; n10	11302
	1.71 yg; ±0.49; n7	11302
Paraxerus poensis	1.3; r1-2; n6	2997, 2998
	2 emb; n1 f	1948
	2 emb; n1 f	2955
Petaurista elegans	mode 1; r1-2	7382, 7383
Petaurista magnificus	1 emb ea; n2 f	7382, 7383
Petaurista petaurista	1 yg	8142
	1 emb; n1 f	8501
	1 emb ea; n2 f	7161
	mode 1; r1-2	7383
	mode 2	9175
Petinomys fuscocapillus	2 yg	8501
Protoxerus stangeri	mode 1 emb; r1-2	8855
	1.5 emb; r1-2; n2 f	2997, 2998
	2 emb; n1 f	187
Pteromys volans	2 yg; n1	11441
	mode 2-3; r1-4	8053
	2-4	4645
Pteromyscus pulverulentus	1.3 emb; r1-2; n14 f	7687
Ratufa bicolor	1; n1	8270
	1 yg ea; 3 lt	11741
	1	40
	1-2 yg	7383
	2-3 yg	10828
Ratufa indica	1 yg ea; n2 lt	10559
Ratufa macroura	mode 1; r1-2	8501
	mode est 3-4	8500
	3-5	8798
	4 yg; n1 f	11507b
Rhinosciurus laticaudatus	1.3; r1-2; n3 lt	4406
Sciurillus pusillus	2 emb; n1 f	8087
Sciurus aberti	2.9 emb, ps	10378
	3-5 yg	1871a
	3 emb; n1 f	6387
	3.20; ±0.65; r2-4; n5 lt	3165
	3.3 emb; r3-4; n3 f	7145
	3.4 yg; ±0.9; r2-5; n8 nests	5618a
Sciurus alleni	3; n1 lt	522
Sciurus aureogaster	2; n1 lt	4228
	2 emb ea; n2 f	2341
Sciurus carolinensis	1.45 wean; r1-2; n11 lt, 10 f	9190
	1.6 wean/season	2088
	2 emb; n1 f	5803
	2.25; n8 lt	8188
	2.5 emb; se0.74; r1-4; n26 f	7297

	2.56 yg; r1-4; n27 nests	4716
	2.56 ps; n43 f	5808
	2.60 yg; n93 lt	5808
	2.68 yg; r1-4; n19 nests	6562
	2.7; n16 lt	3898
	2.7; mode 2-4; r1-7; 2.50; n148 spring; 3.23; n55 fall	9943
	2.74; n27 lt	1361
	2.75 yg; n122 lt	5808
	3 emb	2088
	3.0 ps; r1-4; n16 lt	9214
	3.0; r2-4; n2 lt	1241
	3 ps ea; n3 f	9190
	3.02; n447 lt	589
	3.05 lg emb; r1-5; n21 f	6562
	3.1 emerge; se0.4; n58 f	10791
	3.14 ps; r2-5; n37 f	6562
	3.19 emb; n16 f	5808
	3.43 ps; se0.09; r1-6; n98 f; 2.7; se0.14; winter; 3.49; se0.11; spring	7944
	3.6	2426
	3.8 emb; n5 f	9214
	3.96 sm emb; r1-6; n24 f	6562
	4 emb; n1 f	7498a
	4 yg; n1 nest	6197
	5 yg; n1 nest	7498a
	7 emb; n1 f	9943
	8 yg; 1 nest, <3 d	588
	8.5 yg; r8-9; n2	588
Sciurus colliaei	2.5 emb; r2-3; n2 f	5409
	3 emb; n1 f	5405b
Sciurus deppei	2 emb; n1 f	5423b
	3 emb; n1 f	7624
	5; n1 lt; hybrid with *S. yucatanensis*	3633
Sciurus granatensis	1.5 emb; r1-2; n2 f	8562a
	1.8; n4 lt	2998
	1.9 yg; n12 lt	3801
	mode 2 emb; max 3	3024
	2.6 emb; r2-3; n5 f	7940
Sciurus griseus	2 emb; n1 f	11228a
	3-5	1474
	3-5 yg	5146
Sciurus nayaritensis	2.5 emb; r2-3; n2 f	6387
Sciurus niger	2.19 ps; se0.32; n16 yearlings winter	7943
	2.27; r1-4; n11 lt	7501
	2.58; n110 lt	1361
	2.6; r1?-4; n14 lt	8188
	2.77 yg; n13 lt	170
	2.8 yg	1871a
	2.80 ps; se0.33; n10 yearlings summer	7943
	2.85 ps; se0.11; n52 f summer	7943
	2.97 emb; n38 f	170
	3.0; r2-4; n2 lt	522
	3.01 ps; se0.10; n80 f winter	7943
	3.11 emb; n18 f	6562
	3.23 ps; n22 f	6562
	3.5 yg; r3-4; n4 nests	10445
	5.5 ps; r5-6; n2 f	6298
	7 emb; n1 f	4971
Sciurus richmondi	2.67 emb; r2-3; n6 f	5417
Sciurus vulgaris	2; n1 lt	12139
	3-4	2468
	3-5	8798
	3-6	7433
	3-6; yg f	4645, 8053

	mode 3-7; r3-10	3571
	3.0 emb; r1-4; n5 f	11255
	3.4 emb	2469
	4; n1 lt	3749
	4 yg; n1 lt	11255
	5-10; adult f	4645, 8053
	6; n1 lt	2923
	9; hunted; 4; unhunted	8053
Sciurus yucatanensis	3 emb; n1 f	5423b
	3 yg; n1 f	950
	5; n1 lt; hybrid with *S. deppei*	3633
Spermophilopsis leptodactylus	3-6	8053
	3-6	6273
	4.1; r1-9	10568
	5.2 emb; mode 5; r2-8; n5 f	9774
Spermophilus armatus	4-6 yg	1474
	5.8 emb; r4-8; n22 f	11646a
	6-8 yg	1871a
Spermophilus beecheyi	3 emb; n1 f	11228a
	3 yg; n1 f	5020
	4.14 yg; r3-8; n7 lt	6843
	5 yg; n1 f	2896
	5.14 emb; r3-7; n7 f	2896
	5.46 implantation sites; n13 lt	1725
	5.5-8.1; r4-15	10454
	5.86; n22	1726
	6.06 emb, yg; n34 lt	10862
	6.11 ps; est r1-9; n114 f	10862
	6.25 emerge; n8 nests	3114
	6.86 yg; mode 7; r6-8; n7 lt	3281
	7; r4-15	1474
	7.2 emb; r4-11	4081
	7.5; r4-11	5030
	7.5 emb; r5-12	6489
	9.84 emb; n86 f	5227
Spermophilus beldingi	1-9	4916
	4.0 yg; sd1.0; n5 yearling f	7547
	4.3 emerge; sd2.1; n115 lt	12064
	4.42 emerge; sd2.06; r3-6; n219 lt	9900
	4.57 emb; sd1.07; n20 yearling f	7547
	5.0 ps; n2 yearling f	7547
	5.71 yg; sd1.45; n17 adult f	7547
	5.8 yg; r5-7; n9 lt	6988
	6.33 emb; sd2.01; n21 adult f	7547
	6.88 ps; sd1.22; n17 adult f	7547
	7.1; r3-10; n37 lt	7082
	7.4 emb; ±0.8; n175 f	10541
	8; r4-12	5030
Spermophilus citellus	mode 6-8; r3-11	3571, 7433
	6; n2 lt	1552
Spermophilus columbianus	r1-7; r2.57-5.07; r of 6 populations	12061
	2-7	1474
	2.3-4.6; n12 populations	2661
	2.3 emerge; r1-3; n55 lt	7631
	2.3; se0.09; n61 lt; r1.8-2.5, over 5 yr	7630, 7632
	2.69; n29 lt	2658
	2.7 ps; r1-4; n22 lt	7631
	2.7 emerge; r1-5; n31 lt	7631
	2.78 emerge; se0.83; n23 lt [but n72 yg/23 lt = 3.13]	11469
	2.8 cl; r2-4; n17 lt	7631
	2.9 yg; r1-4; n14	11959
	2.9; se0.12; n62 lt; r2.4-3.3, over 5 yr	7630, 7632
	2.94-4.22 emb, ps; sd0.73-1.45; n6 samples	12063
	3.04 emerge; se0.96; n28 lt [but n86 yg/28 lt = 3.07]	11469

	3.2 emb; r1-5; n28 f	7631
	3.3 cl; r2-4; n29 lt	7631
	3.3 ps; r1-5; n54 lt	7631
	3.36; n121 lt with food added	2658
	3.4 yg; r1-5; n15 f	3242
	3.5 juv; r3-4; n4 lt	7631
	3.5; r2-7	2088
	3.57 emb; r3-4; n7 f	2546
	3.6 yg; r1-6; n11	11959
	3.65 emerge; r1-6; n92 lt	3244
	3.70 yg; mode 3-4; n174	12062
	3.9 cl; r1-7; n12 lt	7631
	~4; r2-7	6798
	4.04; r2.9-6.7 across 5 populations; n37 lt	11959
	4.2 emb; r4-5; n5 f	7631
	4.6 yg; n5 lt	9873
	4.6 ps; r3-7; n8 lt	7631
	5.4 emb; r2-7; n56 f	9872
	6.6 yg; r5-8; n7 lt	5909
Spermophilus dauricus	5-8	8050
	5 emb; n1 f	182
Spermophilus elegans	2.7 emerge; se0.23; n15 lt	8469
	3.6 emerge; se0.75; n12 yearling f	8469
	4.8 emerge; se0.50; n19 adult f	8469
	5.2 emerge; se0.40; n21 lt	8469
	5.3 emb; r4-9; n16 f	1865
	6.0 yg; r5-8; n5 f	1865
	6.1 ps; r4-9; n40 f	1865
	7.0 yg; r5-9; n6 lt	2484
	7.5 yg; r5-11; n8 lt	5909
Spermophilus erythrogenys	mode 7-9 emb; n426 f	8050
Spermophilus franklinii	mode 6-7 yg; r5-8	5210
	7.5 emb; r5-11; n26 f	10235
	9.4 ps, emb, yg; r2-13	5185
	9.7; se0.6; r7-12; n10 lt	10948
	10 emb; n1 f	495
Spermophilus fulvus	6; r4-13; more accurate account: 5-9	8050
	6; mode 5-9; r4-13	4645
	6; r5-9	10845
Spermophilus lateralis	2-8	1474
	3-7; n14 lt	8348
	4.3 yg; ±0.6; r3-9; n3	10055
	4.4 emb; r3-6; n10 f	2415
	4.5 yg; n2 lt	1871
	4.5; r4-5; n13 lt	8489
	5 emb; n1 f	145
	5.0 emb; r2-6; n9 f	4081
	5.0 emb; r3-8; n36 f	7080, 7082
	mode 5 ps, yg, emb; n138 f	7080, 7082
	5.1; ±1.32; r3-9; n35	10055
	5.1; r3-8; n43 lt	10762
	5.2 ps; r2-7; n88 f	7080, 7082
	5.28 emb; 2se0.19; n179 f	1323, 1324
	5.38 emb, ps; 2se0.12; n284 f	1323, 1324
	5.4 emb; ±0.72; r4-6; n8 f	10055
	5.9 ps; ±1.41; r3-9; n24 f	10055
	6.0 ps; r5-7; n3 f	7809
	6.0; r4-9; n25 lt	8489
	6.5 emb; r6-7; n2 f	7593a
	6.5 emb; r6-7; n4 f	6560
Spermophilus madrensis	4 emb; r2-5; n4 f	886
	5 emb; n1 f	886

Spermophilus mexicanus	4-10	1474
	5 yg; r1-10	12035
	5 emb; n1 f	522
	5 yg; n1 lt	2901
	10 emb; n1 f	2901
Spermophilus mohavensis	5-9; n4 lt	8348
	6 emb; n1 f	1471
Spermophilus parryii	6 yg; n1 nest	6993
	6.1 emerge; sd1.00; r4-8; n14 lt	7089
	6.1 yg; sd0.99; n19 lt	7089
	6.5 ps; r6-7; n2 f	6993
	7 emb; n1 f	5030
Spermophilus perotensis	6.5 emb; r6-7; n2 f	2341
Spermophilus pygmaeus	5.43, 6.13, 7.6 emb; n3 authors	8050
	6.7; mode 7; r2-12	9453
	7.7 emb; mode 7; r4-13; n252 f	8050
	7.75 emb; mode 8; r4-13; n302 f	11322
Spermophilus richardsonii	2-10	1474
	3.4; se0.05; n7 yearling f	7284
	3.8 emerge; sd1.3; r1-6; n24 f	7283
	4.3; se0.4; n8 lt, 2 f; 3.5; se0.4; n15 lt, 11 f; 4.9 yg; se0.3; n15 f	7284
	4.4 emerge; se0.5; 11 lt	7284
	4.8; se0.03; n10 adult f	7284
	6.3 emerge; sd2.1; n26 adult f	7288
	6.4 emerge; sd2.1; n55 yearling f	7288
	7.4 emerge; sd2.2; n32 yearling f	7288
	7.4 emerge; sd1.4; n53 yearling f	7288
	7.5; r6-11	5030
	7.9 emerge; sd2.7; n19 adult f	7288
	7.93 emb; n15 f near term	7826
	8.1 emerge; sd1.8; n18 adult f	7288
	8.19 cl; 6.93 live emb; se0.18; n126 f	9890
	8.25 yg; r6-11; n4 lt	5909
	8.75 emb; r4-13; n84 f	7826
	9.85 ps; r6-15; n13 f	7826
Spermophilus saturatus	4.14; sd0.86; r1-5; n37	5658
	4.17 yg; sd0.62; r3-5; n18 f	5663
	7 emb; n1 f	2088
Spermophilus spilosoma	4.57; r3-7; n7 lt	522
	5-12	10496
	5 emb; n1 f	502
	6 yg; n1 f	978
	6 emb; r5-7; n3 f	502
	6.6; r4-11	10496
	6.6 yg; r4-12	1871a
	7 emb; n1 f	145
	8; n1 f	2349
	8 emb ea; n2 f	5405b$_1$
Spermophilus suslicus	mode 4-7; r4-12	4645
	mode 7; r5-11; n120 f	8050
	5.5 emb; r3-9	8424
Spermophilus tereticaudus	1.86 yg; n14 lt	7788
	3.5 emb; r3-4; n2 f	4081
	3-8; n10 lt	8348
	6-12 emb	1474
	6.1 emb; r5-10; n11 f	7789
	6.43; n92	7789
	6.5; r4-9; n26 lt	7789
	6.5 emb; n222 f; r3.3-9.0; different sites	9078
	6.7 ps; r2-9; n35 f	7789
Spermophilus townsendii	4.8 yg; r1-10; n12 lt	9116
	5-10, rarely 15	1474
	7.7 emb, ps; se0.2; n51 yearling f	10110

	7.99 emb, ps; r7.15-9.11 over 6 yr; n257 lt	10110
	8.3 emb; n22 f	9582
	8.3 yg; sd1.7; n135 yearling f	9117
	8.5 yg; r6-13; n20 lt	5343
	8.5 emb; n129 f	6354
	8.6 yg; r6-12; n4 f	9115
	8.6 emb; r4-16; n52	9582
	8.75 emb; r8-10; n4 f	5030
	8.9 emb; r7-10	5030
	9.0 emb; r4-12; n21 f	5343
	9.2 emb, ps; se0.3; n21 adult f	10110
	9.3 emb; n163 f	139
	9.4 yg; sd2.2; n60 adult f	9117
	15 emb; n1 f	5312
Spermophilus tridecemlineatus	5-13	5030
	4.2 emerge; se0.8; yearling f	7023
	4.9 emerge; se0.6; yearling f 1st lt	7023
	5.7 emerge; se1.3; adult f 2nd lt	7023
	mode 6-7; r5-10	2060
	6-10 emb	502
	6.0 yg; r2-10; n2 lt	11379
	6.1 emerge; se0.8; yearling f	7023
	6.9 emerge; se0.8; adult f	7023
	7-10; rarely 14	1474
	7 ps, 8 ca; n1 f	6166
	7.0 emerge; se0.3; adult f 1st lt	7023
	7.4 emerge; se0.4; adult f 1st lt	7023
	7.5 yg; r1-10	1871a
	8.1 emb; n269 f	2140
	8.4; r6-11; n12 lt	1286
	8.57 yg; r5-13; n28 lt	12108
	8.67 emb; sd1.14; r7-11; n18 f	3294
	9 yg	5343
	9 ps; n1 f	3703
	9.1 emb; r6-13; n9 f	495
	9.3 emb; r6-12; n6 f	$5405b_1$
	10.2; r10-11; n5 lt; prob birth	7023
	11 emb; n1 f	1812a
Spermophilus undulatus	1-4 yg	7656
	5-11	10516
	r7-9 emb; n160 f	8050
	7.8 yg; r5-10; n6 f, 6 lt	6992
Spermophilus variegatus	3 emb; n1 f	1649a
	3.8 yg; n6 lt	5347
	4 emerge; r1-7	8019
	4 emb; n1 f	4962
	5-7	1474
	5 emb; r3-9	8019
	5; n1 lt	522
	5 emb; n1 f	7145
	5 emb; n1 f	4914
	6 emb; r5-7; n2 f	502
	7 emb; n1 f	2341
Spermophilus washingtoni	5-11	1474
	8 emb, ps; n11	9582
	8 emb; r5-11; n26	9582
Spermophilus xanthoprymnus	4-6	8050
Sundasciurus lowii	2.5 emb; r2-3; n2 f	2313
Sundasciurus tenuis	3.0; mode 3; r2-4; n4 lt	4406
Tamias amoenus	4-8	2088
	4.9	5662
	5 yg; r2-9	1871a
	r5-7 emb; n5 f	1295
	5.0 emb; n1 f	11646a

	5.0 yg; r4-6; n2 f	1295
	5.14 cl; 4.6 emb; se0.14; n42 f	9889
	5.8 emb; n7 f	1295
	6.1; r4-8; n17 lt	10762
Tamias bulleri	2.5 emb; r2-3; n2 f	523a
Tamias dorsalis	4-6 yg	1871a
	5.0 emb; r2-8; n4 f	4433a
	5.3 emerge; r5-6; n3 lt	4433a
Tamias minimus	2-6	1474
	4 yg; n1 lt	10053
	4 ps; n1 f	3703
	4 emb; r3-5; n2 f	261a
	4.2 ps; r3-5; n5	10839
	4.5 emb; r3-6; n4 f	10954
	4.54 cl; 4.03 emb; se0.18; n59 f	9889
	5; r2-6	2088
	5 ps; n1 f	261a
	5 yg; n1 lt	3390
	5 ca; r3-7; n2	10839
	5.0 emb; mode 5; r3-7; n9 f	279
	5.1 ps; r3-8; n17 lt	10954
	5.4 ps; se0.25; r3-8; n24 f	10053
	5.5 yg; r3-9	1871a
	5.7 yg; se0.52; r5-8; n7	10053
	5.7 emb; se0.15; r3-8; n52 f	11230
	5.7 ps; se0.15; r2-10; n97 f	11230
	6.4 emb; se0.31; r5-9; n11 f	10053
	7 emb; n1 f	10218, 10219
Tamias palmeri	3.75; n4 lt	4825
	4.20 emb; mode 4; r3-6; n15 f	2415
Tamias panamintinus	4.0; n4 lt	4825
	5.75 emb; r4-9; n4 f	2415
Tamias quadrimaculatus	4.4; r2-6; n10 lt	10762
	8 yg; n1 lt	9302a
Tamias quadrivittatus	3.25 emb; r2-4; n4 f	4962
	4-6 emb	502
	5.0 ps; r4-6; n2 f	4914
Tamias ruficaudus	4.65; se0.13; n24 lt	726
	5.16; se0.20; n24 lt	726
Tamias sibiricus	r3.2-6.8; n~12 colonies	982
	mode 4-5; r3-12; rarely 1-2	4645
	4-6	10177
	4.4 emerge; sd1.6; 66% 3-5; r1-8; n176 lt, 56 f	983
	4.8-6.9 emb, yg; varies w/region & yr	9953
	5 yg; n1 lt	9416
	5.57 emb; r3-10	8053
Tamias speciosus	4.3; n3 lt	10762
Tamias striatus	1.8 emerge; r1-4; n44	11989
	2.8-5.3 emerge; over 3 yr	8511
	3 emb; n1 f	1472
	4-5	9164
	4 emb, ps; r1-6; n13 f	12014
	4.67 yg; r4-5; n3 f	8225
	4.7 yg; r4-6; n11 lt	11956
	5 emb; n1 f	10839
	5 emb; n1 f	175
	5; r3-7; n2 lt	175
	5.02 emb; mode 5; r2-7; n41 f	2022
	5.2 ps; r2-7; n5 f	10839
	6 emb; n1 f	9612
	6 emb; n1 f	9639a
	7.0 emb; r6-8; n2 f	10220
Tamias townsendii	3-6	1474
	3-6	2088

	3.75; sd0.45; r3-4; n12	5658
	4.5 yg; r4-5; n2 lt	3391
	4.5; r3-5; n6 lt	10762
Tamias umbrinus	max 7 emb	1871a
Tamiasciurus douglasii	4-8	1474
	5.0; r4-8; n5 lt	10098
	5 yg	5146
	mode 5-6; r3-7	6387
Tamiasciurus hudsonicus	3 emb; n1 f	495
	3 emb; n1 f	6560
	3.0 yg; r2-4; n2 nest	4496
	3 ps ea; n2 f	6166
	3.2; 4.1 cl	2324
	3.2 yg; n9 f	2324
	3.3 emb; mode 3; r2-5; n16 f	2671
	3.4 ps; mode 4; 95% ci 3.1-3.7; r1-5; n43	5641
	3.4; r2-5; n61	10097, 10098
	3.5 yg; r1-10	1871a
	3.7 ps; n34 f	7299
	4.0 ps; n20 f	11894
	4-5 yg; n9 lt	4274
	max 4 emb; 5 ps; n1 f	12049
	4 ps; n4	10839
	4; r1-6	2088
	4.1 emb, ps; n164 f	9374
	4.2 yg; r2-7; n11 nests	6297
	4.2; n12 lt	6158
	4.2 emb; r1-6; n17 f	6297
	4.2 emb; r3-5; n38 f	4274
	4.3 ps; mode 4; 95%ci 3.9-4.7; r2-8; n39	5641
	4.37 cl; r2-7; n83 lt	7299
	4.5 yg; r4-5; n2 f	10585
	4.7 ps; r3-8; n46 f	6297
	4.7 cl; n123 f	9374
	5 ps; r4-6; n3 f	11956
	5.0 emb; r4-6; n4 f	10954
	5.4 yg; r4-8; n7 f	3241
	5.4 yg; r4-8; n8 f	3241, 8745
Tamiops macclellandi	2 emb; n1 f	4813b
	3 emb; n1 f	7161
Tamiops swinhoei	2-5 emb	10825
Trogopterus xanthipes	1.64 yg; mode 1-2; r1-4; n25 lt	11447, 11448
Xerus erythropus	2-4 yg	4354a
	3-4	5277
	3-4	2439
	3.5 emb; r3-4; n2 f	1149
	4; r2-6	5760
Xerus inauris	1-4	12134
	1.9; r1-3	4697
	2.2 emb; r1-3; n27 f	10154, 10158
Xerus rutilus	2 emb; n1 f	8017
	2 emb; n1 f	8017
SEXUAL MATURITY		
Ammospermophilus harrisii	f: vag perf: 10-11 mo	7788
	m: scrotal testes: 14-19 wk	7788
Ammospermophilus leucurus	f: ~1 yr	5661
	m: testes large: ~1 yr	5661
Callosciurus caniceps	f: 1st preg: 230 g	4406
	m: fertile: 50% 210 g; 95% 259 g	4406
Callosciurus nigrovittatus	f: 1st preg: 130 g	4406
	m: fertile: 50% 149 g; 95% 269 g	4406
Callosciurus notatus	f: 1st preg: 150 g	4406
	m: fertile: 50% 113 g; 95% 257 g	4406
Cynomys gunnisoni	f: 1st preg: 1 yr	6561

Cynomys leucurus	f: 1st mate: most 1 yr	531
Cynomys ludovicianus	f: 1st mate: 2 yr; some 1 yr	4958
	f: 1st conceive: yearling	5916
	f: preg: 40% yearlings; 88% 2 yr	5895
	m: 1st mate: 2 yr	4958
	f/m: 1st mate: 2 yr	1474
	f/m: 2 yr	10139
Cynomys parvidens	f/m: 1 yr	10809
Funambulus pennanti	f: 1st mate: 7-10 mo	8735, 9799
	f: 6-8 mo	554
	f: 6-8 mo	9175
	m: spermatogenesis: 7-9 mo	8996
	f/m: 7-11 mo	1764
Lariscus insignis	f: 1st preg: 190 g	4406
Marmota bobak	f/m: ~2 yr	8798
	f/m: 3rd summer	8050
	f/m: 1st mate: 4 yr	5172
Marmota caligata	f: 1st mate: est 2 yr	2088
Marmota caudata	f/m: 3rd yr	8050
	f/m: 4th yr	2361
Marmota flaviventris	f: 1st mate: 2 yr	2708
	f/m: 1st birth: 2 yr; median 3 yr	5328
Marmota marmota	f/m: ~2 yr	8798
	f/m: 3-4 yr; full grown: 2 yr	3571
Marmota menzbieri	f/m: 3rd summer; rarely 2nd yr	6924
Marmota monax	f: 1st mate: 1 yr; n6/21	4270
	m: 1st mate: 1 yr; n15/45	4270
	f/m: 1st mate: mode 2 yr	4083
Marmota sibirica	f/m: 3rd summer	8050
Paraxerus poensis	f: 1st birth: ~126 d; n1	2998
Ratufa macroura	f/m: ~2 yr	8798
Rhinosciurus laticaudatus	f: 1st preg: 170 g	4406
Sciurus carolinensis	f: 1st mate: 124 d	589
	f: 1st birth: yearling	2088
	f: 1st birth: mode 1 yr; 1/99 f <10 mo	10133
	f: fertile: ~1 yr	9214
	f: 1st repro: few 4-6 mo; most 10-14 mo	7297
	f: > 500 g	9943
	m: fertile: ~10 mo	9214
	m: testes: large 9-11 mo	589
	m: testes: large 10-13 mo; spring born	7297
	m: testes: large 14-16 mo; summer born	7297
	f/m: 1st mate: ~10-11 mo	2426
	f/m: 1st mate: 2nd yr	9943
Sciurus niger	f: 1 yr	1474
	f/m: 1st mate: ~1 yr	170
	f/m: 10-11 mo	1871a
	f/m: ~1 yr	8798
Sciurus vulgaris	f: ovulation: ~1 yr; sometimes 7-8 mo	6526
	f: 1st birth: 8-10 mo	4645
	f: 8-12 mo	8053
	f/m: 8-10 mo	3571
	f/m: ~1 yr	8798
Spermophilus beecheyi	f/m: 1st mate: ~1 yr	10862
Spermophilus beldingi	f: 1st repro: 1 yr	9900
	m: mate: 1 yr	4931
	m: 1st repro: 2 yr	9900
Spermophilus citellus	f/m: 9-10 mo	3571
Spermophilus columbianus	f/m: 1st mate: r1-2.4 yr; n82	2658, 2659
	f/m: 1 yr; n4	3242
Spermophilus franklinii	f/m: 1st repro: 10-11 mo	5185
Spermophilus lateralis	f: vag perf: 11 mo	12115
	f: 1-3 yr	1323
	m: ~1 yr	601

	m: 2-3 yr	1323
Spermophilus major	m: 1 yr	8050
Spermophilus richardsonii	f: 1st birth: yearlings	9890
	f/m: 11 mo	7289
Spermophilus spilosoma	f/m: ~1 yr	9723
Spermophilus tereticaudus	f/m: 1st mate: 10-11 mo	7788
Spermophilus townsendii	f: 1st preg: yearlings	9117
Spermophilus tridecemlineatus	f: 1st mate: 1 yr	5345
	m: testes descent: 72 d; n2	5343
	m: sperm: 7.5 mo	11581, 11582
Sundasciurus? lowii	f: 1st preg: 62 g	4406
Sundasciurus tenuis	f: 1st preg: 66 g	4406
	m: fertile: 50% 50.2 g; 95% 88.3 g	4406
Tamias amoenus	f: 1st birth: ~1 yr	1295
	m: 1st mount: 51 d	1295
	f/m: ~1 yr	1295
	f/m: spring after birth	1474
	f/m: yearlings	9889
Tamias minimus	f: 1st mate: yearlings	11230
	f/m: 1st repro: min 8 mo, but not likely in wild	10053
	f/m: 1st mate: yearlings	9889
Tamias quadrivittatus	f: 1st birth: ~12 mo	11385
	f/m: 10-11 mo	11385
Tamias ruficaudus	f: 1st birth: 18-29% 1 yr	726
Tamias sibiricus	f/m: 1st mate: yearlings	983
	f/m: 11 mo	8053
	f/m: ~1 yr	10422
Tamias striatus	f: 1st preg: 3.5 mo; most 1 yr	12014
	f: ~3 mo	1472
	m: testes: large 7-8 mo	12014
Tamiasciurus douglasii	f: 1st repro: 4 mo; n1	5922
Tamiasciurus hudsonicus	f: 10 mo	6158
	f: season of birth	6297
	f/m: 1st mate: 1 yr	2324
	f/m: ~1 yr	8798
Trogopterus xanthipes	f/m: 1st mate: 22 mo; n1 pr	11447, 11448
Xerus inauris	f/m: ~1 yr	4697
CYCLE LENGTH		
Cynomys ludovicianus	estrus: 1 d	4958
Funambulus pennanti	estrus: ~16 h	9175
Sciurus vulgaris	estrus: 2 wk; standing estrus: 3-4 d	8053
Spermophilus beldingi	estrus: 4.7 h; ±0.04; n54	4317
Spermophilus columbianus	14-15 d; interestrous interval	9873
Spermophilus parryii	estrus: 1 d	7090
Spermophilus richardsonii	5.1 d; sd1.5; r3-7; n10	7285
Tamias sibiricus	estrus: 12 h	6736
	12 d ea; n7 cycles, 1 f	6736
	13.6 d; sd1.6; mode 13 37%; 80% 12-14; r11-21; n82 cycles, 3 f	983
Tamias striatus	estrus: 3-10 d	10102
GESTATION		
Ammospermophilus harrisii	min 29 d; n1	7788
Ammospermophilus leucurus	4 wk	5661
	min 29 d; n4 lt	8348
Ammospermophilus nelsoni	est 26 d; field data	4512a
Callosciurus prevosti	47.2 d; r46-48; n5	11975
Cynomys gunnisoni	30 d	8572
Cynomys leucurus	est 27-33 d	10436, 10437
	est 30 d	531
Cynomys ludovicianus	28-32 d	10139
	30-35 d	10954
	32 d	5916
	34.8 d; se0.1; r34-37; n32 lt	3371
	35 d; r33-37	339, 3404

Funambulus palmarum	within 28-40 d; n1 ?	6163
Funambulus pennanti	est ~30 d	9799
	40-42 d	554
	41 d; r40-42	105
	42 d	8720
	<46 d	6588
Funisciurus congicus	~52 d	11302
Glaucomys sabrinus	37 d; n1	7685
	~40 d	5146
	42 d	11588
Glaucomys volans	40 d	10206
Marmota baibacina	40 d	8050
Marmota bobak	40 d	10516
	~40 d	4645
	40-42 d	8798
	40-42 d	8050
Marmota caligata	~30 d	571
Marmota caudata	~30 d	9175
Marmota flaviventris	4 wk	5328
	30 d	3481
	~1 mo	5146
Marmota marmota	33-34 d; n1	8770, 8771
	33-35; 42?; d	3571
	35-38 d	8798
	35-42 d	7433
Marmota monax	~4 wk; field observation	4270
	30 d	2015
	31-32 d; n3	5037, 5038
	31-33 d; n3	4083
	35-40 d	2088
Marmota sibirica	40-42 d	8050
Paraxerus cepapi	56.5 d; r56-58; n4	11295, 11297, 11298
Paraxerus palliatus	59 d; n1	11302
	60-65 d	10158
Pteromys volans	4 wk	8053
Ratufa bicolor	28-35 d	7383
Ratufa indica	2.5-3 mo	10559
Ratufa macroura	28 d	8798
Sciurus aberti	≥38-≤46 d; n1	5618a
	40 d; field data	5618a
	46 d; n3	3165
Sciurus carolinensis	~44 d	11514
	~44 d	9943
	44 d; r43-46	4274
	44-45 d; n1	1241
Sciurus griseus	> 43 d	5143
	est > 43 d	1474
	44 d	5146
Sciurus niger	44 d	1474
	45 d	8798
	~45 d	10809
Sciurus vulgaris	32-40 d	6526
	32-40 d	7433
	35-40 d	8053
	35-40 d	10293
	35-40 d	4645
	38 d	8798
	38 d	2923
	~38 d; r32-40	3571
	38-40 d	12139
Spermophilopsis leptodactylus	1.5 mo	9774
Spermophilus armatus	24 d	2984
Spermophilus beecheyi	implant: 8 d	10862
	25-30 d	1474

	4 wk	10806
	~30 d	10862
Spermophilus beldingi	23-26 d; mating to decrease in f mass	9897
	27-31 d	10953
Spermophilus citellus	25-30 d	1552
	28 d	7433
	28 d; r25-30 d	3571
Spermophilus columbianus	23.75 d; r23-24; n4 lt, 4 f	9872, 9873
	24.2 d; se0.21; r23.5-25.0; n7; sperm in smear to decrease in f mass	7632
Spermophilus dauricus	30 d	8050
Spermophilus elegans	22-23 d; n1	3143
Spermophilus franklinii	~28 d	5210
Spermophilus fulvus	~1 mo	4645
	~1 mo	8050
Spermophilus lateralis	27 d; n1 f; capture to birth	7080, 7082
	est 27-28 d; n1; pairing to birth	1562
	min 33 d; n14 lt	8348
	~35 d	8349
Spermophilus mohavensis	min 24 d; n4 lt	8348
Spermophilus parryii	~25 d	6993
Spermophilus pygmaeus	25-26 d; field data?	11322
	25-30 d	8050
Spermophilus richardsonii	22-23 d; r21 d 15.5 h-23 d 14 h; n28; sperm in vagina to birth	7281, 7282, 7285
	22.08 d; sd0.43; r21.18-23.26; n46	7288
	22.3 d; sd0.5; n22; adult	7288
	22.4 d; sd0.5; n32; yearlings	7288
	22.5 d; sd0.5; n13; adult	7288
	22.7 d; sd0.6; n52; yearlings	7288
	22.8 d; sd0.7; n14; adult	7288
	22.8 d; sd0.6; n29; yearlings	7288
	28-32 d	5030
Spermophilus suslicus	min 3 wk	8050
	22-26 d	4645
Spermophilus tereticaudus	≥26 d; n10 lt	8348
	~27 d	9078
	est 28-29 d	7788, 7789
Spermophilus townsendii	>20 d; est 24 d	139
	est 24 d	9116
Spermophilus tridecemlineatus	27 d; n1	1286
	27 d; n7	3432
	27-28 d	12108
	27-28 d; n1	5343
	27-28 d; n2	11379
	28 d; mode 28; n7	5345, 5346
	4 wk	7493
Spermophilus undulatus	25 d	6992
	~1 mo; from graph	4850
Spermophilus variegatus	est ~30 d	1474
Tamias amoenus	28-30 d	1871a
Tamias cinereicollis	min 30 d	8549
Tamias dorsalis	28-31 d	1871a
Tamias palmeri	min 33 d; capture to birth	4825
Tamias panamintinus	min 36 d; capture to birth	4825
Tamias quadrimaculatus	31 d	9302a
Tamias quadrivittatus	est 30-33 d	11385
Tamias sibiricus	28 d	9416
	31.3 d; sd1.1; mode 31 57%; r28-35; n56 lt, 17 f	983
	35-40 d	4645
	35-40 d	8053
Tamias striatus	~30 d	2981
	31 d; n1 f	1472
	31 d	12014
	31 d	10102

	32 d	175
Tamiasciurus douglasii	36-40 d	5146
	est 40 d	5922
Tamiasciurus hudsonicus	~1 mo; field data	2088
	33 d; r31-35; n4	6159
	35 d; n1	3241, 8745
	~36 d	1871a
	38 d	1474
	40 d	8798
	est 40 d	4274
Trogopterus xanthipes	80.3 d; r74-89 d	11447, 11448
Xerus inauris	est 42-49 d	12134
LACTATION		
Ammospermophilus harrisii	wean: 7 wk	7788
Ammospermophilus leucurus	lact: 2 mo; field data	5661
	wean: min 65 d; 100% survival 67 d; n4 lt	8348
	wean: 2+ mo	5661
Ammospermophilus nelsoni	emergence: ~30 d	4512a
Cynomys spp	emergence: 33-37 d	8798
	wean: 7-8 wk	8798
Cynomys gunnisoni	lact: 1+ mo	8572
Cynomys ludovicianus	solid food: ~40 d	1871a
	solid food: 6th wk	339
	emergence: 6 wk	1474
	emergence: 43.4 d; sd3.5; r38-50; n17 lt	4959
	emergence: 44 d	5895
	wean: 6-7 wk text; 7-8 wk summary	339
	survive without milk: 7 wk	5342
	wean: est 7-8 wk	10139
	wean: 3 mo	4958
Cynomys mexicanus	wean: 41-50 d	1673
Funambulus pennanti	lact: 6 wk	9799
	lact: 2 mo	554
	lact: 2-2.5 mo	8735
	wean: 8 wk	9175
Funisciurus congicus	wean: 50 d	11302
Glaucomys sabrinus	solid food: 47 d	1117
	wean: 55 d	3239
	wean: 65 d	1117
	wean: ~90 d	1871a
Glaucomys volans	wean: 44 d	10211
	wean: 55 d	1672
	wean: ~2 mo	10584
Hylopetes fimbriatus	independent: 2.5 mo	9175
Marmota bobak	wean: 35-40 d	5172
	emergence: soon after birth	5172
Marmota caligata	lact: 3-4 wk; & 1st emergence	571
Marmota flaviventris	emergence: 20-30 d	3481
	wean: 25 d	1871a
	emergence: 30 d	1474
	wean: 3-4 wk	5328
Marmota marmota	emergence: 20-40 d; 2 mo	3571
	emergence: 39-40 d	8798
	emergence: 40 d	8771
	solid food: 2 mo	8771
	wean: 2 mo or longer	8771
	wean: 8-10 wk	8798
Marmota monax	solid food: 32.1 d; sd2.0; r29-38; n22	3240
	solid food: 33 d	570
	emergence: 39-40 d	8798
	emergence: 6 wk	1830
	wean: 4-5 wk	1830
	wean: ~38 d	570
	wean: ~44 d	10187

	wean: 45 d	2015
	wean: 8-10 wk	8798
Paraxerus cepapi	solid food: 18-29 d	11297, 11302
	solid food: 3 wk	11299
	wean: ~5 wk	10158
	wean: 5-6 wk	11297, 11302
Paraxerus ochraceus	nest emergence: 3-4 wk	5760
Paraxerus palliatus	solid food: 28-57 d	11302
	wean: 39-57 d	11302
Petaurista petaurista	wean: 2-2.5 mo	9175
Ratufa bicolor	solid food: 5 wk	33
Ratufa macroura	wean: 8-10 wk	8798
Sciurus spp	solid food: 45-47 d	8798
	emergence: 45-47 d	8798
	wean: 8-10 wk	8798
Sciurus aberti	wean: 10 wk	5618a
Sciurus carolinensis	wean: 7-10 wk	9943
	wean: 8 wk	1361
	wean: 2 mo	1474
	wean: 10 wk	10158
	wean: 82 d	7297
Sciurus niger	solid food: 50-54 d	8798
	emergence: 50-54 d	8798
	wean: 8 wk	1361
	wean: 8-10 wk	8798
	wean: 2-3 mo	1474
	wean: 9-12 wk	10809
	wean: 11-12 wk	7943a
Sciurus vulgaris	solid food: 40 d	8053
	solid food: 40-42 d	8798
	emergence: 40-42 d	8798
	emergence: 42 d	2923
	wean: 52 d	12139
	wean: 56-68 d	3571
	wean: 8-10 wk	8798
	wean: 70-75	8053
	lact: 10-12? wk	2468
Spermophilopsis leptodactylus	emergence: 6 wk	9774
Spermophilus armatus	emergence: 21 d	2984
Spermophilus beecheyi	solid food: 34 d	10862
	solid food: 8 wk	2896
	emergence: 36-50 d; ~6 wk in nest	6489
	emergence: 6 wk	1474
	wean: 5-6 wk	10806
	nurse: max 74 d	10862
Spermophilus beldingi	emergence: 3-4 wk	7548
	emergence: 25-28 d; nurse 1-3 d after emergence	9897
	wean: 24 d	7550
Spermophilus columbianus	solid food: 25-26 d	11959
	emergence: ~30 d	11959
	wean: 25-32 d	11959
	nurse: 30 d	9872, 9873
Spermophilus elegans	wean: 28-35 d	1864
	emergence: 29-33 d	3143
	wean: 5 wk	1871a
Spermophilus franklinii	wean: 28 d	10948
Spermophilus lateralis	solid food: 6 wk	7082
	wean: 28-34 d	8489
	nurse: 7-8 wk	7082
	wean: min 29 d; 100% survival 32 d; n14 lt	8348, 8349
Spermophilus major	solid food: 2 wk	8050
	emergence: 2 mo	8050
Spermophilus mohavensis	wean: min 32 d; 100% survival 40 d; n4 lt	8348

Spermophilus pygmaeus	solid food: 8 d	8050
	emergence: 15 d; 20-22 d	8050
	emergence: 82 d	11322
Spermophilus richardsonii	emergence: 29.0 d; sd1.0; n35; adult f	7282, 7288
	emergence: 29.0 d; sd1.1; n41; yearling f	7282, 7288
	emergence: 29.4 d; sd1.3; n11; adult f	7282, 7288
	emergence: 29.4 d; sd1.6; n51; yearling f	7282, 7288
	emergence: 29.5 d; sd1.3; n25; yearling f	7282, 7288
Spermophilus saturatus	1st food handling: 34.0 d; sd1.8	5663
	1st capture: 6 wk	5658
	lact: 7 wk	5658
Spermophilus suslicus	solid food: ~20 d	4645
	emergence: ~20 d	4645
	wean: 4-5 wk	8050
Spermophilus tereticaudus	wean: 20 d	9078
	wean: 5th wk	7788
	wean: min 35 d; 100% survival 50 d; n10 lt	8348
Spermophilus townsendii	wean: 19-22 d	9116
	solid food: 26-28 d	9116
	nurse: 34-36 d	9115
Spermophilus tridecemlineatus	emergence: ~28 d	1286
	wean: start 18-28 d	1286
	wean: ~26-27 d	9249
	wean: 28 d; n1 lt	11379
	solid food: 30 d	5343
	wean: 5 wk	3432
Spermophilus undulatus	solid food: 22 d	6992
	solid food: 35 d	8050
	emergence: 35 d	8050
	care of young: 11 wk	4850
Spermophilus variegatus	wean: 6-8 wk	5347
Tamias amoenus	solid food: 40 d	5662
	lact: 35-40 d	9889
	lact: 7-8 wk	5662
	wean: 6 wk	1474
Tamias cinereicollis	solid food: 36-40 d	8549
	nurse: 41-45 d	8549
Tamias dorsalis	wean: 30 d	1871a
Tamias minimus	lact: 35-40 d	9889
	wean: 40-60 d	1871a
	lact: 49 d; n1; one other shorter	10053
Tamias palmeri	solid food: by 6 wk	4825
Tamias panamintinus	solid food: by 6 wk	4825
Tamias quadrivittatus	wean: 6-7 wk	11385
Tamias sibiricus	emergence: 33.9 d; sd3.9; mode 30-38; r24-40; n23	982
	wean: 28-30 d	8053
	wean: 1 mo	4645
Tamias striatus	emergence: 5-7 wk; n22	8511
	emergence: 40 d	2981
	emergence: when 2/3 grown	1474
Tamias townsendii	1st capture: 6 wk	5658
	lact: 7-8 wk	5658
Tamiasciurus hudsonicus	solid food: nibbled 34 d, eaten 37-41 d	8745
	solid food: 5 wk	4274
	solid food: ~37 d	7877a
	wean: 69-72 d	8745
	wean: 7-8 wk	6297
	wean: 8-10 wk	8798
	wean: 62-78 d; n3 lt	10097
	wean: 9-11 wk	1871a
Trogopterus xanthipes	wean: 90-120 d	11448
Xerus inauris	wean: begins 50 d	10158

INTERLITTER INTERVAL		
Funambulus pennanti	~5 mo	9799
Funisciurus congicus	60 d; n3	11302
Glaucomys volans	~69 d; n1	8978a
	100 d; n1	10307a
	3.5-12 mo	6497a
	124-143 d	6359a
Marmota bobak	2 yr	5172
Marmota flaviventris	1-1.5 yr	2708, 5328
Marmota marmota	mode 2-3 yr; max 4	3571
Paraxerus palliatus	r61-89 d; n6	11302
Paraxerus poensis	40-65 d	2997
	63 d; r61-65; n2	2998
Ratufa bicolor	149 d; r94-235; n4 yg died 1-2 mo	40
	407 d; n1 yg lived	40
Tamias amoenus	1 yr	9889
Tamias minimus	1 yr	9889
Tamias sibiricus	103.5 d; sd20.0; r77-165; n26 intervals, 15 f	983
LITTERS/YEAR		
Ammospermophilus interpres	1-2	10110a
Ammospermophilus leucurus	1	5661
	1-2	10110a
	max 2	1474
Cynomys gunnisoni	1	8572
Cynomys ludovicianus	1	10954
Funambulus pennanti	2	9799
	3	554
	3	8735
Glaucomys sabrinus	1; sometimes 2	2088
Glaucomys volans	2	8798
	2	7686
	2	10206
	max 2	9164
Hylopetes fimbriatus	2	9175
Marmota baibacina	1	8050
Marmota bobak	0.5	5172
	1	4645
	1	8050
Marmota caligata	0.5	571
Marmota caudata	1	8050
Marmota flaviventris	1	2708
Marmota marmota	1	3571
	1	7433
Marmota monax	1	4083
	1	10187
	1	2015
Marmota sibirica	1	8050
Paraxerus cepapi	2	2411
	min 3	11253
Paraxerus palliatus	1 wild; several captive	10158
Pteromys volans	1-2	8053
Ratufa bicolor	>1	7383
	2	10828
Ratufa macroura	1-2	8798
Sciurus aberti	1-2	7145
	est 2	502
Sciurus carolinensis	1-2	2426
	1 yearling; 2 adult	9943
	2	9164
	2	11956
	2	8188
	2	1474
	2	9190
	max 2	7944

	2; adult f	6562
	mode 2	9214
	mode 2	7297
	2; r1-3	589
Sciurus griseus	1	5146
	1	1474
Sciurus niger	0-2 lt/10 mo season	7943
	1-2	10809
	1-2	8798
	1-2	170
	2	8188
	2 adult; 1 yearlings	1474
Sciurus vulgaris	1-2	8798
	1-3	8053
	est 2	2435
	2-3; r2-5 ?	3571
	4	6526
Spermophilopsis leptodactylus	1	10568
Spermophilus armatus	1	1474
Spermophilus beecheyi	1	6489
	1	3114
	1	4081
	1	10862
	1	10806
Spermophilus beldingi	1	9897, 9900
	1	4081
Spermophilus citellus	1-2	7433
	1	3571
Spermophilus columbianus	1	9872
Spermophilus dauricus	1	8050
Spermophilus erythrogenys	1	8050
Spermophilus franklinii	1	5185
	1	10235
Spermophilus fulvus	1	10845
	1	4645
	1	8050
Spermophilus lateralis	1	10762
	1	4081
Spermophilus major	1	8050
Spermophilus pygmaeus	1	8050
Spermophilus richardsonii	1	5030
Spermophilus spilosoma	1	9723
	est 2	2349
	est 2	1474
	est 2	502
Spermophilus suslicus	1	4645
	1	8050
Spermophilus tereticaudus	1	9078
	rarely 2	7789
Spermophilus townsendii	1	10110
Spermophilus tridecemlineatus	1	11379
	1 est	502
	1; possibly 2nd lt late in summer	1474
	2	7023
Spermophilus undulatus	1	10516
Spermophilus variegatus	1-2	7145
Tamias spp	1	6918
Tamias amoenus	1	1295
	1	9889
	1	10762
	1	1295, 1296
Tamias dorsalis	1	2805
	2	502

Tamias minimus	1	9889
	1	2088
	1	10053
	max 2	1474
Tamias quadrimaculatus	1	10762
Tamias ruficaudus	1	726
Tamias sibiricus	1	10177
	1	10422
	1	4645
	1	8053
	mode 1; 3 rare	982, 983
	1-2	9953
Tamias speciosus	1	10762
Tamias striatus	1	9164
	1	10839
	1-2	1472
	1-2	8511
	2	12014
	2	175
	2 ?	11989
Tamias townsendii	1	10762
Tamiasciurus douglasii	1 mode; 1-2	5922
	2	10098
	2?	5146
Tamiasciurus hudsonicus	0/4 f with 2 lt	10839
	1	2671
	1	10098
	1	3241
	1-2	8798
	1-5; r1-2; n10	6158
	mode 2	4274
	2	1474
	max 2	6297
	2?	10954
Xerus inauris	1	10158
	1	4697
SEASONALITY		
Ammospermophilus harrisii	mate: mid Jan-mid Mar; Mexico, sw AZ	7145
	preg f: Feb-May; s AZ	7789
Ammospermophilus interpres	lact f: Mar; n2; Coahuila, Mexico	522
	repro: Feb; TX	10110a
Ammospermophilus leucurus	testes: peak Nov-Dec; captive	5654, 5657
	testes: peak Feb; CA	5654
	mate: early-mid Mar; CA	5661
	birth: mid Mar-Apr; n4 lt; CA	8348
	birth: early-mid Apr; CA	5661
	birth: mid-end Apr; CA	5661
	birth: est May; NM	502
	birth: peak May; CA	4081
	repro: Feb-July; TX	10110a
Ammospermophilus nelsoni	preg f: Feb-Mar; CA	4512a
Callosciurus erythraeus	preg f: Aug; n1; China	182
	preg/lact f: July; n4; n Vietnam	10826
	repro: Mar-May, Oct-Dec; according to hunters; n Vietnam	10827
Callosciurus nigrovittatus	m without sperm: peak Jan-Mar, July-Sept, low Apr-June, Oct-Dec; Malaysia	4406
	preg f: yr rd; Malaysia	4406
Callosciurus notatus	preg f: mid Apr-July; n3/15; n Bomeo, Indonesia	2313
	preg f: yr rd, peak Apr-June; Malaysia	4405, 4406
	preg f: yr rd, peak Apr-June; Selangor, Malaysia	7161
Callosciurus prevosti	preg f: end Apr-early Aug; n3/17; n Bomeo, Indonesia	2313
Cynomys gunnisoni	mate: Mar-Apr; w central US	1474
	mate: mid Apr-early May; CO	3291
	preg f: Apr-May; n29; CO	145

	preg f: June; n1; CO	8572
	birth: end Apr-early May; AZ	9583
	birth: early May; w central US	1474
	lact f: Apr; n13; CO	145
	lact f: May; n2; AZ	145
Cynomys leucurus	conceive: Mar-Apr; WY	531
	mate: end Mar; CO	10835
	mate: end Mar-early Apr; WY	1871a
	birth: end Apr-early May; WY	10437
	birth: early May; WY	531
	lact f: none after 15 June; WY	531
Cynomys ludovicianus	testes: large Mar, small Apr; MT	5895
	testes: large Dec-Feb, small Mar-Oct; OK	338
	mate: end Jan-mid Feb; KS	10139
	mate: end Feb; CO	10835
	mate: Feb; zoo	10149
	mate: Feb-Mar; SD	10954
	mate: Feb-Mar; SD	4958
	mate: Mar-early Apr; WY	1871a
	mate: Mar-Apr; Great Plains, N America	5916
	preg f: end Mar-early May; Great Plains, N America	5916
	birth: Mar; captive	339
	birth: Mar-Apr; KS	10139
	birth: early Apr; SD, WY	10954
	birth: mid-end Apr; MT	5895
	birth: Apr-May; Great Plains, N America	5916
	birth: May; NM	502
	lact f: June; n2; ND	3703
	pups emerge: May; CO	10835
	wean: May-June; SD	4958
Cynomys mexicanus	preg f: Mar; n1; Coahuila, Mexico	522
	birth: May; n1; captive	8573
Cynomys parvidens	birth: Apr; n1; UT	10809
	birth: Apr; n4; captive	8571
Dremomys lokriah	birth: May-Aug; Nepal	7383
Exilisciurus exilis	preg f: end May-mid Aug; n2/7; n Borneo, Indonesia	2313
Funambulus palmarum	spermatogenesis: yr rd; India	8732
	preg f: Mar-Apr, July-Aug; India	11242
	birth: May, Sept, Jan; India	8732
	birth: yr rd, peak Nov-Apr; ne monsoon; Sri Lanka	8501
Funambulus pennanti	spermatogenesis: Jan-July; n226; India	8996
	testes: large Jan-July, regress Aug-Sept, quiescent Oct-Nov; Delhi, India	8995, 10013
	testes: large Jan-Sept, small Oct-Dec; India	5516
	testes: large Mar, small Sept; India	6588
	mate: Jan-Aug; Delhi, India	9799
	mate: yr rd; Pakistan	105
	preg f: Feb-Sept, peaks Mar, July; n Bangalore, India	8735, 9799
	preg f: Apr-Sept; Pakistan	1764
	preg f: May 40%, Nov 0%; India	6588
	preg f: May-June; n10; Rajahstan, India	8720
	birth: Feb-Sept; w Rajahstan, India	8723
	birth: Mar-Aug; India	8996
Funisciurus anerythrus	preg f: yr rd, peak Aug-Oct; Zaire	8855
	repro: small peak Apr, large peak Aug-Oct; Gabon	2997
Funisciurus carruthersi	lact f: May; n1; e Africa	5760
Funisciurus congicus	birth: Oct, Mar; Namibia	11300
Funisciurus lemniscatus	repro: July-Jan; Gabon	2997
Funisciurus leucogenys	preg f: Feb; n1; Cameroun	2955
Funisciurus pyrrhopus	preg f: end Jan; n1; Budongo, Uganda	5760
	preg f: Jan-Mar, May; n9; Zaire	8855
	birth: yr rd; captive	6755
	birth: est yr rd; n10; zoo	9522
	repro: July-Jan; Gabon	2997

Glaucomys sabrinus	conceive: est Oct; n1; CA	8928
	birth: Mar-June; w US	5146
	birth: May, occasionally Oct; BC, Canada	2088
	birth: May-June; US	1474
	birth: early June; ID	9385
Glaucomys volans	testes: scrotal Feb, Apr, Aug, Oct; few data; Mexico	1672
	mate: Feb-Mar, July; e US	10205, 10206
	mate: mid Feb-Mar, June-July; WI	5210
	preg f: Mar; n2; MN	495
	preg f: July; n1; KY	11588a
	birth: Feb, Sept; n12; captive	6497a
	birth: end Mar-early May; PA	9164
	birth: early Apr-mid Sept, peak Apr, Aug; MI, MA	7686
	birth: Apr; IL, WI	2979
	birth: Apr-May, Aug-Sept; e US	1474
	birth: end Apr; NY	7235
	birth: May, Aug; MI	1472
	birth: end Aug-mid Jan; FL	8978a
	birth: Sept-Feb; n16; AL	6497a
	birth: Oct; n1; LA	3848
	birth: yr rd, peak Jan-Mar; LA	3853
	lact f: Mar; n1; Mexico	1672
	repro: peaks end summer-early fall, spring; VA	10211
Heliosciurus gambianus	young found: Feb; Karamoja, e Africa	5760
	lact f: Nov; Cameroun	2955
Heliosciurus rufobrachium	testes: large Aug, Feb; n2; Africa	5760
	mate: June-Oct; Malawi, Zambia, e Zimbabwe	10153
	preg f: Dec-Jan; n2; e Africa	185
Heliosciurus ruwenzorii	preg f: Mar; n1; Zaire	8856
Hylopetes alboniger	mate: end Feb-early Mar; Nepal	7383
	birth: mid May-June; Nepal	7383
	repro: Apr-mid June; Nepal	7382
Hylopetes lepidus	preg f: erratic; Malaysia	7687
Iomys horsfieldii	preg f: Mar; n1; Malaysia	7161
Marmota baibacina	mate: Apr; USSR	8050
	mate: spring; USSR	9837
	birth: May; USSR	8050
Marmota bobak	mate: Mar-Apr; Mongolia	10516
	mate: ~Apr; USSR	8050
	mate: spring; USSR	4645
	birth: mid June; USSR	5172
	emergence: mid-end May; USSR	8050
Marmota caligata	mate: Apr-May; nw BC, Canada	2088
	mate: mid May; WA	571
	birth: early-mid June; WA	571
	birth: end spring-early summer; w N America	1474
	juveniles: end July; Skeena R, BC, Canada	10597
Marmota camtschatica	birth: 2 wk after hibernation; USSR	5518
Marmota caudata	mate: immediately after hibernation; Pakistan?	9175
	birth: Apr-early July; USSR	8050
Marmota flaviventris	spermatogenesis: Mar-May; Europe	2075
	mate: Mar; w US	5146
	mate: Apr; 2 wk after thaw; CA	7803
	mate: after hibernation; 2 wk; WY	364
	preg f: Mar-Apr; CA	7803
	preg f: Apr; N America	5029
	preg f: Apr; SD	10954
	birth: Mar-Apr; w US	1474
	birth: early May; BC, Canada	2088
	birth: early June; CO	2708
	birth: June; CO	5328
	lact f: May-June; CA	7803
	lact f: June-July; SD	10954

Marmota marmota	mate: Apr; captive	8770
	mate: Apr-early May; Alps, Europe	12083
	mate: Apr-May; n Austria, n Switzerland	7433
	mate: end Apr; Germany	7605
	birth: Apr-May; Germany	3571
	birth: Apr-May; n Austria, n Switzerland	7433
	birth: May; captive	8770
	birth: May-early June; Alps, Europe	12083
Marmota monax	seminal vesicles: large Feb-Mar, low end Apr; PA	1830
	testes: small Aug-Oct; NY	8934
	mate: early Feb; MO	10981
	mate: mid Feb-mid Mar; PA	1830
	mate: end Feb-Mar; MD	4083
	mate: 1st 3 wk Mar; NY	533
	mate: est Mar-Apr; captive	4270
	mate: Mar-mid Apr; WI	5210
	mate: Mar-Apr; e US	1474
	preg f: mid Feb-Apr; PA	10187
	preg f: Apr; n1; NY	9639a
	birth: Feb; lab	10186
	birth: Mar; lab	2016
	birth: Mar-Apr; lab	12043
	birth: end Mar-Apr; lab	2015, 2016
	birth: Apr; lab	570
	birth: Apr; PA	1830
	birth: Apr-mid May; MD	4083
	birth: Apr-mid May; NY	4270
	birth: end Apr; NY	5029
	birth: end Apr-early May; NY	7235
	lact f: Apr-early June; PA	10187
	emergence: mid May; PA	1830
Marmota sibirica	mate: Apr; USSR	8050
	birth: end May-?; USSR	8050
Paraxerus alexandri	mate: not seasonal; central Africa	187
	preg f: Mar-Apr, July, Sept; n5; Zaire	8855
Paraxerus boehmi	preg f: end Dec; n1; Lake Kivu, Zaire	8856
	preg f: yr rd; Zaire	8855
Paraxerus cepapi	testes: large July, small Apr; Transvaal, S Africa	11296
	preg f: peak Oct-Apr, none May, Sept; Botswana	10154
	birth: June, Nov; few data; Transvaal, S Africa	11298
	birth: Sept-Jan; Transvaal, S Africa	11299
	birth: yr rd; Botswana	11299
	birth: yr rd, peak Oct-Apr; S Africa	10158
	lact f: May; n1; Lake Tanganyika, e Africa	186
Paraxerus lucifer	testes: large Aug-Sept; e Africa	5760
	mate: may peak drier half of yr; e Africa	5760
	preg f: Mar-Apr; n1; Tanzania	186
Paraxerus ochraceus	preg f: July, Nov; e Africa	5760
	preg f: Nov-Dec; n2; Kenya	4925
	yg found: Mar, June-July, Sept; e Africa	5760
Paraxerus palliatus	birth: Aug-Mar; S Africa	10158
	lact f: Mar; n1; s Kenya	5760
	lact f: Oct; n1; Mafia I, Tanzania	5760
	juveniles: Apr; central Africa	187
Paraxerus poensis	lact f: Sept; Gabon	2997
Petaurista magnificus	preg f: Nov; n1; Nepal	7382
	birth: Feb-Mar; Nepal	7383
Petaurista petaurista	preg f: Feb; n2; Malaysia	7161
	preg f: Apr; n1; Sri Lanka	8501
	birth: June; n1; Sri Lanka	8501
Protoxerus stangeri	preg f: none dry season; Cameroun	2955
	lact f: Jan-Feb, Nov; Gabon	2997
Pteromys volans	mate: summer; Eur USSR	4645
	birth: Feb-June; USSR	8053

Pteromyscus pulverulentus	birth: nearly yr rd; Malaysia	7687
Ratufa affinis	preg f: June; n2; n Bomeo, Indonesia	2313
	lact f: Aug; n2; n Bomeo, Indonesia	2313
Ratufa bicolor	birth: Apr-May, Aug-Sept; Vietnam	10828
Sciurus aberti	mate: Mar-Apr; sw US	1474
	mate: Apr-May; CO	3166
	birth: Apr-May; WY	1871a
	birth: Apr-May; sw US	1474
	birth: May-Aug; sw US	7145
	birth: mid June-mid July; AZ	5618a
	lact f: July; n1; Durango, Mexico	523a
	repro: mid Apr-June; AZ	10378
Sciurus alleni	preg f: Mar; n1; Coahuila, Mexico	522
	lact f: Apr; n2; Coahuila, Mexico	522
Sciurus anomalus	juveniles: end Apr-early May; Lebanon	6419
Sciurus aureogaster	preg f: July-Aug; n2; Mexico	2341
Sciurus colliaei	preg f: June; n2; Sinaloa, Mexico	5409
	preg f: June; n1; Durango, Mexico	5405b
	lact f: June; n1; Durango, Mexico	5405b
Sciurus carolinensis	anestrus: Aug-Dec; England, UK	2426
	spermatogenesis: Oct-July; ONT, Canada	10015
	testes: scrotal Dec-Feb, July; KE	1241
	testes: large Dec-May, small June-Mar; s England, UK	11516
	testes: large Jan-Aug, small Sept-Nov; BC, Canada	9190
	testes: functional Feb-Mar, June-July; England, UK	164
	testes: large May-Jan, small Feb-Apr; S Africa	7297
	testes: regressed June-Mar; s England, UK	10629
	testes: regressed July-Aug; NC	589
	testes: regressed July-Aug; IL	1361
	testes: inguinal ~50% peak Mar, July, Dec; KY	1241
	mating condition: Aug; n4/15; NY	9214
	mate: Jan-Feb, July; ne US	1474
	mate: peaks Jan-Mar, June-July; UK	9943
	mate: Feb, May-June; KS	8188
	mate: Jan-mid July; IL	1361
	mate: peaks Feb, June, none Aug-Nov; WI, IN	7570
	mate: Mar-June; BC, Canada	2088
	mate: June, Dec; se US	1474
	mate: peak July, Dec; S Africa	10158
	mate: Nov-Feb; S Africa	7297
	mate: mid Dec-mid Jan; NC	11478
	mate: end Dec-Sept, peaks Jan, end May-early June; OH	7944
	mate: est peaks Mar-Apr, June-July; BC, Canada	9190
	preg f: yr rd, peak Jan-Feb, Aug; e TX	3898
	preg f: Jan-July; England, UK	2426
	preg f: Mar 46%, Apr 54%, Aug 75%; KY	1241
	preg f: Mar-Apr, June; n4; KS	8188
	preg f: Apr-June, Sept-Oct; IL	1361
	preg f: June; n1; FL	7498a
	birth: peaks Feb-Mar, July; OH	7944
	birth: peaks Feb-Mar, July-Aug; VA	5808
	birth: mid Feb-Mar; IL	1361
	birth: Mar, July-Aug; NC	11478
	birth: end Mar-early Apr, end July-early Aug; WI	5210
	birth: May, July; BC, Canada	2088
	birth: May, Aug; QUE, Canada	11956
	birth: June, Oct; NY	7235
	birth: June-Jan, peak Oct-Jan; S Africa	2323
	birth: Aug-Oct, Dec-Mar; S Africa	7297
	lact f: 67% Sept; KY	1241
	lact f: Oct; n1; FL	7498a
Sciurus deppei	preg f: Dec; n1; Campeche, Mexcio	5423b
	birth: Feb; n1 hybrid; Yucatan, Mexico	3633
	birth: July; San Luis Potosí, Mexico	2228a

Sciurus granatensis	mate: June; Panama	3024
	mate: Dec-July; Panama	3801
	preg f: Mar, Sept-Oct; Venezuela	7940
	preg f: June; Panama	3024
Sciurus griseus	mate: Jan-May; CA	5143
	preg f: Mar; n1; CA	11228a
	birth: Feb-June; w US	1474
Sciurus niger	testes: functional yr rd; IN	5801
	testes: regress July-Aug; IL	1361
	estrus: Jan, June; WY	1871a
	mate: Jan-Feb, end May-June; WI	5210
	mate: Jan-Feb, June-July; n N America; 1 mo earlier s N America	1474
	mate: peaks Feb-Mar, July-Aug; DE, MD	10809
	mate: peaks Feb, June, none Aug-Nov; WI	7570
	mate: Dec-June; KS	8188
	mate: mid Dec-June; IL	1361
	preg f: Jan-Mar, July; MI	170
	preg f: Feb-Mar, June; n5; KS	8188
	preg f: Feb-June, Aug-Oct; IL	1361
	birth: peak mid Mar, July; WY	1871a
	preg f: May-Aug, Dec-Feb; FL	7501
	preg f: mid June; Coahuila, Mexico	522
	birth: Jan-Feb, June-Aug; FL	7501
	birth: Feb-Apr, Aug-Sept; n 1 mo earlier s N America	1474
	birth: mid Feb-Mar; IL	1361
	birth: Dec-Sept, peaks Dec-Jan, May-June; IL	7943
	lact f: end Mar-early Apr; Coahuila, Mexico	522
Sciurus nayaritensis	lact f: July; n2; Durango, Mexico	523a
Sciurus richmondi	mate: low Feb-Sept; Nicaragua	5417
	preg f: Feb, Apr, June; Nicaragua	5421
	lact f: June, Sept; Nicaragua	5421
Sciurus vulgaris	spermatogenesis: Dec-July; quiescent Aug-Nov; s France	2469
	testes: large Dec-Jan, small Aug; England, UK	9343
	testes: large Jan; Switzerland	4561
	mate: Jan-Feb, ~2 wk; Eur USSR	4645
	mate: Feb-Mar; USSR	8053
	mate: Mar; n1; W Germany	2923
	mate: Dec-July; s France	2468
	mate: Dec-July; Germany	3571
	mate: spring-summer; Germany	7433
	ovulation: peaks Jan-Mar, mid May-June, low mid Mar-Apr, mid Aug-mid Sept; USSR	6526
	preg f: Feb-June; s France	2469, 2470
	preg f: Dec-July; s France	2468
	preg f: Dec-July; n15; England, UK	9343
	birth: ~Jan; n1; zoo	3271
	birth: Jan-Aug; Germany	3571
	birth: Feb-Aug; Denmark	2435
	birth: Feb-June/July; s France	2468
	birth: May; n1; captive	12139
	lact f: Mar-mid Sept, peak July 38.4%; s France	2469
Sciurus yucatanensis	preg f: July; n1; Quintana Roo, Mexico	5423b
Spermophilopsis leptodactylus	testes: large Mar; n1; USSR	8053
	mate: Feb-Mar; Turkmenia	6273
	mate: mid Feb-; several wk; USSR	8053
	mate: mid Apr, June; Turkmenia	9774
	birth: Apr & later; Turkmenia	6273
	birth: end May; Turkmenia	9774
	repro: Feb-June; Karakumy Desert, Turkmenia	9498, 9499
	repro: Apr; Turkmenia	10568
Spermophilus armatus	mate: Mar-Apr; UT	2983, 5893
	preg f: mid Mar-early May, peak early Apr; ID	11646a
	birth: Apr; w US	1474

	birth: May-?; UT	2983
Spermophilus beecheyi	spermatogenesis: Nov-Apr; CA	6489
	testes: large Feb-Apr; CA	10862
	mate: Jan-Mar; CA	3281
	mate: end Jan-mid Mar; CA	6489
	mate: Feb-Apr; CA	3114
	mate: Feb-mid Apr; low elevations, CA	5030
	mate: Feb-June; high elevations, CA	5030
	mate: end Feb-early Mar; CA	10806
	preg f: Apr-May; OR	2896
	preg f: Mar; n1; CA	11228a
	birth: ?-Mar; CA	6489
	birth: Mar-June; CA	10862
	birth: peak Mar-Apr; w US	1474
Spermophilus beldingi	testes: large Mar-May; CA	7079, 7082
	testes: large Apr; Big Bend, CA	7547
	testes: large May; Tioga Pass, CA	7547
	mate: Apr-May; Big Bend, CA	7547
	mate: May-June; Tioga Pass, CA	7547
	mate: May; MI	1480
	mate: end May-early June; CA	4317, 9900
	birth: Apr; CA	5030
Spermophilus citellus	birth: Apr-May; Europe	7433
	birth: Apr-May; se Europe	3571
	birth: May; Russia	1552
Spermophilus columbianus	mate: Apr-May; MT	6798
	mate: f: 3.8 d after emergence; se0.20; r2-8	7632
	preg f: Mar-Apr; w Canada	9872
	preg/lact f: mid Apr-June; sw ALB, Canada	539
	birth: end Mar-early Apr; w N America	1474
	birth: May-June; ALB, Canada	11959
	wean: mid May; w Canada	9872, 9875
	yg emerge: June; 2 wk; ALB, Canada	11469
Spermophilus dauricus	preg f: May; n1; Mongolia	182
	birth: May; USSR	8050
	lact f: end May; USSR	8050
	yg found: July; Mongolia	182
Spermophilus elegans	mate: Apr; CO	3143
	preg f: Apr-early May; WY	1865
Spermophilus erythrogenys	mate: mid Apr-early May; USSR	8050
	birth: end May; USSR	8050
Spermophilus franklinii	testes: scrotal May-July; MAN, Canada	5185
	mate: Feb-mid Apr; low elevations, MAN Canada, ND, MN	5030
	mate: Feb-June; high elevations, MAN Canada, ND, MN	5030
	preg f: May; n1; MN	495
	preg f: mid May-June; MAN, Canada	5185
	birth: May-June; MAN Canada, ND, MN	5030
	birth: end May-mid June; WI	5210
Spermophilus fulvus	mate: Feb; Turkmenia	5545
	mate: before Mar; Siberia, e Europe	9773
	mate: end Mar-early Apr; Europe, USSR	4645
	mate: after hibernation; USSR	8050
	birth: Apr-May; Russia	10845
	birth: Apr-May; USSR	8050
	birth: end Apr-early May; Eur USSR	4645
	yg emerge: Apr; Turkmenia, USSR	5545
Spermophilus lateralis	spermatogenesis: Apr-May; Lassen Co, CA	601
	testes: increase Dec-Apr; Lassen Co, CA	601
	testes: large Mar, small Aug; WY	10055
	testes: large Mar-Apr; CA	7080
	testes: large Mar-June; Lassen Co, CA	7082
	mate: end Apr-early May; WY	10055
	CL present: Apr; Lassen Co, CA	7082
	preg f: Apr; n2; CA	7593a

	preg f: May; n1; AZ	145
	preg f: May-Aug; n9; CA	4081
	birth: May; BC, Canada	2088
	birth: May; captive	602
	birth: May-early June; n14; CA	8348
	birth: end May-early June; WY	10055
	birth: early-mid June; lab	12115
	birth: June-July; AZ	7145
	birth: before July; CO	279
	birth: early spring; w US	1474
	yg emerge: end June; WY	10055
Spermophilus madrensis	preg f: May; n1; Chihuahua, Mexico	866
	preg f: June; n4; Chihuahua, Mexico	866
	lact f: July; n7; Chihuahua, Mexico	866
Spermophilus major	mate: mid May?; USSR	8050
	emergence: mid-end June; USSR	8050
Spermophilus mexicanus	mate: Apr; TX, Mexico	1474
	preg f: Apr; n1; TX	2901
	preg f: June; n1; Coahuila, Mexico	522
	birth: May; n1; TX, captive	2901
	birth: May; TX, Mexico	1474
Spermophilus mohavensis	birth: Mar; CA	1474
	birth: Apr-mid May; n4; CA	8348
Spermophilus parryii	mate: Apr, 3 wk; YUK, Canada	7090
	birth: mid May-mid June; nw BC, Canada	2088
	birth: June-July; n N America	1474
Spermophilus perotensis	preg f: July; n2; Mexico	2341
Spermophilus pygmaeus	mate: mid Mar; Russia	11224
	mate: mid Mar; USSR	9453
	birth: early-mid May; USSR	8050
Spermophilus richardsonii	testes: large Mar, small May; WY	1865
	conceive: mean date: Apr 12; SAS, Canada	9890
	conceive: Apr-early May; ALB, Canada	7826
	birth: Apr-May; SAS, Canada	2872
	birth: May; nw US, sw Canada	1474
	birth: 27 d after hibernation; ALB, Canada	7285
	lact f: May-early June; SAS, Canada	7279
	emergence: June; WY	1865
Spermophilus saturatus	testes: large Mar; captive	600
	mate: end Apr; WA	5658
	birth: end May; n22 lt; WA	5658
Spermophilus spilosoma	testes: scrotal Mar-Aug; TX	9723
	preg f: Apr; n1; NM	145
	preg f: May-July; WY	1871a
	preg f: June; n2; NE	5405b,
	preg f: June; n2; NM	502
	preg f: June-July; few data; Coahuila, Mexico	522
	preg/lact f: end Apr-Sept; TX	9723
	birth: July; San Luis Potosí, Mexico	2228a
	lact f: June; n1; NM	502
Spermophilus suslicus	mate: after hibernation; Eur USSR	4645
	birth: end May-early June; USSR	8050
Spermophilus tereticaudus	fertile m: Jan-Apr; s AZ	7789
	birth: peak mid Mar-mid Apr; s AZ	7789
	birth: end Mar-mid Apr; n10 lt; CA	8348
	birth: Apr, possibly other mo; sw US	1474
Spermophilus townsendii	mate: Jan; WA	9871
	preg f: Mar; NV	139
	birth: Apr; NV	139
	birth: Mar; w central US	1474
	yg emerge: Mar; WA	9871

Spermophilus tridecemlineatus	sperm: Dec-June; IL	7493, 11580, 11582, 11583
	testes: scrotal Jan-July; lab	5345
	testes: large: Dec-May, small July; IL	11579, 11580, 11583
	vag perf: Jan-July, peak Mar-Apr; lab	5345
	ovulate: Mar-Apr; MN	2729
	mate: Mar-Apr; MN?	2729
	mate: Mar-June; TX	7023
	mate: Apr; 2 wk; MI	9681
	mate: Apr; lab	1286
	mate: Apr; n1; lab	5346
	mate: Apr-May; MAN, Canada	2140
	mate: Apr-May; WI?	3432
	mate: peak end Apr; NB	11379
	preg f: Apr-May; n9; MN	495
	preg f: Apr-May; n6; NE	5405b$_1$
	preg f: May; n1; KS	1812a
	birth: est Apr-May; WI	9249
	birth: May; n1; lab	5346
	birth: May; lab	1286
	birth: May; KS	5343
	birth: May-early June; MAN, Canada	2140
	birth: May-June; N America	5030
	birth: May-June; IL	7493, 11581
	birth: May-June; IL, WI	2060
	birth: mid-end May; KS	1812a
	birth: end May; WI?	3432
	lact f: June; n1; ND	3703
	lact f: June; NE	5405b$_1$
	emergence: mid June; MAN, Canada	2140
	repro: May; ne CO	3294
Spermophilus undulatus	sperm: May; AK	7381
	mate: early-mid May; AK	4850
	mate: May; AK	6992
	birth: early May-early June; USSR	8050
	birth: early-mid June; AK	4850
	birth: June; AK	6992
	wean: end June-early July; USSR	8050
Spermophilus variegatus	mate: Mar-Apr; TX	5347
	mate: Mar-July; N America	1474
	preg f: Apr; n1; Coahuila, Mexico	522
	preg f: May; n1; CO	1649a
	preg f: June; n2; NM	502
	birth: Apr-Aug; N America	1474
	birth: peak May-June; NM	502
Spermophilus washingtoni	birth: Feb, Mar; Columbia R Basin, WA	5030
	birth: Mar; WA, OR	1474
	emergence: Apr; WA, OR	1474
Spermophilus xanthoprymnus	lact f: June; USSR	8050
	emergence: end June; USSR	8050
Sundasciurus hippurus	preg f: none end Apr-Aug; n15; n Borneo, Indonesia	2313
Sundasciurus lowii	preg f: end Apr-Aug; n3/16; n Borneo, Indonesia	2313
	lact f: Aug; n1; n Borneo, Indonesia	2313
Syntheosciurus brochus	lact f: Apr; n1; Panama	11585
Tamias spp	birth: early June; Yellowstone, MT	6918
Tamias amoenus	testes: large Apr, small June; SAS, Canada	9889
	mate: Apr; WA	1295
	mate: Apr-May; SAS, Canada	9889
	preg f: end Apr-mid May; CA	10762
	preg f: May; n1; ID	11646a
	preg f: May-June; SAS, Canada	9889
	birth: ~early May; WA?	5662
	birth: May; nw US sw Canada	1474
	birth: May; WA	1295
	birth: May or later; sw BC, Canada	2088

	birth: May-June; SAS, Canada	9889
	birth: May-June; WY	1871a
	lact f: June; n1; ID	11646a
	weanling emergence: June; nw US, sw Canada	1474
Tamias bulleri	preg f: June-July; n2; Durango, Mexico	523a
	lact f: June-July; n≥6; Durango, Mexico	523a
	lact f: July-Aug; n2; Coahuila, Mexico	522
Tamias cinereicollis	birth: peak June; AZ	8549
Tamias dorsalis	m fertile: Jan-June; AZ	2805
	mate: May-early June; AZ	2805
	preg f: none July; n23; UT	2340
	birth: May-Sept; NM	502
	birth: end June-early July; AZ	2805
	lact f: July; Sonora, Mexico	1471a
	wean: Aug; AZ	2805
Tamias minimus	testes: large Mar, small mid summer, scrotal Mar-June; WY	10053
	testes: large May, small mid June; no Apr data; Canada	9889
	mate: Apr; WI	5210
	mate: Apr-Aug; MI	6796
	mate: May; no Apr data; Canada	9889
	mate: after hibernation; CO	11230
	preg f: Apr-June; WY	10053
	preg f: May; n2; SD	261a
	preg f: May-June; no Apr data; Canada	9889
	preg f; May-mid July; CO	11230
	preg f: mid June-early July; n9/20; CO	279
	preg/lact f: June; SD	10954
	birth: May; BC, Canada	2088
	birth: May-early June; WY	1871a
	birth: May-June; no Apr data; Canada	9889
	lact f: Apr; NV	6488
	lact f: May; n1; SD	261a
	lact f: May-Aug; WY	10053
	lact f: June-July; n2; se MT	6166
	lact f: June-July; n5; ND	3703
	lact f: ?-Aug; MI	6796
Tamias quadrimaculatus	preg f: end May-mid June; CA	10762
Tamias quadrivittatus	mate: est end Feb-Mar; UT	11385
	mate: early-mid June; Mt Taylor Valencia Co, NM	4962
	birth: May; n1; NM	502
Tamias ruficaudus	mate: end Apr-May; MT	726
	birth: end May-June; MT	726
Tamias sibiricus	estrus Apr-May; Amur-Zeya R, Siberia, Russia	10177
	mate: Apr; USSR	10422
	mate: Apr-early May; USSR	8053
	mate: Apr-May; zoo	2654
	mate: peak Apr-May; Germany	3662
	mate: end Apr-early May; Amur-Zeya R, Siberia, Russia	10177
	mate: early spring; Europe, w USSR	4645
	birth: end May; Amur-Zeya R, Siberia, Russia	10177
	birth: end May-early June; 20-30 d later mts; Siberia, Russia	9953
	birth: May-early June; USSR	10422
	birth: May-early June; USSR	8053
	emergence: early July; USSR	8053
Tamias speciosus	repro: seasonal; CA	10762
Tamias striatus	spermatogenesis: peak Mar; WI	8224
	testes: large mid Feb-early Oct; NY	12014
	conceives: mid Mar-early Apr, mid June-mid July; NY	12014
	mate: Feb-Apr, June-Aug; NY	2981
	mate: mid Feb-early Mar, July; OH	11989
	mate: mid Mar-?; ONT, Canada	10102
	mate: mid Mar-mid Apr, July; WI	5210
	mate: Apr, July-Aug; e US	1474
	mate: Apr-Sept; MI	6796

	preg f: Mar; n1; MN	495
	preg f: Apr-Sept; MI	1472
	preg f: Mar, June; NY	175
	preg f: Mar-Oct; OH	2022
	preg f: July; n1; NY	9639a
	birth: Mar-May, July-Sept; NY	2981
	birth: Apr; July; NY	175
	birth: mid Apr-mid May, mid July-mid Aug; NY	12014
	birth: end Apr-early May, Aug; s QUE, Canada	11956
	birth: peaks May, Aug; OH	2022
	birth: May, Aug-Sept; e US	1474
	lact f: end May-Aug; few data; MI	6796
Tamias townsendii	mate: Apr; BC, Canada	2088
	mate: Apr; w US	1474
	mate: end Apr-early May; WA	5658
	birth: May; BC, Canada	2088
	birth: end May-early June; n11 lt; WA	5658
Tamiasciurus douglasii	testes: scrotal Mar-July; n50%; BC, Canada	10543
	mate: early spring; Mexico	6387
	mate: Mar-July; CA	5922
	birth: June, Oct; w US	1474
	lact f: May-Sept; n50%; BC, Canada	10543
Tamiasciurus hudsonicus	epididymal sperm: end Jan-Sept; NY	4274
	testes: large Apr-early Sept; QUE, Canada	11956
	testes: inactive Sept-Nov; NY	6297
	mate: mid Jan-Sept; WI	5210
	mate: Feb-Mar, June-July; n N America	1474
	mate: mid Feb-Sept; NY	4274
	mate: peaks Mar-Apr, June-July; QUE, Canada	6158
	mate: Mar-May; n SAS, Canada	2324
	mate: Apr-June; CO	2671
	mate: end Apr; BC, Canada	2088
	preg f: Feb-Sept; NY	6297
	preg f: Apr; n1; MN	495
	birth: Mar-May; WY	1871a
	birth: Mar-July; SD	10954
	birth: ~Apr 1; NY	7235
	birth: Apr-May, Aug-Sept; n N America	1474
	birth: early May; Canada	5882
	birth: May-early June; BC, Canada	2088
	lact f: May-July; n5/7; se MT	6166
Tamiops macclellandi	preg f: Apr; n1; Malaysia	7161
Tamiops swinhoei	repro: spring-fall; n Vietnam	10825
Trogopterus xanthipes	birth: Mar-mid Apr; captive	11448
Xerus erythropus	yg found: peak Aug-Oct; s Uganda	5760
	repro: yr rd; Senegal	1149
Xerus inauris	preg f: July-Nov; Transvaal, S Africa	12134
	preg f: Aug, Oct; n2; sw Transvaal, S Africa	8965
	preg f: yr rd; Transvaal, S Africa	4697
	preg f: yr rd, none Apr, Sept; few data; Botswana	10154
	birth: Aug-Nov; Transvaal, S Africa	12134
	repro: yr rd; S Africa	10158
Xerus rutilus	preg f: Aug; n1; Ethiopia	8017
	preg f: Sept; n1; Kenya	8017
	preg f: Sept, Nov; n2; e Africa	5760
	juveniles: yr rd; e Africa	5760

Order Rodentia, Family Geomyidae

Pocket gophers (*Geomys* 100-450 g; *Orthogeomys* 500-800 g; *Pappogeomys* 110-900 g; *Thomomys* 45-545 g; *Zygogeomys* 450 g) are solitary, subterranean herbivores from the United States south to northern South America. We found no reproductive information on *Zygogeomys*. Males are larger than females (*Geomys* 5670, 11728, 11729; *Thomomys* 1474, 3740, 5021, 10100). The hemochorial, discoid placenta has a partially interstitial, antimesometrial implantation into a bipartite uterus (7563, 7569, 7574, 10310, 10924). A postpartum estrus may occur (7488, cf 1367) after the birth of naked and blind neonates whose eyes open at 22-23 d (*Geomys* 10534), or 26 d-5 wk (*Thomomys* 258, 2138, 4743, 8798, 9575, 9646, 11671). Females have 4-14 mammae (499, 502, 1474, 1475, 2344, 4745). Males have a baculum, seminal vesicles, coagulating glands, inguinal testes, and scrotal epididymides (1473, 4157a, 5669, 10924). Details of anatomy (3128, 5739), endocrinology (4157), and conception (7574) are available.

NEONATAL MASS		
Geomys bursarius	4.3 g; n1	11891
	5.1 g; r4.9-5.4; n4	11229
	5.125 g; n4	10534
	6-7 g	5210
Pappogeomys castanops	6.07 g; r3.8-7.9; n3	4565
Thomomys bottae	2-3 g	9646
	4.1 g; r4-4.2; n2	4743
Thomomys talpoides	2.77 g; se0.14; n6 yg, 1 lt	258
	3.58 g; se0.09; n20 yg, 4 lt	258
Thomomys bulbivorus	6.12 g	11671
NEONATAL SIZE		
Geomys bursarius	CR: 38 mm; n1	11891
	CR: ~40 mm; n4	11229
	TL: 47 mm; n4	10534
Pappogeomys castanops	40.97 mm; r38-43; n3	4565
Thomomys bulbivorus	50 mm	11671
WEANING MASS		
Geomys arenarius	CR: max 30 mm	11727
Geomys bursarius	22.5 g; nibble food	11891
Pappogeomys spp	m: 90 g; lowest wgt 1st capture	10160
	f: 110 g; lowest wgt 1st capture	10160
Thomomys bulbivorus	86.0 g	11671
WEANING SIZE		
Thomomys bulbivorus	16.4 cm	11671
LITTER SIZE		
Geomys arenarius	4.7 emb; r4-6; n3 f	5424a
Geomys bursarius	1.7 ps; se0.23; n34 f	11719a
	mode 2-3, r1-5	495
	2.0 yg; r1-3; n2 f	10534
	2.3 emb; se0.35; n16 f	11719a
	2.36 emb, yg; r1-3; n25 f	3042
	2.5 emb; r2-4; n10 f	5670
	2.5 cl; se0.30; n61 f	11719a
	2.52 ps; r1-4; n23 f	11890
	2.66 yg; r1-4; n6 nests	11890
	2.70 emb; r1-5; n31 f	11890
	3-4; r2-6	5210
	3-5; r1-8	1474
	3 yg ea; n3 nests	11891
	3.40 emb; ±0.13; mode 3-4 68%; r1-8; n73 f	11229
	3.46 ps; ±0.03; mode 3-4 67%; n155	11229
	3.6 emb; r2-6; n13 f	5405b,
	3.9 emb; r3-6; n20 f	11957a
	mode 4-5; r3-6	2060
	4 yg; n1 lt	11229
	4.0 ps; r2-8; n19 f	11957a
	4.15 emb; mode 4; r1-6; n34 f	9574a
Geomys personatus	3.18 emb; r2-4; n11 f	5670

Geomys pinetis	1-3	1474
	1.52 emb; ±0.11; r1-2; n19 f	11816
	1.73 ps; ±0.09; r1-5; n60 f	11816
	1.74 emb; sd0.51; mode 2; r1-3; n64 f	1367
	2.10 cl; sd0.54; n64 f	1367
Orthogeomys hispidus	2; n1 lt	4228
	2 emb; n1 f	2341
Pappogeomys bulleri	1 emb; n1 f	3702
Pappogeomys castanops	1-3	1474
	1.8 emb; r1-3	522
	2.08 emb; r1-4; n25 f	2305
	3.8 emb, ps, yg; r2-5; n22 f	4565
Pappogeomys merriami	2 emb; n1 f	2341
Pappogeomys tylorhinus	2.0 emb; r1-3; n4 lt	2050
Thomomys bottae	2.5; r1-4; n2	9646
	3; r2-4; n4 lt	2700
	3.2 ps; 6 lt	552
	3.33 emb; r2-4; n3 f	522
	3.8 emb; r1-7	145
	3.91 emb; n>50 f	9580
	4.33 lg viable emb; se0.47; n21 f	6542
	4.5 ps; r1-9; n59 f	6294
	4.6 yg; se0.11; n36? lt	5021
	4.75 med viable emb; se0.34; n40 f	6542
	4.92 emb; se0.15; r1-9; n110 f	7329
	5.53 sm viable emb; se0.24; n70 f	6542
	5.74 emb; r1-10; n74 f	6294
	5.85 emb; n>50 f	9580
	6 emb; n1 f	502
	6.223 emb; mode 6; r3-12 (16?); n220 f	1074
	6.3 emb; n10 f	9580
	8 emb; n1 f	7488
Thomomys bulbivorus	3-9	9575
	4-9 yg	11671, 11672
	4.2 emb; n8 f	9580
	4.68 ps; se0.17; mode 5; r2-8; n59 f, 10 locations	11280
	5 yg ea; n3 lt	11671
	8.33 emb; r8-9; n3 f	11671
Thomomys clusius	mode 6-7	499
Thomomys monticola	mode 3-4	5145
Thomomys talpoides	2 emb; n1 f	261a
	2.89; r1-5; n9 f	2138
	3-9	9575
	3.3 emb, ps; n400? f	4336
	3.65 emb, ps; n670 f	4336
	3.67 emb; r2-5; n3 f	279
	4 emb; n1 f	4962
	4 emb; n1 f	4745
	4 emb; n1 f	6488
	4 yg; r3-10	1871a
	4.0 emb; n2 f	10954
	4.0 emb; se0.53; r2-7; n12 f	11230
	r of x 4.3-5.2 emb, ps; r of se 0.2-0.4	4337
	4.4 emb, ps; se0.13	10924
	4.4 emb, ps; se0.21; n33	4336
	4.7 emb, ps; se0.24; n33	4336
	4.8 emb, ps	11830
	4.9 emb, ps; se0.14; n12	4336
	4.9 large ps; 5.6 all ps; r3-8; n12 f	260
	5 emb; n>50 f	9580
	5.1 ps; se0.16; r2-8; n66 f	11230
	5.2; r5-6; n5 lt	258
	5.5 emb; r5-6; n2 f	10219
	6 ps; r5-7; n3 f	3703

	6.32 emb; r4-9; n25 f	11673
	6.4 emb, ps; se0.18; n86 lt	4336
	mode 6-7	499
	9.7 ps; r6-13; n3 f	6166
Thomomys townsendii	5 emb ea; n2 f	11646a
	5.7 emb; r5-6; n3 f	11646a
	6.83; mode 7; r3-10; n12 f	4983
	7.1 emb; r5-9; n7 f	11646a
Thomomys umbrinus	2 emb; n1 f	522
	2 emb; n1 f	2341
	2-3	499
SEXUAL MATURITY		
Geomys bursarius	f: loss of pubic symphysis: 100-127 g	11706
	m: testes scrotal: 140 g	11706
	f/m: 3 mo	11890
Geomys pinetis	f/m: 6 mo	11816
Pappogeomys spp	f: 1st birth: min 1st yr	10160
Thomomys bottae	f: 1st birth: 3? mo	5021
	f: season of birth	2232
	m: yr after birth	2232
	f: 1st mate: min 1st yr	2700
Thomomys monticola	f: 1st mate: yr after birth	5145
	f/m: yr after birth	5144
Thomomys talpoides	f: 1st mate: yearling	11230
CYCLE LENGTH		
Geomys bursarius	estrus: 29.5 h	2500
	8 d; r7-9; n3	2500
GESTATION		
Geomys bursarius	18 d; n1; capture to birth	5334
	18-19 d	1474
	min 51 d; n1; capture to birth	10534
Thomomys bottae	19 d; n2	9646
	est 30 d	5021
Thomomys talpoides	~28 d	2088
LACTATION		
Geomys bursarius	food in cheek pouches: 32 d	10534
	leave den: 5-6 wk	11706
Thomomys talpoides	solid food: 17 d	258
Thomomys bulbivorus	solid food: 3 wk	
	wean: 6 wk	11671
INTERLITTER INTERVAL		
Thomomys talpoides	1 yr	11230
LITTERS/YEAR		
Geomys bursarius	1	3042
	1	11229
	1; north; 2+; south	1474
	est 2; 1.31; est 1.31-1.70	11890
	perhaps 2	11706
Geomys pinetis	min 2	1474
	min 2	11816
Pappogeomys castanops	max 4	2305
Thomomys bottae	est 1	502
	1; 2 occasional	5021
	2.0	7329
Thomomys bulbivorus	mode 1	9575
Thomomys monticola	1	5144, 5145
Thomomys talpoides	mode 1	9575
	1	502
	1	4336
	1	11230
	1	11673
	1-2	1474
Thomomys townsendii	2	1474

SEASONALITY

Geomys arenarius	preg f: June, Aug; n3; TX	5424a
Geomys bursarius	fertile m: Jan-June; CO	11229
	mate: Jan/Feb-July; TX	3042
	mate: end Mar-May; WI	5210
	preg f: Jan-Mar, June, none Apr-May; Bee Co, TX	5670
	preg f: Jan-May; KS	9574a
	preg f: Feb-Aug, TX	11890
	preg f: peak end Mar-Apr; TX	11229
	preg f: Apr-May; MN	495
	preg f: Apr-July; n13; NE	5405b$_1$
	preg f: end Apr-July; MAN, Canada	11957a
	birth: Feb-Aug; s US	1474
	birth: Apr; IL, WI	2060
	birth: peak Apr-May; CO	11229
	birth: Apr-July; n US	1474
	lact f: Jan, June-July; few data; TX	11706
	lact f: Apr, June; NE	5405b$_1$
	lact f: Oct; n2; KS	1812a
	repro f: peak Jan-Apr; TX	11719a
Geomys personatus	preg f: Dec-Mar, May; Bee C, TX	5670
Geomys pinetis	epididymal sperm: low Oct-Jan; FL	11816
	epididymal sperm: yr rd; FL	1367
	mate: est yr rd; FL	1367
	repro: yr rd; FL	11816
Orthogeomys hispidus	preg f: July; n1; Mexico	2341
	lact f: May; n1; San Luis Potosí, Mexico	2228a
	repro: est seasonal; Veracruz, Mexico	4228
Pappogeomys castanops	mate: June-Aug, Dec-Mar; Coahuila, Mexico	522
	preg f: June-Aug; TX	2305
	preg f: yr rd; TX	2305
Pappogeomys merriami	preg f: Aug; n1; Mexico	2341
Pappogeomys tylorhinus	sperm: peak Sept-Oct; Mexico	2050
Thomomys spp	preg f: June; n high altitude; spring-summer, s N Am	499
Thomomys bottae	uteri: large Apr, Aug-Sept; Mesa Verde, CO	2700
	mate: Mar-Sept; Mesa Verde, CO	2700
	mate: yr rd, low fall; CA	7329
	preg f: Jan-mid Apr; Sierra Nevada, San Joaquin Valley, CA	5021
	preg f: Feb-Mar; NM	145
	preg f: peaks Mar-Apr, Oct-Nov; CA	1074
	preg f: end Apr-May; Mesa Verde, CO	2700
	preg f May; n1; NM	502
	preg f: June-July, Dec; n3; Coahuila, Mexico	522
	birth: Feb-mid Apr; Sierra Nevada, San Joaquin Valley, CA	5021
	birth: Oct-June; w US	1474
	lact f: May, not Aug-Sept; Mesa Verde, CO	2700
	juvenile caught: May-Aug; n AZ	552
	repro m: Feb-July; n AZ	552
	repro f: Mar-June; n AZ	552
Thomomys bulbivorus	mate: Apr-June; Willamette Valley, OR	9575
	birth: Apr-July; Willamette Valley, OR	11671, 11672
Thomomys monticola	birth: July-Aug; CA	5145
Thomomys talpoides	testes: large Mar-May; MT	10924
	fertile m: July-Sept; Grand Co, CO	11230
	mate: Mar-July; OR	11673
	mate: Apr-June; sw WA	9575
	mate: May-June; CO	11230
	preg f: end Mar-mid June; CO	4336
	preg f: Apr-June; MT	10924
	preg f: mid May-June; WY	11830
	preg f: May-July; WY	1871a
	preg f: June; n1; SD	261a
	preg f: June; n1; Mirabel Spring, Valencia Co, NM	4962
	preg f: June-July; Grand Co, CO	11230

	preg f: July; n2; Black Hills, SD, WY	10954
	birth: Apr; BC, Canada	2088
	birth: mid May-mid June; MT	10924
Thomomys townsendii	preg f: Apr-June; n12; ID	11646a
	birth: Mar-Apr; s ID, n NV	1474
Thomomys umbrinus	preg f: Jan; n1; Coahuila, Mexico	522
	preg f: July; n1; Mexico	2341

Order Rodentia, Family Heteromyidae

Pocket mice and kangaroo rats (*Dipodomys* 30-180 g; *Heteromys* 35-85 g; *Liomys* 30-90 g; *Microdipodops* 10-25 g; *Perognathus* 5-50 g) are nocturnal, terrestrial to semifossorial herbivores in the New World. Males are often larger than females (4913, 5651, 6488, 6978). The hemochorial, endotheliochorial, or hemoendothelial, labyrinthine, discoid, deciduous placenta has a mesometrial or antimesometrial, interstitial implantation into a duplex uterus (5737, 6355, 7563, 7896, 10820). As simultaneously lactating and pregnant females are observed, a postpartum estrus is possible (1120, 2928, 4518, 4913, 5661, 6978, 8007; cf 2236). Neonates are blind and naked (1785, 4913, 8798, 9221, 10197, 11364). Eyes open at 9-17 d (*Dipodomys* 1474, 1503, 1782, 2164, 2693, 2927, 2935, 3282, 6148, 9075, 9076, 9221a, 10398a, 10670; *Perognathus* 2927, 2935, 4517) or 17-21 d (*Heteromys* 3321, 9252; *Liomys* 2927, 2935; cf 3321). Females have 6 mammae (502, 1120, 1474, 1475, 2236, 3024, 4080, 4494, 4745, 11364). Males have a baculum (884a, 1473, 6431, 9224). Details of anatomy (2768, 8472, 8475), intersex (671), copulation (162a, 737, 5652), milk composition (5958, 10524), and light or green vegetation (882, 1197, 5657, 5660, 6978, 8145, 9010a, 9353, 10197a, 11097) are available.

NEONATAL MASS

Dipodomys deserti	3.04 g; r2.2-4.6	881
	5-6 g; from graph	1503
Dipodomys heermanni	2.8 g; r2.3-3.2; n4 premature	10670
	3.65 g; r3.4-3.9; n4	10670
	4.3 g; sd0.46; r3.8-4.8; n6	9221, 9221a
	max 5 g	3282
Dipodomys merriami	3.0 g	5661
	~3 g; n6 lt	10197
	3.04 g; r2.2-4.6; n9 lt	1782
	3.5 g; r3.2-3.8; n3	2693
	4.0 g; n6	9076
Dipodomys microps	4.0 g	5661
Dipodomys nitratoides	2.2 g; r1.5-3.5; n7 yg, 4 lt	2164
	4.0 g; n8	2927
Dipodomys ordii	est 5 g; 1 d; from graph	10111
Dipodomys panamintinus	4.5 g; n3	2927
Dipodomys spectabilis	7.7 g; r6.0-8.7; n7	4913
	7.8 g; r5.3-9.3; n3	502
Dipodomys stephensi	4.4 g; r3.8-4.7	6148
Heteromys anomalus	3.3 g	9252
Heteromys desmarestianus	3.0 g	3321
Liomys pictus	2.5 g; n5	2927
Liomys salvini	1.9 g	3321
Microdipodops pallidus	1.0 g; n5	2927
Perognathus californicus	1.5 g; n1, 2 d	2927
Perognathus formosus	1.0 g	5661
Perognathus longimembris	0.5 g	5661
	~1.3 g; n26	4517

NEONATAL SIZE

Dipodomys deserti	TL: ~52 mm	881
Dipodomys merriami	42.5 mm; n6	9076
Dipodomys ordii	TL: est 65 mm; 1 d; from graph	10111
Dipodomys spectabilis	HB: 40 mm; n1	502
	CR: 60 mm; largest emb	877
Dipodomys stephensi	HB: 40.0 mm; r37-44	6148

WEANING MASS

Dipodomys heermanni	36.2 g; sd1.02; r29.3-47.5; n24, 25-32 d	9221a
Dipodomys merriami	10 g; 2 wk	9076
	emergence: 17 g; 4 wk	5661
	1st capture: 20-23 g	1784
Dipodomys microps	emergence: 21 g; 4 wk	5661
Dipodomys nitratoides	20 g; 20 d; from graph	2927
Dipodomys stephensi	25-28 g; 18-20 d	6148
Liomys pictus	20-25 g; 25-30 d; from graph	2927
Perognathus californicus	10 g; 20 d; from graph	2927
Perognathus formosus	emergence: 8 g; 4 wk	5661
Perognathus longimembris	emergence: 4 g; 4 wk	5661

LITTER SIZE

Dipodomys agilis	2.6; ±0.7; r2-4; n15 f	6708
Dipodomys deserti	3.29; mode 3; r2-5; n7 lt	1503, 1504
	3.43 emb; mode 3; r1-6	881
	3.5 emb; r3-5; n4 f	11044a
Dipodomys elator	2-4 emb	1642
Dipodomys californicus	2.65 emb; r2-4; n20 f	2224
Dipodomys compactus	2 emb; n1 f	666
	2 ps; n1 f	666
Dipodomys heermanni	2-5	5146
	2-5	1474
	2.64 yg; r2-3; n14 lt	3282
	2.65; n20 f	3282
	2.76 yg; n25 lt	9221a
	3 emb; n1 f	11044a
	3.0 emb; r3-4; n4 f	10398a
	3.5 yg; r3-4; n2 lt	10398a
	3.5 emb; r3-4; n2 f	10670
Dipodomys ingens	3-4; max 6	1474
	4.6 emb, yg, ps; n7	11715
	5.25 emb; r5-6; n4 f	4080
Dipodomys merriami	1.67; n3 lt; n5	522
	2 emb; n1 f	502
	2.02 emb; r1-3; n133 f	9075, 9076
	2.11; r2-3; n9 lt	2693
	2.25; r2-3; n4 lt	2927, 2935
	2.46 emb, ps, yg; sd0.72; r2-5; n70 f	5661
	2.5; r1-6; n129 lt, 51 f	2236
	2.6 ps; n157 f	1197, 6978
	2.6 emb; r1-5; n163 f	1197, 6978
	2.67 emb; r2-3; n6 f	4897
	2.67 emb; r1-5; n127 f	6430
	2.83 yg; n12 lt	1782
	3 yg; n1 f	1786
	3.1 emb; r2-4; n32 f	140
Dipodomys microps	2.4 yg; r1-4; n49 lt, 43 f	2236
	2.42 emb, ps, yg; sd0.53; r1-3; n65 f	5661
	2.5 emb; r2-4; n6 f	6488
	3 emb; n1 f	7489
	4 emb ea; n2 f	6560
Dipodomys nelsoni	2 emb; n1 f	523a
	2 emb; n1 f	878
	2 emb; n2 f	878
Dipodomys nitratoides	est 2; r1-3	2164
	2.3 yg; mode 2; r1-3; n12 lt	2927, 2928, 2935
Dipodomys ordii	1-6 yg	1871a
	1.67 emb; r1-2; n3 f	145
	2.2 emb; n33 non adult	882
	2.35 emb; n40 f	7041
	2.37 emb; r2-3	5372
	2.5 cl; max 5	3611
	2.5 emb; r1-5	3611

	2.5 emb; r2-3; n2 f	1812a
	2.5 ps; n38 f	4873
	2.5 emb; n133 adult	882
	2.8 emb; n18 f	4873
	2.87 emb; sd0.56; r2-4; n31 f	3294
	3 emb; n1 f	3703
	3.0 emb; r2-4; n2 f	4745
	3.0; r2-4; n2 lt	522
	3.0 yg; r2-4; n10 lt, 6 f	2370
	3.1 emb; r2-5; n13 f	5405b$_1$
	3.1 cl; n49 f	4873
	3.25 emb; n4 f	502
	3.25 emb; n73 f	7041
	3.6 emb; r2-5; n50 f	140
	3.8 emb	9353
	4 emb; n1 f	502
	4 emb; n1 f	1470
	4 emb; n1 f	1119
	5 emb; n1 f	4962
Dipodomys panamintinus	3.5; r3-4; n2 lt	2927, 2935
	4.2 emb; r4-5; n5 f	5158
Dipodomys phillipsii	2.67 emb; r2-3; n3 lt	3702a
Dipodomys spectabilis	1 emb ea; n2 f	2228a
	1.81 emb, ps, yg; mode 2; r1-3; n69 f	1120
	2.75 emb, ps, yg; r2-4; n12 f	4913
	3-4 yg	502
Dipodomys stephensi	2.67 yg; n6 lt	6148
Dipodomys venustus	1 yg; n1 nest	4512
Heteromys anomalus	2; r1-3; n2 lt	9252
	4.14 emb; r3-5; n7 f	11044a
Heteromys australis	2.83 emb; r2-4; n6 f	11044a
Heteromys desmarestianus	2.57; r1-4; n14 f	11044a
	2.7 weaned; mode 3; r3-5; n6 lt	3321
	3.1 emb	3321
	3.3 yg; n6 lt	3321
	3.67 emb; r3-4; n3 f	7624
	4.0 emb; r3-6; n4 f	2590
Heteromys gaumeri	2 emb; n1 f	4494
	3 ps; r1-5; n2 f	950
	3.0 emb; r2-4; n3 f	950
	4 emb; r3-5; n2 f	5423b
Heteromys oresterus	3 emb; n1 f	9223a
	4 yg; r3-5	3990$_1$
Liomys adspersus	3.2 emb; r2-4; n5 f	3319
	4.8 emb; r4-6; n9 f	11044a
Liomys irroratus	3.5 emb; r3-4; n4 f	4228
	4.3 emb; r1-6; n6 f	217
	4.39 emb; mode 4; r2-8; n52	2707
	5 emb; n1 f	2707
Liomys pictus	3.0	9176
	3.5; r2-5; n6 lt	2927, 2928, 2935
	3.80 emb; mode 3; r2-6; n49 f	7060
	4.29 emb; r3-6; n17 f	11388
	4.5 emb; r4-5; n2 f	4228
Liomys salvini	2.5; r2-3; n2 lt	3209
	3 emb ea; n2 f	1475
	3.55 emb; mode 3; r2-6	1634
	3.8 emb	3321
	4 yg; n1 lt	3321
	5 emb; n1 f	11044a
Microdipodops megacephalus	1-7 emb	1474
	3.9 emb; r1-7; n54 f	4228b
	4.0 emb; r1-6; n37 f	4229
	5.0 emb; r4-6; n2 f	11044a

Microdipodops pallidus	3.5 emb; n6 f	11044a
	4.0 emb; r1-6; n56 f	4228b
Perognathus anthonyi	4.2 emb; r3-5; n5 f	11044a
Perognathus arenarius	2 emb; n1 f	6151
	4.5 emb; r2-6; n6 f	11044a
Perognathus baileyi	3-4 emb	1474
	3.5 emb	8271
	3.6 ps	8271
	mode 4 emb	9077
Perognathus californicus	4.0; r2-5; n3 lt	2927, 2928, 2935
Perognathus fallax	3 emb; n1 f	11044a
Perognathus fasciatus	4 cl; n1 f	6166
	5.1 emb; r2-7; n9 f	3703
	5.42 emb; r2-9	6770
	5.64 ps; r2-12	6770
	5.7 emb; r3-9; n7 f	6166
	6 ps; n1 f	3703
	6 ps; n1 f	$5405b_1$
	6 ps; r3-9; n6 f	6166
	6.0 cl; r4-9	6770
	6.0 ps; r3-11; n14 f	11957a
	6.1 emb; r4-10; n8 f	11957a
Perognathus flavescens	3.67 emb; r2-5; n3 f	$5405b_1$
	4 emb ea; n3 f	495
	5 yg; n1 f	495
	6 emb; n1 f	495
Perognathus flavus	2-6	1474, 1871a
	2.75 emb; r2-3; n4 f	$5405b_1$
	3 emb; n1 lt	4962
	3 emb; r2-4; n2 f	502
	3-6	1474
	4; n1 lt	2935
	4 emb; n1 f	4745
	4 emb; n1 f	522
	4.5 emb; r4-5; n2 f	11044a
	6 yg; n1 lt	502
Perognathus formosus	5.34 emb, ps, yg; sd0.94; r4-7; n35 f	5661
	5.38; mode 5	2773
Perognathus hispidus	2-6	1474
	2-9	8272
	5 emb; n1 f	2547
	5 ps; n1 f	$5405b_1$
	5.5 emb; r5-6; n2 f	$5405b_1$
	6 emb; n1 f	10954
Perognathus inornatus	~4	2928
Perognathus intermedius	3-6	1474
	7 emb; n1 f	11044a
Perognathus longimembris	3-7	1474
	4; r1-6; n57 lt	4518
	4.25; r3-6; n4 lt	1785
	4.75 emb, ps, yg; sd0.77; r4-6; n16 f	5661
	5.38; mode 5	2773
Perognathus nelsoni	3-5 yg	2228a
	3.0; r2-5; n5 lt	522
Perognathus parvus	3.7; n3 lt	8007
	4 emb; n1 f	2544
	4.67; r4-6; n3 lt juv f	10262
	4.77 ps; r4-6; n16 f	5184
	4.91 emb; r4-6; n23 f	5184
	4.93 emb; n11 f	11281

	5.16 emb; mode 4; r2-8; n132 f	9579, 9581
	5.38; mode 5	2773
	5.5 emb; r5-6; n4 f	1119
	5.7 emb; mode 6; r4-7; n6 f	6488
	8 emb; n1 f	2544
Perognathus penicillatus	r3-4 emb; n3 f	523a
	3.4 emb; r2-4	9077
	4.38 emb; r3-6; n16 f	377
	4.5 emb; r3-6; n2 f	522
Perognathus spinatus	3.25 emb; r3-4; n4 f	11044a
	4 emb ea; n≥2 f	4079a
SEXUAL MATURITY		
Dipodomys deserti	f: vag perf: 24-33 d	881
	m: scrotal testes: 85 d	881
Dipodomys heermanni	f: 1st estrus: 40-43 d	9221a
	f: 1st mate: 1st yr	1474
	m: scrotal testes: 35-38 d	9221a
	f/m: 1st yr	5146
Dipodomys merriami	f: vag perf: 24-33 d	1782
	f: vag perf: 24-33 d	1782
	f: 1st conceive: 64 d	2236
	f: 1st birth: 97 d	2236
	f: 1st yr	5661
	m: scrotal testes: 84 d	1782
	m: scrotal testes: 85 d	1782
	m: yearling	5661
Dipodomys microps	f: 1st estrus: 34.2 d; ±2.9; r29-39	2236
	f: 1st yr	5661
	m: testes large; est ~1 yr	5661
Dipodomys nitratoides	f: 1st conceive: 84 d	2928, 2935
Dipodomys ordii	f: 1st estrus: 41 d; n1	3611
	f: 1st mate: 3 mo	3611
	f: 1st yr	7041
Liomys adspersus	f: 1st conceive: min 3-4 mo; most 1 yr	3319
	m: fertile: 1 yr	3319
Liomys pictus	f: 1st conceive: 98 d	2928, 2935
Perognathus fallax	f/m: 166 d	4518
Perognathus formosus	f: 1st conceive: ~1 yr	5661
	m: testes large: ~1 yr	5661
Perognathus longimembris	f: 1st conceive: 41 d	4518
	f: 1st conceive: ~1 yr	5661
	m: testes large: ~1 yr	5661
Perognathus parvus	f: 1st birth: 1st yr	10262
CYCLE LENGTH		
Dipodomys heermanni	mode 17-18 d; r6-43; n~100 cycles	9221a
Dipodomys merriami	mode 13 d; median 13.4; n106 cycles, 13 f	2236, 11764
Dipodomys microps	mode 12 d; median 12.5; n422 cycles, 27 f	2236, 11764
Dipodomys ordii	4-5 d	5374
	5-6 d	2928
Perognathus inornatus	5-6 d	2928
Perognathus longimembris	~10 d	4518
GESTATION		
Dipodomys deserti	29-30 d; n4	1504
	29-32 d	8798
Dipodomys heermanni	30 d	10809
	31.3 d; r30-33; n19	9221, 9221a
Dipodomys merriami	17-23 d; n1	1786
	4 wk	5661
	mode 33 d; r32-35; n65	2236
	33 d; ±2	1782
Dipodomys microps	4 wk	5661
	mode 31 d; r30-34; n33	2236
Dipodomys nitratoides	32 d; n2	2935

Dipodomys ordii	est 29-30 d	2370
	29-30 d	8798
Dipodomys panamintinus	29-30 d; n1	2927, 2935
Dipodomys spectabilis	17 d; n1	502
	22-27 d	877
	~27 d	1474
Heteromys desmarestianus	est 27-28 d; r23-30; n4	3321
Liomys pictus	25 d; r24-26; n2	2927, 2928, 2935
Liomys salvini	27-28 d; n1	3321
Perognathus californicus	25 d; n2	2927, 2928, 2935
Perognathus fallax	r24-26 d; n4	4518
Perognathus fasciatus	~4 wk	1474
Perognathus flavus	≤26 d; n1	2935
Perognathus formosus	4 wk	5661
Perognathus longimembris	3 wk	5661
	mode 22-23 d 74%; r21-31; n31; 31 d not successful	4518
Perognathus parvus	21-25 d	5184
	est 21-28 d	9581
LACTATION		
Dipodomys deserti	solid food: 15 d	1503
	wean: 15-25 d	1503
Dipodomys heermanni	solid food: 18-24 d	9221a
	solid food: 23 d	3282
	wean: 25 d	3282
	wean: 25-32 d	9221a
	wean: 26 d	10398a
	wean: 6 wk	10809
Dipodomys merriami	solid food: 9 d	1782
	solid food: 10 d	9076
	solid food: 11 d	9075
	solid food: 13 d	1503
	nurse: 3 wk	2236
	nurse: 3 wk	9075
	wean: 17-22 d	1503, 1782
	wean: 19-22 d	2935
	wean: 20-25 d	1784
Dipodomys microps	wean: 3 wk	5661
Dipodomys nitratoides	solid food: 9 d	2927
	solid food: 20 d; n1	2164
	wean: 21-24 d	2927, 2935
Dipodomys panamintinus	wean: 27-29 d	2927, 2935
Dipodomys spectabilis	wean: 20-25 d	877
	wean: 3-4 wk	4913
Dipodomys stephensi	solid food: 15 d	6148
	nurse: 18 d; n14; 22 d; n2	6148
Liomys pictus	solid food: 13-14 d	2927
	wean: 24-28 d	2927, 2935
Perognathus californicus	solid food: 13 d	2927
	wean: 22-24 d	2927, 2935
Perognathus formosus	wean: 3 wk	5661
Perognathus hispidus	independent: 1 mo	1871a
Perognathus longimembris	solid food: 10 d	4517
	f died, young survived: 14 d	4517
	self-sufficient: 18 d	4517
	wean: 30 d	4518
	wean: 3 wk	5661
Perognathus parvus	wean: est 3 wk	5184
INTERLITTER INTERVAL		
Dipodomys merriami	min 63 d	2236
Dipodomys ordii	4.7 mo; r3-8	3611
Dipodomys spectabilis	5-7 wk	4913

LITTERS/YEAR

Dipodomys agilis	2 rare	6708
Dipodomys deserti	2-5	1474
Dipodomys californicus	2/season	2224
Dipodomys heermanni	1-3	5146
	1-3	1474
Dipodomys merriami	>1 in good yr	5661
	1-2	1474
Dipodomys microps	>1 in good yr	5661
Dipodomys ordii	est 1	502
	est 2	5372
	2	7041
	2	3611
Dipodomys spectabilis	1-3	4913
Heteromys goldmani	3-4	11388
Liomys adspersus	1.44; n19 f	3319
Perognathus fallax	1	6708
Perognathus fasciatus	1	1474
Perognathus flavescens	est 2	1474
Perognathus flavus	2	1474
	2+; irregular breeding	502
Perognathus formosus	1	5661
Perognathus hispidus	est 2; in north	1474
Perognathus longimembris	1	5661
	1-2	1474
Perognathus parvus	1.1; r1-3; n50	8007
	1-2	10262
	2	9581

SEASONALITY

Dipodomys agilis	testes: large Dec-July, small Aug-Nov; San Gabriel, CA	6708
	spermatogenesis: Jan-July; San Gabriel, CA	6708
	preg/lact f: Jan-July; CA	6708
	repro f: peak Mar-Aug; CA	7028
Dipodomys californicus	preg f: Feb-Sept; CA	2224
Dipodomys compactus	preg f: Aug; n1; TX	666
Dipodomys deserti	mate: Feb-June; sw US	1474
	preg f: Feb-Apr; n4; AZ	11044a
Dipodomys elator	preg f: Feb, June-July, Sept; TX	1642
	lact f: July; n2; TX	11044a
Dipodomys heermanni	mate: Feb-Aug; CA	10670
	mate: Feb-Oct, peak Apr; CA, OR	1474
	mate: Mar-Aug; w US	5146
	mate: low Nov-Jan; CA	3282
	preg f: Mar-Aug; CA	10670
	preg f: June; n1; CA	11044a
	preg f: July; n6; CA	10398a
	birth: yr rd, peak Feb-Aug; Morro Bay, CA	10809
Dipodomys ingens	preg f: Feb, May; n4; CA	4080
	repro: est Jan-May; CA	4080
Dipodomys merriami	testes: large yr rd; NV	6978
	mate: Feb-Oct; sw US	1474
	mate: Mar-May or later; Alamogordo, NM	979
	mate: peak Mar-June; se AZ	1784
	preg f: Jan-July; Coahuila, Mexico	522
	preg f: Feb-Aug; NV	1197
	preg f: peak Mar-May; CA	1783
	preg f: Mar-Oct, not Nov-Feb; s AZ	9076
	preg f: Mar-Oct, peaks May, Sept; AZ	9075
	preg f: Aug; n1; NM	502
	preg f: Dec-May; CA	1783
	birth: mid Feb-mid June, peak Mar ; CA	5661
	lact f: Feb-Aug, except July; small July sample; NV	6978
	repro f: peak Mar-May; CA	7028
	repro f: June-Oct, Feb; AZ	9010

	repro: yr rd, peak m June-Sept, peak f Sept-Oct; AZ	12089
Dipodomys microps	conceive: Feb-mid Mar; Owens Valley, CA	5661
	preg f: Apr-June; no data other mo; NV	6488
	birth: Mar-mid Apr; CA	5661
	birth: May-June; w US	1474
Dipodomys nelsoni	preg f: July; n2; Coahuila, Mexico	878
	preg f: July; n1; Durango, Mexico	523a
	preg f: July; n1; Zacatecas, Mexico	878
	lact f: June, Sept; San Luis Potosí, Mexico	2228a
Dipodomys nitratoides	mate: Dec-Aug; captive	2927
	mate: yr rd; CA	2164
Dipodomys ordii	sperm: yr rd, peak June-Feb, low Mar-June; TX	3611
	sperm: yr rd; central OK	4873
	testes: large yr rd; NM	882
	estrus: Feb-early July; lab	5374
	mate: peaks Jan-Mar, end Aug-Oct; UT	2769
	mate: Feb-Sept; w US	5146
	mate: Aug-Mar; OK, TX	7041
	mate: Dec-Oct; captive	2370
	mate: spring, fall; few data; Alamogordo, NM	979
	CL present: Aug-May, peaks Sept-Oct, Jan-Feb; TX	3611
	preg f: Jan; n1; AZ	145
	preg f: Feb; n2; Coahuila, Mexico	522
	preg f: May-Sept; n13; NE	5405b,
	preg f: June, Oct; n2; NM	502
	preg f: July; n1; ND	3703
	preg f: Aug-Apr; TX	3611
	preg f: Sept-Oct; n2; KS	1812a
	preg f: Oct; n2; NM	145
	preg f: yr rd, low Mar-Apr; NM	882
	birth: mid-Feb-mid-July; NM	5372
	birth: May-June; w US	1474
	lact f: May; n1; SD	261a
	lact f: Nov; n4; NM	502
	repro: Feb-Aug; ne CO	3294
Dipodomys panamintinus	preg f: apr-May; n5; CA	5158
	repro: peak Feb-Mar; CA	5158
Dipodomys phillipsii	preg f: June; n1; Durango, Mexico	3702a
	preg f: Oct; n2; Jalisco, Mexico	3702a
Dipodomys spectabilis	mate: Jan-Aug; AZ	1120
	preg f: Aug; n2; San Luis Potosí, Mexico	2228a
	birth: Jan-Aug; sw US	1474
	lact f: Apr, Oct; n4; Socorro, NM	9648
	repro: Dec-Aug; NM	4913
Dipdomys stephensi	preg f: June-July; CA	6148
Heteromys anomalus	scrotal testes: yr rd; Venezuela	451
	preg f: Jan; n2; Colombia	11044a
	preg f: Mar-Apr; n4; Tobago	11044a
	preg f: Apr; n1; Venezuela	11044a
	preg f: May-June, Sept, Dec; 45 km s of Calabazo, Venezuela	451
	lact f: Feb, Oct; 45 km s of Calabazo, Venezuela	451
	lact f: Apr, June-July; n5; Venezuela	11044a
	repro: yr rd, peaks at seasonal transitions; Venezuela	7996
Heteromys australis	preg f: Jan-Mar; n6; Panama	11044a
	lact f: Jan-Feb; n3; Panama	11044a
Heteromys desmarestianus	preg f: Apr; n1; Mexico	11044a
	preg f: Mar-Apr; n7; Panama	11044a
	preg f: Mar, May-June; n4; Guatemala	11044a
	lact f: Mar, June; n5; Panama	11044a
Heteromys gaumeri	preg f: end Apr-early May; n3; Yucatan peninsula, Mexico	950
	preg f: July; n1; Quintana Roo, Mexico	5423b
	preg f: Dec; n1; Campeche, Mexico	5423b

Heteromys goldmani	testes: large yr rd; s Mexico	11388
	ovaries: regress dry season; s Mexico	11388
	birth: May-Dec; Mexico	11388
Heteromys oresterus	preg f: July or Aug; n1; Costa Rica	9223a
Liomys adspersus	testes: large Oct-Apr, small June-July; Panama	3319
	mate: peak Mar-Apr; Panama	3319
	preg f: Feb; n9; Panama	11044a
	preg f: Dec-Apr; Panama	3319
	lact f: Mar-Apr; n2; Panama	11044a
	yg found: Dec-May; Panama	3317
Liomys irroratus	preg f: Aug; n1; Mexico	2341
	preg f: July-Jan, Mar, May, peak Aug-Nov; n52; Mexico	2707
Liomys pictus	preg f: Feb,-Sept, Nov-Dec; n49; Mexico	7060
	preg f: Mar; Villa Flores, s Mexico	11388
	preg f: dry season; n8/10; Oaxaca, Mexico	9176
	repro: Sept-May; Veracruz, Mexico	4228
Liomys salvini	preg f: July; n1; Guatemala	11044a
Microdipodops megacephalus	preg f: May; n2; NV	11044a
	birth: May-early July; w US	1474
	birth: peak May-June; NV	4229
	lact f: May; n1; NV	11044a
Microdipodops pallidus	preg f: end Mar-Sept; NV	8004
	preg f: May; n6; NV	11044a
	lact f: May; n1; NV	11044a
Perognathus amplus	repro f: June-Aug; AZ	9010
Perognathus anthonyi	preg f: May-June, Oct; n5; Baja CA, Mexico	11044a
	lact f: May; n2; Baja CA, Mexico	11044a
Perognathus arenarius	preg f: Mar; n1; Cerralvo I, Baja CA, Mexico	6151
	preg f: Mar-Apr; n6; Baja CA, Mexico	11044a
	lact f: Apr; n1; Baja CA, Mexico	11044a
Perognathus baileyi	mate: June-Oct, peak June, Sept; AZ	9010
	preg f: Apr-Aug; AZ	9077
	birth: Apr-May; sw US	1474
Perognathus fallax	mate: spring; few data; CA	6708
	preg/lact f: Oct; n3; CA	11228a
Perognathus fasciatus	preg f: June-Aug; n9; ND	3703
	preg f: July; n7; se MT	6166
	preg f: July-mid Aug; MAN, Canada	11957a
	birth: Apr-Aug; WY	1871a
	lact f: July; n1; se MT	6166
Perognathus flavescens	preg f: May; n1; MN	495
	preg f: May, July; n3; NE	5405b$_1$
	birth: May; n1; MN	495
	lact f: July; n1; NE	5405b$_1$
	repro: Apr-July; central US	1474
Perognathus flavus	mate: Apr-Nov; w central US	1474
	preg f: Mar-Oct; NM	1871a
	preg f: May; n1; Valencia Co, NM	4962
	preg f: May-June; n2; NM	502
	preg f: June; n1, none Mar-Apr; Coahuila, Mexico	522
	preg f: July; n1; NM	11044a
	preg f: July-Aug; n2; TX	11044a
	birth: June; n1; NM	502
	lact f: July; n1; TX	11044a
	lact f: Aug; n1; San Luis Potosí, Mexico	2228a
	lact f: Dec; NM	502
Perognathus formosus	mate: early Apr; CA	5661
	preg f: peak June; UT	2773
	birth: 1st 3 wk May; Owens Valley, CA	5661
	birth: May-July; CA, NV, UT	1474
Perognathus hispidus	preg f: Aug; n1; SD, WY	10954
	preg f: Aug-Sept; n2; NE	5405b$_1$
	birth: yr rd; s US; seasonal; n, mid US	1474

Perognathus intermedius	preg f: Apr; n1; az	11044a
	birth: May-July; sw US	1474
	repro f: June-Sept, peak June; AZ	9010
Perognathus longimembris	mate: end Apr-early May; CA	5661
	preg f: peak June; UT	2773
	birth: Apr-July; sw US	1474
Perognathus nelsoni	preg f: end Mar-July; Coahuila, Mexico	522
	birth: Aug; San Luis Potosí, Mexico	2228a
Perognathus parvus	testes: large spring; e WA	5657
	estrus: Apr-June; WA	8007
	preg f: May-early June; n6; central NE	6488
	preg f: May-Aug, peak June-July; WA	8007
	preg f: peak June; UT	2773
	preg/lact f: May-Sept; no data other mo; ID	10262
	birth: May-June; WY	1871a
	birth: June-Aug; n39; BC, Canada	5184
	birth: end spring-early summer; w US	1474
	lact f: May-Sept peak June-July; WA	8007
	repro m: Mar/Apr-June/July; WA	8007
Perognathus penicillatus	mate: end Apr-May; no data June-Feb; Alamogordo, NM	979
	preg f: Feb, June; few data; Coahuila, Mexico	522
	preg f: May-Sept; AZ	9077
	preg f: June-July; n3; Durango, Mexico	523a
	birth: May-Sept; sw US, n Mexico	1474
Perognathus spinatus	preg f: Apr; n≥2; lower Colorado R, CA	4079a
	preg f: Apr; n4; Baja CA, Mexico	11044a

Order Rodentia, Family Castoridae, Genus *Castor*

Beavers (*Castor canadensis*, *C. fiber*) are 5-45 kg, semiaquatic herbivores of temperate North America and Eurasia. Males have been reported to be larger than, smaller than and equal to females (4570, 4815, 8050). Urogenital and digestive products exit via a cloaca (8050, 11791, 12065). A vaginal plug occurs (6056). The hemodichorial, labyrinthine, reniform placenta with a subplacenta has a superficial, mesometrial implantation into a Y-shaped, duplex uterus (2003, 2649, 3268, 3766, 7563, 7565) and is eaten after birth (4564, 8255, 9817). A postpartum estrus (2109) follows the birth of furred and open-eyed neonates (323, 1474, 3571, 4124, 4564, 5146, 7433, 8050, 8133, 8255, 8798, 9817, 10922; cf 11045). Females have 4 teats (1474, 3571, 4032, 7433, 8050, 9387). Males are ascrotal (testes descend into a subpelvic diverticulum during the breeding season) and have a baculum and seminal vesicles (1473, 2648, 2649, 3524, 8050, 11791). Details of anatomy (2645, 2646, 3766, 8133, 8763), birth (9817), and milk composition (12137, 12138) are available.

NEONATAL MASS		
Castor canadensis	361.7 g; r265-490; n5	9817
	~380 g; r365-390; n2	4124
	387.9-614.5 g; term emb	8133
	490 g; r390-700; n10, 3 lt	1203
Castor fiber	300-700 g	11791
	380-620 g	8050
	475 g; n1 stillborn	4564
	f: 548.5 g; r527-570; n2, 1 d	2645
	555 g; n9, 3 lt	12138
	700 g; n1	11791
NEONATAL SIZE		
Castor fiber	245 mm; n1 stillborn	4564
WEANING MASS		
Castor fiber	4.2 kg; n8, n2 lt, 90 d	12138
LITTER SIZE		
Castor canadensis	1-6 yg	4011
	1.5 emb; r1-2; n2 f	7325a
	1.7; r0.9-2.5; f <2 yr	8248
	2 yg; n1	4818
	2.6 emb; se0.16; mode 2; n40 f	11675, 11676

	2.8 ps; se0.18; mode 3; n26 f	11675, 11676
	2.8 cl; sd1.3; n89 f	8280
	2.9 emb; sd1.1; n60 f	8280
	2.9 ps; sd1.3; n243 f	8280
	2.91 emb; mode 2; r1-5; n22 f	8133
	3 emb ea; n2 f	3037
	mode 3 ps, emb; 76% 2-4; r1-7; n504 f	6360
	3.0 cl; se0.16; mode 2; n48 f	11675, 11676
	3.02 emb; n176 f	9387
	3.1 emb	1168
	3.1 ps; n232 f	6360
	3.2 ps; r1-8; n43 lt	6679
	3.37 ps; r1-6; n19 f	8133
	3.4 ps, emb; n22 f	8408
	3.4 emb; n272 f	6360
	3.5; r3-4; n6 f	9577
	3.6 cl; n27 f	8408
	3.66; mode 3-4; r2-5; n9 f	8133
	3.72 yg; mode 4; r1-8; n65	1202
	3.8 emb	4011
	3.9 ps; n28 f	4634
	4.0; n2 lt	8254
	4.0; r3.3-4.7; f 2.5 yr	8248
	4.08 emb; r2-7; n11 f	10922
	4.4 cl; n48 f	4634
	4.5 emb; r3-6; n4 f	5405b,
	4.81 emb; 5.19 cl; n16 f	4634
	5.3 ps; r4-7; n18 f	11956
	5.50 emb; 5.04 without resorption; r1-9; n24 f	1278
	5.6 emb; r3-7; n17 f	11956
	6 yg; n1 f	9817
	7 emb; n1 f	11658
	8 emb, 9 cl; n1 f	4514
	10 emb; n1 f	5210
Castor fiber	1.7 yg; sd0.67; mode 2; r1-4; n120 yg, 71 sites	12136
	mode 2-5; r2-7	3571
	2.57 emb; r1-4; n18 f	4570
	2.7; r1-6; n117 lt	2649
	3.0; r1-4; n3 lt	12138
	3.0 ps; r1-5; n5	4570
	3.0; r2-5; n10 lt	2109
	4.7; r4-5; n3 lt	4564
	4.7 emb; max 8; n9 f	6157
SEXUAL MATURITY		
Castor canadensis	f: 1st cl, ps: yearling	6679
	f: 1st mate: 2nd season after birth	8133
	f: 1st mate: 2.5 yr	1474
	f: 1st preg: 1-2 yr	500
	f: 1st preg: 1.5-2 yr	8248
	f: 1st preg: est ~2 yr	1278
	f: parous: by 2.5-3 yr thus mature 1.5-2 yr	6221
	f: 1st repro: 1.5 yr	11675
	m: sperm: 1.5-2 yr	6221
	m: epididymal sperm: yearling	6679
	m: 1st repro: est ~2 yr	1278
	f/m: ~21 mo	5210
Castor fiber	f: 1st cl: 6 mo	2645
	f: 1st estrus: 2.5 yr	8050
	f: 1st mate: 1.5-2 yr	11791
	f: 1st birth: 2 yr 29%	2649
	f: 1st birth: 3 yr	4645
	f/m: 1st mate: 3 yr	6056
	f/m: mode 3-4 yr	3571

CYCLE LENGTH

Castor fiber	estrus: 3 h	6056
	estrus: 10-12 h	4815
	estrus: 12-24 h	2649
	estrus: 1-2 d	2647
	r7-9 d; n19 f	2647
	7-12 d	2649
	14 d	4815
	14-15 d	8050

GESTATION

Castor canadensis	~90 d	2088
	~120 d	5210
	est 128 d; n1	1203
Castor fiber	90 d	2109
	~90 d; max 107	3571
	3.5 mo	6056
	105-107 d	4645
	105-107 d	8050
	105-107 d	4815
	106.3 d; r104-111; n3	4564
	107 d; sd2.5; n39	2649, 12135
	128 d	7433

LACTATION

Castor canadensis	den emergence: 4 wk	4032
	nurse: 50 d	4818
	wean: ≤2 mo	1871a
Castor fiber	solid food: ~4 wk	3571
	solid food: after 30 d	12138
	nurse: 8 wk	3571
	wean: 2 mo	11791
	nurse: 90 d	12135, 12138

LITTERS/YEAR

Castor canadensis	1	1202
	1	11675
	1	8254
Castor fiber	1	2109
	1	3571, 4645
	1	7433
	1	8050

SEASONALITY

Castor canadensis	spermatogenesis: peak Jan-May; SAS, Canada	9201
	mate: Jan-Apr; BC, Canada	2088
	mate: end Jan-Feb; WI	5210
	mate: end Jan-Feb; PA	1278
	preg f: Feb; n2; WA	7325a
	preg f: end Feb-?; WY	4011
	preg f: Mar-Apr; n4; NE	5405b$_1$
	birth: Feb-July; MS	11675
	birth: Apr-Mar; w US	5146
	birth: Apr-July; N America	1474
	birth: end Apr-early June; WY	8133
	birth: end Apr-early July; BC, Canada	2088
	birth: May; IL, WI	2060
	birth: May; n2; QUE, Canada	8255
	birth: May-June; WY	1871a
	birth: May-Aug; n25; end Sept; n1; zoo	12128
	birth: mid May-June, sometimes July; CO	9387
	birth: end May; MAN, Canada	4032
	birth: June-?; WY	4011

Castor fiber	estrus: Jan-Mar; Europe	7433
	mate: Jan-Mar; Europe, USSR	4645
	mate: Jan-May; Russia	4815
	mate: Feb-Mar; zoo	4564
	mate: Feb-Mar; Europe	3571
	mate: Dec-May, peak mid Jan; Poland	2649
	birth: Apr-May; Europe	7433
	birth: Apr-May; Europe, USSR	4645
	birth: Apr-Aug, peak May; Poland	2649
	birth: end Apr-early May; Europe	3571
	birth: May; according to local farmers; Ukraine	9555
	birth: May; Lapland, Finland	6056
	birth: May-June; Finland	6157
	birth: end May; Mongolia	8512
	birth: end May-early June; n Europe	11791

Order Rodentia, Family Anomaluridae

The west and central African, scaly-tailed, flying-squirrels (*Anomalurus*, *Idiurus*, *Zenkerella*) are arboreal herbivores that range from 15 g (*Idiurus*) to 1050 g (*Anomalurus*) in tropical and subtropical forests. The natural history of the flightless *Zenkerella* has not been recorded and that of *Idiurus* is poorly known. *Anomalurus* is reported to be either social (5760) or solitary (331). A copulatory plug is present (2439, 5760). The hemomonochorial, villous, discoid placenta has a superficial, mesometrial attachment (6625, 7563). Neonates are active with fur and open eyes (5760). Females produce copious milk from 2 (5760) or 4 (331) mammae. Both sexes bring food to young (5760, 9468).

NEONATAL MASS		
Anomalurus derbianus	~160 mm	5760
LITTER SIZE		
Anomalurus beecrofti	1 emb; n1 f	2439
	1 yg ea; n2 f	4354a
Anomalurus derbianus	1	9468
	1	331
	1 yg	2446a
	1 emb; n1 f	2446a
	mode 1; r1-3	5760
	1-2	11253
Anomalurus peli	mode 2-3; max 4; hunter's reports	65
Idiurus macrotis	1 emb ea; n3 f	8854
Idiurus zenkeri	mode 1	5760
LITTERS/YEAR		
Anomalurus derbianus	2	11253
Anomalurus peli	2	65
SEASONALITY		
Anomalurus beecrofti	lact f: Jan, Oct; n2; Nigeria	4354a
Anomalurus derbianus	large testes: Feb, July, Dec; Uganda, w Kenya	5760
	birth: Jan; n4; Uganda, w Kenya	5760
	birth: Sept-Nov; British Cameroun	9468
	birth: dry season; n1; wet season; n1; Zaire	11253
	lact f: Feb, May; Rio Muni	5388a
	f with juvenile: Oct; Zambia	331
	juvenile alone Oct-Nov; Zambia	331
Anomalurus peli	birth: Sept & other mo; Gold Coast	65
Idiurus macrotis	preg f: Aug; n3; Lake Kivu, Zaire	8854
Idiurus zenkeri	testes: small Nov-Dec; n2; Camerouns, central Forest Refuge	5760
	no preg f: Nov-Dec; n5; Bwamba, Camerouns, central Forest Refuge	5760

Order Rodentia, Family Pedetidae, Genus *Pedetes*

Springhares (*Pedetes capensis* 2-4 kg) are nocturnal, semi-fossorial, saltatorial, solitary herbivores in arid and semi-arid regions of central and southern Africa. A vaginal plug is present after mating (1508, 1509). The labyrinthine, endotheliochorial, discoid placenta implants into a bicornuate uterus (1508, 1509, 7565, 7568, 8173, 8247, 10154) and is eaten after birth (9295). The precocious neonate is furred with eyes which open 0-3 d after birth (1507, 1508, 1950, 4563, 5760, 9295, 9945, 11102). Females have a postpartum estrus, an erectile clitoris, and 2-4 teats (1509, 1951, 5760, 8247). Males have visible teats, semi-external testes, seminal vesicles, penial spines, and a baculum (1508, 1509, 1951, 3823, 9945, 10096). Details of endocrinology (11142) are available.

NEONATAL MASS		
Pedetes capensis	145 g; n1; est premature	1950
	240-280 g	5760
	250 g; n1	4563
	252 g; ±19; r222-319; n12	1508, 1509, 1510, 1512
	255 g; n1	1950
	289 g; r278-300; n2	9295
	~300 g	1507
WEANING MASS		
Pedetes capensis	1.3 kg; den emergence, independent	1507, 1509
LITTER SIZE		
Pedetes capensis	1	11102
	1 emb	4925
	1; n1 lt	4563
	1 emb; n1 f	8247
	1 yg ea; n2 lt	9295
	1 yg ea; n2 lt	1950
	1.01 emb; r1-2; n152 f	1507, 1508, 1509
SEXUAL MATURITY		
Pedetes capensis	f: 1st conceive: ~2.7 kg	1508, 1509
	m: spermatogenesis: ~1 yr	10096
	m: sperm: 2.3-2.8 kg	1508
	m: 2.9 kg	1509
	f/m: 2.6-2.8 kg	1507
GESTATION		
Pedetes capensis	est ~2 mo	5760
	~77 d	1509
	~79 d; high progesterone	5923
	est 80-82 d	9295
LACTATION		
Pedetes capensis	den emergence: 6-7 wk	1509, 1510
	wean: min 46 d	1508, 1509
INTERLITTER INTERVAL		
Pedetes capensis	est 101 d	1509
SEASONALITY		
Pedetes capensis	testes: active yr rd; Botswana	1508, 1512
	mate: yr rd; Botswana	1507, 1512
	preg f: Mar, Oct-Nov; Transvaal, S Africa	8965
	preg f: Dec-Feb; Zimbabwe	5923
	preg f: yr rd; Botswana	10154
	preg f: yr rd, except Jan, Mar; S Africa	11142
	preg/lact f: yr rd; Botswana	1508
	lact f: Mar, Nov; Transvaal, S Africa	8965
	lact f: yr rd; Botswana	1509
	juveniles appear: May; Zimbabwe	5923

Order Rodentia, Family Cricetidae

Cricetids (*Abrawayaomys* ~200 mm; *Aepeomys* 100-120 mm; *Akodon* 10-60 g; *Ammodillus* 85-105 mm; *Andalgalomys* 45 g; *Andinomys* 80-91 g; *Anotomys* 100-120 mm; *Auliscomys* 45-105 g; *Baiomys* 4-13 g; *Beamys* 50-150 g; *Bibimys* 23-34 g; *Blarinomys* 130-160 mm; *Bolomys* 25-63 g; *Brachiones* 80-95 mm; *Brachytarsomys* 200-250 mm; *Brachyuromys* 85-105 g; *Calomys* 17-59 g; *Calomyscus* 15-30 g; *Chilomys* 85-100mm; *Chinchillula* 120-180 mm; *Cricetomys* 0.6-1.6 kg; *Cricetulus* 20-40 g; *Cricetus* 112-908 g; *Daptomys* 47 g; *Delanymys* 5-6 g; *Dendromus* 5-27 g; *Dendroprionomys*; *Deomys* 40-70 g; *Desmodilliscus* 11 g; *Desmodillus* 39-70 g; *Dipodillus* 16-58 g; *Eligmodontia* 10-32 g; *Eliurus* 35-103 g; *Euneomys* 78-86 g; *Galenomys* 105-120 mm; *Gerbillurus* 11-50 g; *Gerbillus* 18-81 g; *Graomys* 105-165 mm; *Gymnuromys* 125-160 mm; *Habromys* 19-48 g; *Hodomys* 290-450 g; *Holochilus* 280 g; *Hypogeomys* 300-350 mm; *Ichthyomys* max 120 g; *Irenomys* 30-65 g; *Isthmomys*; *Juscelinomys*; *Kunsia* 160-185 mm; *Leimacomys* ~120 mm; *Lenoxus* 150-170 mm; *Lophiomys* 590-920 g; *Macrotarsomys* 80-100 mm; *Malacothrix* 7-22 g; *Megadendromys* 49-66 g; *Megadontomys* 77-79 g; *Meriones* 30-275 g; *Mesocricetus* 90-258 g; *Microdillus* 60-80 mm; *Microxus* ~90 mm; *Myospalax* 150-288 g; *Mystromys* 75-185 g; *Neacomys* 18-20 g; *Nectomys* 160-420 g; *Nelsonia* 120-130 mm; *Neotoma* 94-585 g; *Neotomodon* 37-66 g; *Neotomys* 63-69 g; *Nesomys* 185-230 mm; *Nesoryzomys* 100-200 mm; *Neusticomys* 40 g; *Notiomys* 25-95 g; *Nyctomys* 39-61 g; *Ochrotomys* 15-30 g; *Onychomys* 9-60 g; *Oryzomys* 10-120 g; *Osgoodomys* 48-68 g; *Otomys* 47-255 g; *Otonyctomys* 29-36 g; *Ototylomys* 53-120 g; *Oxymycterus* 46-125 g; *Pachyuromys* 40 g; *Parotomys* 89-155 g; *Peromyscus* 11-110 g; *Petromyscus* 11-28 g; *Phaenomys* ~150 mm; *Phodopus* 20-53 g; *Phyllotis* 20-100 g; *Platacanthomys* 75 g; *Podomys* 27-44 g; *Podoxymys* ~100 mm; *Prionomys* 60 mm; *Psammomys* 88-190 g; *Pseudoryzomys* 30-56 g; *Punomys* 130-155 mm; *Reithrodon* 77-95 g; *Reithrodontomys* 5-28 g; *Rhagomys*; *Rheomys* 90-140 mm; *Rhipidomys* 68-100 g; *Rhombomys* 140-430 g; *Saccostomus* 30-96 g; *Scapteromys* 150-200 mm; *Scolomys* ~90 mm; *Scotinomys* 12-15 g; *Sekeetamys* 41-61 g; *Sigmodon* 57-275 g; *Steatomys* 5-70 g; *Tatera* 30-227 g; *Taterillus* 42-62 g; *Thomasomys* 35-61 g; *Tylomys* 63-370 g; *Typhlomys* 90-100 mm; *Wiedomys* 100-130 mm; *Xenomys* 155-165 mm; *Zygodontomys* 50-104 g) are primarily terrestrial herbivores from the New World, Asia, and Africa.

Males are larger than females in *Akodon* (2221, 2223, 7213), *Calomys* (667), *Cricetulus* (7503, 8457), *Gerbillus* (4713, 8134), *Meriones* (8961, 9175, 11905), *Neotoma* (217, 3087, 8962), *Oryzomys* (2221), *Phodopus* (9516), *Sigmodon* (1795, 7254; cf 3850), *Tatera* (9175; cf 4925, 8501), while females are larger than males in *Eligmodontia* (8307). Male *Gerbillus* have an additional chromosome (3492, 6972).

Copulatory ties occur (*Baiomys* 3096; *Calomys* rarely, 667); *Neotoma* 2507; cf 3087, 3096; *Ochrotomys* 2507; *Onychomys* 2533, 6209) as do vaginal plugs (*Calomys* 5474; *Dipodillus* 4675, 6234, 6237; *Meriones* 4459, 4460, 7972, 9451; cf 6237, 9468; *Mesocricetus* 1382, 2419, 4460; *Neotoma* 11696; *Pachyuromys* 4675, 6234, 6237; *Peromyscus* 4459, 4460, 12005; *Tatera* 7146; *Zygodontomys* 11373).

The hemochorial or hemoendothelial, labyrinthine, discoid placenta has an antimesometrial, interstitial implantation into the duplex or bicornuate uterus (987, 993, 1138, 2419, 3011, 3269, 3270, 3752, 5735, 7563, 9451, 9707) and is eaten after birth (1853, 4991, 6237, 6302, 6803, 6807, 10573, 11319, 11696). Implantation be delayed by lactation (6238, 8805, 9114). Estrus occurs postpartum (*Akodon* 2409; *Baiomys* 5064; *Calomys* 2410, 4853, 8430; *Cricetomys* 3130; *Cricetus* 11319; *Deomys* perhaps 5760; *Desmodilliscus* 8696; *Desmodillus* 5684; *Dipodillus* 6235, 6237, 6238; *Gerbillus* 4713; *Malacothrix* 10154; *Meriones* 56, 1035, 4645, 6236, 6237, 6803, 6867, 7150, 7971, 7972, 8263, 9175, 9451, 9692; *Mystromys* 4220; *Neotoma* 2415, 2909, 3087; cf 3262; *Onychomys* 2907, 10697; *Oryzomys* 10572, 11388; *Ototylomys* 4606, 4607; *Peromyscus* 1236, 1363, 2527, 2529, 2806, 3897, 4119, 5797, 6149, 6150, 6493, 6708, 7072, 10573, 10574; *Psammomys* 2234; *Reithrodontomys* 9107; cf 10587a; *Rhombomys* 7778; *Scotinomys* 4967; *Sigmodon* 4898, 7253; *Tatera* 987, 1815; cf 7146; *Tylomys* 4606, 10755; *Zygodontomys* 87, 11373) except in *Beamys* (2912), *Mesocricetus* (2419), *Phyllotis* (7241), and *Saccostomus* (11082).

Some cricetids are furred at birth (*Lophiomys* 5760) and open their eyes at 0-2 d (*Sigmodon* 4898, 7030, 7253, 9572, 10570), 0-4 d (*Otomys* 331, 2338, 2339, 2571, 5760, 10158), 6 d (*Ototylomys* 4606, 7624), 14 d (*Akodon* 7212), or 18-21 d (*Saccostomus* 2861, 4329, 10158). Others are nearly naked (few hairs or sparse coat) and open their eyes at 6-9 d (*Calomys* 5474, 7189, 8430), 6-10 d (*Oryzomys* 1474, 7190, 10572, 11305, 11931), 11-16 d (*Eligmodontia* 6807), 12-20 d (*Podomys* 6301), 2 wk (*Beamys* 298, 2912), or 15-18 d (*Nyctomys* 946). *Neotoma* and *Tylomys* are born either furred or naked and open their eyes in 2-15 d (*Tylomys* 4606, 4607, 4608, 10755) or 11-21 d (*Neotoma* 1474, 2679, 2906, 2909, 3041, 3087, 3201, 4281, 6490, 7030, 8315, 8657, 8659,

8859, 8860, 9106, 9689, 10581, 11888, 11930). Other cricetids are naked and blind at birth (*Bolomys* 2371; *Calomys* 1502; *Calomyscus* 9175; *Delanymys* 5760; *Dendromus* 187; *Gerbillus* 8727; *Holochilus* 7192; *Malacothrix* 10158; *Meriones* 7098; *Peromyscus* 1474, 5886, 7885; *Reithrodontomys* 1474, 10587; *Tatera* 8500, 8501). The eyes of these altricial cricetids open at 6-8 d (*Zygodontomys* 87, 11373, 11931), 7-9 d (*Calomys* 4853), 8-13 d (*Reithrodontomys* 4915, 6300, 6391, 8189, 9107, 10099, 10587a; cf 5609), 8-23 d (*Desmodillus* 5684, 7824, 10158), 10-14 d (*Otomys* 11710a), 10-15 d (*Cricetulus* 2731, 4695, 6602, 7503, 9175), 10-20 d (*Onychomys* 504, 1474, 4986, 4991, 9366, 10588), 10-28 (*Peromyscus* 1236, 2727, 2910, 3897, 4125, 4609, 4780, 5748, 6149, 6150, 6302, 6493, 7303, 7304, 7311, 8698, 8907, 9114, 9603, 9802, 10573, 10574, 10575, 10576, 10577, 11288), 11-12 d (*Phodopus* 9516, 11285, 11548), 11-14 d (*Ochrotomys* 6497), 12-14 d (*Petromyscus* 2481), 12-15 d (*Baiomys* 977), 12-16 d (*Mesocricetus* 1071, 1382, 3571, 4098, 6656), 12-22 d (*Meriones* 1035, 2233, 2917, 6237, 6803, 7096, 7754, 7778, 8428, 8961, 9175, 9692, 11252, 11905; *Scotinomys* 4967), 13 d (*Calomyscus* 3587), 13-15 d (*Psammomys* 2233, 2234, 8731), 14 d (*Cricetus* 3571, 7433, 8457), 14-22 d (*Gerbillus* 109, 4348, 4351, 4713, 5760, 5785, 9175, 11397), 14-24 d (*Gerbillurus* 403, 2481), 14-25 d (*Tatera* 987, 5051, 7168, 8725, 9175), 14-30 d (*Cricetomys* 103, 1147, 3130, 5051, 5760, 7528, 10158), 16-20 d (*Mystromys* 4246, 7168, 10158), 22 d (*Malacothrix* 5887), 22-24 d (*Dendromus* 2572), or 28 d (*Desmodilliscus* 8696).

Females have 4 mammae (*Deomys* 7983; *Eliurus* 7983; *Mystromys* 4220, 10158; *Neotoma* 502, 1474, 1563, 2679, 3041, 4281, 4745, 8659, 10581, 11888; *Otomys* 331, 4495, 9945, 10158; *Otonyctomys* 4494; *Ototylomys* 4494, 4606; *Parotomys*, 9945, 10158), 4-6 mammae (*Deomys* 4495, 5760; *Peromyscus* 502, 1474, 1475, 1475, 2910, 3323, 4494, 9612; *Petromyscus* 9945; *Rheomys* 1475, 4965), 6 mammae (*Brachyuromys* 7983; *Calomyscus* 9175; *Gerbillus* 4497; *Onychomys* 502, 1474, 4745; *Podomys* 1474; *Reithrodontomys* 502, 1474, 4745, 4963; *Rhipidomys* 5111), 6-8 mammae (*Cricetomys* 331, 3130, 4495, 7528; *Gerbillurus* 2381, 9945), 8 mammae (*Cricetulus* 7814, 9175, 10516; *Cricetus* 7433, 7814, 7815, 8457; *Dendromus* 331, 2439, 4495, 5760, 9945, 10158; *Holochilus* 5111; *Malacothrix* 5887, 9945; *Meriones* 1138, 4497, 8961; *Myospalax* 8050; *Nectomys* 5111; *Oryzomys* 1474, 1475, 3024, 5111, 10572; *Oxymycterus* 596; *Pachyuromys* 6237; *Phodopus* 3334; *Scapteromys* 596; *Tatera* 331, 4495, 9945, 10158; *Zygodontomys* 5111), 8-12 mammae (*Sigmodon* 502, 1470, 1474, 7253, 10570), 8-16 mammae (*Steatomys* 331, 5760, 9945, 10158), 10-12 mammae (*Saccostomus* 331, 5760, 9945, 10158, 11082, 11607), or 12-16 mammae (*Mesocricetus* 278, 3571, 7814, 7815). Males have seminal vesicles, preputial glands, ampullary glands, penial spines, and a baculum (352, 1473, 1818, 2731, 3752, 4105, 4964, 6495, 8457, 9603).

Details of anatomy (1138, 2419, 2732, 3331, 7877, 7965, 8243, 9878, 10244), endocrinology (4166; *Gerbillus* 5707; *Mesocricetus* 642, 2270, 6324, 8085; *Neotoma* 1884; *Peromyscus* 114, 751a, 3253, 4120, 5800, 10795; *Phodopus* 10768; *Psammomys* 5706, 5708), conception (3433, 8510), spermatogenesis (94, 989a, 7797c, 7843), copulation (2507, 2540, 3096, 5676; *Calomys* 667, 2188; *Cricetomys* 3130; *Meriones* 84, 1452a, 2528, 4928, 4929, 8858; *Neotoma* 2504, 2508, 3310; *Onychomys* 2519, 2533, 4991, 6209, 9366; *Oryzomys* 2502; *Peromyscus* 1898, 1899, 2503, 2509, 2510, 2511, 2512, 2515, 2516, 2517, 2526, 2534, 2535, 6607, 10119; *Phodopus* 9516; *Psammomys* 2234; *Reithrodontomys* 2331; *Tatera* 8728; *Tylomys* 4460), birth (998a, 2481), pregnancy failure (83, 1318, 2520, 2521, 2523, 2966, 4205, 4610, 5673, 7962, 7967, 9229, 10202, 11608, 11969, 11970), birth (1853, 5001, 8096, 8698, 10573), geographic variation (1563a), reproductive energetics (3812, 3814, 6975, 6976, 7030, 7303, 7305, 8147, 8915, 8915, 9424, 9619, 10322, 10843, 11548), and light (629, 10947, 12114; *Cricetulus* 6951; *Cricetus* 1571; *Meriones* 2614, 9451; *Mesocricetus* 1645, 4948, 4951, 4952, 7486, 8516, 8635, 10818a; *Onychomys* 3533; *Peromyscus* 684, 2265, 2706, 3412, 3804, 3805, 4002, 4554, 5019a, 5367, 5369, 5370, 5371, 5894, 6659, 6661, 6810, 8455, 8455a, 8756, 10245, 10751, 10751a, 11547, 11648, 11649, 11650, 11651, 11652; *Phodopus* 1183, 1643, 1644, 2803, 3864, 4885, 4890, 10252, 10382, 12008, 12009; *Sigmodon* 1837, 5368, 7254; *Zygodontomys* 4573a) are available.

NEONATAL MASS

Akodon dolores	f: 3 g; from graph	7213
	m: 3 g; from graph	7213
Akodon longipilis	3.3 g	
Akodon molinae	m: 3 g; from graph	7213
	f: 3 g; from graph	7213
Baiomys musculus	1.4 g; n3 emb, 1 lt	2341

Baiomys taylori	1.1 g; n21; <12 hr old	977
	1.20 g; sd0.08; n47	5064
	f: 1.2 g; n94	7545
	m: 1.2 g; n100	7545
Beamys hindei	3.2 g; r2.1-4.3; n13	2912
Bolomys lasiurus	3 g	2371
	f: 3.65 g; ±1.1; n10, 1 d	7193
	m: 3.70 g; ±1.1; n10, 1 d	7193
Calomys callosus	1.3-2.5 g	8430
	f: 2.2 g; n21, 1 d	7189
	m: 2.4 g; n21, 1 d	7189
	2.5 g	5474
	m: 2.7 g; n39	7545
	f: 2.8 g; n31	7545
Calomys lepidus	f: 2.0 g; n85	7545
	m: 2.1 g; n90	7545
Cricetomys gambianus	r14-36.4 g; n33	103
	19.5 g; r19-20; n2	4329
	f: 22.5 g; r22-23; n2	3130
Cricetulus barabensis	2 g	4695
Cricetulus griseus	1.5-2.5 g	7503
Cricetulus triton	≤5.25 g	6602
Cricetus cricetus	3.18 g; r2.4-3.8; n9	9440
	~7 g	3571
Desmodilliscus braueri	0.9 g	8696
Desmodillus auricularis	1.84 g; n5	7824
	4.4 g; n19	5684
Eligmodontia typus	1.725 g; r1.5-2.8; n16 yg, 4 groups	6807
Gerbillurus paeba	1.9 g; n23	2481
	2.3 g; n≥5	403
Gerbillurus tytonis	1.9 g; n22	2481
Gerbillus gleadowi	1.9 g; r1.8-2.0; n4	8727
Gerbillus perpallidus	2.24 g; sd0.29; n19 lt	11397
Gerbillus pyramidum	2-3 g; from graph	4348
	2.5 g	4713
Holochilus brasiliensis	4-6 g	7192
Malacothrix typica	1.1 g; sd0.32	5887
	1.3 g; r1.1-1.5; n2	10154
Megadontomys thomasi	4.5 g	6302
Meriones crassus	2.5-4.5 g	8428
	3.2 g; sd0.13	6803
Meriones libycus	4-7 g	8428
	5 g	11905
Meriones persicus	5 g	8428
Meriones shawi	3-5 g	9451
	3.5-6.0 g	8428
Meriones tamariscinus	~2.5 g; n6; from graph	8961
	3.55 g; mode 4.0-4.5; r2.5-6.5; n68	1035
Meriones tristrami	3 g	8428
Meriones unguiculatus	r2.5-3.3 g; means by lt sz & sex; n1220, 357 lt	7964
	2.8 g; sd0.28; n22 lt	7096, 7098
	3 g	7778
	3.0 g	6867
Meriones vinogradovi	3 g	8428
Mesocricetus auratus	1.9 g	4830
	m: 2.179 g; sd0.5074; n39	1071
	f: 2.225 g; sd0.4797; n40	1071
	2.5 g	4830
	m: 2.92 g; r2.0-4.0; n54, 1 d	8643
	f: 2.96 g; r2.0-3.8; n56, 1 d	8643
Mystromys albicaudatus	6.5 g; r5.0-7.8; n39	4246, 7168
Neotoma albigula	10.9 g; r9.3-12.5; n>9	9106
	f: 11.6 g; sd1.3; n17	9689
	m: 12.0 g; sd1.3; n10	9689
Neotoma cinerea	13.5 g; r12.0-17.3	2909

Neotoma floridana	10.4 g; r9.8-11.0; n5	3285, 8859
	r10.5-14 g; n6	10581
	12.25 g; r12-2-12.3; n2	7649
	14 g; r11.8-14.8; n21	4281
	14.8 g; ±0.3	7030
	~15 g	8659
	15.3 g; n16	5892
Neotoma fuscipes	10 g; n5	2679
	r12.5-13.8 g; n6 yg, 3 lt	11888
	~14 g	6490
Neotoma lepida	4.6 g; r4.4-4.8; n2; est premature	8860
	6.1 g; se0.05; litter size 5	1563
	f: 9.5 g; sd1.5; n10	9689
	m: 9.6 g; sd0.9; n8	9689
	10.4 g	2906
	10.5 g; se0.24; litter size 1	1563
Neotoma mexicana	f: 10.0 g; r8.9-11.9; n6	8105
	m: 10.8 g; r10.0-11.7; n6	8105
Neotoma micropus	10-13.4 g	1242
Neotoma stephensi	10.6 g; se1.1; n14	11232
Neotomodon alstoni	3.4 g; sd0.67	8096
Nyctomys sumichrasti	4.7 g ea; n2	946
Ochrotomys nuttalli	2.7 g; r1.8-3.6; n47	6497
Onychomys leucogaster	2.2 g; r1.7-2.7; n22 yg, 7 lt	10588
	2.8 g	4986
Onychomys torridus	2.34 g; r2.1-2.6; n7	4991
Oryzomys capito	3.7-4.0 g	11931
Oryzomys palustris	3.14 g; r2.35-4; n13	10572
Oryzomys subflavus	5.89 g; r4.42-7.51	11305
Otomys denti	~20 g	2571
Otomys irroratus	12.5 g; r9.6-15.5; n28	2338, 2339
	14-17 g	2446a
	f: 14.5 g; r13-16; n2	2571
Otomys sloggetti	10.6-12.2 g; n12	11710a
Otomys typus	~12.5 g	5760
Otomys unisulcatus	12.3 g; n1 emb possible near term	11267a
Ototylomys phyllotis	10.2 g; se0.3; n86	4606, 4607
Peromyscus boylii	2.2 g; n1	502
Peromyscus californicus	4.31 g	7013
	4.92 g; ±0.78; r3.6-5.8; n19	10573
Peromyscus crinitus	2.2 g; r1.8-2.6	2910
Peromyscus eremicus	2.23 g; r1.85-2.60	1236
	2.54 g; ±0.65; r2.1-2.9; n9	10573
Peromyscus gossypinus	2.19 g; ±0.33; n45	8698
Peromyscus hooperi	2.62 g; n16	9603
Peromyscus interparietalis	2.74 g; r2.55-3.00	1236
Peromyscus leucopus	1.5-2 g	5210
	1.73 g; se0.03; n66	3323
	1.79 g; se0.02; n232 yg, 52 lt	7304
	1.80 g; sd0.25; n101	4125
	1.8 g; n312	7303
	1.87 g; ±0.03; r1.4-2.4; n114	10573
	2.5 g; sd0.38; 2 d	4780
Peromyscus maniculatus	r0.9-1.6 g; n16	10574
	1.295 g; se0.027; n37 near term	9225
	1.46-1.91 g; n1835, 389 lt; r across litter size	7697
	1.48 g; n11	7013
	1.58 g; n73	7698
	1.60 g; n296	7699
	1.62 g; ±0.13; r1.3-1.9	10576
	1.63 g; ±0.12; r1.1-2.3	10577
	1.65 g; se0.1	7305
	1.65 g; sd0.178; n14	4218
	1.65 g; n103	7698
	1.7 g	5658

	1.70 g; sd0.178; n18	4218
	1.70 g; se0.16; n165	7308
	1.73 g sib pairs; se0.04; 1.85 g non sib pairs; se0.03	4749
	1.73 g; r0.8-2.4; n248	10573
	1.74 g; n57	7698
	1.77 g; sd0.87; n24	4218
	1.80 g; ±0.68; r1.3-2.2; n≥13	10575
	1.8 g; se0.02; n80	6493
	1.86 g; se0.039; n18	5751
	1.95 g; se0.16; n257	7308
	1.98 g; se0.082; n18	5751
	1.98 g; se0.05; n35, lt size 5; varies w/lt size	7312
	f: 2.0 g; se0.1; n37, 1 d	2727
	m: 2.1 g; se0.1; n55, 1 d	2727
	f: 2.6 g; n77	7545
	m: 2.6 g; n117	7545
	m: 2.7 g; n104	7545
	f: 3.1 g; n102	7545
Peromyscus megalops	3.9 g; r2.9-5.0; n3	6302
Peromyscus melanocarpus	4.5 g; se0.1; n41	9114
Peromyscus melanotis	1.67 g; ±0.20; r1.3-2.0; n44	10573
Peromyscus mexicanus	4.4 g; se0.2; n17	9114
Peromyscus perfulvus	3 g	4609
Peromyscus polionotus	1.2 g	6302
	1.36 g; se0.029; n37	9225
	1.41-1.70 g; across lt sz	5576
	1.59 g; se0.04; n44	5577
	1.70 g; sd0.158	11725
	1.74 g; sd0.268	11725
	m: 2.10 g; sd0.30; n50, 1 d	1603
	f: 2.19 g; sd0.28; n60, 1 d	1603
Peromyscus truei	2.31 g; ±0.42; r1.7-3.0; n21	10573
	2.32 g	7013
Peromyscus yucatanicus	2.5 g; se0.0; n92	6149
Petromyscus collinus	2.1 g; n5	2481
Phodopus sungorus	1.67 g; ±0.18; n many	11548
	2.5 g; n15	11285
Podomys floridanus	2.4 g; r1.9-2.9; n15	6301
Psammomys obesus	5.5 g; sd0.6; n9	8731
	~7 g	3513
Reithrodontomys fulvescens	1.11 g; r1.07-1.15; n2	10587
Reithrodontomys humulis	r0.82-1.12 g; n8	2793
	1.2 g; r0.8-1.6	6300
	1.2 g; n27	5609
Reithrodontomys megalotis	1-1.5 g	5210
	1.5 g	10587a
Reithrodontomys montanus	0.91 g; r0.88-0.94; n2 dwarfs	6391
	1.1 g; r1.0-1.3	6391
Rhombomys opimus	4.5-5 g	7778
Saccostomus campestris	2.4 g; sd0.47; n173	11607
	2.8 g	2861
Scotinomys teguina	~0.5 g	4967
Scotinomys xerampelinus	~0.5 g	4967
Sigmodon hispidus	6.19 g; sd0.58; n10 1 d	9572
	6.3 g; sd0.82; n10 1 d	9572
	6.6 g; ±0.4	7030
	6.6 g; se0.1	8915
	6.7 g; se0.12	6975
	6.8 g; se0.07	6975
	7.23 g; r6.5-8; n19	10570
Sigmodon ochrognathus	4.5-6.6 g	4898
Steatomys pratensis	1.55 g; r1.5-1.6; n2	3693
Tatera afra	~4.5 g	7146
Tatera brantsii	4.4 g; n1	7168

	~5 g	2401
	6 g; n1 fullterm	7146
Tatera indica	3 g	8725
Tatera leucogaster	2.8 g; r2.6-3.0; n7	7797a, 7797b
Tylomys nudicaudus	10.2 g; ±0.3; n86	4607
	19.9 g; r15-24; n47	10755
	20.2 g; se0.3; n95	4606, 4608
Zygodontomys brevicauda	3.1-4.4 g	11931
	f: 3.5 g; ±0.15; n10	87
	m: 3.6 g; ±0.12; n15	87
	3.6 g; r2.8-4.8; n76 lt	11373
NEONATAL SIZE		
Beamys hindei	35.5 mm	298
Bolomys lasiurus	42 mm	2371
Cricetomys gambianus	HB: 69-87 mm	5051
	f: 89 mm; r89-89; n2	3130
Cricetulus griseus	30 mm	2731
Desmodillus auricularis	HB: 28.2 mm; n5	7824
Gerbillurus paeba	HB: 29.6 mm	403
	HB: 31.8 mm; n10	2481
Gerbillurus tytonis	HB: 34.2 mm; n17	2481
Gerbillus pyramidum	HB: 33 mm	4348
Malacothrix typica	HB: ~26 mm	5887
Meriones meridianus	30.5 mm; se0.5	11252
Meriones unguiculatus	37.1 mm; n22 lt	7096
Myospalax myospalax	80-100 mm	8050
Mystromys albicaudatus	HB: 53 mm	4246
Neotoma floridana	87.4 mm; n5	8859
Neotoma lepida	36.4 mm; r34-38.9; n2 est premature	8860
Neotomodon alstoni	CR: 29 mm; largest emb	11730
	40.5 mm; sd5.3	8096
Nyctomys sumichrasti	f: HB: 52.2 mm; r50-54.4; n2	946
Ochrotomys nuttalli	TL: 50.8 mm; r38-58; n47	6498
Otomys irroratus	HB: 71.1 mm; r64-76	2339
Ototylomys phyllotis	CR: 41-45 mm	4607
	HB: 67 mm; ±3.2; n86	4606
	TL: 119.5; ±0.9; n86	4606
Peromyscus gossypinus	TL: 47.16 mm; ±0.22; n45	8698
Peromyscus leucopus	32.1 mm; sd2.7; n101	4125
Peromyscus maniculatus	23 mm; se0.2; n37	9225
Peromyscus megalops	39.5 mm; r38-41.5; n2	6302
Peromyscus melanocarpus	44.8 mm; se0.2; n41	9114
Peromyscus mexicanus	43.9 mm; se0.7; n17	9114
Peromyscus polionotus	23.4; se00.2; n37	9225
Petromyscus collinus	HB: 33.7 mm; n3	2481
Reithrodontomys megalotis	7-8 mm	5210
Saccostomus campestris	47 mm w/tail	2861
Scotinomys teguina	~32 mm	4967
Scotinomys xerampelinus	~32 mm	4967
Steatomys pratensis	r73-75 mm; n3	4329
Tatera brantsii	33.8 mm; n1 fullterm	7146
Tatera inclusa	HB: 45 mm; n1	5050
Tylomys nudicaudus	CR: 50-60 mm	4607, 4608
	HB: 84.3 mm; se0.7	4606, 4608
WEANING MASS		
Beamys hindei	59.2 g; n10	2912
Calomys callosus	f: 9.4 g; n21, 16 d	7189
	m: 9.6 g; n21, 16 d	7189
	18 g	5474
	~20 g; 3 wk	5474
Cricetulus barabensis	14 g	4695

Cricetulus griseus	10.2 g; r7.9-15.4; 21 d	8685
	f: 15.1-16.8 g	468
	m: 15.9-17 g	468
Cricetus cricetus	f: 101.2 g; ±13; n63	9085
	m: 108.9 g; ±11.97; n60	9085
Gerbillurus paeba	12-15 g; ~20 d; from graph	403
Gerbillus perpallidus	17.23 g; n16	11397
Gerbillus pyramidum	11 g; 19-23 d	4351
	~20 g; from graph	4348
Meriones libycus	45 g; 30 d	11905
Meriones shawi	20-30 g; depends on lt sz	9451
Meriones tamariscinus	11.5 g; r10-13; n2	1035
	20 g; 21 d	8961
Meriones unguiculatus	12-15 g; 20 d	9692
	12.9 g; sd2.29	7098
	13 g; 21 d	6867
	f: 17.8 g; se0.3; r12-23; n82	7969
Mesocricetus auratus	22.93 g; sd6.9; n43, 7 lt, 21 d	4098
Mystromys albicaudatus	40 g	4220
Neotoma albigula	103.3 g; r86.1-113.9; n4, 66 d	9106
Neotoma cinerea	75 g	3087
	f: 85-132.3 g	2909
	m: 117.7-150.2 g	2909
Neotoma floridana	45.9 g; ±2.3	7030
Neotoma lepida	34.4 g; se0.16; lt sz 5	1563
	35.5 g; se1.64; lt sz 1	1563
Neotoma stephensi	35 g	11231
Ochrotomys nuttalli	12.5 g; r6.8-13.1; n40, 3 wk	6497
Onychomys leucogaster	13.1 g; n3, 23 d	4986
Onychomys torridus	9.2 g; n1, 20 d	4991
Oryzomys palustris	22 g; from graph	8231
Oryzomys subflavus	~35 g	11305
Otomys sloggetti	25-30 g; n10, 2 wk; from graph	11710a
Otomys irroratus	~30 g; n18; from graph	2338
Peromyscus californicus	f: 20.5 g	4119
Peromyscus crinitus	13.2-15.0 g; 4 wk	2910
Peromyscus hooperi	10.07 g	9603
Peromyscus leucopus	7.9 g; n100, 18 d	7303
	8.9 g; sd1.61; 20 d	4780
	8.9 g; se0.12; r6.2-13.1; n102	2492
	9.0 g; se0.4; n8; leave nest	11288
	m: 11.5 g; n45	8455
Peromyscus maniculatus	f: r7.10-8.80; n3 samples	4218
	7.3 g; se0.3; n8; leave nest	11288
	8 g	5658
	8.00 g; se0.492; n16,; 20 d	5751
	8.08 g; se0.310; n17, 20 d	5751
	m: 8.47 g; se0.25; n26, 20 d	10795
	f: 8.6 g; se0.1; n18, 21 d	2727
	8.65-11.76 g; n433 lt, 23 d; by litter size	7697
	f: 8.95 g; r8.8-9.1; n2, 21 d	10723
	m: 9.3 g; se0.2; n33, 21 d	2727
	m: 11.1 g; se0.3; n22, 21 d	2727
	f: 11.2 g; se0.4; n19, 21 d	2727
	11.6 g; se0.25; n64, 4 wk	6493
Peromyscus polionotus	5.29-6.57 g; by litter size	5576
	m: 5.56-8.41 g; 17-25 d	1603
	f: 5.71-8.76 g; 17-25 d	1603
Peromyscus yucatanicus	13.1 g; n50, 28 d	6149
Phodopus sungorus	f: ~10 g	3249
	m: ~17 g; 21 d	3249
	17.4 g; r11.7-21.5	5443
Podomys floridanus	~10-15 g; 3-4 wk	6301
Psammomys obesus	~40 g; 28 d	8731

Reithrodontomys fulvescens	3.0-3.5 g; n4 lt	8189
Reithrodontomys montanus	5.4 g; r4.3-7.5; 15 d	6391
Sigmodon hispidus	10-20 g	3850
	17.1 g; ±0.6	7030
	18.2 g; se0.46	6975
	21.7 g; se0.32	6975
	~27 g; 18 d; from graph	9572
Tatera leucogaster	16-22 g; 18-28 d	7797a, 7797b
Tatera valida	30 g	7794
Tylomys nudicaudus	20-60 g	10755
Zygodontomys brevicauda	~22 g; r15.9-27.6	11373
WEANING SIZE		
Beamys hindei	HB: 105-122 mm	4330
Gerbillurus paeba	HB: 48-58 mm; 20 d; from graph	403
Peromyscus leucopus	144.4 mm; n45	8455
LITTER SIZE		
Akodon azarae	3.5 yg; ±0.10; mode 3; r1-7; n222	2409
	4.1 emb; r3-7; n13 f	596
	4.6 yg	2221
	5.0 emb; r3-7; n12 f	596
Akodon boliviensis	5 emb; n1 f	8302
Akodon dolores	3.9 yg; r2-6	7213
Akodon longipilis	3.7; r2-5; n9 lt	4054
	3.78 emb; se0.11; r2-5; n51 f	8305a
	5 emb; r4-6; n2 f	7241
Akodon molinae	3.65 yg; mode 4-5; n275 lt	7212
	3.9 yg; r2-6	7213
Akodon olivaceus	5.08 emb; se0.38; r4-8; n12 f	8305a
	5.5 emb; se0.27; n9 f	3551
	5.5; r3-8; n18 lt	4054
	5.6 emb; sd1.1; n11 f	7241
Akodon urichi	5.0; r4-6; n9	7995, 7998
Andinomys edax	3 emb; n1 f	8302
Anotomys leander	1 emb; n1 f	11372a
Anotomys trichotis	1 emb; n1 f	11372a
Auliscomys boliviensis	3-4 emb	8302
Auliscomys micropus	4.07 emb; se0.26; r1-7; n27 f	8305a
	4.5 emb; r4-5; n2 f	4054
Baiomys musculus	1-4 emb	4228
	2.0	9176
	2 emb ea; n4 f	1475
	2.92 emb; n26 f	8188a, 8190
	3 emb; n1 f	2341
	3.9 emb; n9 f	11388
Baiomys taylori	1-5	1474
	2.2 weaned	7544
	2.49 yg; sd0.072; r1-5; n58 lt	8805
	2.5; r2-3; n4 lt	1785
	2.6 emb; r2-4; n8 f	8963
	2.72 yg; se0.13; r1-5; n39 lt	977
	2.8 yg	7544
	2.8; r1-5	217
Beamys hindei	2.8 yg; mode 3; r1-5; n39 lt	2912
	3.5; r3-4; n2 lt	298
	mode 4 yg; 4-7	4330
Blarinomys breviceps	1.25 emb; r1-2; n4 f	21
Bolomys lasiurus	3.3; sd0.5; r3-4; n7 lt	10494
	3.62 yg; mode 4; r1-6; n69 lt, 20 f	2371
	5.6; r2-9	7193
	mode 6.0 emb; max 13; many f	5540
Bolomys obscurus	3.5 emb; r3-4; n2 f	596
	5 ps; n1 f	596
	6.2 emb; se0.3; n32 f	7347

Calomys callosus	4.3; n324 lt	7191
	4.4; sd2.3; r2-8; n5 lt	10494
	4.5 yg; ±1.8; r2-9; n27 lt, 13 f	7188, 7189
	5.0 weaned	7544
	5.5 yg	7544
	5.5; r1-10; n65 lt	8430
	6.21 yg; n112 lt	5474
Calomys laucha	3.75 emb; r3-4; n4 f	1502, 4682
	4.0 yg; r3-5; n2 f	4682
	5.3 yg	7347
	5.7 or 6.0 emb; r4-8; n7 f	596
	6.2 emb; se0.1; n179 f	7347
Calomys lepidus	3.9 weaned	7544
	4.4 yg	7544
Calomys musculinus	5.4 yg; ±0.11; r1-15	2410
	7.5 emb; se0.2; n114 f	7347
Calomyscus bailwardi	2; n1 lt	9175
	3-5	3587
	4.29 ps; r2-7; n7 f	6293
Calomyscus mystax	3-7	9488
Cricetomys emini	1 emb; n1 f	2446a
Cricetomys gambianus	1-5	228
	1 emb; n1 f	2452
	2-4 juveniles	8078
	mode 3-4 yg	7528
	3 yg; n1 f	337
	3 emb, yg; n2 f	4329
	3.0; r2-4; n2 lt	1147
	3 yg; r2-4; n2 lt	2446a
	4; n1 lt	2439
	4 emb; n1 f	331
	4; r1-5	103
Cricetulus barabensis	5.33; r1-9; n12 lt, n5 f	4695
	7.1 emb; n106 f	3334
Cricetulus curtatus	5.0 emb; r4-6; n2 f	10516
	5.7 emb; n10 f	3334
Cricetulus eversmanni	4-6	4645
Cricetulus griseus	4.8 yg; 4.7 born live; mode 5; r1-8; 4.2 wean; n500 lt	468
	4.8 born alive, 4.35 wean; n31 lt, 10 pr	8685
	5.0	1693
	5.52 yg; mode 6; r1-9; n190 lt	7503
	5.53 yg; mode 6; r2-10; n39 lt	2730
	6.1 yg; r1-11; n>500	1554
	7.6 ovulations; ±0.2; r5-10	8510
	7.92 cl; ±0.41	2731
Cricetulus longicaudatus	5-6	3335
	6.1 emb; n36 f	3334
Cricetulus migratorius	2.3; r1-4; n3 lt	12128
	4-6	9175
	5.57 ps, emb; r4-7; n7 lt	6293
	6; r3-10	4645
Cricetulus triton	3; r1-5	12128
	mode 6; most 4-6	6602
Cricetus cricetus	3.4 emb; sd0.3; n25 f	3951
	5.83; r3-11; n6	9440
	mode 6-12; r4-21	8457
	6.75 yg; n6 lt	7436, 9085
	6.8 yg; sd1.6; n5 lt	3951
	mode 7-8; r5-12	7815
	7.6; r4-10; n27 lt	11319
	8.79 yg; ±1.35; n14 lt	7436, 9085
Delanymys brooksi	3 emb ea; n2 f	2446a
	3-4	5760
Dendromus kahuziensis	3.5; r3-4; n2 lt	187
Dendromus melanotis	mode 2-4; max 7	10158

	3-6	2449
	3-8 yg	5760
	3.5 yg; n4 lt	2572
	3.67 emb; r3-5; n9 f	2444, 2446a, 2452
	3.8 emb; n5 f	2572
	4.4 emb; r4-6; n5 f	331, 10154
	8 yg; n1 f	186
Dendromus mesomelas	2.75; max 4; n14 lt	8855
	3-8 yg	5760
	4 yg; n1 f	9945
	4.0 emb; n3 f	2572
	4.0 emb; r2-6; n3 f	4329
	5 emb; n1 f	2955
Dendromus messorius	3; n1 lt	187
	3 emb; n1 f	4495
Dendromus mystacalis	2.6 emb; mode 2-3; r2-4; n13 f	2754
	mode 3; max 4	2452
	mode 3-4; r3-7	5760
	mode 3-4; max 8	10158
	3 emb; n1 f	4925
	3.5 emb; r3-4; n2 lt	2572
	4 emb; r4-6	331
	6 yg; n1 lt	2572
	7 yg; n1 lt	187
Dendromus nyikae	3-8	5760
	4 emb; n1 f	4329
Deomys ferrugineus	1.59 emb; mode 2; r1-2; n32 f	2446a
	1.69 emb; mode 2; r1-2; n67 f	2573, 2574
	1.75; r1-2; n24 lt	8855
	mode 2 yg; r1-3	5760
Desmodilliscus braueri	2.6 yg; r2-3; n8	8696
Desmodillus auricularis	2-6	9945
	2 emb; n1 f	7823
	2 yg; r1-3; n5 lt	7822, 7824
	2.3 yg; r1-3; n3 lt, 1 f	5684
	3.9 emb; r1-7; n13 f	10154, 10158
Dipodillus simoni	4.68 yg; r1-7; n25 lt	6235, 6237, 6238
Eligmodontia typus	4.0; r2-7; n4	6807
	5.90 emb; se0.39; r3-9; n20 f	8307
	7.0 emb; r6-8; n2 f	4054
Gerbillurus henleyi	4.0; n2 lt	8134
Gerbillurus paeba	3-4	2381
	3.0 yg	7822
	3.5 yg; sd1.4; n11 lt	403
	3.7 emb; r2-5; n39 f	10154, 10158
	4 emb; n1 f	9945
	4.1 emb; n20 f	7823
	4.6 yg; n7	2481
Gerbilurus tytonis	4.4 yg; n7	2481
Gerbillurus vallinus	5 yg; n1 nest	10158
Gerbillus spp	3 emb; n1 f	299
Gerbillus andersoni	3.9; r3-7; n10 lt	8134
Gerbillus campestris	4.0 emb; mode 4; n2 f	7906
	5.0 emb; r3-6; n3 f	8134
Gerbillus cheesmani	8 ea; n2 lt	9175
Gerbillus dasyurus	3 ps; n1	10616
	4 emb; n1 f	4497
	4.4	109
Gerbillus gerbillus	3-5	12128
	4.3; r3-6; n7 lt	8134

Gerbillus gleadowi	2 yg ea; n2 lt	8727
	3; r2-5	9175
	3.5 emb; r3-4; n2 f	8727
	5.5; r5-6	8723
Gerbillus latastei	5 emb; n1 f	7906
Gerbillus nanus	3-7	5785
	3.25 ps, emb; r2-4; n4 f	6293
Gerbillus perpallidus	4.46; sd1.25; r1-8; n109 lt	11397
	5 emb; n1 f	8134
Gerbillus pyramidum	2 yg ea; n2 lt	4344
	3 ps; 5 emb	8134
	3.12; r1-6; n498 lt	4713
	3.5; r2-5; n2 lt	12128
	4.75 yg; r4-7; n4 lt	4348
Graomys griseoflavus	6.0; r3-8; n237 lt	2388
	8.1; r7-10; n7 lt	2387
Gymnuromys roberti	2 emb	7983
Habromys lepturus	1.9	9176
Hodomys alleni	1 emb; n1 f	3700
	2 yg; n1 lt	3700
Holochilus brasiliensis	1-5	7192
	4 emb; n1 f	596
	5 emb; n1 f	5111
Holochilus magnus	3 ps; n1 f	596
Holochilus sciureus	3.12 emb; r1-8; n354 f	10983
	3.47; r2-5; n30	10982
Hypogeomys antimena	1 yg	7983
Ichthyomys tweedi?	2 emb; n1 f	11372a
Irenomys tarsalis	3.5 emb; r3-4; n2 f	8305a
Isthmomys pirrensis	2.5 emb; r2-3; n4 f	11044a
Lophiomys imhausi	1 yg ea; n2 lt	2446a
	mode 2-3 yg	5760
Macrotarsomys bastardi	2.45 yg; r1-3; n11 lt, 5 f	6402
Malacothrix typica	3.9; r2-6; n10 lt	10154, 10158
	4.2 yg; r2-8; n9 lt	5887
Megadontomys thomasi	2.0	9176
	2.6; se0.19; max 4; n21 lt	2728
	3.5	6302
Meriones crassus	1 yg; n1 lt	2109
	3.3 emb, ps, nestlings; n10 lt	8134
	3.5 yg; r3-4; n4 lt	2233
	3.7; mode 4; r1-8	12128
	4-5	9175
	4.4 yg; sd0.23; 3.6 wean; sd0.33; n15 lt	6803
	4.56 yg; r3-6; n9 lt	6236, 6237, 6238
	max 8	8428
Meriones hurrianae	3-5 emb; n13 f	8720
	4.0 emb; r3-5; n2 f	10616
	4.0 emb; r3-5; n4 f	79
Meriones libycus	3-4	9175
	3.7 yg; r1-7; n3 lt	2233
	4 emb; n1 f	6293
	4 emb; n1 f	7906
	mode 5-6	7778
	5.3 yg; r2-8	7778
	8.25; r8-9; n4 lt	11905
Meriones meridianus	4-7	4645
	4.6 yg; r1-9	7778
	5.8 yg; r1-9	7778
	6.4 yg; r4-12	7778
Meriones persicus	2 ea; n2 lt	9175
	6 emb; n1 f	2404a
	max 7 yg	7778
Meriones sacramenti	max 6	8428

Meriones shawi	4.91 yg; r2-7; n11 lt	6236, 6237, 6238
	5.79 yg; n159 lt	9451
	5.85; r1-10; n159 lt	9451
	6.5; mode 6; r1-16	12128
	max 8	8428
Meriones tamariscinus	3.64; mode 4; r1-7; n66 lt	1035
	mode 4-5; r1-8	7778
Meriones tristrami	2-7 yg	7778
	5.8; se0.3; n9 f	2528
	max 6	8428
Meriones unguiculatus	2-5	10596
	3.2 wean; late maturing f	1863
	3.2; r1-7; n48 lt	4588
	3.42; mode 3-7; r1-9; n357 lt	7964
	4.0 emb; 47 d gestation	7971
	4.1-5.2; varies with f age	56
	4.1; r1-11	9692
	4.5 born, 4.4 wean; r1-9; n187 lt	6867
	4.7 wean; early maturing f	1863
	4.83 yg; n272 f	7973
	4.89 yg; r2-9; n27 lt	7754
	4.9 yg; r2-8; n35 lt	7969
	5.0 yg; r1-9; n284 lt	382
	5.04 yg; r1-9; n757 lt, 103 f	1768
	5.12 yg; sd1.62; r1-9; n198	84
	5.2 emb; 26 d gestation	7971
	5.3 yg; se0.58; late maturing f	1863
	6.0 yg; se0.31; early maturing f	1863
	6.3 yg; r1-9; n40 lt; 5.35 wean	2917
	6.4; r2-11	7778
	6.6 oocytes	6866
	7-9 cl	1138
Meriones vinogradovi	mode 6-8	7778
	max 8	8428
Mesocricetus auratus	1-12	4830
	5.8 emb; se0.8; left; 9.2 emb; se1.9; right; n15 f	7989
	mode 6-12; r3-15	3571
	6.1 wean; sd3.8; 21 d	4098
	6.41 yg; max 12; n34 lt	1382
	6.5 cl; se0.3; left; 9.3 cl; se0.1; right; n70 f	7989
	6.93 yg; se0.23; sd2.414; r2-15; n107 lt	1071
	8.6; se0.63; n24	6210
	9.26; sd2.86; mode 8-10; r2-16; n128 lt	278
	r10.3-14.3 cl; n5 wgt gps; 2-16 implantations	9160
	11.1 yg; se0.3; n25 multiparous	5061
	11.6 yg; se0.8; n25 primiparous	5061
	11.7 yg; sd2.1; n7 lt	4098
	12.5 emb; se0.6; 15.7 cl; se0.7; n11 f	6210
	13-15 emb	2419
Mesocricetus brandti	5.97; r1-13; n171 lt	6656
	14.0 emb; r13-15; n2 f	7815
Mesocricetus raddei	8-10	4645
Myospalax myospalax	mode 1-2 yg; r1-4	8050
	4-5 yg; hunters reports	8050
	6 emb ea; n4 f	8050
Myospalax psilurus	2 emb; n1 f	12105
	4-6	1817
Mystromys albicaudatus	2-5	10158
	2.9; n51 lt	7168
	3 yg	4220
Neacomys tenuipes	3.8; r2-5; n12	7995, 7998
Nectomys squamipes	2.5 yg; r2-3; n2 f	3077
	3 emb; n1 f	3077
	4-5 yg; n3 f	3077

	5 yg	3077
	6 emb; r5-7; n2 f	3077
Neotoma albigula	1.5 emb; r1-3; n14 f	522
	2 yg; n1 lt	6173
	2.1 emb; r1-3; n13 f	3262
	2.1 yg; mode 2; r1-3; n29 lt	3201
	2.25 emb; r1-3; n8 f	502
	2.5 yg; r2-3; n12 f	9689
	2.7 emb; r2-3; n3 f	11462
Neotoma alleni	1 emb; n1 f	3700
Neotoma cinerea	2.75 emb; r1-4; n4 f	3262
	3 emb; n1 f	4745
	3.2 emb; r3-4; n5 f	2415
	3.4 yg; mode 3; r1-6; n34 lt	2909
	3.6 emb; r3-4; n8 f	10954
	3.75 emb; mode 4; r1-5; n8 f	2615
	4; n1 lt	5001
	4 yg; n1 lt	5405b$_1$
	4.7 emb; r4-6; n3 f	11462
Neotoma chrysomelas	1 emb ea; n2 f	11044a
Neotoma floridana	1 emb; n1 f	6298
	mode 2-3; r1-4	11930
	2-4	1474
	2.0; r1-3; n10 lt	8659
	2.3 yg; mode 2; r1-5; n9 lt	8859
	2.6; ±0.2	7030
	2.8; mode 3; n5 lt	949
	3 yg; n1 f	8657
	3; r2-4; n7 lt	8315
	mode 3-4; r1-4; n5 lt	10581
	3.2; r2-5; n8 lt	5892
	3.2 emb, ps; r2-7; n50 f	3851
	3.41 yg; r2-4; n12 f	949
	3.5 yg; r2-5; n4 lt	8859
	3.67 emb; r3-4; n6 f	949
	4 emb; n1 f	1812a
	4 emb; n1 f	3262
	4 emb; n1 f	5405b$_1$
	4 yg; n2 lt	5405b$_1$
	4.25 ps; r3-6; n4 f	6298
Neotoma fuscipes	2 yg ea; n3 f	11888
	2.6 yg; r1-4; n18 nests	11289
	2.667 emb; r2-3; n3 f	3041
	2.8 yg; r2-3; n7 f	3041
	3 emb; n1 f	3593a
	3.0 yg; mode 2; r1-3; n10 lt	6490
	3.5 emb; r3-4; n2 f	11289
Neotoma lepida	2 emb; n1 f	3262
	2.29 yg; mode 2; r1-5; n100 lt	2906
	2.6 emb; r1-5	1563
	2.64 yg; r2-4; n14 f	9689
	2.7; ±0.7; r1-4	6708
	2.8 yg; r1-5 yg; n211 lt	2907
	2.86 emb; r2-3; n7 f	6560
	3 emb ea; n2 lt	6488
	3.11 emb; r2-4; n9 f	2415
	4 yg; n1 lt	10453
	4 yg; n1 lt	8860
	5 emb; n1 f	11462
Neotoma mexicana	1-2 emb	4745
	r1-4 emb; n3 f	523a
	1.33 emb; r1-2; n3 f	522
	2 emb ea; n2 lt	4962
	2.1 yg; r1-3 yg; n8 lt	8105

	2.44 emb; sd0.27; n14 yg f	1366
	2.52 cl; sd0.31; n19 yg f	1366
	2.57 emb; r2-4; n7 f	3262
	2.67 emb; r2-4; n3 f	11462
	3 emb; n1 f	502
	3.43 emb; sd0.35; r2-5; n19 f	1366
	3.47 ps; sd0.39; r2-5; n22 f	1366
	3.66 cl; sd0.31; r2-5; n45 f	1366
Neotoma micropus	2 emb; n1 f	522
	2 yg ea; n4 f	8962
	2 emb; mode 2; n5 f	4962
	2 yg ea; n11 lt	3201
	2.12 emb; r1-3; n8 f	949
	2.24 emb; n42 f	8962
	2.4 emb; r2-3; n5 f	3262
	2.63 emb; se0.77; mode 2; r1-4; n35 f	11696
	2.74 yg; se0.83; mode 2; r1-4; n38 lt	11696
	2.81 ps; se0.81; mode 3; r1-4; n63 f	11696
	2.85 yg; r1-4; n27 f	949
	3 emb; n1 f	502
	3 emb ea; n2 f	11462
Neotoma phenax	2 emb; n1 f	1471a
	mode 2 yg; r1-2	5423
	2.7 emb; r2-3; n3 f	5423
Neotoma stephensi	1.08; se0.07; n159	11232
	2 emb ea; n4 f	4900
Neotomodon alstoni	2 emb; n1 f	4228
	3.1 yg; mode 3; r1-6; n137 lt	8096
	3.4 emb; r2-5; n12 f	2341, 2346, 2351
Neusticomys monticolus	2 emb; n1 f	11372a
Notiomys macronyx	4.5 emb; r4-5; n2 f	8305a
Notiomys valdivianus	3.5 emb; r3-4; n2 f	8305a
Nyctomys sumichrasti	2.0	9176
	2 emb; n1 f	11388
	2.0 yg; r1-3; n11 lt	946
	2.5 emb; r1-4; n2 f	3704
	2.5 emb; r2-4; n4 f	2590
Ochrotomys nuttalli	2 yg; n1 lt	2863a
	2.63 emb, yg; ±0.10; r1-5; n71 lt, several sources 2.7 emb; n10 f; 2.54 yg; n61 lt	1027
	2.65 yg; mode 2-3; r1-4; n85 lt	6497
	2.7 yg	6498
	3.5 emb, yg; r3 yg, 4 emb; n1 f	5187
Onychomys leucogaster	r2-5; n10 lt	10588
	3 ps; n1 f	4525
	3.54 or 3.6 yg; r1-6 yg; n205 lt	2907
	3.7; r1-6; n153 lt	8559
	4 emb ea; n1 f	1812a
	4 yg ea; n3 lt	4986
	4 emb; r2-5; n10 f	11957a
	4.25 emb; r4-5; n4 f	502
	4.58 emb; sd1.02; r3-7; n36 f	3294
	4.7 ps; r4-6; n3 f	11957a
	4.7 emb; r3-8; n9 f	5405b$_1$
	5.0 emb; n2 f	522
Onychomys torridus	2.4; r1-5; n213 lt	8559
	2.6 emb, yg; r1-5; n105	10697
	3.0 emb; r2-4; n2 f	5409
	3.2 emb; r3-4; n5 f	522
	3.95 emb; r2-7; n21 f	10697
	4 emb; n1 f	502
	4 emb; n1 f	6560
	4 emb; r3-5; n4 f	1470
	7; n1 lt	2109

Oryzomys albigularis	3-4 emb	3024a
Oryzomys alfaroi	3 emb; n1 f	2590
	3.2-4.0; n2 locations	9176
	3.5 emb; r3-4; n4 f	11388
	4 emb; n1 f	5414
	4.0 emb; n2 f	3210
	4.0 emb ea; n2 f	4228
	4.7 emb; r4-6; n3 f	2228a
	7 emb; n1 f	7624
Oryzomys argentatus	1-5	10809
Oryzomys bicolor	3 emb; n1 f	3024
	3.5 emb; r3-4; n2 f	5111
Oryzomys bombycinus	3.33 emb; r2-4; n3 f	8550
Oryzomys caliginosus	1.5 emb; r1-2; n2 f	3317
Oryzomys capito	2 emb; n1 f	5111
	2.8 yg; max 5; n8 lt	3122
	2.9 yg; max 4; n11 lt	3122
	3.5 wean; max 6; n33 lt	11931
	3.68 yg; sd0.17; n35 f	3316
	3.75 emb; r3-4; n4 f	3024
	3.8; r2-6; n10	7995, 7998
	3.92 emb; 95%ci 0.56; r2-5; n14 f	3319
Oryzomys concolor	2.5 emb; r1-4; n2 f	5111
	3.6; r2-5; n7	7995, 7998
Oryzomys couesi	3.0 emb; r2-4; n3 f	3210
	3.3 emb; r1-6; n3	2590
	3.5 emb; r2-5; n2 f	11388
	4 emb; n1 f	2228a
	4 emb; n1 f	9542a
	4.125 emb; r1-6; n8 f	7624
	5 emb; n1 f	2351
Oryzomys delicatus	3.0 emb; r2-4; n2 f	5111
Oryzomys delticola	3.4 emb; r2-4; n5 f	596
Oryzomys flavescens	3.4 yg	7347
	5.1 emb; r3-7; n8 f	596
	5.2 emb; se0.3; n22 f	7347
Oryzomys fulvescens	3 emb ea; n2 f	1475
	3.75 emb; r3-4; n4 f	4228
	4 emb; n1 f	217
	4 emb; n1 f	5423b
	4 emb; n1 f	5423b
	4.0 emb; r3-6; n4 f	11388
	4.0 emb; r2-6; n8 f	5414
Oryzomys longicaudatus	4.5 emb; r3-6; n4 f	3551
	4.9 emb; r3-9; n44 f	4054
	4.93 emb; se0.23; r2-11; n44 f	8305a
	7.29 emb; r6-9; n7 f	8556
Oryzomys melanotis	3.3 emb; r3-4; n3 f	4228
	4 emb; n1 f	5423b
	4.6 emb; r3-6; n3 f	2590
Oryzomys nigripes	2-8 emb	5540
	3.0; ±1.3; r1-5; n18 lt	7190
	3.6 yg	2221
Oryzomys palustris	2-4 yg	11930
	3.0-3.8; n2 locations	9176
	3 emb; n1 f	5423b
	3.6 wean; mode 1-2; n149 lt	8231
	3.7 emb; r3-4; n3 f	1475
	4 emb; n1 f	5188
	4.2 emb; r3-7; n6 f	4228
	4.8 emb; r2-7; n20 f	7810
	5 emb; n1	8907
	5 emb; n1 f	950
	5 emb; n1 f	217

	5 emb; r3-7	1290
	5 yg; ±0.2; r4-6; n10 lt	2002
	5.9 cl; r3-12; n20 f	7810
Oryzomys subflavus	1-8 emb	5540
	4.1 yg; r3-6; n14 lt	11305
	5.5; n2 lt	10494
Osgoodomys banderanus	3 emb; n1 f	2341
Otomys angoniensis	2 emb; n1 f	337
	2.0 emb; r1-3; n3 f	4329
	2.5; r2-4; n10 f	10158
	2.8 emb; r1-5; n109 f	10712
	2.9 emb; r2-5; n14 f	8965
	3 emb; n1 f	4925
	3 emb; n1 f	334
	3.5 emb; r3-4; n2 f	2454
	3.7 emb; r3-5; n3 f	10154
	4 yg; n1 f	5051
Otomys denti	1.07; mode 1; r1-2; n14 lt	2571
	2 emb; n1 f	5051
	2.5; r1-4; n2 lt	4329
Otomys irroratus	1 emb ea; n2 f	4925
	1.25; max 3; n23 lt	8855
	1.33 emb; r1-2; n3 f	4925
	1.37 emb; mode 1; r1-3; n73 f	2571
	1.48; ±0.61; n81	1312
	1.59 yg; n44 lt	8384
	1.7 emb; r1-2; n3 f	2444, 2452
	2 emb; n1 f	4495
	2; n1 lt	185
	2.33 emb, yg; mode 2; r1-4; n39 f	2338, 2339
	3.2 emb; r1-7; n9 f	8965
Otomys maximus	3.7; r3-5; n3 f	10158
Otomys sloggetti	1.44 yg; se0.17; r1-2; n9 lt, 5 f	11710a
Otomys typus	1-2; r1-4	5760
	2 emb; n1 f	337
Otomys unisulcatus	1 emb; n1 f	11267a
Ototylomys phyllotis	1.75; r1-2; n8 f	1475
	2 emb; n1 f	6105
	2 ps; n1 f	950
	2 emb ea; n2 f	4494
	2 emb ea; n4 f	11044a
	2.0; r1-3; n5 f	7624
	2.0 emb; r1-3; n5 f	7624
	2.15; r1-5; n40	6270
	2.25 emb; r1-3; n16 f	5423b
	2.29 emb; mode 2; r1-4; n75 f	2590
	2.4 yg; r1-4; n36 lt	4606, 4607
	2.43; r1-4; n7 f	950
Oxymycterus rutilans	2.1 emb; r1-4; n7 f	596
	3.1 yg	2221
Pachyuromys duprasi	3.36; r1-6; n11 lt	6237
	3.5; r3-4; n2 lt, n1 f	4674
	5; r1-9	12128
Parotomys brantsi	2.1 emb; r1-3; n9 lt	10154
Parotomys littledalei	3 emb; n1 f	9945
Peromyscus attwateri	3 emb; n1 f	9601
	3.3 emb; r3-4; n3 f	11044a
	4.0 emb; r3-5; n3 f	9601
Peromyscus aztecus	3.0 emb; r1-4; n4 f	11044a
	3.5	9176
	4 ps; n1 f	11044a
Peromyscus boylii	2-6	1474
	2 emb; n1 f	217
	2 emb; n1 f	11228a

	2-3 emb; n4 f	502
	2.2-2.3; n2 locations	9176
	2.80 yg; sd0.41; mode 3; r1-4; n12	1363
	3-5 emb	2228a
	3 emb; n1 f	4962
	3 emb; n1 f	4228
	3 emb; r1-4; n8 f	523a
	3.1 emb; se0.12; n42 f	5243
	3.28 ps; sd0.35; n99	1363
	3.39 emb; sd0.81; n23 f	1363
	3.40 cl; sd0.42; n179	1363
	3.5 emb; r3-4; n4 f	6560
	4 emb; n1 f	1470
	4 yg; n1 lt	502
	5 emb; n1 f	11388
Peromyscus californicus	1.79; r1-2; n34 lt	7013
	1.87 yg; se0.09; r1-3; n15 lt	10573
	1.99 yg; r1-3; n190	9255
	1.9; se0.04; max 4; n493 lt	2728
	2.11; r1-3; n19 lt	7013
	2.5; ±0.8; r2-4; n6 f	6708
Peromyscus crinitus	2.9; se0.07; max 6; n206 lt	2728
	2.96 yg; r1-6; n122	9255
	3 emb; n1 f	2415
	3.07 yg; mode 4; r1-5; n130 lt	2910
	3.6 emb; r3-5	4226
	4.0 emb ea; n3 f	6488
Peromyscus difficilis	2.5 emb	2341
	2.8; se0.11; max 5; n88 lt	2728
	3 emb ea; n2 f	523a
	3 emb ea; n2 f	2341
	3.3 emb; r2-4; n3 f	2228a
	3.5 emb; r3-4; n2 f	522
	4 emb; r3-5	4962
	5.0; r4-6; n2 lt	502
	4.19 cl; sd0.92; r2-7; n57	1841
	4.24 ps; sd0.84; r2-6; n24	1841
	4.25 emb; sd1.38; r2-6; n31 f	1841
Peromyscus eremicus	1 emb; n1 f	5405b
	2-3 emb	2228a
	2.22 yg; r1-4; n14 lt	1236
	2.42 emb; se0.15; n12 f	3812, 3813
	2.42 yg; sd0.95; n404 lt	2319
	2.53 emb; r1-4; n13 f	6414
	2.55 yg; r1-5; n153	9255
	2.60 yg; ±0.24; r2-4; n5 lt	10573
	2.7 emb; r1-5; n37 f	522
	2.8; se0.03; max 6; n372 lt	2728
	2.9; ±1.2; r2-6; n21 f	6708
	3 emb; n1 f	1470
	3 emb; r1-4; n7 f	502
Peromyscus furvus	2 emb ea; n2 f	4228
Peromyscus gossypinus	3.5; r3-4; n4 lt	8907
	3.7 yg; ±0.14; mode 4; r1-7; n72 lt	8698
	3.9 emb; ±0.16; mode 4; r2-6; n32 f	8698
	5.67; r4-6 emb, 7 yg; n3 f	5187
Peromyscus grandis	2.5 ps; r2-3; n2 f	11044a
Peromyscus guatemalensis	2 emb ea; n2 f	11044a
Peromyscus hooperi	2.9 emb; n10 f	9603
	3.5 emb; n4 f	9603
Peromyscus interparietalis	2.4 yg; mode 3; r1-4; n40 lt	1236
Peromyscus leucopus	2.75 emb; r2-3; n4 f	950
	3 emb; n1 f	2341
	3 emb; r2-4; n2 f	5423b
	3.33 yg; r3-4; n3 f	9802

	3.33 emb; r3-4; n3 f	2025
	3.43 yg; ±0.13; r1-5; n21 lt	10573
	3.48 emb; r2-5; n83 f	10564
	3.5 emb, yg; se0.21; r1-6; n35 lt	11877
	3.5 yg; mode 3; r1-6; n60 lt	4125
	3.58 emb, ps; sd1.31; n12 f	67
	3.7; r1-7	9164
	3.74 emb; sd0.51; r1-6; n42 f	1363
	3.8 emb; r2-6; n24 f	6298
	3.86 emb; r3-5; n7 f	2548
	3.9 emb; r1-6; n10 f	522
	3.91 emb; se0.34; n11 f	3812, 3813
	4 emb; n1 f	575
	4 emb ea; n3 f	2547
	4 ea; n6 f	6487
	4 emb; r3-5; n2 f	1812a
	4.0 yg; mode 4; r2-7; n91 lt	5460
	mode 4-5; r1-8	6321
	4.12 ps; r2-6; n8 f	950
	4.2 yg; se0.22; n39 lt	3978
	4.2; mode 4; r1-7; n137 lt	7533
	4.25 emb; r4-5; n4 f	495
	4.26; r2-6; n39 f	1472
	4.29; se0.10; mode 4; r1-8; n128 lt	6150
	4.3 ps; r1-7; n9 f	6298
	4.32 ps; sd0.97; n85	1363
	4.33 emb, ps; sd1.35; n21 f	67
	4.36 yg; ±0.10; r1-6; n53 lt	10573
	4.4 emb; r2-6; n43 f	5405b$_1$
	4.46 cl; sd1.24; r2-8; n177	1363
	4.5 emb; r4-5; n2 f	576
	4.5 emb; se0.8; n12 f	654
	4.59 emb?; sd0.98; n116 f	2318
	4.60 yg; se0.13; mode 4.5; r2-7; n107 lt	7304
	4.67 yg; r1-9; n229	9255
	4.7; r3-6; n9 lt	11956
	4.7; r1-8; n400 lt	4779
	4.7; se0.06; max 9; n871 lt	2728
	4.76 yg; se0.21; mode 5; r2-8; n59 lt	3323
	4.76; se0.10; mode 5; r2-8; n110 lt	6150
	4.77 emb; ±0.18; mode 5; r3-7; n31 f	6555
	4.8 yg; se0.6; n12	2529
	4.8 emb; r4-6; n5 f	4228
	4.8-4.9 emb	4205
	4.89; mode 5; r4-5; n9 lt	7169
	4.94 emb; se0.34; mode 5; r2-7; n90 f	7303, 7304
	5 yg; n1 nest	4267
	5.0 emb; r4-6; n2 f	6797
	5.04 yg; ±0.08; r3-7; n50 lt	2082
	5.1 emb; r4-9; n40 f	10954
	5.20 emb; se0.36	758
	5.3 emb; r4-7; n3 f	502
	5.3; se0.2; n7 lt	684
	5.52 emb; se0.30	758
	6 emb; n1 f	261a
	6 emb; n1 f	3703
	6 emb; n1 f	9612
Peromyscus madrensis	3 emb; n1 f	11044a
Peromyscus maniculatus	r2-7; n19 f	7311
	3.00-4.80 ps; n26 lt, 4 samples	2806
	3 emb; n1 f	4228
	3.28 weaned; n72 lt, 21+ d	5019
	3.29 emb; r2-4; n7 f	575
	3.4 emb, yg; se0.07; r3-4; n52 lt	11877
	3.5 wean	7544

3.5 emb; r2-4; n14 f	6493
3.7; ±0.3; n25 lt, 7 f	2727
3.85 yg; n27 lt	9225
3.91 lg viable emb; se0.19; n55 f	6542
4.00-4.84 emb; n40 f, 4 samples	2806
4 emb; n1 f	11228a
4.0 emb; n2 f	502
4.0 emb; r3-5; n2 f	2548
4.0 emb, yg; r2-5 emb, 5 yg; n4 f	2544
4.08 yg; r1-9; n152 lt	10573
4.1 emb; r3-8; n7 f	522
4.1 emb; ±0.21; n14 f	6814
4.1 yg; n98 lt	4987
4.15 yg	7544
4.16 yg; r1-7; n32 lt	6493
4.25 yg; n25 lt, 0-2 d	5019
4.29 emb; r1-7; 4.38 ps; n405 f	9235
4.30 emb; se0.42; n10 f	3812, 3813
4.3; ±1.3; r2-7; n11 f	6708
4.3; n568 lt	3418
4.33 yg; se0.14; n80 lt	3150
4.4 emb; n71 lt	7081
4.4 ps; se0.06; mode 4; r1-9; n252	3624
4.4 emb; se0.02; mode 4; r1-9; n427 f	3624
4.42 med viable emb; se0.07; n325 f	6542
4.5; mode 4; r2-8	2088
4.5; ±0.31; n21 lt	2727
4.5 emb; sd1.12; n57 f	9423
4.5 emb; mode 5; r2-9; n62 f	5405b,
4.5 yg; r1-9; n186 lt	2907
4.5 emb; n217 f	722
4.55; sd1.10; r1-7; n65 lt	5658
4.575; mode 5; r2-7	7305
4.58 emb; mode 4; r1-9; n140 f	10256
4.6 emb; se0.11; n96 f	5243
4.6; se0.01; max 10; n711 lt	2728
4.6 cl; se.05; mode 4; r1-11; n747	3624
4.65 emb; se1.67; r1-8	2699
4.67; r4-5; n3 lt	1785
4.7; se0.4; n11 lt	5673
4.70 emb; sd1.14; r2-8; n71 f	3294
4.7; se0.04; max 11; n2285 lt	2728
4.71; mode 5; r1-9; n69 lt	4218
4.72 yg; r2-9; n389 lt	7697
4.73 sm viable emb; se0.04; n808 f	6542
4.75 cl; ±1.12; n57 f	9423
4.75 yg; r1-11; n610	9255
4.8 yg; se0.2	4749
4.8 emb; r3-6; n5 f	2415
4.8; r4-7; n6 lt	10574
4.8 yg; r1-7; n64 lt	5797
4.84 yg; r1-9; n70	9255
4.89 emb; r4-6; n9 f	6560
4.89 yg; mode 5; r2-7; n37 lt	7013
4.9; r1-7; n22 lt	9722
4.9 ps; n82 f	7081
4.96 emb; r1-7; n59 f	3703
4.98 yg; n804 lt	7312
mode 5; r3-6	2060
5.0; r3-7	5748
5 emb; n1 f	738
5 emb; n1 f	7487
5 emb ea; n2 f	1812a
5 emb ea; n2 f	6796
5.0 emb; r4-6; n3 f	4962

	5.0 ps; r3-6; n3 f	261a
	5.0 emb; r4-6; n7 f	261a
	5.0 emb; r4-6; n7 f	11646a
	5.0 emb; r3-7; n22 f	6166
	5 emb; mode 5; r3-9; n48 f	9576
	5.03; se0.18; r1-9; n98	7306
	5.06 yg; mode 5; r3-7; n52 lt	7013
	5.1 emb; ±0.34; n11 f	6814
	5.14; se0.14; n104	7308
	5.15 live emb; r2-9; n285 f	723
	5.2 emb; se0.54; r3-7; n6 [5 in table]	10839
	5.2 yg; n465 lt	7698
	5.21 emb; mode 5; r4-6; n14 f	5798
	5.25; se0.09; n102	7308
	5.25; se0.07; n132 lt	7310
	5.27 emb; r4-7; n11 f	1119
	5.28 ps; r2-9; n288 f	723
	5.3 ps; r4-7; n18 f	11956
	5.3 yg; n332 lt	7699
	5.31 emb; ±0.49; n13 f	1364
	5.33 ps; ±0.41	1364
	5.33 yg; r3-7; n3 f	1853
	5.35 emb	551
	5.38 yg; ±0.13; r2-8; n50 lt	2082
	5.4 ps; r4-9; n14 f	6166
	5.5; mode 5-6; r2-9; n53 lt	9891
	5.57 cl; ±0.46; n25	1364
	5.6 emb; r3-7; n17 f	11956
	5.6 emb; se0.12; r2-9; n111 f	11230
	5.71 emb; n62 f	6015
	5.74 cl; r1-12; n288 f	723
	5.9 ps; se0.31; r2-10; n34 f	11230
	6 emb; n1 f	10219
	6.0; se0.7; n12	2517
	6.1; r5-9; n21 lt	9891
	6.2; se0.6; n13 lactating f	2517
	6.5 emb; r6-7; n2 f	10218, 10220
	8.3 ps; 5.2 emb; ±0.81; n10	4610
Peromyscus megalops	2.0	9176
	3 emb; n1 f	2341
Peromyscus melanocarpus	1.8 yg; mode 2; r1-3; n28 lt	9114
	2.3 emb, ps; mode 3; r1-3; n30 f	9114, 9176
Peromyscus melanophrys	2 emb ea; n2 f	523a
	2.6 yg; se0.3; r1-4; n24 f, 24 lt	3228
	3 ps; n1 f	2351
	3 emb ea; n2 f	2228a
	3.3; se0.16; max 4; n33 lt	2728
	3.5 emb; r3-4; n2 f	217
Peromyscus melanotis	3.7 emb; r1-5	2341
	3.8 emb; r1-5; n13 f	522
	4.26 yg; se0.18; r1-8; n27 lt	10573
Peromyscus melanurus	2.7 emb; r2-3; n3 f	6149
	2.9	9176
Peromyscus merriami	3.5 emb; r3-4; n2 lt	5409
Peromyscus mexicanus	1 emb; n1 f	2341
	mode 2	6149
	2.1 yg; r1-3; mode 2; n9 lt	9114
	2.4 emb; r2-3; n10 f	4228
	2.5 emb; r1-4; mode 3; n17 f	11388
	2.6 emb; r2-3; mode 3; n20 f	9114, 9176
	2.7 emb; r2-3; n6 f	3211
	2.7 yg; mode 2-3; r1-4; n20	6149
	3.0 emb; r2-4; n2 f	2228a
	3 emb ea; n4 f	1475
Peromyscus ochraventer	5 ps ea; n2 lt	217

Peromyscus pectoralis	2.9 emb; r2-5; n20 f	522
	4 emb; r3-5; n3 f	3108
	4.6 ps; r3-9; n8 f	3108
	5.0 emb; r3-7; n5 f	2343
Peromyscus perfulvus	2.6 yg; r1-3; n11 lt, 4 f	4609
Peromyscus polionotus	2.67; n27 lt	2262
	3.1 emb; n63 f	1547
	3.135 emb; sd0.86; 52% mode 3; r1-5; n172 f	1548
	3.35 yg; n27 lt	9225
	3.59 yg; sd1.10; mode 3-4; r1-7; n290 lt	11725
	3.64 yg; se0.39; n11 lt	3812, 3813
	3.8; se0.09; max 9; n636 lt	2728
	3.89 yg; r1-9; n217	9255
	3.97 yg; n32 lt	8907
	4.0 emb; n13 f	8907
	4.68; mode 4; r1-8; n179 lt	5576
	5.5 emb, yg; r5 emb, 6 yg; n2 f	5187
Peromyscus sitkensis	6	1474, 2088
Peromyscus stirtoni	2.0 emb; r1-3; n8 f	11044a
	3 emb; n1 f	5408
	3.0 ps; r2-5; n3 f	11044a
Peromyscus truei	2.67 emb; r2-5; n6 f	522
	2.84 yg; ±0.14; r2-5; n19 lt	10573
	3 emb ea; n2 f	6560
	3 emb ea; n3 f	4962
	3.32; r2-5; n22 lt	7013
	3.5 emb; r3-4; n2 f	11228a
	3.52; r1-5; n27 lt	7013
	3.6 yg; r1-6; n110 lt	2907
	4.29 emb; r2-6; n7 f	2415
	4.65; se0.912; r1-8	2699
Peromyscus yucatanicus	2 emb; n1 f	4494
	2.8 emb; mode 3; n9	6149
	3.0 emb ea; n5 f	6269
	3.33 emb; r3-4; n3 f	950
	3.5 yg; mode 3-4; r1-5; n63	6149
	5 ps; r3-8; n5 f	950
Peromyscus zarhynchus	2.0 emb; r1-3; n14 f	6149
Petromyscus collinus	2 emb; n1 f	9945
	2-3	11864
	2-3 emb ea; n12 f	9945
	2-3; n13 f	10158
	2.8 yg; n7	2481
Phodopus campbelli	9 emb; n1 f	12105
Phodopus roborovskii	6.3 emb; r3-9; n13 f	3334
Phodopus sungorus	2.6 wean	5443
	3.6 yg	5443
	4.99 yg; r1-9; mode 5; n281 lt	3249
	5.6 yg; ±1.7; r2-10; n173 lt, 30 pairs	11548
	8.0 emb; r5-12; n25 f	3334
Phyllotis darwini	3 emb; n1 f	8556
	3.94 emb; r2-7; n16 f	4682
	4.0 emb; n6 f	8302
	4.14 emb; r2-7; n14 f	4682
	4.17; n70 f	11407
	5.1 emb; sd1.0; n11 f	7241
	5.2 emb; se0.58; n5 f	3551
Phyllotis osilae	4.38 emb; r2-6; n8 f	4682
Podomys floridanus	1.72 yg; r1-3; n25	9255
	2.1 yg; r1-3; n10 lt	2551
	2.25 emb; se0.48; n4 f	3813
	2.4; se0.05; max 5; n239 lt	2728
	2.6 yg; r1-4; n≤19	6301
	3.1 yg; r2-4; n8 lt	6301
	3.2 emb; r2-5; n57 f	6301

Psammomys obesus	mode 3-4; max 5	10481
	3.2; r1-8; n10 lt	8134
	3.3; r2-5; mode 3; n6 lt	2233, 2234
	3.6 yg; sd1.3; r1-6; n56 lt	3513, 9825
	4.1; sd1.4; n91 lt	8731
Punomys lemminus	2 emb; n1 f	8302
Reithrodon physodes	mode 2; r1-4	12128
	3.5 emb; r3-4; n2 f	596
	4.53 emb; r1-8; n17 f	8306
Reithrodomtomys brevirostris	3.5 emb; r3-4; n2 f	5413
Reithrodontomys chrysopsis	3.5 emb; r3-4; n2 f	4963
Reithrodontomys creper	2 emb; n1 f	4963
	2 emb ea; n2 f	3024a
	2.6 emb; r2-4; n10 f	11044a
Reithrodontomys darienensis	4 emb; n1 f	11044a
Reithrodontomys fulvescens	2 yg; n1 f	10587
	3 emb; n1 f	523a
	3.0 yg; r2-4; n9 lt	1563a
	3.5 emb; r3-4; n4 f	4228
	4-5 emb	2228a
	4.0; r3-5; n2 f	4963
	5 emb; n1 f	1475
	6 emb; n1 f	10261
Reithrodontomys gracilis	3.0 emb; r2-4; n2 f	12037
	3.2 emb; r3-4; n5 f	5423b
	3.3 emb; r3-4; n3 f	12037
Reithrodontomys hirsutus	4 emb; n1 f	4963
Reithrodontomys humulis	1.88; n8 lt	6300
	2 yg; n1 f	4915
	3.2; mode 3; r2-4; n9 lt, 9 f	5609
	3.4 yg; mode 3; r1-8; n25 lt	2794
Reithrodontomys megalotis	2.6 yg; r1-7; n25 lt	10587a
	3 yg; n1 f	10099
	3 emb ea; n2 f	2341
	3.5 emb; r3-4; n2 f	6560
	3.5 emb; r2-4; n4 f	10954
	3.83; r1-7; n198 lt	9107
	3.84 yg; n114 lt, 32 f	550
	4 emb; n1 f	4228
	4 emb ea; n4 f	1812a
	4.3 emb; r2-8; n75 f	$5405b_1$
	4.3 emb; r1-7; n7 f	502
	4.4 emb; r2-6; n39 f	3399
	4.67 emb, ps; r5,5 emb, 4 ps; n3 f	11628
	5 emb; n1 f	6487
	5.17 emb; r5-6; n6 f	3703
	5.27 emb; r3-8; n10 f	551
	6 emb; n1 f	9642
	6 emb; r5-7; n4 f	261a
	7 emb; n1 f	4745
	8.5 emb; r8-9; n2 f	6552
Reithrodontomys mexicanus	2.5 emb; r2-3; n2 f	4963
	3.0	9176
	3 emb; n1 f	2228a
	3 emb ea; n2 f	11388
Reithrodontomys montanus	2.86 yg; r2-4; n14 lt	6391
	3 emb; n1 f	261a
	3-7 yg	1871a
	~4; r1-9	11700
	4 emb; r3-5; n3 f	$5405b_1$
	5 emb; n1 f	1812a
	5 yg; n1 f	$5405b_1$
Reithrodontomys raviventris	3.7-4.0	9887
	~4	10809

Reithrodontomys sumichrasti	2.7 emb; r2-4; n10 f	4963
	3.5	9176
	4 emb; n1 f	11044a
Rheomys raptor	1 emb; n1 f	11372a
Rheomys thomasi	1 emb; n1 f	1475
	1 emb; n1 f	11372a
Rhipidomys mastacalis	3.33 emb, yg; r3-5 emb, 2 yg; n3 f	5111
	3.8; r3-5; n5	7995, 7996, 7998
Rhombomys opimus	3.84; n32 lt	10202
	5-6	10516
	mode 5-6; most 4-7; r1-14	7778
Saccostomus campestris	4; n1 lt	11935
	4.8; r2-8	2860, 2861
	5 emb; n1 f	9893
	5.1 emb, yg; r2-9; n10 f	4329
	5.1 ps; r3-9	4329
	6.5 emb; n2 f	331
	6.7; r1-10; n7	10158
	7 emb; n1 f	2452
	7.1 yg; sd3.99; r3-13; n18 primiparous	11082
	7.4; r5-10; n8 lt	10154
	7.9 yg; sd2.34; r3-11; n27 multiparous	11607
	8 emb; n1 f	9945
	10 emb; n1 f	9892
	11 yg, 4 wean; n1 lt	8385
Saccostomus mearnsi	5.0 emb; r4-6; n2 f	4925
	7 emb; n1 f	2443
Scapteromys tumidus	3.4; r2-4; n7 lt	4683
	4.0 emb; r3-5; n11 f	596
Scotinomys teguina	2 emb ea; n4 f	3024a
	2.28 emb; sd0.643; r1-3; n21 f	4967
	2.73 yg; sd0.845; mode 2-3; r1-5; n156 lt	4967
	3 emb; n1 f	1475
Scotinomys xerampelinus	2.67 emb; sd0.707; r2-4; n9 f	4967
	3.12 yg; sd0.669; mode 3; r2-4; n83 lt	4967
Sekeetamys calurus	2.9 yg; max 6; n47 lt	7983
Sigmodon alstoni	5.0; r2-6; n6	7995, 7996, 7998
Sigmodon fulviventer	mode 7-8	1527
Sigmodon hispidus	r2-8 emb; n10 f	1475
	2.79 emb, yg; n14 lt	1155
	3 emb; n1 f	11507a
	3 yg; n2 lt	6303
	3.3 emb; r2-7; n14 f	4228
	3.5 emb, yg; n2 lt	1155
	3.5 emb; r2-5; n2 f	8907
	3.5 emb; r3-5; n4 f	3317
	3.5 emb; r2-4; n4 f	11388
	3.52 emb; mode 3; r2-6; n27 f	7624
	3.65 emb; r2-6; n11 f	8316
	3.85 emb; r1-6; n25 f	6303
	3.86 emb; r3-7; n7 f	950
	3.9 emb; r2-6; n27 f	7624
	4.0 emb; r2-6; n4 f	6303
	4.0 emb, yg; n24 lt	1155
	4 cl; n1	1852
	4.2 yg; se0.1; n63 lt	8146
	4.5 emb; n2 f	2228a
	4.6 ps; r3-7; n5 f	6303
	4.6 emb, yg; n14 lt	1155
	4.7 yg; se0.52; n7	6975
	4.7 yg; se0.28; 4.4 at 12 d; se0.27; n16 lt	6976
	4.7 emb; r2-10; n56 f	2590
	4.75 yg; r3-6	10570
	4.8; ±0.3	7030

	5 emb; n1 f	523
	5 emb; n1 f	8196
	5 emb; n1 f	9199
	5 emb; r3-9	522, 523
	5.0 emb; r3-10; n7 f	5423b
	5.0 yg; se0.4; r2-7; n16 lt	8915
	5.0; r2-8; n41 lt	1563b
	5.1; r1-8; n25 lt	1563b
	5.12 emb; r3-8; n26 f	8026a
	5.3 yg; se0.2; n52 lt	8146
	5.3; r1-13; n56 lt	1563b
	5.39; se0.10; r1-15; n626 table, n627 text	1563b
	5.47 yg; r2-11; n60 lt	5806
	5.5 yg; r1-8; n10 lt	1563a
	5.6 emb, yg; n30 lt	1155
	5.6 yg; r2-10; n44 lt	7253
	5.7 emb, yg; n3 lt	1155
	5.81 emb; r1-11; n75 f	3849
	5.9-6.3 emb; 8-9 cl	4213, 4213a
	6 emb; r3-9	1290
	6 ps; r5-7; n2 f	950
	6.10 yg; r2-10	9572
	6.1; r4-12; n11 lt	1563b
	6.1 yg; se0.2; n73 lt	8146
	6.2 emb; r1-15; n67 f	4212
	6.3 emb, yg; n6 lt	1155
	6.5 yg; r6-7; n2 nests	10492
	6.53 yg; r1-12	9572
	6.6 emb; r4-10; n16 f	9710
	6.6 yg; r2-15; n28 lt	5726
	6.7	3308a
	6.9 yg; se0.41; 6.6 at 12 d; se0.35; n17 lt	6976
	7 emb; n1 f	1812a
	7.0 yg; se0.39; n15	6975
	7.4; n27	1163a
	7.5 emb; r4-11; n2 f	502
	7.5 emb; r5-10; n6 f	3211
	7.6 emb; r6-9; n9 f	551
	8.3 ps; r3-30; n35 f	3849
	8.5 emb; r6-11; n2 f	2351
	9.04 emb; ±0.31; r3-14; n70 f	7029, 1563b for range
	10 emb; n1 f	3850
	12 emb; r8-17; n6 f	1470
	14.3 emb; r14-15; n3 f	5806
Sigmodon ochrognathus	3.0 emb; r2-4; n4 f	4898
	3.0 yg; r2-6; n8 lt	4898
	5.67 emb; r2-9; n6 f	522
Steatomys krebsii	4.4 yg; n5 lt; 2-8 d	331
	5 emb; n1 f	331
Steatomys minutus	5; n1 lt; ~12 d	331
Steatomys pratensis	3 emb; n1 f	5760
	3.0 emb; n2 f	331
	3.2 emb; r1-9; n5 f	10158
	3.5; r3-4; n2 lt	4329
	4.5; r2-7; n2 lt	10154
	5 emb; n1 f	4495
	6.0; n2 lt	9893
Tatera afra	3.98 emb; mode 4; r2-6; n62 f	7146

Tatera boehmi	3.5 ps; r2-5; n2 lt	4329
	5 emb; n1 f	2446a
Tatera brantsii	2.0 yg	7822
	2; n3 lt	7168
	2.64 emb; mode 3; r1-4; n42 f	7146
	2.94 cl; mode 3; r1-4; n49 f	7146
	3.2 emb; n6 f	7823
	3.3 emb; r1-5; n12 f	10154, 10158
	4.0; n2 lt	12128
Tatera indica	2-4 yg	8500, 8501
	4; r1-9; n13 lt	8725
	4.76 yg [text], 5.11 yg [tables]; r2-7; n27 lt	987
	4.78 emb; mode 5; r1-8; n74 f	5230, 5231
	5-8 yg; n>9	8720
	5 emb; n1 f	6293
	5.25 ps/emb; r3-4 ps, 5-7 emb; n4 lt	10616
	6 ps; n1 f	7382
	6.33 emb; ±0.318; mode 5-7; r2-10; n36 f	729
	6.38 ps; ±0.304; mode 5-8; r2-13; n50 f	729
Tatera leucogaster	mode 4-5; r2-8	9945
	4.2 emb; n13 f	9893
	4.38 emb; ±1.2; r2-6; n69 f	8384, 8386
	4.44 ps; ±1.63; r2-6; n148 lt	8386
	4.49 emb; r2-9; n53 f	10154, 10158
	4.6; r3-7; n7	4329
	4.6; r3-7; n21 lt	8965
	4.8 emb; ±0.36; r3-6; n10 f	7797a, 7797b
	5.6 emb; ±0.29; r3-8; n25 f	7797a, 7797b
	5.67 emb; r4-7; n3	9892
	5.67 emb; n6 f	331
Tatera nigricauda	4 emb; n1 f	2443
	5.22 live emb; se0.24; r1-8; n37 f	7794
Tatera robusta	4.6; r2-7; n5 lt	7794
	5 emb ea; 2 f	4925
	5.7 yg; r5-6; n3 f	5051
	8 emb; n1 f	5051
Tatera valida	2-6 yg	5058
	3.7 ps, emb; r2-5; n9 lt	10799
	3.8 emb; r3-4; n5 f	7794
	4 emb; n1 f	2443
	5 emb; n1 f	331
	5.0 nestlings; r4-6; n2 lt	4495
Taterillus emini	3.5 nestlings; r3-4; n2 lt	4495
Taterillus gracilis	2-6 yg	4354a
	3 emb; n1 f	10799
	4-6 yg	5058
Taterillus pygargus	3-5	5058
	4 yg	8695a
Tylomys nudicaudus	2.0 yg; n3 lt	9176
	2.3 yg; mode 2; r1-4; n47 lt	4606, 4608
	2.67 yg; r1-4; n24 lt	10755
	4.0 emb; n1 f	2590
Tylomys watsoni	3 emb; n1 f	3024a
Wiedomys pyrrhorhinos	3.8; sd1.8; r1-6; n5 lt	10494
Zygodontomys brevicauda	2 emb; n1 f	5111
	3.0 emb, yg; r2 yg, 3-4 emb; n4 f	3024
	3.7 wean; max 7; n33 lt	11931
	3.7 emb; r3-5; n7 f	3317
	4.1; r3-7; n44	7995, 7996, 7998
	4.5 emb, yg; r4 emb, 5 yg; n2 lt	451
	4.64 yg; ±0.27; n91	87
	4.85 yg; sd1.72; r1-11; n283 lt	11373

SEXUAL MATURITY

Akodon azarae	f: 75.3 d	2409
	m: 83.7 d	2409
	f/m: 60 d	2221
	f/m: min 2 mo	8305
Akodon dolores	f: 1st birth: 33.3% < 3 mo	7213
Akodon longipilis	f: 1st preg: 1-2 mo	8305a
	f: 1st birth: 112.3 d; sd3.3; n6 f	7241
	m: 47.5 g	7241
Akodon molinae	f: vag perf: 30 d: r25-30	7212
	f: 1st birth: < 3 mo	7212, 7213
	m: testes descend: 25 d; r21-30	7212
Akodon olivaceus	f: 1st preg: 2 mo	8305a
	f: 1 yr	7241
	f/m: 22-24 g	7241
Akodon urichi	f/m: 2.7 mo; n9	7995, 7998
Auliscomys micropus	f: 1st preg: ~2 mo	8305a
Baiomys taylori	f: 1st conceive: 28 d	5064
	f: 1st conceive: 60-70 d	8805
	f: 1st birth: 81.5 d; r64-101; n8 f	977
	f: 40 d	10110a
	m: testes descend: 70-80 d	8805
Beamys hindei	f: 1st repro: 5 mo; majority 6.5-8 mo	2912
	f/m: 136 mm	5760
Bolomys lasiurus	f: 1st birth: 4-5 mo	2371
	f: 40.9 d	7193
	m: 34.9 d	7193
Calomys callosus	f: vag perf: 20 d	8430
	f: vag perf: 40.1 d; ±7.6	7188, 7189
	f: vag perf: 42-46 d	5474
	f: most mate: 2 mo	8430
	f: 1st preg: 50 d	5474
	f: 1st birth: 6 wk	8430
	m: testes descend: 30 d	5474
	m: fertile: 42-45 d	5474
	m: 19.6 d; ±6.6	7188
Calomys callidus?	m: testes descend: 14.9 d; r14-16; n43	4853
Calomys musculinus	f: 72.5 d, 16.88 g	2410
	m: testes descend: 14.6 d; r14-19; n77	4853
	m: 50% 82.03 d, 19.8 g	2410
Cricetomys gambianus	f: vag perf; 21.63 wk; sd1.38; n4	103
	f: 1st estrus: 22.63 wk; sd1.75; n4	103
	f: 1st birth: 7 mo	5760
Cricetulus griseus	f: 1st mate: 48 d	1693
	m: testes descend: 1 mo	7503
	f/m: 8-12 wk	1693
Cricetus cricetus	f: vag perf: 60-90 d	11319
	f: 1st birth: 3 mo	11319
	f: 1st birth: 298.17 d; ±92.59; n23	7436, 9085
	f: 12 wk	7436, 9085
	m: 8 wk	7436, 9085
	f/m: 2nd yr	4645
Dendromus mesomelas	f: 67 mm	4329
	m: 74 mm	4329
Dendromus mystacalis	m: fecund: 60-63 mm	4329
Dendromus nyikae	f: 1st preg: 71 mm; n1	4329
	m: fecund: > 73 mm	4329
Dipodillus simoni	f: 1st mate: 2 mo	6237
	f: 1st birth: 5 mo	6235
	m: 1st mate: 3.5 mo	6237
	m: 1st sire: 9.5 mo	6235
Eligmodontia typus	f: 1st conceive: 6-8 wk	8307
Gerbillus pyramidum	f: vag perf: 70 d	4713
	m: 1st mate: 45 d	4713
	m: 1st mate: 75 d; n1	4348

	m: 1st sire: 93 d; n1	4348
	f/m: 75-80 d	4351
Gerbillurus paeba	f: 1st mate: 84 d; n1	403
	m: 1st sire: 63 d; n1	403
Holochilus brasiliensis	f: 2-4.5 mo; 107-165 g	7192
	m: 2-3 mo; 115-205 g	7192
Holochilus sciureus	f: 1st preg: 20-40 g	10983
	m: 1st sperm: most 30-50 g	10983
Malacothrix typica	f: 1st birth: 111 d; r102-120; n2	5887
	f/m: 1st mate: 51-72 d	10154
	f/m: 1st mate: >70 d	5887
Megadontomys thomasi	m: spermatogenesis: 75 d	6302
Meriones crassus	f: 1st mate: end of 2 mo	6237
	m: 1st sire: 82 d	6237
Meriones libycus	f/m: 3 mo	11905
	f/m: est ~3 mo	2235
Meriones shawi	f: vag perf: 59.6 d; mode 54-63; r44-83	9451
	f: 1st birth: 2.5-6 mo	9451
	f/m: > 4 mo	6237
Meriones unguiculatus	f: vag perf: 26.8 d; r13-43	1862, 1863
	f: vag perf: 40-60 d	6867
	f: vag perf: 41.3 d; ±4; r33-53; 27.7 g; ±0.4; r21-36; n82	7969
	f: vag perf: ~44-54 d	7964
	f: vag perf: 45 d	9692
	f: vag perf: 48.7 d; r40-76; n29	7754
	f: ovary/uterus wt: 90 d	7965
	f: 1st mate: 127 d; ±7; r60-272	56
	f: 1st birth: 72 d	1768
	f: 1st birth: 62 d	1862
	f: 1st birth: 110 d; ±9; n40	7969
	f: 1st birth: 118.7-161 d	1863
	f: 63-84 d	6867
	m: testes descend: 35 d	9692
	m: testes descend: 36.3 d; r28-45; n492	7754
	f/m: 1st mate: 10-12 wk	9692
Mesocricetus auratus	f: vag perf: 8-14 d; 8 g	1071
	f: 1st birth: 73.6 d; se1.50; sd8.99; r59-90	1071
	m: ~7-8 wk; 60-70 g	2419
	f/m: 36-40 d	3571
	f/m: 7-8 wk	1382
Mesocricetus brandti	f: 1st estrus: 46 d; se1; n60	4952
	f: 1st birth: 50 d	6656
	m: testes enlarge: 5-6 wk	4952
Mystromys albicaudatus	f: 1st birth: 146 d	7168
	f: 1st birth: 4.8 mo	4220
Neacomys tenuipes	f/m: 1 mo; n12	7995, 7998
Neotoma albigula	f: vag perf: 83.5 d; r80-87; n2	9689
	m: testes descend: 101 d; n4	9689
Neotoma cinerea	f/m: ~1 yr	3087
	f/m: 2 yr	1871a
Neotoma floridana	f: 5-6 wk	3285
	f: ~5 mo, 160 g	4680
	m: testes descend: ~200 g	4680
Neotoma lepida	f: 2-3 mo	2906
	f/m: 60 d	1474
	f/m: 1 yr	6708
Neotoma mexicana	f: ~2 mo	1366
Neotoma micropus	f: 5 mo	8962
	f: yr of birth	3262
Neotoma stephensi	f: 9-10 mo	11232
Neotomodon alstoni	f: vag perf: 49.0 d; sd10.1; r35-76; n78	8096
	f: 1st repro: 174.5 d; r65-241	8096

Onychomys leucogaster	f: 1st preg: 95 d	10588
	f: 1st birth: min 4 mo; most 5-6	8559
	f: 1st birth: 191 d; n1	2907
	m: 1st sire: 4 mo	8559
Onychomys torridus	f: 1st birth: 4 mo	8559
	m: 1st sire: 3-4 mo	8559
	f/m: 6 wk	4991
	f/m: 7 wk; field bred; 4-5 mo; lab	10697
Oryzomys bicolor	f/m: 3? mo	7995, 7996
Oryzomys capito	f: 1st ovulate: 50 d	11931
	f: 1st preg: 45-50 d	3316
	m: 1st sire: 46 d	11931
	f/m: <2 mo	3319
	f/m: 2 mo; n10	7995, 7998
Oryzomys concolor	f/m: 3 mo; n7	7995, 7998
Oryzomys longicaudatus	m: 1st mate: 1 mo	8305a
Oryzomys nigripes	f: vag perf: 57.6 d; ±8.2; r46-66	7190
	m: testes descend: 37.1 d; ±7.8; r32-43	7190
	f/m: next breeding season	2221
Oryzomys palustris	f: vag perf: 40-45 d	7810
	f: 1st mate: 7 wk	4280
	f/m: 50 d	1474
Otomys angoniensis	f: 139 mm	4329
	m: 158 mm	4329
Otomys denti	f: fecund: 142 mm	4329
	f/m: 90 g	2571
Otomys irroratus	f: 9-10 wk; 76 g	10158
	f: >5 mo; 90 g	2571
	m: 13 wk; 96.0 g	10158
	m: >5 mo; 80-100 g	2571
	f/m: 1st mate: 3-4 mo	2338, 2339
Otomys sloggetti	f: 11 wk, 90 g	11710a
	m: 16 wk, 119 g	11710a
Otomys typus	f/m: 1st mate: < 3 mo	5760
Ototylomys phyllotis	f: vag perf: 21-70 d; n40	4606, 4607
	f: 1st mate: 29 d	4606, 4607
	m: testes descend: 15-38 d	4606, 4607
	m: 1st mate: 175 d	4606, 4607
Oxymycterus rutilans	f/m: 90 d; based on wt	2221
Pachyuromys duprasi	m: 1st sire: 2 mo	6237
Peromyscus boylii	f: 1st estrus: 50.93 d; ±1.92; min 37; n33	1854
	m: 1st sperm: 60 d	1363
	m: 1st epididymal sperm: 65 d	1363
Peromyscus californicus	f: vag perf: 38.7 g; se0.7; n30	4119
	f: 1st estrus: 44.1 d; se1.0; n30	4119
Peromyscus crinitus	f/m: 1st mate: 70 d min, most 4-6 mo	2910
Peromyscus eremicus	f: 1st estrus: 39.21 d; ±1.54; min 28; n61	1854
	f: 1st birth: 10 mo; min 3-4	2319
	f: 1st repro: 103.75 d; sd23.61; r71-168; n87 f excludes 13 f >180 d	10054
Peromyscus gossypinus	f: vag perf: 43 d; n27	8698
	f: 1st conceive: 73 d	8698
	m: epididymal sperm: ~45 d	8698
Peromyscus hooperi	f: 1st mate: 69 d; sd6.99; r67-76; n3	9603
Peromyscus leucopus	f: vag perf: 38-40 d	4205
	f: vag perf: 38.6 d; se2.6; r27-75; n20	9226
	f: 1st estrus: 42-44 d	4205
	f: 1st estrus: 46.22 d; ±3.18; min 28; n32	1854
	f: 1st estrus: 51.0 d; se3.6; r30-101; n20	9226
	f: 1st mate: 10-11 wk	1474
	f: 1st birth: 3.5 mo	1472
	f: median 13.15 g; 95%ci 9.11-18.99; island	67
	f: median 17.20 g; 95%ci 13.98-21.16; mainland	67
	m: median 14.06 g; 95%ci 8.90-22.23; island	67

	m: median 22.85 g; 95%ci 17.21-30.33; mainland	67
Peromyscus maniculatus	f: vag perf: 32 d 50%	6550
	f: 1st estrus: 48.72 d; ±1.23; min 28; n186	1854
	f: 1st conceive: min 31-42 d; across generations	7312
	f: 1st birth: 60-74 d; n4 groups	4749
	f: 1st birth: 3-4 mo	11956
	f: yearlings	11230
	m: testes enlarge: 40-60 d	10795
	f/m: 1st mate: ~5-6 wk	1474
	f/m: 1st repro: 84.1 d; sd20.38	4204
	f/m: breed: most 1 yr	3774
	f/m: breed yr of birth rare	7310
	f/m: 4.5-9 wk (see section f)	5019
	f/m: 7 wk	3150
	f/m: ~50 d	2806
	f/m: 7-11 mo	6796
	f/m: yr of birth	2699
Peromyscus megalops	f: 46-48 d	6302
Peromyscus melanocarpus	f: vag perf: 79.3 d; r70-98; n6	9114
	m: scrotal swell; 124.4 d; r86-154; n9	9114
Peromyscus mexicanus	f: vag perf: 46 d; n1	9114
	m: scrotal swell; 66.5 d; r49-77; n4	9114
Peromyscus polionotus	f: vag perf: 26 d	8907
	f: 1st estrus: 29.64 d; ±0.47; min 23; n121	1854
Peromyscus truei	f: 1st estrus: 50.09 d; ±2.44; min 28; n34	1854
	f/m: yr of birth	2699
Peromyscus yucatanicus	f: vag perf: 51.5 d	6149
Phodopus roborovskii	f/m: 3 wk; r14-24 d	3334
Phodopus sungorus	f: 1st birth: 139 d; r109-233	5443
Phyllotis darwini	f: vag perf: 60.0 d; sd10.2; 40.8 g; sd6.5; n3	7241
	m: scrotal testes: 50.3 d; sd7.2; 46.3 g; sd6.4; n6	7241
Podomys floridanus	f: vag perf: 35 d; r18-48	6301
Psammomys obesus	f: vag perf: 53 d	2234
	f: vag perf: 57 d	8731
	f: 1st conceive: 92-106 d	2234
	f: 1st birth: 8-11.5 mo	9825
	m: testes descend: 54 d	8731
Reithrodon physodes	f: 1st preg: est 2 mo	8306
Reithrodontomys megalotis	f: 1st mate: 4.5 mo	1474
	f: 1st birth: 62 d; n1	9107
	m: 1st mate: 59 d	9107
	f/m: 1st repro: 4 mo 8 d	10587a
	f/m: 2 mo	1871a
Reithrodontomys montanus	f: 1st birth: 12 wk; n1	6391
	f/m: 2 mo	1871a
Rhipidomys mastacalis	f/m: 3? mo; n5	7995, 7996, 7998
Rhombomys opimus	f/m: 3-4 mo	7778
Saccostomus campestris	f: vag perf: 34.6; sd2.27; n62	11082, 11608
	f: 1st estrus: 44.6 d; sd3.17; n19	11082, 11608
	f: 1st conceive: 56.4 d; sd8.12; r44-71; n21	11082
	f: 1st birth: 96 d; min 67 d	2861
	f: >110 mm; wild; ~102 mm	4329
	m: 100-109 mm	4329
Scotinomys teguina	f: vag perf: 33.8 d; r28-39; n41	4967
	m: epididymal sperm; 6-8 wk	4967
Scotinomys xerampelinus	f: vag perf: 51.8 d; r44-60; n15	4967
Sigmodon alleni	f: 1st birth: 87 d	9959
Sigmodon alstoni	f/m: 2 mo; n6	7995, 7998
Sigmodon fulviventer	f: 1st birth: 77 d	524
Sigmodon hispidus	f: vag perf: 30-40 d	7254
	f: vag perf: 40-47 d	3110
	f: 1st estrus: r43-56 d; n9 (assorted treatments)	3110
	f: 1st preg: 57 g	522
	f: 1st birth: 65 d; spring; 84 d; Sept	1795

	f: 32 g (Mar-June), 60 g (July-Oct), 105 g (Nov-Feb)	7029
	m: testes descend: 20-30 d	7254
	m: epididymal sperm: 2 mo 2/5; 3 mo 100%	1795
	m: 40 g (Mar-June), 60 g (July-Oct), 100 g (Nov-Feb)	7029
	f/m: 40 d	1474
Sigmodon ochrognathus	f: 1st breed: 45 d	4898
	f: 1st birth: 71 d	525
Tatera afra	f: 3 mo; born early in season; 5-7 mo; born later	7146
	m: 6-8 wk born spring-early summer; 6-7 mo born later	166
Tatera brantsii	f: 3 mo; born spring; 8 mo; born summer-fall	7146
	m: 122 mm; testes wt 1.59 g; full sperm activity; n1	166
Tatera inclusa	f: 1st birth: 125 d; n1	5050
Tatera indica	f: 10-12 wk	5230
	m: 12-14 wk	5230
Tatera leucogaster	f: preg f: 120-129 mm 12%; 130-139 mm 75%	4329
	m: fecund 137-138 mm	4329
Tatera valida	f/m: 11-15 wk	5058
Taterillus gracilis	f/m: 12 wk	5058
Taterillus pygargus	f/m: 12 wk	5058
Tylomys nudicaudus	f: 1st mate: 90 d	4606, 4608
	m: 1st mate: 144 d	4606, 4608
	f/m: 10-11 wk	10755
Zygodontomys brevicauda	f: 1st ovulate: 50 d	11931
	f: 1st preg: 21 d	11373
	f: 25.6 d; ±1.21; n7	87
	m: abundant sperm: 40 d	11373
	m: 1st sire: 51 d	11931
	m: 42.3 d; ±1.94; n6	87
	f/m: 2 mo; n44	7995, 7996, 7998
CYCLE LENGTH		
Baiomys taylori	4.9 d; r3-6; n33 cycles, 30 f	5064
	7.54 d; se0.23; n19 f	8805
Bolomys lasiurus	4-8 d; n9 f	7193
Brachytarsomys albicauda	8-9 d; n1 f	7477
Calomys callosus	proestrus: 1.3 d; ±0.4; estrus: 1.5 d; ±0.8	7189
	metestrus: 2.0 d; ±1.7; diestrus: 2.1 d; ±0.82	7189
	6 d; n6 f	5474
	6.6 d; ±1.4; n17 f	7189
Calomys musculinus	5.7 d	2410
Cricetomys gambianus	4.4 d; ±1.9	312
	5-6 d	103
	6 d; n1 f	7477
Cricetulus griseus	proestrus: 0.5, estrus: 1.5, metestrus: 2.5, diestrus: 2.5 d	8243
	4 d	468
	4-5 d	1693
	4.46 d; r3-7; n69 cycles, 5 f	8243
Cricetus cricetus	estrus: several hr-2 d; diestrus: 6 d	9085
	7.5 d	9085
Dipodillus simoni	estrus: ~1 d	6237
	10 d; r9-14; n4 cycles, 2 f	6237
Gerbillurus paeba	6.2 d; sd1.2; n8 cycles	2482
	7.2 d; sd1.5; n9 cycles	2482
Gerbillurus tytonis	6.2 d; sd1.2; n5 cycles	2482
Gerbillurus vallinus	11.3 d; sd1.2; n3 cycles	2482
Holochilus brasiliensis	6-8 d	7192
Meriones crassus	estrus: ~1 d	6237
Meriones libycus	5 d; smears	11905
Meriones shawi	estrus: ~1 d	6237
	4.74 d; r2-35; mode 4 (603/671); n671 cycles, 39 f	9451
Meriones tristrami	4.2 d; mode 4; median 4.0; se0.1; n775 cycles, 41 f	2527
Meriones unguiculatus	estrus: 5-23+ hr; n36, 9 f	583
	4.6 d; r4-6; n14 cycles	583
	14.6 d; r10-18; n10 cycles; pseudopregancy?	583
Mesocricetus auratus	4 d	9160
Mesocricetus brandti	4 d	6656

Neotoma floridana	mode 4-6 d; r3-8; n13 f; no true comified cells	1703
Neotomodon alstoni	4.5 d; sd0.4; mode 4; r3-7	8096
Onychomys leucogaster	4.9; se0.3; n73 cycles	3533
	6.3; mode 4; median 4.4; se0.6; n402 cycles, 32 f	2527
Oryzomys capito	5-6 d	11931
Oryzomys nigripes	proestrus: 1.6 d; ±0.9; estrus: 2.6 d; ±1.1	7190
	metestrus: 2.3 d; ±1.3; diestrus: 2.9 d; ±1.1	7190
	10 d; ±1.5; n14 f	7190
Oryzomys palustris	7.62 d; ±0.19; n21 cycles, 7 f	2002
Ototylomys phyllotis	8.8 d; sd0.5; n14 f	4606, 4607
Pachyuromys duprasi	7.11 d; r3.5-11.5; n15 cycles, 1 f	6237
Peromyscus californicus	6.4 d; se0.2; mode 6; median 6.2; n161 cycles, 12 f	2527
	9 d; se0.4; mode 6,10; median 9; r5-20; n59 cycles, 18 f	4119
Peromyscus eremicus	5.3 d; se0.3; mode 4; median 4.7; n939 cycles, 58 f	2527
Peromyscus crinitus	6.1 d; se1.3; mode 5; median 4.9; n30 cycles, 7 f	2527
Peromyscus gossypinus	4.7 d; se0.2; mode 4; median 4.2; n733 cycles, 22 f	2527
	5.26 d; r3.5-10; n13 cycles, 6 f	8698
Peromyscus leucopus	6.0 d; se0.2; mode 4; median 4.9; n780 cycles, 59 f	2527
Peromyscus maniculatus	4-6 d	2806
	4.8 d; r4-5.8 d; n18 cycles, 39 f	1851
Peromyscus melanophrys	4.6 d; se0.3; r4.4-4.9; n40 cycles, 10 f	3229
Phodopus campbelli	estrus: ~10 h	11971
	4 d; mode 4; r4-5 (15% of f)	11971
Psammomys obesus	4-5 d	3513
Saccostomus campestris	4.1 d; sd1.35; n38 cycles, 8 f	11607
	4.1 d; sd7.5; n71 cycles, 19 yg f	11607
Scotinomys teguina	8.2 d; mode 6; median 6.4; se0.4; n13 cycles, 4 f	2527
Sigmodon hispidus	proestrus: 14.5 h; r12-21; estrus: 46.6 h; r21-123	1852
	metestrus: 145 h; r9-21; diestrus: 116 h; r42-156	1852
	estrus: 2.6 d; r1-12; n106 cycles, 21 f	7253, 7254
	6.2 d; ±0.45; r4-8; n9 cycles, 6 f	10656
	8 d; r5-9	1852
	8 d; r4-20; n106 cycles, 21 f	7253, 7254
Tatera brantsii	proestrus: 16-18 h; estrus: 20-28 h	7146
	metestrus: ~24 h; diestrus: 2 d	7146
	r4-6; one case 3 d	7146
Tatera indica	r4-5 d; n250 cycles	987
Tatera valida	7 d	1997
Tylomys nudicaudus	6.8 d; se0.4; n15 f	4606, 4608
Wiedomys pyrrhorhinos	1-6	10493
Zygodontomys brevicauda	r4-7 d; n11 f	11373
	r4-14 d; n10 f	11373
	r5-6 d; n16 cycles	11931
GESTATION		
Akodon azarae	22.7 d; may delay implantation	2221
	24.5 d; sd0.86	2409
Akodon molinae	23 d; r21-25	7212
Baiomys taylori	≤20 d; min ibi	977
	20 d	8805
	20 d	10110a
	22 d with lactation	8805
	23 d; r21-26; n7; ppe to birth	5064
	25.5 d; r22-29; n31; pairing to birth	5064
Beamys hindei	22-23 d; n7 lt; pairing to birth	2912
	30 d	5049
Bolomys lasiurus	21 d; min ibi	2371
	~23 d	7193
Calomys callosus	19-22 d	8430
	21.8 d; ±1.04; r20-23; n14	7188, 7189
	24.7 d; mode 25; r21-29; n70; pairing to birth	5474
Calomys laucha	25 d; n1	4682
Calomys musculinus	20.96 d; sd1.07; n15	2410
	21 d; ±1	4853
Cricetomys gambianus	27-36 d	103
	31.7 d; r31-32; n3; pairing to birth	3130

	32.3 d; se2.8; r30-38; n7	227, 312
	42 d; n2	2439, 2446a, 2449, 7528
Cricetulus griseus	implant: 5-6 d	8510
	implant: 6 d	993
	19.5 d; r19-21	468
	20 d	7503
	20-21 d	1693
Cricetulus migratorius	11-13 d	4645
Cricetus cricetus	15.5-21 d	7436, 9085
	19-20 d	3571
	19.32 d; mode 17; r17-37; n25	11319
	20 d	8457
	20 d	7433
	4-5 wk	4645
Dendromus melanotis	23-27 d	2572
Desmodilliscus braueri	26 d	8696
	35 d with lactation	8696
Desmodillus auricularis	21 d	5684
	35 d with lactation	5684
Dipodillus simoni	20.4 d; r20-21; n11 lt, 6 f	6235, 6237, 6238
	32 d; r29-36; n6 with lactation	6238
Eligmodontia typus	18 d; n1	6807
Gerbillurus paeba	26 d; n1	10528
Gerbillus dasyurus	r24-26 d; n4	109
Gerbillus gleadowi	~20 d	9175
Gerbillus nanus	~20 d	9175
	21-23 d	5785
Gerbillus perpallidus	20.22 d; sd0.55; r19-21; n18 lt	11397
Gerbillus pyramidum	21 d	4713
	~22 d	4344, 4348, 4351
	37-42 d; with lactation	4713
Holochilus brasiliensis	28.4 d; r26-30	7192
Macrotarsomys bastardi	>24 d; separate m from f	6402
Malacothrix typica	23.11 d; sd1.59; n19 lt; without lactation	5887
	~26 d	10158
	26.75 d; r25-35; n6	10154
	27.45 d; sd0.69; n11 lt; with lactation	5887
Meriones crassus	21 d	8428
	21 d; n3	6236, 6238
	21.6 d; r21-24	6803
	23.5 d; mode 21; r21-31; n4	6237
	31 d; n1 nursing 3 yg	6236, 6238
	32.6 d; nursing ≥ 2 yg	6803
Meriones hurrianae	28-30 d	9175
Meriones libycus	22.3 d; r22-23; n3	11905
	29 d; r23-35	7778
Meriones meridianus	20-21 d	11252
	28 d; r22-30	7778
Meriones persicus	~28 d	9175
Meriones shawi	implant: 4 d	9451
	20 d	9451
	20.5 d	6237, 6238
	24.8 d; nursing 2 yg	9451
	31.85 d; r30-32; n4 nursing 4-7 yg	6237, 6238
	36.3 d; nursing 8 yg	9451
Meriones tristrami	24 d	8428
Meriones unguiculatus	implant: 6 d	11960
	implant: 8 d	3269
	24-26 d	7754
	24-26 d	6867
	r24-26 d; n86	56
	24 d + 8-30 h	9692
	25 d	82
	25.2 d; ±0.1; r24-26; n67 lt	7970

	min 25-26 d; lactation prolongs gestation	1768
	29-30 d	7778
Meriones vinogradovi	21.5 d	8428
Mesocricetus auratus	15 d 21 h; n28 f	1071
	16 d	3571
	16 d; r15-17; n9	1382
Mesocricetus brandti	15 d	6656
Mystromys albicaudatus	37 d	10158
	38 d; r36-39	4220
Neotoma albigula	<30 d	3201
Neotoma cinerea	25-35 d; ibi	3087
	r27-31 d; n16	2909
Neotoma floridana	30-36 d	8659
	33-39 d; n1 f	8315
	33-41 d; recaptured f	8859
	34 d; r33.3-35	5892
	34.7 d; r32-37	7030
	est 6 wk	4281
Neotoma fuscipes	est 23 d; ≥16 to ≤29; n1	2679
	33 d; n2	11888
Neotoma lepida	30-36 d	2906
Neotoma mexicana	32.7 d; r31-34; n8, 5 f	8105
Neotoma micropus	30-39 d; pairing to birth	11696
	<33 d	3201
	mode 33-35 d	1242
Neotoma stephensi	31 d	11232
Neotomodon alstoni	27.3 d; sd1.1; mode 27; r26-30; n50 primiparous f	8096
Nyctomys sumichrasti	30 or 31 d; 1 f	946
Ochrotomys nuttalli	29-30 d	3897
Onychomys leucogaster	26 d; 30 d with lactation	8006
	27 d	4986
	27-32 d	1871a
	29-32 d	2907
	32 d; n1; pairing to birth	10588
	32-38 d; with lactation	2907
	32-47 d; with lactation	1871a
	39.7 d; r33-47; n3 with lactation	10588
Onychomys torridus	27-29 d	10697
Oryzomys argentatus	25 d	10809
Oryzomys capito	27.6 d; r25-31; n20; pairing to birth	11931
Oryzomys palustris	mode 21-24 d; r21-28; pairing to birth	8231
	25 d; n5	10572
Otomys irroratus	est 35-40 d	2339
	max 42 d	1312
Otomys sloggetti	38 d; min ibi	11710a
Ototylomys phyllotis	51 d; sd0.6; r50-52; n6	4606, 4607
	53.6 d; sd7.6; r49-69; n9 with lactation	4606, 4607
Pachyuromys duprasi	20 d; n2 lt, 1 f	4674, 4675
	20.4 d; mode 20; r19-22; n7	6234, 6237, 6238
	24 d	4582
Peromyscus boylii	23 d; n1	1363
	29.3 d; r26-32; n6 with lactation	1363
Peromyscus californicus	23.6 d; ±0.36; r21-25; n5	10573
	31.6 d; se0.2; r31-33; n18	4119
	34.5 d; se0.6; n15 with lactation	4119
Peromyscus crinitus	24-25 d	1871a
Peromyscus eremicus	21 d; n1	10573
	28-32 d	2526
Peromyscus gossypinus	22.86-23.34 d; n7 non lact	8698
Peromyscus hooperi	33.50 d; sd0.71; r33-34; n2	9603

Peromyscus leucopus	23.05 d; r22-25; n19	10573
	25.95 d; se0.5; n62	684
	29.56 d; r23-37; n25 with lactation	10573
Peromyscus maniculatus	3-4 wk	5019
	21-40 d	2806
	~23 d	10750
	23-24 d	2517, 2523
	23.2 d; se0.2; n11	5673
	23.51 d; r22-27 d; n43	10573, 10576, 10577
	mode 24 d (10/19); r23-26 d; n19 lt	7311
	25-26 d; with lactation	2517, 2523
	26-35 d; n1	10574
	27.14 d; r22-35; n21 with lactation	10573, 10576, 10577
	28.54 d; se0.81	6493
Peromyscus melanocarpus	30 d; n1	9114
	37.0 d; r31-41; n9 with lactation	9114
Peromyscus melanophrys	24.7 d; se0.1; r23-26; n24 lt, 24 f	3228
Peromyscus melanotis	23.55 d; ±0.11; r23-24; n9	10573
	26.57 d; ±0.73; r23-32; n7 with lactation	10573
Peromyscus mexicanus	30 d; n1 lact f	9114
	35 d; r31-39; n2; min ibi	9114
Peromyscus perfulvus	43.25 d; r39-46; n4	4609
Peromyscus polionotus	implant: 5 d	6302
	23.8 d; r23-24	6302
Peromyscus truei	26.20 d; ±0.26; r25-27; n5	10573
	40 d; n1 with lactation	10573
Peromyscus yucatanicus	30 d; pairing to birth	6149
	31-33 d; n5 with lactation	6149
Phodopus sungorus	18 d; min ibi	5443
	19 d; mode 19; r18-19; n29; pairing to birth	3249
	20-22 d	3334
	21-22 d	11285
Phyllotis darwini	33.6 d; n59; pairing to birth	11407
Psammomys obesus	23 d; r23-31; pairing to birth	8731
	24 d; min pairing to birth	9825
	24.8 d; n1; ibi	2234
	36 d; n1 lactating f	2234
Reithrodontomys humulis	21-22 d; n1; pairing to birth	5609
	≤25 d	6300
Reithrodontomys megalotis	22 d; min ibi	9107
	23-24 d; n4 lt, 2 f	10587a
	23-24 d	1474
Reithrodontomys montanus	21 d; min ibi	6391
Rhombomys opimus	23-32 d	7778
Saccostomus campestris	implant: ext 4-5 d	11607
	20-21 d	2861
	21.2 d; sd0.45; n36	11082
Sigmodon leucotis	~35 d	9960
Sigmodon hispidus	26-27 d	6975, 7030, 8915
	~27 d	7253
Sigmodon ochrognathus	34.5 d; r33-36; n2	4898
Tatera brantsii	22.5 d	7146
Tatera inclusa	23 d; n2	5050
Tatera indica	22.3 d; r18-30; mate at ppe: 36 d; r34-40; delayed with lactation of 3+ yg	987
	28.22; r27-30 d; ibi	8725
Tatera valida	~25 d	5058
Taterillus gracilis	26 d	5058
Taterillus pygargus	3 wk	8695a
	30 d	5058
Tylomys nudicaudus	implant: 4 d	4606
	min 39 d	10755
	40.6 d; se0.5; r35-51; n47 lt	4606, 4608

Zygodontomys brevicauda	25.0 d; ±0.19; n29	87
	25.1 d; sd1.0; r24-27; n10 f	11373
	26.0 d; r25-32; n16; pairing to birth	11931
	27.9 d; sd2.4; n78; pairing to birth	11373
LACTATION		
Akodon azarae	wean: 14-15 d	2221
Akodon molinae	wean: 26 d; r21-30	7212
Baiomys taylori	teat attachment: 18-22 d	977
Beamys hindei	solid food: ~3 wk	2912
	wean: 5-6 wk	2912
Calomys callosus	wean: 15-17 d	7189
	wean: 21 d	5474
Cricetomys gambianus	solid food: 17 d	3130
	solid food: 17-18 d	10158
	solid food: 26-28 d	103
	wean: 26-28 d	103
	wean: 5 wk	7528
	den emergence: 43-52 d	3130
Cricetulus barabensis	wean: 20 d	4695
Cricetulus griseus	wean: 21 d	8685
	wean: 25 d	468
	wean: < 4 wk	2730
	den emergence: 4 wk	2730
Cricetulus migratorius	wean: 3rd-4th wk	9175
Cricetulus triton	solid food: 15 d	6602
Cricetus cricetus	solid food: 5 d	8457
	solid food: 8 d-3 wk	3571
	solid food: 10 d	9440
	wean: 25 d	7436
	den emergence: 25 d	8457
Desmodillus auricularis	solid food: 18 d	5684
	solid food: 21 d	7824
	wean: 33 d	7824
	nest emergence: 10 d	5684
Gerbillurus paeba	wean: 19 d	10158
	wean: 20-30 d	403
	wean: 28-30 d	2481
Gerbillurus tytonis	wean: 28-30 d	2481
Gerbillus nanus	wean: end 3 wk	9175
	mother leaves yg: 4 wk	5785
Gerbillus pyramidum	wean: 25-30 d	4348
	wean: ~4 wk; occasional 2 mo	4713
Holochilus brasiliensis	wean: 3 wk	7192
Lophiomys imhausi	independent: 40 d	2446a
Malacothrix typica	lact: max 32 d	5887
Megadontomys thomasi	wean: 30 d	2728
Meriones crassus	solid food: 16 d	2233
	wean: 20 d	6236
Meriones hurrianae	wean: 21 d	9175
Meriones libycus	solid food: 17 d	2233
	wean: 30 d	11905
	independent: 1 mo	7778
Meriones meridianus	independent: 20-22 d	11252
Meriones persicus	wean: 18-20 d	9175
Meriones shawi	solid food: 3 wk	9451
	wean: 3 wk	9451
Meriones tamariscinus	wean: 14-15 d	1035
	wean: 21 d or more?	8961
Meriones unguiculatus	solid food: 17-20 d	7098
	wean: 20-25 d	7778
	wean: 21 d	7098
	wean: 23-28 d	1768
	wean: 32 d	7964

Mesocricetus auratus	solid food: 8 d	1071
	independent: 20-25 d	3571
Mesocricetus brandti	wean: <20 d	6656
Mystromys albicaudatus	solid food: 20 d	4246
	teat attachment: 15-20 d	4246, 7168
	teat attachment: to 3rd wk	4220
	wean: 21-28 d	4220
	wean: 32 d	7168
	wean: 38 d	10158
Neotoma albigula	lact: 20-25 d	3201
	1st trapped: 28-35 d	9689
	wean: 27-40 d	9689
	wean: min 37 d; max 62-72	9106
Neotoma cinerea	1st trapped: 3 wk	6895
	wean: 26-30 d	2909
	wean: 5 wk	3087
Neotoma floridana	solid food: 16-20 d	8859
	solid food: 19-22 d	7030
	solid food: 24 d	4281
Neotoma fuscipes	wean: ~3 wk	3041
	wean: 36 d; n1	2679
Neotoma lepida	solid food: 3 wk	2906
	wean: 27-40 d	9689
	wean: 4 wk	2906
	1st trapped: 35-41 d	9689
Neotoma stephensi	solid food: 2 wk	11231
	wean: 5-6 wk	11232
Nyctomys sumichrasti	teat attachment: 2 wk	946
Ochrotomys nuttalli	wean: 17-18 d	6497
Onychomys leucogaster	nest emergence: 5 d	9366
	solid food: 10 d	9366
	solid food: 23 d	4986
	practically wean: 19-20 d	10588
	wean: 24 d	9366
	wean: ≤1 mo	1871a
	wean: 31 d; n1 lt of 4	8006
Onychomys torridus	wean: 20 d	4991
Oryzomys nigripes	wean: 14-15 d	2221
Oryzomys palustris	wean: 11 d	10572
	wean: 20 d	8231
Oryzomys subflavus	wean: 4 wk	11305
Otomys irroratus	wean: 12 d	2446a
Otomys sloggetti	solid food: 12 d	11710a
	wean: 16 d	11710a
Otomys irroratus	solid food: 2 d	2338
	teat attachment: 7-14 d	10158
	wean: 13 d	2338
	wean: 4-5 wk	2571
Ototylomys phyllotis	teat attachment: 30 d	4606, 4607
Oxymycterus rutilans	wean: 14 d	2221
Pachyuromys duprasi	solid food: 20 d	6237
	wean: 29 d	6237
Peromyscus californicus	wean: 30 d	2728
Peromyscus crinitus	wean: 23 d	2728
	wean: 4 wk	2910
Peromyscus difficilis	wean: 26 d	2728
Peromyscus eremicus	wean: 20-22 d	1236
	wean: 21 d	2728
Peromyscus hooperi	wean: 22.67 d; sd2.89, n3	9603
Peromyscus interparietalis	wean: 16-21 d	1236
Peromyscus leucopus	leave nest: 14-16 d	11288
	solid food: 13-16 d	7304
	solid food: 18 d	7303
	independent: 20 d	7303

	wean: 21 d	2728
	wean: 22 d	7304
	wean: 22 d	4780
	wean: 24-30 d	1871a
Peromyscus maniculatus	leave nest: 14-16 d	11288
	1st emergence: 16.0 d; se0.4; r15-17; n4	7308
	1st emergence: 16.6 d; se0.3; r16-18; n8	7308
	wean: r18.01-18.93 d; n3 areas	4218
	wean: 3 wk	5658
	wean: 21-23 d	2728
	wean: 21-25 d	7843
	wean: 22-37 d	2806
	wean: 23-25 d; begins at eye opening	10575, 10576, 10577
	wean: 30-40 d	2699
	no wgt loss after 24 hr: 18-24 d; n2 subspecies	5750
	zero growth: 19.2 d	7308
	zero growth: 25.3 d	7308
	1st trapped: 4 wk	5658
Peromyscus megalops	wean: 21-23 d	6302
Peromyscus melanocarpus	no wt loss after 24 hr: 4-5 d after eye opening	9114
Peromyscus melanophrys	wean: 26 d	2728
	wean: 33-35 d	3228
Peromyscus mexicanus	no wt loss after 24 h: 4-5 d after eye opening	9114
Peromyscus polionotus	teat attachment: 2 wk	8907
	wean: 21 d	2728
	wean: 28 d; r25-31	6302
Peromyscus truei	wean: ≤30 d	1871a
	wean: 30 d	2699
Petromyscus collinus	wean: 30-33 d	2481
Phodopus sungorus	solid food: 10 d	11548
	solid food: 16 d	11285
	den emergence: 16 d	11285
	wean: 20 d	5443
	lact: 20-21 d	11285
Podomys floridanus	wean: 3-4 wk	6301
	wean: 26 d	2728
Psammomys obesus	solid food: 11 d	2233, 2234
	solid food: 14 d	8731
	den emergence: 11-12 d	2234
	wean: 28 d	8731
	wean: 4 wk	10481
Reithrodontomys fulvescens	wean: 13-16 d	8189
Reithrodontomys humulis	solid food: ~9 d	5609
	wean: 3 wk	5609
Reithrodontomys megalotis	solid food: 11-12 d	10587a
	wean: 16 d max	9107
	wean: ≤20 d	1871a
	wean: 21 d	10797
	wean: ~24 d	5210
Reithrodontomys montanus	wean: 14 d	6391
	leave nest: 3-4 wk	1871a
Rhombomys opimus	solid food: 12-13 d	7778
	wean: 22 d	7778
Saccostomus campestris	coprophagy: 5 d	2861
	solid food: 13 d	2861
	solid food: 17 d; sd1.22; n71	11082
	solid food: 25 d	4329
	leave nest: 2.5-3 wk	2861
	1st use of cheek pouch: 3 wk	2861
	wean: 25 d; n181	11082
	lact: 35 d	4329
Scotinomys teguina	solid food: 16-18 d	4967
	wean: 3 wk	4967

Scotinomys xerampelinus	solid food: 21-24 d	4967
	wean: 3.5 wk	4967
Sigmodon hispidus	solid food: 10 d	9572
	wean: 10 d	10570
	wean: 10-12 d	6975, 7030
	wean: 12 d	6975, 8915
	wean: 18-21 d	9572
	wean: 20 d	7253
Sigmodon ochrognathus	solid food: 8 d	4898
	wean: 15 d	4898
Tatera brantsii	teat attachment: 17 d	7168
	wean: 22-24 d	7146
	wean: 29 d	7168
Tatera inclusa	wean: 22 d; n1 lt	5050
Tatera indica	wean: 20 d	987
Tatera leucogaster	wean: 28-28 d	7797a
Taterillus pygargus	wean: 3 wk	8695a
Tylomys nudicaudus	teat attachment: 2-3 wk	10755
	wean: 2-3 wk	10755
Zygodontomys brevicauda	solid food: 9-10 d	11373
	can survive without mom: 10 d	87
	wean: 16-21 d	11373
INTERLITTER INTERVAL		
Akodon longipilis	min 16-18 d	7241
Akodon molinae	30 d	7212
Baiomys taylori	20.5-21 d	7544
	23 d; r21-26; n7	5064
	27.6 d; se1.8; r20-64; n26 intervals, 6 f	977
Beamys hindei	min 62 d	2912
Calomys callosus	20.5-21 d	7544
	mode 22 d; r20-23; n28	8430
	min 21 d	5474
Calomys lepidus	20.5-21 d	7544
Cricetulus griseus	min 19 d	8685
Cricetus cricetus	131.40 d; ±67.77; n11	9085
Dipodillus simoni	46.3 d; r30-57; n10	6237
Meriones crassus	59-75 d; n4	6237
Meriones shawi	31.25 d; r30-32; n4	6237
Meriones tamariscinus	25-29 d; r28-84	1035
Meriones unguiculatus	38.6 d; se0.46; r25-123	1768
	54.7 d; mode 30-39; n192 lt	382
	40-52 d; depending on age	56
Mystromys albicaudatus	min 36 d	7168
Neotoma albigula	49 d; n1	9106
Neotoma cinerea	25-35 d	3087
Neotoma floridana	83 d; n1	8659
Neotoma lepida	60 d	2906
Neotoma stephensi	130 d est	11232
Nyctomys sumichrasti	min 38 d	946
Ochrotomys nuttalli	min 25 d; n3; r25-28; n31 young live	6497
	min 25 d; n1; r25-28; n5 no lactation	6497
Onychomys leucogaster	26-37 d no lactation	8559
Onychomys torridus	27-30 d	10697
	28-30 d; r26-35	8559
	1 mo	979
Otomys sloggetti	49 d; r38-74; n4 intervals, 2 f	11710a
Peromyscus californicus	24 d	9255
Peromyscus crinitus	25 d	9255
	min 27 d; most 29-31; 24-25 no lactation	2910
Peromyscus eremicus	27 d	9255
	50 d; min 28	2319
Peromyscus interparietalis	31.9 d; r27-37; n25; est lactational gestation	1236

Peromyscus leucopus	20 d	9255
	mode 24 d; r22-24; n182 lts	6150
	29.2 d; r23-42; n151 intervals, 27 f	4779
	29.56 d; r23-37; n25	10573
Peromyscus maniculatus	20.5-21 d	7544
	21-23 d	9255
	27.14 d; r22-35; n21	10573, 10576, 10577
	1 mo; young removed 2 wk	5797
	35.2 d; min 18; n489	7312
Peromyscus melanocarpus	30 d; n1 young die	9114
	37.1 d; n8 young live	9114
Peromyscus melanotis	26.57 d; se0.73; r23-32; n7	10573
Peromyscus polionotus	21 d	9255
	23-24 d; no lactation	11725
	24-26 d	8907
	29.43 d; ±3.81; r23-41; with lactation	11725
Peromyscus truei	40 d; n1	10573
Peromyscus yucatanicus	27 d; n1	6149
	28 d; n4; young die	6149
Phodopus sungorus	min 18 d; n4	5443
	mode 19-23 d 35%; r18-87	11548
	32.0 d; r18-80	7255
	35.6 d; r16-120	7255
	93 d; r22-360	7255
Podomys floridanus	33 d	9255
Psammomys obesus	38.3 d; r35-44 d; n6, 1 f	2234
Reithrodontomys humulis	31.6 d; r24-49 d; n5	6300
Reithrodontomys megalotis	27.6 d; r22-38; n26, 2 f	550
Saccostomus campestris	51.8 d; sd9.06; r46-74; n14	11082
Scotinomys teguina	30.8 d; r27-36; n16 young die	4967
	31.1 d; r29-36; n32 young live	4967
Scotinomys xerampelinus	31.9 d; r30-38; n10 young die	4967
	34.3 d; r29-38; n17 young live	4967
Tatera brantsii	131 d; n1	7168
Tatera inclusa	2 mo; n1	5050
Tatera indica	28.22 d; r27-30	8725
	67 d	1689
Tatera nigricauda	40-44 d; wet season	7794
	97-107; dry season	7794
Tatera valida	109 d	7794
Tylomys nudicaudus	min 39 d	10755
Zygodontomys brevicauda	~25 d	87
	mode 25 d 75%; r24-28; n71	11373
LITTERS/YEAR		
Akodon azarae	mode 1; r1-4	2221
Akodon urichi	3.6	7995, 7998
Baiomys taylori	10	10110a
Beamys hindei	5 max	2912
Calomys callosus	3 max	5474
Calomyscus bailwardi	0-2	3587
Calomyscus mystax	2	9488
Cricetomys gambianus	2	11253
	5-6	103
Cricetulus eversmanni	2	4645
Cricetulus migratorius	2; in the south perhaps more	4645
Cricetus cricetus	1-2	9440
	1-2	4645
	1-3	11319
	2; r2-3	3571
	2-3	7433
	2-3	9085
	3 max	8457
Gerbillus nanus	3	5785
Gerbillus pyramidum	5	4713

Malacothrix typica	>1	10158
Meriones crassus	2	9175
Meriones meridianus	2-3	7778
	3-4	4645
Meriones tamariscinus	2; r2-3 ?	4645
Meriones unguiculatus	mode 5; r1-10; n657; 13 dead/2160 born	56
	3; r2-3	10516
	3 max	7778
	3.0; se0.30; late maturing	1863
	5.2; se0.84; early maturing	1863
Mesocricetus auratus	3 lt/6 mo; n1 f	1382
	3-4	2419
	3-5; r3-8	3571
Mesocricetus brandti	3 max	6656
Mesocricetus raddei	2	4645
Myospalax myospalax	1	8050
Neacomys tenuipes	6.1	7998, 7995
Neotoma cinerea	1	2615
	1	1474
	1	2088
	2 lt	6895
	2-3	3087
	3.0; max 7	2909
Neotoma floridana	2-3	8659
	2-3	3285
Neotoma fuscipes	1	11289
	1-5	6490
Neotoma lepida	1-4 lt/season	6708
	4	2906
Neotoma mexicana	2	1366
Neotoma micropus	1	3201
	2+; r1-5	8962
	2+; min 2-3	11696
Neotoma stephensi	1.9; sd0.86; r0-5; n51 [note includes 0]	11232
Nyctomys sumichrasti	5 lt/7 mo; n1 f	946
Onychomys leucogaster	est >1	502
	2-3	1474
Onychomys torridus	2-3	1474
	min 2	2109
	juv: 3.8 lt/yr; 9.8 yg/yr; adult: 1.6 lt/yr; 4.0 yg/yr	10697
Oryzomys capito	5.8	7995, 7998
	6.06; n34 f	3316, 3319
Oryzomys concolor	4.2	7995, 7998
Otomys irroratus	3-5	2338
	mode 4-5; 7 poss	10158
Otomys typus	max 5	5760
Peromyscus californicus	5.5; r5-6; n2	10573
Peromyscus crinitus	2.05; max 8	2910
Peromyscus eremicus	max 3	6708
	est 3-4	1474
Peromyscus gossypinus	est 4+	1474
Peromyscus leucopus	1-4/5 mo season	4376
	2-4	1474
	2.0	11877
	4-5 lt/season	1472
	5; r4-6; n2	10573
	5.3; r5-6; n3	10573
Peromyscus maniculatus	1.5-5.2; field	4218
	1.9-2.3	5151
	2 max	6708
	2	11230
	2 lt/season not uncommon; no f w/3 n35f	10839
	2 or more	11956
	2-4	1474

	2-4 or more	2088
	2.1-2.9; lab	4218
	2.5-2.9; poor vs good seed crop	3624
	5.5; r4.7; n2	10573
	5; r4-6; n2	10573
	6; r5-7; n2	10573
	max 8 lt/8 mo; yg removed 2 wk	5797
	1: yg not removed 13 lt/370 d & 12 lt/512 d	5797
	13; n1 pr	6493
Peromyscus melanotis	7.5; r7, 8	10573
Peromyscus pectoralis	est > 1	1474
Peromyscus polionotus	est 2+	1474
Peromyscus sitkensis	2	2088
	est 2	1474
Peromyscus truei	est > 1	1474
	5; r3-7; n4	10573
Petromyscus collinus	1	11864
Phodopus sungorus	mode 3,4	11285
Reithrodontomys megalotis	4-5 lt/season max	9107
Reithrodontomys raviventris	1	10809
Rhipidomys mastacalis	3.5	7995, 7998
Rhombomys opimus	2-3	10516
	max 4; most 2-3	7778
Sigmodon alstoni	5.0	7995, 7998
Sigmodon hispidus	2	6303
	max 9	1474
	9 lt/10 mo	5146
Tatera afra	6-7	7146
Tatera brantsii	4; r4-5 or 5-6*	7146
Tatera indica	5.5	1689
Tatera nigricauda	5; 4 wet season, 1 dry season; r4.9-5.4	7794
Tatera valida	2	7794
Tylomys nudicaudus	6 lt/13 mo; n1 f	10755
Zygodontomys brevicauda	5.5	7995, 7998
SEASONALITY		
Akodon azarae	mate: June-Aug or yr rd; few data; Uruguay	596
	preg f: peak Nov-Apr; central Argentina	7347
	birth: Oct-Apr; Buenos Aires, Argentina	8305
	birth: Nov-Apr; Argentina	2221
Akodon boliviensis	testes: large Dec; few data; Peru	8302
	preg f: Dec; n1; Peru	8302
Akodon dolores	birth: Jan-Mar, July-Aug, Nov-Dec; captive	7213
Akodon longipilis	testes descend: July-Sept; n Chile	7241
	preg f: June-Sept, none Nov-June; n Chile	7241
	preg f: end Oct-Dec; Patagonia, Argentina	8305a
	preg/lact f: Feb-Mar; n12/113 f; Chile	4054
	repro: Oct-May; s Chile	7242
Akodon molinae	birth: Jan-May, July-Sept, Dec, peaks Mar-Apr, July-Aug, Dec; lab	7213
Akodon olivaceus	testes descend: June-Jan; n Chile	7241
	mate: Sept-Mar; Santiago, Chile	3551
	mate: Nov-Dec; Fray Jorge, Chile	3551
	preg f: Sept-Nov; Chile	7241
	preg f: Oct-Dec; Patagonia, Argentina	8305a
	preg/lact f: Nov-Dec, Feb; n27/147; Malleco, Chile	4054
Akodon sanborni	repro: Oct-May; Chile	7242
Akodon urichi	preg f: est yr rd; n Venezuela	7995, 7998
Auliscomys boliviensis	preg f: Oct; n1; Caccacharo, Peru	8302
Auliscomys micropus	preg f: Mar; n2/15; Malleco, Chile	4054
	preg f: Nov-Dec; Patagonia Argentina	8305a
	lact f: Feb-Mar; n2/15; Malleco, Chile	4054
Baiomys musculus	preg f: July; n1; Mexico	2341
	preg f: yr rd; Veracruz, Mexico	4228
	preg f: wet season; s Mexico	11388

Baiomys taylori	mate: Jan-Oct; s TX, n Mexico	1474
	preg f: yr rd, peak Oct-Nov; n54; s TX	8963
Beamys hindei	repro: wet season; Nyasaland	4330
	mate: coincides with rains; Malawi	5760
	preg f: Nov-May; Nyasaland	4330
	birth: yr rd; captive	2912
Bolomys lasiurus	preg f: peak Apr-June; ne Brazil	5540
Bolomys obscurus	preg f: Oct-Feb; Uruguay	596
	preg f: est yr rd; central Argentina	7347
Calomys callosus	preg/lact f: Feb, Apr, Sept, Nov; ne Brazil	10494
	repro: yr rd; captive	8430
Calomys laucha	preg f: summer; n4; grasslands, S America	4682
	preg f: yr rd, low July-Aug; central Argentina	7347
Calomys musculinus	preg f: est yr rd, low July-Sept; central Argentina	7347
Calomyscus bailwardi	mate: est spring-fall; Pakistan	9175
	birth: Mar-June; Armenia USSR	3587
	lact f: June; n1; Iran	2404a
	lact f: Aug, Oct, Dec; Iran	6293
Calomyscus mystax	mate: mid Mar-mid June; USSR	9488
Cricetomys gambianus	epididymal sperm: peak Oct-Jan; Kenya	312
	mate: est rainy season; Nyasaland	7528
	conceive: Apr; captive, Malawi	4329
	preg f: Jan, Mar-Apr, Aug; n4; Kenya	312
	preg f: Sept; Malawi	4329
	preg/lact f: Feb; n2; e Africa	185
	birth: Dec-Mar; s Africa	10153
	lact f: Jan; n2; Zambia	331
	lact f: Apr; n1; Tanzania	186
	repro: twice/yr; Zaire	11253
Cricetulus barabensis	repro: Apr-Oct; est Transbaikal, Siberia, Russia	3334
Cricetulus curtatus	birth: Apr-Sept; Mongolia	3334
Cricetulus eversmanni	birth: summer; EurUSSR	4645
Cricetulus griseus	preg f: Feb-Nov; no data Dec-Jan; Peking, China	2730
Cricetulus longicaudatus	mate: early Mar; USSR	3335
	repro: end Mar-Aug/Sept; Transbaikal, Russia	3334
Cricetulus migratorius	testes: large Dec; Lebanon	6419
	testes: large spring-mid May; Pakistan	9175
	preg f: Mar; n1; Lebanon	6419
	preg f: Aug, Nov-Jan; no data other mo; Iran	6293
	birth: spring-summer; Pakistan	9175
	lact f: May; n1; Iran	2404a
	lact f: Oct; n1; Lebanon	6419
Cricetulus triton	mate: Jan-May; Manchuria, China	6602
	preg/lact f: June, Aug; Korea	5424_1
Cricetus cricetus	birth: May-Aug; Germany	3571
	mate: mid Apr-July; lab	11319
	mate: end Apr; Germany	8457
	mate: end Apr-July; Germany	3571
	mate: May-July; Europe, USSR	7433
	mate: yr rd; lab	9085
	birth: May-July; Eur, USSR	7433
	repro: see DC; Yugoslavia	6050
Dendromus kahuziensis	birth: Feb; n2; central Africa	187
Dendromus melanotis	birth: Jan-Feb, Oct; Zambia	331
	lact f: May; n1; Tanzania	186
Dendromus mesomelas	preg f: Mar, May-June; n3; Malawi	4329
	preg f: Nov; n1; Cameroun	2955
	juveniles: end Feb; Tanzania	186
Dendromus messorius	preg f: Apr; n1; central Africa	4495
	juveniles: Aug; central Africa	4495
Dendromus mystacalis	preg f: Jan; n1; Kenya	4925
	birth: peak Nov-Jan; e Zaire, Uganda	5760
	lact f: Apr; n1; central Africa	187

Dendromus nyikae	testes large: Jan, Aug; n2; Tanzania, Angola	5760
	preg f: Nov; n1; Malawi	4329
	lact f: May; n1; Malawi	4329
	lact f: mid Aug; n1; Tanzania, Angola	5760
Deomys ferrugineus	estrus: yr rd; Zaire	2573
	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	preg f: Apr, June, Dec; Cameroun, Uganda	5760
	preg f: est yr rd, peak May-Nov; Zaire	8855
Desmodillus auricularis	preg f: Jan; n1; Kalahari Gemsbok Nat Park, S Africa	7823
	birth: Mar, Sept; n3; lab	5684
	juveniles: Jan, May, July-Aug, Oct-Nov; Botswana	10154
Dipodillus simoni	birth: yr rd; lab	6237
Eligmodontia typus	preg f: Oct-Apr; Patagonia, Argentina	8307
Gerbillurus paeba	preg f: Jan-Feb, Nov; n44; Kalahari Gemsbok Nat Park, S Africa	7823
	preg f: Oct; n1; Namibia	9945
	preg f: Dec-Mar, June-Aug, Oct; Botswana	10154
	birth: Oct-June; Cape Province, S Africa	404
	birth: yr rd; S Africa	2381
Gerbillus andersoni	mate: Feb-May; Israel	20
	mate: Sept-June; Egypt	8134
Gerbillus campestris	mating condition: Nov; n1 m; Sudan	4345
	preg f: Mar, Dec; n3; Egypt	8134
	lact f: May, Dec; n2; Egypt	8134
Gerbillus cheesmani	mate: est yr rd; Pakistan	9175
	lact f: Feb, Nov; Pakistan	9175
Gerbillus dasyurus	testes: large Mar-Apr; lab	712
	mate: Feb-Apr; Ngev Desert, Israel	715
	mate: Feb-July; Saudi Arabia	109
Gerbillus gerbillus	testes: large Jan-May; Egypt	8134
	testes: large Mar-Apr, small Sept-Oct; Sahara	5707
	preg/lact f: Jan-May; Egypt	8134
Gerbillus gleadowi	preg f: May-June, Nov-Jan; w Rajasthan, India	8723
	birth: peak post monsoon & winter; Pakistan	9175
Gerbillus nanus	mate: summer, winter; w Rajasthan, India	8723
	mate: yr rd, peak summer, winter; Pakistan	9175
	preg f: Nov; n1; no data other mo; se Iran	6293
	lact f: Dec; no data other mo; se Iran	6293
	juveniles: Jan; no data other mo; se Iran	6293
Gerbillus pyramidum	mate: June-Mar; Sudan	4348
	birth: June-Feb; Sudan	4344, 4348, 4351
	repro: yr rd; captive	4713
Graomys griseoflavus	preg f: peak Dec, Feb-Mar, none June-Aug; n Mendozo, Argentina	2388
Gymnuromys roberti	preg f: June-July; Madagascar	7983
Hodomys alleni	preg f: Feb; n1; Nayarit, Mexico	3700
	lact f: Sept; n1; Sinaloa, Mexico	3700
Holochilus brasiliensis	mate: yr rd; Brazil	4681
Holochilus magnus	mate: yr rd; e Uruguay	4681
Holochilus sciureus	preg f: yr rd; British Guyana	10983
Irenomys tarsalis	preg f: Nov; n2; Patagonia, Argentina	8305a
Isthmomys flavidus	lact f: Feb, June; n2; Panama	11044a
Isthmomys pirrensis	preg f: Feb; n4; Panama	11044a
	lact f: Feb-Mar; n6; Panama	11044a
Malacothrix typica	birth: Aug-Mar; Botswana	10154
Meriones crassus	mate: Nov-June; few data; Egypt	8134
	birth: yr rd; n35; zoo	12128
	birth: peak spring-summer; Pakistan	9175
Meriones hurrianae	scent gland: large July-Sept, small Oct-Nov; India	6097
	mate: Aug-Oct; Rajasthan, India	8721
	mate: end Oct-Dec; India	79
	preg f: Sept-Nov; Rajasthan, India	8720
	preg f: yr rd, peak Feb, low Dec-Jan; India	6097, 8723
Meriones libycus	mate: yr rd; USSR	7778
	mate: spring, fall; Pakistan	9175

	preg f: Oct; n1; Iran	6293
	juveniles: end Oct; Iran	6293
Meriones meridianus	repro: Feb-Nov; USSR	8275
	mate: Mar-Sept; EurUSSR	4645
	preg f: peaks Sept, Apr; USSR	8275
	preg f: yr rd; USSR	7778
	repro: mid Feb-mid June; USSR	9499
Meriones persicus	large follicles: Sept; n1; Iran	6293
	preg f: May; n1; Iran	2404a
	juveniles: Sept-Oct; Iran	6293
Meriones shawi	birth: yr rd; n51; zoo	12128
Meriones tamariscinus	repro: Feb-Nov; USSR	8275
	birth: yr rd, peak Apr-Sept, 46/67; lab	1035
	birth: Apr-Oct, peak May, Aug; USSR	7778
Meriones tristrami	juveniles: Sept; n1 lt; Iran	6293
Meriones unguiculatus	preg f: mid May-Sept; Mongolia	10516
	preg f: June-July; Mongolia	10516
Mesocricetus auratus	mate: Mar-Oct; Germany; yr rd; captive	3571
	birth: yr rd, peak May-Oct; captive	1382
	birth: yr rd; lab	1382
	repro: yr rd; captive	1071
Mesocricetus brandti	mate: none Nov-Mar; lab	6656
Mesocricetus raddei	birth: summer; Europe, USSR	4645
Myospalax myospalax	birth: end Apr-early June; USSR	8050
Myospalax psilurus	birth: Mar-Apr; Mongolia	1817
Neacomys tenuipes	preg f: est yr rd; n Venezuela	7995, 7998
Nectomys squamipes	preg f: Aug, Oct-Nov; Brazil	3078
	preg f: Oct-Nov; Argentina	3077
Neotoma albigula	mate: Jan-Aug; sw US	1474
	mate: yr rd; Coahuila, Mexico	522
	preg f: Mar, Sept-Oct; n8; NM	502
	preg f: Apr-June; no data other mo; CO	3262
	preg f: Mar-May, July; n14; Coahuila, Mexico	522
	preg f: June-July; n3; Durango, Mexico	523a
Neotoma bryanti	lact f: May; n1; Baja Califomia	11044a
Neotoma cinerea	mate: Feb-July; captive	3087
	mate: May-Sept; w US, BC, Canada	1474
	preg f: May-Sept; LA	2615
	preg f: end May-July; n1; CO	3262
	preg f: June; few data; SD	10954
	birth: mid Mar-July; lab	3087
	birth: mid Apr-mid Aug; NV	3087
	birth: early May-June; BC, Canada	2088
	lact f: end May-July; n6; CO	3262
	lact f: June; n2; CO	1649a
	lact f: July; n1; NE	$5405b_1$
Neotoma chrysomelas	preg f: Feb, Sept; n2; Honduras	11044a
Neotoma floridana	testes: peak Jan, low Sept-Oct; KS	8859
	estrus: yr rd; lab	1703
	mate: end Jan-fall; KS	3285
	mate: Feb-Aug, peaks early Feb, end Mar, early Apr, May; KS	8859
	mate: yr rd, peak summer; FL	4680
	mate: yr rd, low winter; OK	3851
	preg f: June; n1; KS	1812a
	preg f: July; n1; NE	$5405b_1$
	birth: Mar-Apr; central US	949
	birth: Mar-Sept; PA	9164
	birth: mid Mar-early Sept; PA	8659
	birth: yr rd; s US	1474
Neotoma fuscipes	mate: Mar-Aug; CA	11289
	mate: end Mar-?; CA	2679
	conception: Jan-Sept, peak Feb-Apr; CA	6490
	preg f: Dec; n1; CA	3593a
	birth: Feb-May; OR	3041

	birth: Apr-early May; n6; CA	2679
	birth: peak May-June, few Jan-Oct; OR, CA	1474
	lact f: Mar, Oct; CA	11228a
Neotoma goldmani	lact f: July-Aug; San Luis Potosí, Mexico	2228a
Neotoma lepida	testes: peak Aug-May; CA	6708
	preg/lact f: Nov-May; CA	6708
	lact f: Sept; n1; CA	11228a
Neotoma mexicana	testes: small Aug-Nov; CO	1366
	ovulation: Mar-July, peak Mar-May; CO	1366
	preg f: Apr, June; few data; CO	3262
	preg f: Apr, Aug; n3; Coahuila, Mexico	522
	preg f: May; n1; NM	502
	preg f: July; n3; Durango, Mexico	523a
	lact f: end May-early June; n2; CO	3262
Neotoma micropus	epididymal sperm: yr rd; TX	11696
	mate: Mar-May; no data other mo; NM	979
	preg f: Mar-Sept; TX	11696
	preg f: June; n1; NM	502
	preg f: Dec; n1; Coahuila, Mexico	522
	preg/lact f: May-June; no data other mo; CO	3262
	birth: Feb-June; central N America	949
	birth: yr rd; s TX	8962
Neotoma phenax	preg f: Feb; n1; Sonora, Mexico	1471a
	preg f: Apr; n1; Mexico	5423
	preg f: Dec-Jan; Sinaloa, Mexico	5423
Neotoma stephensi	birth: most Feb-July, peak Mar-May; AZ	11232
Neotomodon alstoni	repro: mid June-Sept; Mexico	2346
	repro: Mar-Sept; Cerrodil, Ajusco, Mexico	9232
Notiomys macronyx	preg f: Nov; n1; Patagonia, Argentina	8305a
	lact f: Feb; n1; Malleco, Chile	4054
Notiomys valdivianus	preg f: Nov; Patagonia Argentina	8305a
Nyctomys sumichrasti	preg/lact f: July-Aug; n3/25; nw Nicaragua	3704
Ochrotomys nuttalli	mate: mid Mar-early Oct, peaks end spring, early fall; TN	6496
	mate: Sept-Apr; e TX	7022
	mate: yr rd, peak spring-summer; TN	6497
	preg f: Mar; n1; TX	8189a
	preg f: Apr; n1; MO	2863a
	preg f: Apr-June, Aug-Sept; NC	1290
	birth: est Mar-Oct; KE	3897
	lact: Apr; n1; KY	11588a
	lact f: Dec; n1; TX	8189a
Onychomys leucogaster	repro: Mar-Aug; ne CO	3294
	mate: Jan-Feb, Apr; few data, none after May; NM	979
	preg f: Mar-Apr; n2; Coahuila, Mexico	522
	preg f: Apr, June, Aug; MAN, Canada	11957a
	preg f: Apr, June-Aug; n9; NE	5405b,
	preg f: Apr, Aug; n2; NM	502
	preg f: Apr-Dec; TX, AZ	504
	preg f: May-Aug; n US	504
	preg f: June; n2; KS	1812a
	birth: yr rd, peak May-Sept; lab	8559
	birth: peak Feb-Aug; UT	2907
	birth: Feb-Sept; sw US	1474
Onychomys torridus	mate: Mar-May; NM	979
	preg f: May-Aug; n US	504
	preg f: Apr-Dec; TX, AZ	504
	preg f: June-July; n5; Coahuila, Mexico	522
	preg f: June-July; n3; Durango, Mexico	523a
	preg f: Nov; n2; Sinaloa, Mexico	5409
	birth: Feb-Sept; sw US	1474
	birth: July, Sept; n2; captive	2109
	birth: yr rd, peak Jan-July; captive	8559
	repro: yr rd; lab	10697

Oryzomys alfaroi	preg f: Nov; n3; San Luis Potosí, Mexico	2228a
	preg f: yr rd; Oaxaca, Mexico	9176
Oryzomys argentatus	mate: Feb-Nov; LA	10809
Oryzomys capito	mate: yr rd; Panama	3316, 3317, 3319
	preg f: Jan, July; n4; Panama	3024
	preg f: peak Aug-Jan, Oct-Nov; Panama	3319
	birth: yr rd; Trinidad	11931
Oryzomys concolor	preg f: est yr rd; Venezuela	7995, 7996
Oryzomys couesi	preg f: Feb, Sept, Nov; n3; El Salvador	3210
	preg f: Nov; n1; San Luis Potosí, Mexico	2228a
Oryzomys delticola	mate: end summer-early fall; few data; Uruguay	596
Oryzomys flavescens	mate: Jan-May; n12/14; Uruguay	596
	preg f: Sept-May; central Argentina	7347
Oryzomys fulvescens	preg f: Dec; n1; Campeche, Mexcio	5423b
Oryzomys longicaudatus	preg f: Mar; n5/7; Santiago, Chile	3551
	preg f: Apr, Nov-Dec; Patagonia, Argentina	8305a
	preg/lact f: peak Dec-Feb; Malleco, Chile	4054
	repro: Oct-May; Chile	7242, 7666
Oryzomys melanotis	preg f: July; n1; Yucatan, Mexico	5423b
Oryzomys nigripes	preg f: Jan-May; Balcarce, Argentina	2221
Oryzomys palustris	mate: Feb-Oct; captive	10572
	mate: yr rd; se US	1474
	preg f: Mar-Apr, June-Sept, Nov; NC	1290
	preg f: Mar-Sept; Veracruz, Mexico	4228
	preg f: May; n1; Qunitana Roo, Mexico	950
	preg f: Aug; n1; Cozumel, Quintana Roo, Mexico	5423b
	preg f: yr rd, peak end spring-fall; MS	11871
	repro: yr rd; LA	7810
Oryzomys subflavus	preg f: peak Apr-May; lab	11305
Osgoodomys banderanus	lact f: Mar; n1; Mexico	11044a
Otomys angoniensis	mate: yr rd; Transvaal, S Africa	7200
	preg f: May, July, Nov; Malawi	4329
	preg f: June; n1; Kenya	4925
	birth: perhaps Oct-Mar; S Africa	10158
	birth: summer; Transvaal, S Africa	8965
	birth: yr rd, peak Apr-Sept; nw Kenya	10712
Otomys denti	preg f: May, June, Nov; Malawi	4329
	preg f: yr rd; Zaire	2571
Otomys irroratus	mate: yr rd, except June-July; captive	2338, 2339
	mate: yr rd; Zambia	331
	preg f: Jan, Oct; few data; e Africa	4925
	preg f: yr rd; Zaire	2571
	birth: Mar-Apr; no data other mo; central Africa	4495
	birth: Aug-Apr; Transvaal, S Africa	8965
	birth: yr rd except June-July; captive	2338
	lact f: Dec, Feb; few data; e Africa	185
	repro: June-Aug; Namibia	9945
Otomys maximus	birth: est Aug-Mar; S Africa	10158
Otomys typus	preg f: May-Aug; few data; Ethiopia	7600
	birth: yr rd; Kenya	5760
Ototylomys phyllotis	preg f: peak Jan-June; Belize	2590
	preg f: end Apr-early May; n7; Yucatan peninsula, Mexico	950
	preg f: May-June; n5; Guatemala	11044a
	preg f: Sept; n1; Honduras	11044a
	lact f: Jan; n1; Guatemala	11044a
	repro: Dec-Apr, July-Aug; Yucatan peninsula, Mexico	5423b
	repro: yr rd; Central America	6270
Oxymycterus rutilans	preg f: yr rd; Balcarce, Argentina	2221
Pachyuromys duprasi	birth: summer; captive	6237
	birth: yr rd; n23; captive	12128
Parotomys littledalei	preg f: Aug; n1; Namibia	9945
	juveniles: Dec; Namibia	9945

Peromyscus attwateri	preg f: Mar; n3; OK	11044a
	preg f: Dec; n1; KS	9601
	lact f: Apr, Oct; AR	9601
Peromyscus aztecus	preg f: Feb, May, July; n4; Guatemala	11044a
	preg f: dry season; n2/7; Oaxaca, Mexico	9176
Peromyscus boylii	ova: yr rd; CA	5243
	preg f: Apr-Oct; CA	5243
	mate: yr rd, peak spring-summer; sw US	1474
	preg f: May, Aug, Oct, Dec; n4; NM	502
	preg f: June-July; n8; Durango, Mexico	523a
	preg f: Nov; n1; CA	11229a
	preg f: Nov; San Luis Potosí, Mexico	2228a
Peromyscus californicus	testes: peak Apr-mid Oct; CA	7013
	preg f: Feb; n2; CA	11228a
	preg f: mid Apr-mid Oct; CA	7013
	preg f: yr rd; captive	9255
	preg/lact f: Feb-Oct; n20, 4 f; CA	6708
	birth: Apr-mid Oct; CA	7013
	lact f: Feb, Oct; CA	11228a
Peromyscus crinitus	mate: est Mar-Oct; CA	2807
	preg f: May; n3; NV	6488
	preg f: yr rd; captive	9255
	birth: Jan-Aug; 109/135 lt; lab	2910
Peromyscus difficilis	testes: peak June, low Jan; n CO	1841
	mate: Apr-Aug, peak May-June; n CO	1841
	preg f: June; n2; Durango, Mexico	523a
	preg f: July-Aug; n2; NM	502
	preg f: July, Dec; n2; Coahuila, Mexico	522
	preg f: Aug, Oct; n3; San Luis Potosí, Mexico	2228a
	birth: Spring-Oct; sw US, n Mexico	1474
	lact f: Dec; n3; Coahuila, Mexico	522
Peromyscus eremicus	testes: peak May-Sept; CA	6708
	nonfecund m: Oct-Nov; CA	6708
	preg f: Jan-Apr; n7; NM	502
	preg f: July-Aug, Oct, Dec; San Luis Potosí, Mexico	2228a
	preg f: Dec; n2; CA	11228a
	preg f: yr rd; Coahuila, Mexico	522
	preg f: yr rd; captive	9255
	preg/lact f: Jan, Feb, June, Sept; est yr rd; AZ	6414
	preg/lact f: yr rd; CA	6708
	birth: yr rd; captive	1236
	lact f: Sept; n1; CA	11228a
	repro f: Jan, June-July; n9; Durango, Mexico	523a
	repro f: June-Nov; AZ	9010
Peromyscus gossypinus	mate: Aug-May; se US	1474
	mate: yr rd; FL	10091
	preg f: Jan, July; n2; FL	6303
	preg f: Aug-May; FL	8698
Peromyscus guatemalensis	preg f: July; n2; Guatemala	11044a
Peromyscus hooperi	birth: June-Dec, none Jan-May; n14; Coahuila, Mexico	9603
Peromyscus interparietalis	birth: yr rd; captive	1236
Peromyscus leucopus	fertile m: Mar-Nov; MA	7715
	testes: peak end May-mid Sept; ONT, Canada	4376
	sperm present: yr rd, low Dec-Jan; TX	5460
	mate: Mar-June, Sept-Nov; n US	1474
	mate: yr rd; s US	1474
	mate: Mar-Oct; MI	7885
	mate: Mar-Dec; MA	67
	mate: end Mar-mid Nov, peaks Apr-June, Sept-Oct; sw VA	11877
	mate: Apr-May, Nov-Dec; MD	7169
	mate: Apr-May; data only for Mar-May; NM	979
	mate: May-Nov; MA	7715
	conceive: mean 1st Mar 6-Apr 15 across yr; ONT, Canada	7309
	preg f: Jan, Mar-Apr, July, Nov; NC	1290

	preg f: May; n1; Yucatan, Mexico	5423b
	preg f: Mar-Sept; WI	6555
	preg f: Mar-Oct, peaks May-June, Aug-Sept; MA	2724
	preg f: Mar-Dec; Coahuila, Mexico	522
	preg f: mid Mar-mid Nov; MI	1472
	preg f: end Mar-early Oct; PA	9164
	preg f: Apr-May; n4; MN	495
	preg f: Apr-Aug; no data Sept-Nov, Jan; QUE, Canada	758
	preg f: Apr-Sept; ONT, Canada	2082
	preg f: Apr-Oct; n43; NE	$5405b_1$
	preg f: end Apr-early May; n4; Yucatan peninsula, Mexico	950
	preg f: May-Sept; QUE, Canada	11956
	preg f: June; n1; SD	261a
	preg f: June; n1; SD	10954
	preg f: July; n1; ND	3703
	preg f: July; n1; Campeche, Mexico	5423b
	preg f: Oct; n3; NM	502
	preg f: Oct; MI	6797
	preg f: Sept, Dec; n2; KS	1812a
	preg f: yr rd, peak July-Aug, low Dec-Mar; TX	5460
	preg f: yr rd; captive	9255
	birth: Jan-Oct; TX	4125
	birth: Apr-Nov; NY	7235
	lact f: Mar-Oct; WI	6555
	lact f: June; n1; SD	10954
	lact f: June, Aug; n2; SD	261a
	lact f: Aug; n1; Quintana Roo, Mexico	5423b
Peromyscus madrensis	preg f: Mar; n1; Mexico	11044a
	lact f: Mar; n1; Mexico	11044a
Peromyscus maniculatus	testes: peak Mar-June; BC, Canada	9423
	testes: peak mid Mar-mid Oct; CA	7013
	testes: large mid Apr-early Nov; plains, CO	4218
	testes: large end Apr-early Sept; mountains, CO	4218
	testes: large May-mid Aug; tundra, CO	4218
	testes: scrotal ?-Sept; CO	2699
	repro m: end Apr-mid July; YUK, Canada	6015
	ova: yr rd; CA	5243
	mate: mid Mar-Aug; BC, Canada, WA	9891
	mate: end Mar-mid Nov, peaks Apr-June, Sept-Oct; sw VA	11877
	mate: Apr-Aug; NY	1364
	mate: Apr-early Oct; CO	2699
	mate: Apr-Oct; MI	6796
	mate: mid Apr-mid Sept; s QUE, Canada	11956
	mate: May-Aug; CO	11230
	mate: May-Sept; ALB, Canada	7310
	mate: May-Sept; YUK, Canada	3774
	mate: spring; only studied Mar-May; NM	979
	preg f: Jan; n2; CO	9001
	preg f: Jan-May; ID	11646a
	preg f: Feb-Dec; CA	5243
	preg f: Mar-Oct; WA	5658
	preg f: Mar-Oct; MN	722
	preg f: Mar-Nov; n62; NE	$5405b_1$
	preg f: Mar-Nov; Coahuila, Mexico	522
	preg f: Mar-Dec; CA	7081
	preg f: mid Mar-mid Nov, Jan (n1); CA	7013
	preg f: Apr-Aug, peak May-July, none Mar, Sept, Oct; MA	2724
	preg f: mid Apr-Aug; QUE, Canada	11956
	preg f: May; n7; SD	261a
	preg f: May, Sept; n3; NM	502
	preg f: May-June; n2; KS	1812a
	preg f: May-mid July; YUK, Canada	6015
	preg f: peak May-Aug; BC, Canada	9423
	preg f: May-Aug; CO	11230

	preg f: June-July; n6; Durango, Mexico	523a
	preg f: June-Aug; n59; ND	3703
	preg f: July; n22; se MT	6166
	preg f: peak Sept-Nov, Mar-Apr; KS	551
	preg f: Oct, Dec; CA	11228a
	preg f: Oct; n1; CO	738
	preg f: yr rd; WA	9576
	preg f: yr rd; captive	9255
	birth: Mar-Nov; WI, IL	2060
	birth: Apr-Oct, peak Sept-Oct; MI	5019
	birth: Apr-Oct; CA	7013
	birth: peak mid June-mid July; ALB, Canada	7310
	birth: yr rd; captive	6493
	lact f: May, July-Aug; SD	261a
	lact f: July; n2; se MT	6166
	lact f: Nov; n1; CA	11228a
	repro: Feb-Sept; BC, Canada	4667
	repro: Feb-Nov; ne CO	3294
Peromyscus megalops	preg f: Aug; n1; Mexico	2341
Peromyscus melanocarpus	birth: Dec, none Apr-May; captive, Oaxaca, Mexico	9114
Peromyscus melanophrys	preg f: June; n2; Durango, Mexico	523a
	preg f: Dec; n2; San Luis Potosí, Mexico	2228a
	lact f: Mar; n1; Coahuila, Mexico	522
Peromyscus melanotis	preg f: Jan, Apr, June-Aug; n13; Coahuila, Mexico	522
	preg f: June-Aug; n6; Mexico	2341
	repro f: June-July; n5; Durango, Mexico	523a
Peromyscus melanurus	preg f: seasonal; Oaxaca, Mexico	9176
Peromyscus merriami	preg f: June; n1; AZ	11044a
	preg f: Nov; n2; Sinaloa, Mexico	5409
Peromyscus mexicanus	m fertile: Jan-Mar, Aug; few data; El Salvador	3211
	preg f: Mar-Apr; few data; El Salvador	3211
	preg f: July; n1; Mexico	2341
	preg f: July; n2; San Luis Potosí, Mexico	2228a
	preg f: Aug; n1; Oaxaca, Mexico	523
	preg f: yr rd; Oaxaca, Mexico	9176
	preg f: yr rd; Oaxaca, Mexico	11388
	preg f: yr rd; Veracruz, Mexico	4228
	lact f: Feb, Sept; few data; El Salvador	3211
	lact f: May, Sept; Costa Rica	4574
	juveniles: Jan-Feb, May, Nov-Dec; El Salvador	1475
Peromyscus pectoralis	mate: Apr-Oct; w TX, n Mexico	1474
	preg f: Mar-Dec, none Jan; Coahuila, Mexico	522
	preg f: June-Aug; n6; Durango, Mexico	523a
	repro: est yr rd; TX	3108
Peromyscus perfulvus	preg f: Feb, Aug-Nov; w, central Mexico	4609
Peromyscus polionotus	mate: yr rd, peak winter; se US	1474
	preg f: yr rd; captive	9255
	repro: yr rd; FL	8907
Peromyscus simulus	lact f: Oct; n1; Mexico	11044a
Peromyscus stirtoni	preg f: Feb-Mar, June, Sept; n8; Honduras	11044a
	preg f: Apr; n1; Nicaragua	5408
	lact f: Sept; n1; Honduras	11044a
Peromyscus truei	testes: peak Mar-Sept; CA	7013
	mate: June-Sept; captive	2699
	mate: est not winter; NV	6512
	birth: May-Oct; CA	7013
	preg f: Mar, June-July, Sept; n6; Coahuila, Mexico	522
	preg f: May-June; NM	502
	preg f: June-July; n9; Durango, Mexico	523a
	preg f: Oct; n2; CA	11228a
	repro: Feb-?; CO	2699
Peromyscus yucatanicus	preg f: Apr, July-Aug, Dec; n5; Mexico	6269
	preg f: end Apr-early May; n3; Yucatan peninsula, Mexico	950
	preg/lact f: Apr, July-Aug, Dec; Yucatan peninsula, Mexico	5423b

	lact f: July-Aug; n2; Mexico	6269
	repro: yr rd; captive	6149
Petromyscus collinus	preg f: Feb; Namib Desert, Namibia	11863
	preg f: Sept; n1; Namibia	9945
	birth: perhaps summer; S Africa	10158
Phodopus campbelli	preg f: Aug; n1; China	12105
Phodopus sungorus	testes: peak May-Aug, low Nov-Feb; captive	4885
	testes: peak summer, low winter; lab	3249
	mate: end Apr-Sept; Mongolia	3334
	birth: peak Mar-Aug; none Dec-Jan, inside; none Nov-Feb; outside	3249
	birth: yr rd, peak spring-summer; Siberia, Russia	11285
Phyllotis darwini	testes descend: July-Nov; n Chile	7241
	testes scrotal: Aug; Fray Jorge Nat Pk, Chile	3551
	perforate vaginae: 27% of f Aug; Fray Jorge Nat Pk, Chile	3551
	preg f: Mar; n1; Malleco, Chile	4054
	preg f: May, Aug, Oct-Feb; n14; Peru	4682
	preg f: end July-early Aug; Santa Rose, Peru	8302
	preg f: 30-80% Sept-Nov, <10% Jan-July; n Chile	7241
	lact f: Mar; n1; Malleco, Chile	4054
Phyllotis osilae	preg f: Feb, Apr-June; n8; Peru	4682
Podomys floridanus	mate: yr rd; captive, FL	1474
	preg f: June-Mar, peak Sept-Oct; FL	6301
	preg f: yr rd, except Apr; captive	9255
Psammomys obesus	testes: peak Dec-Mar, low June; Algeria	5706
	mate: Sept-May; few data; Sinai Penninsula	8134
	mate: Dec-July, peak Mar; captive	9825
	mate: yr rd; Algeria	2234
	birth: yr rd, peak Mar; captive	3513
Reithrodon physodes	preg f: Jan, May; n2; Uruguay	596
	preg f: peak Nov; Patagonia	8306
Reithrodontomys brevirostris	preg f: June; n2; Nicaragua	5413
Reithrodontomys creper	preg f: Mar, June-July; n10; Panama	11044a
Reithrodontomys darienensis	preg f: Feb; n1; Panama	11044a
	lact f: Feb; n2; Panama	11044a
Reithrodontomys fulvescens	gonads: regressed Mar-Apr; south Mexico	11388
	mate: est Feb-Oct; sw US, n Mexico	1474
	mate: peaks Mar, July; TX	8189
	preg f: July; n1; Durango, Mexico	523a
Reithrodontomys gracilis	preg f: Mar; n1; Guatemala	12037
	preg f: July; n1; El Salvador	12037
	preg f: July; n2; Nicaragua	12037
	preg f: July-Aug; n5; Yucatan peninsula, Mexico	5423b
Reithrodontomys humulis	testes: large Feb, May, June, Aug, Dec; FL	6303
	preg f: Feb; n1; FL	6303
	preg f: peak May-Nov, some winter; TN	2794
	preg f: May, Nov; NC	1290
	birth: May-Nov; se US	1474
	birth: July; n1; FL	6303
	repro: yr rd, except Feb, May, Nov; lab	6300
Reithrodontomys megalotis	mate: begins Mar; BC, Canada	2088
	mate: almost yr rd, except Jan-Mar; CA; w US	1474
	preg f: none Jan, Oct; CA	10099
	preg f: none Feb, Nov; Latin America	4963
	preg f: peak Mar, Sept; KS	551
	preg f: Apr-Nov; n75; NE	5405b,
	preg f: May-June, Sept-Oct, Dec-Jan; n7; NM	502
	preg f: June; n4; SD	261a
	preg f: June-July; n6; ND	3703
	preg f: June-July; n4; KS	1812a
	preg f: July-Aug, Oct-Nov; CA	997
	preg f: July-Aug; n3; Mexico	2341
	preg f: yr rd; CA	3279
	birth: nearly yr rd; El Salvador	1475

Reithrodontomys mexicanus	preg/lact f: Nov; n2; San Luis Potosí, Mexico	2228a
Reithrodontomys montanus	preg f: Mar, June-July; n4; NE	5405b,
	preg f: June; n1; SD	261a
	preg f: Oct; n1; KS	1812a
	birth: yr rd; s US	1474
Reithrodontomys raviventris	fertile m: yr rd; CA	3279
	mate: Mar-Nov; n CA	10809
	preg f: yr rd; CA	3279
	repro f: Mar-Nov; n CA	9887
Reithrodontomys sumichrasti	preg f: Jan; n1; Guatemala	11044a
	preg f: Mar; n1; Panama	11044a
Rhipidomys mastacalis	preg f: est yr rd; n Venezuela	7996, 7998
Rhombomys opimus	birth: end Apr-early Sept; Mongolia	10516
	repro: Jan-mid June; USSR	9499
	repro: peak Mar, none Nov; lab	10202
Saccostomus campestris	mate: summer; S Africa	2861
	conceive: Apr-May, Aug-Sept, Dec-Jan; Malawi	4329
	preg f: Jan-Apr; Botswana	10154, 10158
	preg f: Feb, Apr, July, Oct; captive; Malawi	4329
	preg f: Dec; n1; Namibia	9945
	birth: Oct-Apr; Transvaal, S Africa	8965
	juvenile: yr rd; e Africa	5760
	juvenile: yr rd; Zambia	331
Saccostomus meamsi	preg f: July; n2; e Africa	4925
Scapteromys tumidus	sperm present: yr rd; Uruguay	596
	preg f: Apr-May, Oct-Dec; Uruguay	596
	preg f: peak Dec; few data; Uruguay	4683
Scotinomys teguina	birth: yr rd; captive	4967
Scotinomys xerampelinus	birth: yr rd; captive	4967
Sekeetamys calurus	birth: yr rd; captive	7983
Sigmodon alstoni	preg f: est yr rd; n Venezuela	7995, 7996, 7998
Sigmodon fulviventer	birth: yr rd; captive	1527
Sigmodon hispidus	testes: peak Feb-Aug, low small Nov-Dec; n175 TX	4212
	uteri: large Feb-Oct, small Nov-Jan; TX	4212
	ovaries: peak Mar, low Dec; TX	4212
	ovulation: yr rd, few Oct-Jan; TX	4212
	mate: peaks Feb-July, Sept-Nov; AR	9710
	mate: Apr-Nov, peak May-Sept; ne KS	7029
	mate: yr rd; captive	7253
	preg f: Jan-Mar, July; n7; Yucatan peninsula, Mexico	5423b
	preg f: Jan, May, July, Oct; NC	1290
	preg f: Apr, Aug; Coahuila, Mexico	522
	preg f: Apr-Oct; KS	7029
	preg f: Apr-Nov; FL	6303
	preg f: end Apr-early May; n7; Yucatan peninsula, Mexico	950
	preg f: July; n1; Veracruz, Mexico	11507a
	preg f: July; n1; Oaxaca, Mexico	523
	preg f: July-Aug, none Jan-Feb, June, Oct, Dec; few data; El Salvador	3211
	preg f: Sept; n1; KS	1812a
	preg f: Sept; n2; NM	502
	preg f: yr rd; s Prusia Valley, Mexico	11388
	preg f: yr rd; Veracruz, Mexico	4228
	preg f: yr rd; Belize	2590
	birth: yr rd; Panama	3317
	birth: yr rd, except May-June; El Salvador	1475
	lact f: peaks spring, fall, winter; few data; FL	6303
Sigmodon ochrognathus	preg f: Apr, Oct; n6; Coahuila, Mexico	522
Steatomys krebsii	preg/lact f: Oct; Zambia	331
Steatomys parvus	birth: est summer; S Africa	10158
Steatomys pratensis	preg f: Nov; n1; subSaharan Africa	5760
	birth: Feb; n1; Zambia	331
	birth: July; n2; Malawi	4329
	birth: Oct-May; S Africa	10158

Tatera afra	testes: large June-Dec; S Africa	166
	preg f: Aug-Mar; S Africa	7146
Tatera brantsii	preg f: Jan, July, Nov; n6; Kalahari Gemsbok Nat Park, S Africa	7823
	preg f: yr rd; n63; S Africa	7146
	birth: Apr-June 40%, July-Sept 4%, Oct-Mar 13-17%; Zaire	8564
	birth: yr rd, peak summer; Namibia	9945
	repro: yr rd; S Africa	10158
Tatera indica	fertile m: low Nov-Jan; Pakistan	729
	scent gland: peak Dec-June, low July-Nov; India	6097
	sperm present: July-Apr; India	8733
	mate: Sept-Oct, Jan-Feb; Rajasthan, India	8721
	mate: yr rd; Pakistan	729
	preg f: Jan; n1; Iran	6293
	preg f: Mar-Nov, peak Mar-May, July-Sept; Pakistan	729
	preg f: yr rd; India	6097
	preg f: yr rd, peak monsoon; Rajasthan, India	5230
	birth: yr rd, peak Oct-Apr; Sri Lanka	8500, 8501
	lact f: Apr; n2; Nepal	7382
	lact f: May-Dec; n161; Rajasthan, India	6775
	repro: yr rd; India	8723
	repro: yr rd; Mysore, India	1689
Tatera leucogaster	mate: Sept-Apr; Transvaal, S Africa	8386
	mate: peak rainy season; Zambia	9893
	mate: yr rd; Zambia	331
	preg f: peak Feb-Mar; S Africa	10158
	preg f: 80-100% Oct-May, <20% July-Sept; Transvaal, S Africa	8384
	preg/lact f: Mar, May-June, Oct; Malawi	4329
	birth: summer; Zimbabwe	1815
	birth: yr rd; Botswana	10154
	birth: yr rd, peak summer; Namibia	9945
Tatera nigricauda	testes: large yr rd, peak rains; Kenya	7794
	mate: low end dry season; Kenya	7796, 7797
	preg f: peaks Nov-Feb, Apr-May; Kenya	7794
Tatera robusta	preg f: July-Oct; Senegal	5057
	preg/lact f: peak July-Aug, Nov-Jan; Kenya	7794
Tatera valida	testes: large yr rd; Uganda	7794
	mate: yr rd; Zambia	331
	preg f: 50% June, Nov, near 0% other mo; Uganda	2453
	preg f: July-Oct; Senegal	5057
	preg f: yr rd, peaks Apr-May, Nov-Dec; Uganda	7794
	young found: Mar, Nov-Dec, none Feb, May; no data Jan, Apr, June-Oct; central Africa	4495
Tatera vicina?	preg f: Dec; n2; Kenya	4925
Tatenillus gracilis	preg f: July-Oct; Senegal	5057
Tatenillus pygargus	birth: Mar, Nov; Senegal	8695a
Tylomys nudicaudus	gonads: regressed Mar-Apr; s Mexico	11388
	preg f: Mar; n1; Guatemala	11044a
Wiedomys pyrrhorhinos	lact f: May, Aug-Nov; Caatinga, Brazil	10494
Zygodontomys brevicauda	preg f: Jan, Mar-Apr, Nov; Panama	3024
	preg f: June, Sept-Nov, est yr rd; Venezuela	451
	preg f: est yr rd; n Venezuela	7996, 7998
	birth: est yr rd; Panama	3317

Order Rodentia, Family Spalacidae, Genus *Spalax*

Spalax giganteus, *S. leucodon*, and *S. microphthalmus* are 150-550 g, subterranean, solitary herbivores that tunnel under grasslands from the eastern Mediterranean to Mongolia. Neonates are naked (7860) and suckle from 6 mammae (8050). Males have a baculum and seminal vesicles (6002, 8050).

NEONATAL MASS		
Spalax leucodon	~5 g	7860
LITTER SIZE		
Spalax leucodon	2-3	6419
	mode 3-4; r1-6	8050, 10479
	mode 3-4; r1-6	7860
	4-5; 3 rare	9452
Spalax microphthalmus	2-4	4645
	2.83; r1-5	8154
GESTATION		
Spalax leucodon	>28 d; n1	7860
LACTATION		
Spalax leucodon	wean: est 1-2 mo	10479
LITTERS/YEAR		
Spalax leucodon	1	7860
	1	9452
	1	10479
Spalax microphthalmus	1	8050
	1; r1-2?	4645
SEASONALITY		
Spalax leucodon	birth: peak mid Jan-mid Feb 67%; Israel	7860, 7863
	birth: early Feb-?, peak early Apr; Lebanon	6419
	birth: end Feb-Mar; Odessa, Crimea	9452
	birth: Mar-Apr; Sofia, Bulgaria	10479
Spalax microphthalmus	mate: est Mar; USSR	8050
	mate: spring; Europe, w Ukraine, w Kazakstan	4645
	preg f: ?-Apr; USSR	8050
	birth: end Feb-mid May; Voronesh, Russia	8154

Order Rodentia, Family Rhizomyidae

Bamboo (*Cannomys badius* 500-800 g; *Rhizomys* 1-4 kg) and mole (*Tachyoryctes macrocephalus* 330-930 g; *T. splendens* 160-280 g) rats are fossorial herbivores from tropical and subtropical regions of east Africa (*Tachyoryctes*) and Nepal through southeast Asia (*Cannomys*, *Rhizomys*). Males are larger than females (*Tachyoryctes* 4925, 5760). *Tachyoryctes* are solitary (5268). A copulatory plug is present (*Cannomys* 2938). *Tachyoryctes* probably has a postpartum estrus (5267). Neonates are naked and open their eyes at 3-4 wk (*Cannomys* 2938; *Rhizomys* 182, 7161, 8045). *Rhizomys* have 8-10 teats (182, 7161).

NEONATAL MASS		
Tachyoryctes splendens	~15 g	2446a
	15-25 g; n10	8857
NEONATAL SIZE		
Tachyoryctes splendens	HB: 55-60 mm; n10	8857
LITTER SIZE		
Cannomys badius	mode 2; n5 lt	2938
Rhizomys pruinosus	2.0 emb; r1-3; n3 f	10830
	3 yg; n1 f	10830
	5-6 yg; hunter reports	10830
Rhizomys sinensis	3 yg; n1 nest	182
Rhizomys sumatrensis	2-4 yg	10830
	3-5 yg	6369
	5; n1 lt	7161
	5 yg; n1 lt, 1 wk	8045

Tachyoryctes splendens	1 emb ea; n2 f	4925
	1.20 emb; n38 f	2446a, 5267
	1.25; max 4; n322 lt	8855
	1.3; r1-2	5268
	1.46 emb; mode 1; r1-4; n356 f	2446a, 2449
	1.51 emb; mode 1; r1-4; n82 f	8857
	1.64 emb; n37 f	2446a, 5267
	2 yg; n1 f	4925
SEXUAL MATURITY		
Tachyoryctes splendens	f: 120 d	5268
GESTATION		
Cannomys badius	r40-43 d; n3	2938
Rhizomys sumatrensis	≥22 d; n1	7161
Tachyoryctes splendens	37-40 d	2449
	38.5 d; r37-40	5268
	44-53 d; n6	8857
LACTATION		
Cannomys badius	solid food: 23 d	2938
	wean: 8 wk	2938
Rhizomys pruinosus	independent: 4 mo	10830
Rhizomys sumatrensis	solid food: 29 d	7161
	nurse: 3 mo	7161
Tachyoryctes splendens	wean: 4-6 wk	5267
	wean: 43 d; r35-50	5268
INTERLITTER INTERVAL		
Tachyoryctes splendens	173 d	5268
SEASONALITY		
Rhizomys pruinosus	birth: Feb-Apr; Vietnam	10830
Rhizomys sinensis	birth: Apr-May; central Asia	182
	lact f: May; n1; China	182
	young in nest: June-July, Nov; China	182
Rhizomys sumatrensis	birth: Feb-Apr, June-Aug; Vietnam	10830
Tachyoryctes splendens	preg f: Oct; n1; e Africa	4925
	preg f: yr rd, except May; Kenya	5267
	preg/lact f: Feb; n3; e Africa	4925

Order Rodentia, Family Arvicolidae

Voles and lemmings (*Alticola* 22-69 g; *Arborimus* 28 g; *Arvicola* 70-310 g; *Clethrionomys* 10-59 g; *Dicrostonyx* 20-112 g; *Dinaromys* 30-82 g; *Ellobius* 30-88 g; *Eolagurus*; *Eothenomys* 27 g; *Hyperacrius* 22-60 g; *Lagurus* 11-51 g; *Lemmus* 21-113 g; *Microtus* 10-170 g; *Neofiber* 156-357 g; *Ondatra* 330-2400 g; *Phenacomys* 16-56 g; *Pitymys* 13-58 g; *Proedromys*; *Prometheomys* 60-88 g; *Synaptomys* 15-50 g) are terrestrial to fossorial herbivores from North America and north Eurasia. Females are reported larger than males in *Lemmus* (9762), while males are reported larger in *Microtus* (1933, 2466, 3112, 5511, 5794, 6632; cf 5486, 9894, 10954, 11878), *Neofiber* (944), and *Ondatra* (9503). Each sex has been reported larger in *Clethrionomys* even within the same species (716, 2524, 5501, 5798, 7238, 11679). The sexes are of equal size in *Pitymys* (6199, 6881, 9273, 10954; cf 7904), *Prometheomys* (10275), and *Synaptomys* (2026).

Ovulation is induced (289, 462, 481, 794, 1248, 1249, 1250, 1251, 1267, 1268, 1269, 1270, 1271, 1638, 1749, 1750, 1807, 1875, 1876, 1880, 1881, 1918, 1919, 1964, 2330, 3220, 3610, 3968, 4014, 4015, 4052, 4053, 4169, 4264, 4475, 5000, 5281, 5283, 5284, 5672, 5674, 5675, 5676, 5809, 6348, 6389, 6692, 7291, 7334, 7335, 7336, 7339, 7341, 7342, 7343, 7595, 8021, 8251, 8958, 9009, 9110, 9111, 9478, 9517, 9530, 10421, 11603, 11604, 11605).

The usual sex determination does not obtain in *Dicrostonyx torquatus* (3780), *Ellobius lutescens* (2389, 6651, 6973, 7736), *Lemmus schisticolor* (3493, 3494, 5502, 10372, 11375), and *Microtus oregoni* (1370, 3492, 6973, 8062, 8557, 8998a). Phenolic plant extracts may alter reproduction (793, 795, 796, 797, 1501, 1837, 2113, 4813, 5963, 7812, 8558, 8562, 9462, 9538). A copulatory plug is present (*Clethrionomys* 4172, 5594, 8425; *Lemmus* 3466; *Microtus* 1218, 4278, 4459, 4460; *Pitymys* 783, 9111).

The 5 g (*Ondatra*), hemochorial, labyrinthine, discoid placenta has an interstitial, mesometrial implantation into a bicornuate, bipartite, or duplex uterus (1222, 2464, 2467, 3081, 3082, 5735, 7271, 7563, 7567, 9479,

10421) and is eaten after birth (4278, 4893, 5240, 9503). Estrus occurs postpartum (501, 660, 944, 981, 1218, 1222, 1250, 1407, 1806, 1875, 2026, 2045, 2077, 2087, 2529, 2562, 2563, 3061, 3457, 3459, 3462, 3465, 3466, 3468, 3571, 3622, 3652, 3745, 4014, 4053, 4172, 4264, 4275, 4278, 4469, 4645, 4660, 5039, 5135, 5240, 5593, 6041, 6347, 6348, 6397, 6693, 6782, 6806, 7271, 7904, 8051, 8052, 8920, 8921, 9013, 9110, 9111, 9536, 9656, 9762, 10383, 10446, 10569, 10589, 10681, 11225, 11227, 11362, 11647, 12049, 12070; cf 3558). Lactation may delay implantation (*Clethrionomys* 7271).

Arvicolids are born blind and naked (1474, 2243, 4219, 5311, 8179, 9656, 11467, 12007). A few papers report *Ondatra* (3081, 8051, 10580) and *Pitymys* (783, 3283, 6051) as furred at birth. Eyes open at 8-11 d (*Arvicola* 2045, 3571, 7433, 8052; cf 4660), 8-13 d (*Microtus* 501, 1965, 2087, 3457, 3465, 3468, 3571, 4275, 4278, 4490, 4970, 4999, 5148, 5313, 6128, 6347, 7433, 8052, 9009, 9013, 9269, 9355, 9724, 9747, 9967, 11225, 12049, 12105), 9-13 d (*Clethrionomys* 3571, 5148, 7002, 7238, 7433, 7546, 8052, 8425; cf 10569; *Lagurus* 1474, 3516, 5240; cf 8634; *Lemmus* 2386, 3462, 5135, 6806, 8051, 10210; cf 716; *Pitymys* 783, 3283, 3571, 4273, 6199, 7433, 7904, 9111; cf 6051), 10-12 d (*Synaptomys* 2026, 10329), 10-14 d (*Dicrostonyx* 4335, 8051, 8810), 11-20 d (*Ondatra* 3081, 3082, 3571, 7433, 8051, 9503; cf 7291), 14 d (*Neofiber* 944), 14-16 d (*Phenacomys* 3429, 5150), 18-23 d (*Arborimus* 4264, 5026), or 21 d (*Ellobius* 6403).

Females have 4 mammae (*Arborimus* 1474; *Eothenomys* 182), 4-6 mammae (*Pitymys* 1474, 2376, 3571, 4645, 6199, 8052, 9175, 9530, 9536), 4-8 mammae (*Synaptomys* 1474, 2026, 6486, 10329), 6 mammae (*Hyperacrius* 9175; *Neofiber* 944, 1474), 6-10 mammae (*Ondatra* 1474, 3571, 7433, 8051, 10580, 11934), 8 mammae (*Alticola* 8052; *Arvicola* 3571, 7433, 8052; *Clethrionomys* 502, 1474, 2548, 2564, 3547, 3571, 4035, 7433, 8052, 9612, 10516, 10569, 11362; *Ellobius* 182; *Lagurus* 1474, 8052; *Lemmus* 655, 8051; *Microtus* 502, 1218, 1474, 1965, 2376, 2562, 3571, 4278, 4745, 7433, 8052, 8986, 9612, 9724, 9967; *Phenacomys* 1474, 5551; *Prometheomys* 8051). Males have ampullary glands, preputial glands, seminal vesicles, penial scales, and a baculum (352, 2460, 2464, 2467, 4891, 5091, 6805, 6897, 8051, 8052, 8984, 12081; cf 10323, 10327).

Details of anatomy (668, 1878, 2463, 4468, 8402, 9479, 11606), endocrinology (7707; *Clethrionomys* 3424, 6055a, 7849, 10625, 11602; *Dicrostonyx* 1441, 1779, 9521; *Ellobius* 10328, 10335, 10336; *Lemmus* 304; *Microtus* 64, 724, 1746, 1748, 1749, 1882, 2266, 2657, 3424, 4015, 7048, 7339, 7341, 7344, 8409, 8410, 9152, 11274, 11932; *Ondatra* 719, 723a; *Pitymys* 9537), oogenesis (8402), parthenogenesis (9478), spermatogenesis (4091, 4092, 8517), copulation (1637, 1639, 2385, 2506, 2514, 2531, 4016, 4017, 4018, 4019, 5675), implantation (7850), pregnancy failure (1336, 1916, 1917, 5250, 5594, 5673, 6760, 6760a, 6822, 7337, 7338, 7340, 8958, 9531, 9533, 9539, 10073, 10332a, 10333, 10457, 10458, 11486), milk (7065a), reproductive energetics (5149; *Clethrionomys* 5477; *Microtus* 7292, 10911; *Pitymys* 6533), and light (*Clethrionomys* 3193, 6052, 10622, 10624, 10748; *Dicrostonyx* 4471, 4472, 8809; *Microtus* 519, 1747, 1873, 1882, 1883, 2087, 2115, 2116, 2220, 4087, 4088, 4089, 4090, 4998, 4999, 5139a, 5731, 6339, 6340, 6341, 6356, 6357, 6358, 6359, 6900, 6901, 6910, 6911, 7842, 9087, 9352, 10249, 10250, 10251, 11275; *Pitymys* 2493, 7840, 7841, 7844, 10074) are available. A monograph on New World *Microtus* (including some *Pitymys*) is available (10648a).

NEONATAL MASS

Arborimus longicaudus	2.33 g; n35	4264
	2.58 g; r2.56-2.60; n2	1362
	3.1 g	5026
Arvicola terrestris	m: r3.19-7.84 g; n55	4660
	3.2-7.8 g	3571
	f: r3.28-7.32 g; n55	4660
	5.0 g	2045
	5-6 g	8052
Clethrionomys gapperi	1.18 g	5148
	1.75; se0.03; n229	5149
	1.9 g; r1.7-2.3	10569
Clethrionomys glareolus	f: 1.55 g; se0.032; r1.3-1.8; n15	12080
	m: 1.59 g; se0.051; r1.2-1.9; n17	12080
	1.6-1.7 g	1032
	1.77 g; spring-summer; 2.03 g; fall	7002
	1.8 g; n4, 18 d emb	8179
	1.815 g; 18 d emb	8425
	1.91 g; r1.67-2.50; n80	11362
	2.7 g	1222, 2077
Clethrionomys rufocanus	~2.0-2.5 g; n13, 1 d	8287

Clethrionomys rutilus	f: 2.9 g; n98	7545
	m: 3.0 g; n100	7545
Dicrostonyx groenlandicus	3.8 g; r2.7-4.8; n34	4335
	4.4 g	4469
	f: 4.6 g; n66	7545
	m: 4.6 g; n39	7545
Dicrostonyx torquatus	2.5-3 g	8810
	3.4 g; r2.0-4.2; n23	8051
Dicrostonyx unalascensis	5 g; from graph	2560
	m: 5.2 g; n112	7545
	f: 5.4 g; n122	7545
Eothenomys smithi	2.13 g; se0.06; r1.98-2.26; n16	294a
Lagurus curtatus	0.85 g	6925
	1.47 g; n8	5240
Lagurus lagurus	1.2 g; n1	11647
	1.5 g	3516
Lemmus lemmus	2 g	11679
	f: 3.6 g; r2.2-4.3; n41	3462
	3.67 g; r2.74-4.59; n45; from graph	6806
	3.9 g	4830
	m: 3.9 g; r2.6-4.6; n32	3462
	4.5-5 g	8051
	f: 4.6 g; n55	7545
	m: 4.9 g; n74	7545
Lemmus schisticolor	3.2 g; r2-4	5135
Lemmus sibiricus	3 g	716
	3.6-4.0 g	8051
	f: 4.0 g; n47	7545
	m: 4.3 g; n59	7545
Microtus breweri	3.5 g	7714a
Microtus abbreviatus	f: 3.3 g; n70	7545
	m: 3.6 g; n91	7545
Microtus arvalis	f: 1.09-1.29 g	2462
	1.1-3.2 g	3571
	1.22-2.77 g	9269
	m: 1.35-1.51 g	2462
	1.75 g	7292
	1.79 g; r1.26-2.64	11867
	1.9 g	8153
	1.94 g; n29	6361
	f: 1.95; se0.0015; n1418	9013
	1.99 g; se0.0006; n2738	9013
	m: 2.03; se0.0015; n1320	9013
	2.53 g; r1.5-3.4; n348	3459, 3460, 3465
Microtus brandti	1.9 g; r1.4-2.0; n140	12105
Microtus californicus	2.8 g; n19 yg, 5 lt	4490
	3.18 g; n10, 2 lt	9747
	3.4 g; r2.7-4.8; n35; from graph	6041
Microtus canicaudus	2.4-2.5 g; se0.03-0.06; n4 groups	4196a
Microtus guentheri	~3 g	8115
Microtus miurus	2.29 g; r1.65-3.0; n17	12049
	f: 2.8 g; n88	7545
	m: 3.0 g; n120	7545
Microtus montanus	2.2 g; n18	9724
	3.9 g	7714a
Microtus montebelli	2.5 g; sm lt: 2.7; lg lt: 2.4 g; n95 babies	7762
Microtus nivalis	3.7 g	3571
	3.7 g; r3.6-3.8; n2	5486
Microtus oeconomus	1.86 g; ±0.13; 1 d	9355
	1.9-3.1 g	3571
	f: 2.06 g; se0.03; n32	5313
	m: 2.10 g; se0.03; n36	5313
	2.3 g; r1.9-2.6; n27	3468
	2.4 g; se0.1; n46	3655

	f: 2.95 g; n124	2563, 7545
	m: 2.95 g; n143	2563, 7545
Microtus oregoni	1.56 g	4970
	1.7 g; r1.59-2.2; n41	2087
Microtus pennsylvanicus	2-3 g	6347
	2.07 g; r1.6-3.0; n24 lt	4272, 4278
	2.26 g; se0.05; n197	5149
	2.35 g	5148
	2.4 g; r1.4-3.5; n776	2660
	f: 2.4 g; n368	2660
	m: 2.5 g; n408	2660
	2.92 g; sd0.35; n59 yg, 15 lt	9283
	~3 g	501
	3.2 g; n212	2562, 7545
Microtus richardsoni	5.05 g; n6 emb, 1 f	4969
Microtus socialis	2.1 g	8052
Microtus transcaspicus	2.8 g	8153
Microtus xanthognathus	3.5 g; r2.7-4.2; n28	12049
Neofiber alleni	7.2 g; n3	4282
	12.1 g; r10.0-14.9; n7	944
Ondatra zibethicus	16-20 g	6267
	21 g	11467
	21.3 g; median 21; r16-28; n41	3081, 3082
	23.0 g; sd2.56; n70, 1 d	2695
	25 g	3061
	28 g	9503
Phenacomys intermedius	1.89 g; se0.07	5150
	2.05 g; n14, 3 lt	4219
	2.36 g; n18, 4 lt; 1 d	4219
	2.4 g; r2.0-2.7	3429
Pitymys duodecimcostatus	f: 2.2 g; ±0.03; n50	4123
	m: 2.2 g; ±0.02; n39	4123
	2.24 g; r2-2.8; n21	6361
Pitymys lusitanicus	1.7 g; r1.5-1.9; n3	11820
Pitymys ochrogaster	2.8 g; se0.3; n16	6881
	2.84 g; n264	9111
	2.9 g; ±0.05; r2.0-3.8; n67 yg, 27 lt	3283
	3.1 g; sd0.27; se0.05; r2.4-3.5; n34	6051
Pitymys pinetorum	2.0 g; se0.1; n19 lt	6533
	2.2 g	4273
	2.3 g; n15, n4 lt	3852
Pitymys subterraneus	f: 1.42 g; n1	6199
	m: 1.43 g; r1.41-1.45; n2	6199
	1.58 g; n59	3654
	1.95 g; se0.26; n28	9656
Synaptomys cooperi	2.55 g; r2.4-2.7; n2	10329
	3.9 g; r3.1-4.3; n11	2026
NEONATAL SIZE		
Clethrionomys gapperi	CR: 30 mm	7238
Clethrionomys glareolus	22.0 mm; se0.53; r18-26; n32	12080
	25-35 mm	8052
	CR: 26.5 mm; 18 d emb	8425
	38.5 mm; n32	7002
Dicrostonyx groenlandicus	CR: 20-27 mm	4335
	HB: 35 mm; n1, 1 d	2434
Dicrostonyx torquatus	33-38 mm	8051
Lagurus curtatus	HB: 27 mm; n8	5240
Lagurus lagurus	32 mm; n1	11647
Lemmus lemmus	15-18 mm	8051
Lemmus sibiricus	32-40 mm	8051
Microtus montanus	33 mm; n13	9724
Neofiber alleni	TL: 82 mm; n3	4282

Ondatra zibethicus	28.5 mm; sd2.15; n70	2695
	96 mm	9503
	100.4 mm; median 102; r85-115; n41	3082
Phenacomys intermedius	CR: 30 mm	3429
Pitymys ochrogaster	CR: 30-35 mm	6051
	40-50 mm	3283
Pitymys pinetorum	CR: 27-29 mm	8268
	TL: 39-48 mm	10159
Pitymys sikimensis	27.5 mm; largest emb	7382
Pitymys subterraneus	28.25 mm; se4.33	9656
Prometheomys schaposchnikowi	36 mm	12007
WEANING MASS		
Arborimus longicaudus	17-25 g	4264
Clethrionomys gapperi	9.16 g; 14 d	5148
	11.5-12 g; 16-18 d	7238
Clethrionomys glareolus	8.43 g; n563 lt	4173
	9.1; sd1.9; n219	4343
	9.44 g; n134 lt	4173
	10.0; sd1.5; n77	4343
	12 g; 20 d	1835
Clethrionomys rufocanus	10 g	5501
Dicrostonyx groenlandicus	20 g	4469
	26.0 g; r16.5-34.6; 22 d	4335
Dicrostonyx torquatus	r23-28 g; n> 30, 19 d	5466
Lagurus lagurus	5.5-6.0 g	11647
Lemmus schisticolor	m: 15.6 g; r11.2-21.2; n60, 20-22 d	5135
	f: 15.7 g; r9.6-23.8; n60, 20-22 d	5135
Lemmus sibiricus	r24-30 g; n> 30, 19 d	5466
Microtus agrestis	7.64 g; se0.13; n37 lt, 170 yg	1805
	10.23 g; se0.18; n19 lt, 66 yg	1805
Microtus arvalis	6.2-8.2 g	6910
	8.90 g; 3 wk	9013
	9.18 g; 17 d	11867
	9.69 g; 3 wk	9013
	9.87 g	7292
Microtus californicus	17 g; r12.5-23; n29, 16 d	6041
	21 g; n5, 20 d	9747
Microtus canicaudus	m: 10.7-16.8 g; se0.4-1.0; n4 groups, 18 d	4196a
	f: 12.0-16.4 g; se0.2-0.8; n4 groups, 18 d	4196a
Microtus guentheri	~15 g; from graph	1965
Microtus montanus	9.33 g; n18, 4 lt, 15-16 d	9967
Microtus montebelli	17.6 g	7762
Microtus oeconomus	12 g; 11-12 d; first outside nest	10681
	15.6 g; r14.7-16.4	3468
	r17-19 g; n> 30, 19 d	5466
Microtus pennsylvanicus	11.52 g; 14 d	5148
	14-16 g	4278
Ondatra zibethicus	109.3 g; 4 wk	8051
	median 150 g	3082
	~150-165 g	2695
	197 g; n9, 31 d	3081
Pitymys ochrogaster	19.0 g; sd2.61; se0.51; r13.9-24.1; 21 d	6051
Pitymys pinetorum	7.5-9.5 g; 3 wk	8268, 10159
	8.7 g; se0.3; n19 lt, 18 d	6533
	10-11 g; 16 d	4273
	11.1 g; se0.4; n19 lt, 21 d	6533
Pitymys subterraneus	9.44 g; se1.23; n37	9656
Synaptomys cooperi	~20 g; 3 wk; from graph	2026
WEANING SIZE		
Lagurus lagurus	40 mm	11647
Pitymys pinetorum	TL: 69-98 mm; 3 wk	8268, 10159

LITTER SIZE

Alticola roylei	4.3; r4-5; n3 lt	9175
	5.6 emb; n64 f	10516
	6 ps; n1 f	8052
Alticola stoliczikanus	8 emb; n1 f	8052
Alticola strelzowi	3.6 emb; r2-6 (2-3 early season, 2-6 late season)	8020a
	7-11 emb	8052
	7.1; r3-13	3088
Arborimus albipes	2-3 yg	5245
Arborimus longicaudus	2; r1-3	5026
	2.5 yg; r1-4; n14 lt	4264
	3 yg; n1 nest	5311
	3 emb; n1 f	1362
	3 emb; n1 f	782
	4 yg; n1 nest	11672a
Arvicola sapidus	3.31 emb; r2-5; n19 f	11248
Arvicola terrestris	2-7	7433
	mode 2-8; r2-14	3571
	3.3; r1-8; n107 lt	4660
	3.42; se0.098; sd1.294; n173 f	11668
	3.82 yg; mode 4; r2-5; n17 lt	11204
	4.2; r2-7	981
	4.48 emb; sd1.23; se0.13; mode 4; r2-9; n80 f	11249, 11250
	4.5 emb; r4-5; n2 f	1590
	4.52; se0.120; n119 lt	11668
	4.87 emb; mode 5; r1-9; n116 f	11204
	5.25; se0.396; n20 lt	11668
	5.5; se0.2; r1-10	7772
	5.7 emb; n14 f	8392
	5.7 emb; n122 f	6007
	mode 6-8; r6-14	4645
	6.4 cl	8392
	6.4 emb; by palpation	10447
	6.9; r3-14	8052
Clethrionomys spp	3.12-4.91; r for 7 yr	8256
Clethrionomys californicus	4 emb; n1 f	6708a
Clethrionomys centralis	1-7	634
	4-8 emb	8052
Clethrionomys gapperi	2.3 ova; r1-3; n6 f	3220
	3.33 emb; r3-4; n6 f	575
	3.6 cl; r1-7; n9 f	3220
	4 emb; n1 f	502
	4 emb; n1 f	4035
	4 yg; n1 nest	495
	4 emb ea; n2 f	2548
	4.17 yg; n6 lt, 5 f	10569
	4.33 emb, yg; r3 yg, 5,5 emb; n3 f	6796, 6799
	4.5 emb; r4-5; n2 f	4280
	4.6 yg; r3-7; n20 lt	11956
	mode 5 yg; r3-8	5210
	5 yg; n1 f	9882
	5 emb; n1 f	11504
	5; r2-6	9164
	5.2 emb; se0.22; r3-7; n20 f	10839
	5.29 emb; r3-7; n7 f	279
	5.3 emb; r3-7; n6 f	2025
	5.3 emb, ps; ±1.6; n22 f	8713
	5.38 emb, ps; r4-8; n21 f	10101
	5.47 yg; ±0.13; r3-8; n49 lt	2082
	5.58; se0.19; mode 6; r2-8; n41 lt	5149
	5.7 emb; r5-6; n3 f	279
	5.9 emb, ps; ±0.6; n15 f	8713
	6 emb; n1 f	9612
	6 emb; n1 f	10218, 10220
	6.0 emb; r5-7; n6 f	10144

	6.0 yg; ±1.68; n10 lt	8383
	6 ps; r5-7; n3 f	3703
	6.01 emb, ps; n115 f	7895
	6.1 emb; se0.50; r4-8; n10 f	11230
	6.33 emb; r6-7; n3 f	7809
	6.4 emb stated, [5.86 from data]; r4-7; n14 f	5798
	6.65 cl; se0.127; r4-13; 6.19 ps; se0.118; r3-10; 6.07 emb; se0.123; r3-10; n107 f	723
	7 emb; n1 f	4388
	7.0 emb; r6-8; n4 f	3703
	8 emb; n1 f	5333
Clethrionomys glareolus	2-10	10589
	2.5 emb; r2-3; n2 f	10274
	3 emb; n1 f	7888
	3 emb; n1 f	6548
	3.0 yg; ±0.1; n62 non lactating f	1881
	3.2; ±0.1; n189 primiparous	1874
	3.3 emb; se0.4; 6.7 cl; se0.7; n12 f	1877
	3.4-6.2; n3 locations	8052
	3.42 yg live; ±1.51; r1-7; n83 lt	8021
	3.44; se0.04; mode 4; r1-7; n140 lt	10383
	3.52 yg; n1712 lt	1874
	3.6; n26 lt	7002
	3.62 emb; r1-6; n21 f	12076
	3.65; ±0.1; n299 multiparous	1874
	3.66 yg; mode 4; r1-10; n782 lt, 264 f	1408
	3.71; n14 lt	5307
	3.77 emb; mode 4; r1-6; n90 f	2077
	3.795; r1-8; n83 lt	8021
	3.97 yg; se0.11; n65 lt, 13 f	157
	4 emb; mode 3-5	516
	4.0 wean; r3-6	5477
	4 ea; n2 lt	3555
	4.00 yg; mode 4; r2-6; n20 lt	11362
	4.0 emb; r2-7; n58 f	10175
	4.04 cl; r2-7; n23 f	12076
	4.06 emb; r2-6; n49 f	1844
	4.09 emb; se0.104; mode 4; r2-8; n94 f	12079
	4.11; mode 4; r1-6; n70 lt	1222
	4.2 yg; ±0.2; n43 lactating f	1881
	4.3-5.5	453
	4.3; se0.1; n104 lt	1874
	4.31 ps; se0.155; mode 4; r1-11; n94 f	12079
	4.45 emb; n11 f	955
	4.5 yg; sd1.4; n85 lt	4343
	4.6-6.1 cl	7271
	4.73 emb; mode 5; r3-7; n44 f	10339, 10340, 10344
	4.84 yg; mode 5; r1-10; n799 lt	4172, 4173
	4.85; r1-9; n54 lt	5103
	4.88 emb; se0.055; n546 f	12082
	4.9 yg; sd1.2; n141 lt	4343
	4.902 emb; se0.065; mode 5; r1-10; n388 f	12077, 12079
	4.980 ps; se0.085; mode 4-5; r1-11; n347 f	12077, 12079
	5.0 yg; r3-7	5477
	5.01 cl; ±0.2; n65	1874
	5.03; n726 lt	1874
	5.2 emb; n110 f	9404
	5.22 emb; mode 6; r3-9; n45 f	11670
	5.24 emb; mode 5-6; r3-7; n29 f	11679
	5.32 emb; n368 f	10373
	5.82 yg; mode 6; r1-13; n342 lt	4173
	5.96 emb; n82 f	1076
	6.0-6.8 emb	223
	6; r1-10	4645
Clethrionomys rufocanus	3	5493

	3 emb; mode 5; r2-9	7760
	3.50 yg; se1.5; max 8	7760
	4 emb; n1 f	10516
	5.0 emb; r4-6; n4 f	11679
	5.16 emb, ps; mode 5; r3-8; n125 f	5140
	5.3 emb; mode 5,6; 88% 4-7; r1-10; n747 f	3547
	5.9; r4-11; n11 lt	12105
	6.0 emb; r5-7; n6 f	1913
	6 emb; mode 6; r2-10; n285 f	5501
	6.1 cl; r2-14; n439 f	3547
	6.2 emb; r3-13	8052
Clethrionomys rutilus	3-4	5493
	3.6 wean; ±3.7	2564, 7544
	4 ps; n1 lt	10516
	4.9 yg; ±2.1	2564, 7544
	5 emb; n1 f	10598
	5.0 emb; r4-6; n2 f	2545
	5.2 emb; r4-6; n5 f	12105
	5.3 ps; r4-6; n3 f	6782
	5.50 emb; n78 primiparous	6015
	6.05 emb; r4-9; n18 f	6782
	6.1 yg; r5-8; n8 lt	6782
	6.25 emb; n94 multiparous	6015
	7 emb; n1 f	8808
	7.0 emb; r3-9; n4 f	1913
	7.6 emb; r5-9	8052
	8.3 emb; r6-10; 6 f	5039
Dicrostonyx groenlandicus	1.50-3.33; r1-6; varies w/age, parity	4469
	2.35 wean; n52 lt	4724
	2.52 yg; se0.09; n52 lt	4724
	3-4; max 7	6327
	3.1 yg; se0.13; r1-6; n116 lt	6760, 8958
	3.1; sd1; n8 lt	9697
	3.3; r1-8	4467a
	3.41; se0.21; r1-7; n56 lt	6781
	4-8.5 cl	6013
	4.1 yg; r2-7; n8	2434
	4.78 emb, yg, ps; r3-9; n9 f	10101
	5.47 emb, ps; se0.33; r2-11; n30 f	6781
	5.5; r3-8; n17 lt	6710
	5.7 emb, ps; n47 f	3561
Dicrostonyx hudsonius	3-4; r1-7	1474
Dicrostonyx rubricatus	3.0 emb; n3 f	177
	7.2 emb; r5-11; n5 f	177
Dicrostonyx torquatus	2.63; r1-4; n8 lt	8810
	3.7-6.1; r1-11; varies age & parity	8051
	3.83; r2-6; n9 lt	8810
Dicrostonyx unalascensis	2.8 yg; sd1.2; 2.1 yg survive; sd1.0; survive; n188 lt	2560, 7544
Dinaromys bogdanovi	2-3 yg	7983
Ellobius fuscocapillus	3 yg; n1 lt	8052
	3-5	9175
Ellobius lutescens	4.8 emb; ±0.38; n24 f	6651, 11367
	12.9 ovulations; ±1.7; n17 f	11367
Ellobius talpinus	2-4	8052
	3.0	12111
	mode 4-5; r1-7	4645
Eolagurus luteus	mode 6-9 yg; r4-10	9951
Eothenomys inez	2 emb; n1 f	182
Eothenomys melanogaster	1.5 emb; r1-2; n2 f	182
	2.67 emb; r2-3; n3 f	182
Eothenomys smithi	2.56 emb; mode 2-3; r1-5; n120 f	10665
	4.1; r2-7; n13 lt	12026
	4.45 yg; se0.12; r1-9; n188 lt	294b
Hyperacrius fertilis	2-3	9175
Hyperacrius wynnei	2.5; r2 emb, 3 ps; n2 f	9175

Lagurus curtatus	3 emb; n1 f	4223
	4.0 emb; r3-5; n2 f	1119
	4.4 emb; r2-13; n42 f	6925
	5 ps; n1 f	3703
	5.26 emb; mode 6; r1-11; n281 f	5240
	5.6 emb; se1.2; n11 f	7606
	6 emb; n1 f	7325
	6.11; r4-10; n66 f	7491
	7.0 emb; r6-8; n2 f	10218, 10219
Lagurus lagurus	1.2 weaned	11647
	3.11 yg; 73% 2-5; r1-8; n448 lt	3652
	3.3 yg	11647
	3.3; r1-5; n14 lt	11647
	4.0; r2-7; n88 lt	3516
	4.5-8.1; r1-12; n>100	9950
	most 5-6; r3-7	8052
	mode 5-6; r4-10	8634
	6.17 emb; r1-12; n80 f	9950
Lemmus lemmus	3.5 weaned	7544
	3.75-6.62	5967
	4.0 yg; sd1.87; r1-16; n630 lt	9762
	4.1; r1-7; n77 lt	3462
	4.5; r1-7	4830
	4.8 yg	7544
	5	5493
	5.29 yg; mode 5-6; r1-9; n80 lt	6806
	6.4; mode 5; r2-13; n50 lt	11679
	7.5; r6-9; n2 lt	10210
	max 9 yg	2386
Lemmus schisticolor	4.1 emb; se0.36; r2-6; n35 f	10016
	5.3 emb; r3-7	8051
	5.5 emb, ps; sd1.25; n95 f	1077
	7.0 emb, ps; sd1.6; n30 f	1077
Lemmus sibiricus	1.7 weaned	7544
	3.1 yg	7544
	3.8-8.1 cl	6013
	4.7-8.1; varies with season	716
	5.5; r3-9	9071
	6-13 emb	8051
	6.2 yg; r2-9; n25 lt	7595
	6.9 emb; r3-10; n14 f	6710
	~7 emb; r1-13; n286 f	7595
	7.89; r6-12; n9 lt; 2 d	655
Microtus spp	4.6 cl; ±0.48	5674
Microtus abbreviatus	2.2 weaned	7544
	3.0 yg	7544
Microtus agrestis	2.5-4.6	1872
	r3.4-5.9 yg; n133 lt; varies with parity	1877
	3.5 weaned; se0.30; n19 lt	1805
	3.50 yg; n490 lt, 140 f	6397
	3.5-3.6	519
	3.7 emb; r3-4; n3 f	8986
	3.73; r1-7; n389 lt	8921
	3.9 cl	1248
	mode 4-7; r3-11	3571
	4-7 emb; n6 f	1590
	4.0 yg; r3-6; n9 nests	3835
	4.0-7.5 emb; n2 locations	7703
	4.1; r2-7; n60 lt	6008
	4.3 emb; se0.4; 5.8 cl; se0.6; n11 f	1877
	4.6 weaned; se0.25; n37 lt	1805
	4.66; mode 5; r1-7; n32 lt	12070
	4.7 cl; ±0.50	7339
	4.7; r4-6; n7 lt	10340

	5.0 emb; 5.5 ova ovulated	1218
	5.07 emb; n329 f	10373
	5.1; se0.3; n23 lt	1250
	5.61 emb, ps; se0.35; r4-8; n13 f	8336
	6.1; n80 lt; small islands; 5.3; n140 lt; large islands; r3-10	8645
	max 8	10345
	max 8	8920
Microtus arvalis	mode 1-12; r1-13	3571
	2.1; r1-3; n6 lt	8153
	2.67; r1-8; n508 lt	6398
	2.7 wean; n12 lt	11427
	3-7	2466
	3-8	6902
	3.25 emb; n8 f	955
	3.73; median 5; r1-11; n2518 lt	3459
	3.79 yg; mode 3-4; r1-9; n1016 lt	9013
	mode 4-9 emb; r2-15; mode 4-7 yg	8052
	4-5 emb; n4 f	1590
	4 ps; n1 f	6293
	4 ps; n1 f	6293
	4.0 emb; r3-5; n2 f	6293
	4.1 emb; r3-7; n18 f	2465
	4.2; r1-8; n19 lt	9269
	4.25; r3-5; n4 lt	6237
	4.25; n17 lt	7292
	4.31 emb; r1-8; n16 f	11593
	4.5 emb; r4-5; n2 f	11255
	4.8; n111 lt	9269
	4.92; n67 light f	11867
	5-10	11225
	mode 5-6; most 4-7; r4-15	4645
	5.17 emb; r3-9; n30 f	10344
	5.24 emb; r3-9; n105 f	741
	5.39; mode 5; r2-9 n44 lt	6361
	5.41 emb; se0.039; n1567 f	12082
	5.47; r1-9; n75 lt	3465
	5.49; r2-9; n84 f	10343
	5.86; r4-10; n7 lt	7899
	5.96; n32 heavy f	11867
	6.0 emb; r5-7; n5 f	7905
	7; max 12	3460
	8.0 emb; r5-11; n2 f	10516
Microtus brandti	5.0; max 8; n22 lt	9012
	5.0; r2-8; n28 lt	12105
	r5-9; r6-12; n2 samples	8052
Microtus breweri	3.4	10648
	mode 4-5 yg	7326
Microtus californicus	2-7 yg	4490
	4.17 old ps; n35 f	4893
	4.17; se0.20; r1-10; n220	6040
	4.20 emb; se0.11; sd1.41; mode 4; r1-9; n154 f	4052, 4053
	4.24; se0.11; sd1.8; mode 4; n295	6041
	4.25; se0.28; r1-10; n65	6040
	4.29 new ps; n42 f	4893
	4.69 reared; r1-9; n42 lt	1995
	4.8 yg; r4-5; n5 lt	9747
	4.94 emb; r1-10; n73 f	4893
	5 emb; n1	9747
	5.05 emb; 4.82 healthy emb; 280 f	6433
	5.57 ps; 108 f	6433
	9.29 cl; max 29; n82 f	4053
	11.6 cl; se2.4; 5.0 live emb; se0.3; n10 f	5675
	11.6-12.2 cl	5675
	13.12 cl; r2-50; n526 episodes	6433

Microtus canicaudus	3.23; r2-10; n44 lt	11011
	4.1-5.2 yg; se0.2-0.4; n4 groups	4196a
Microtus chrotorrhinus	2.88 emb; mode 3; r2-5; n8 f	5794
	3.5 emb, cl, ps; n13 f	10838, 10842
	3.56 yg; ±0.19; r2-5; n9 lt	2082
	3.71 emb; se0.19; mode 4; r1-7	6896
	3.91 emb; mode 4; r2-5; n11 museums	5794
	5.32 emb; r3-8; n16 f	5796
Microtus fortis	3-7 yg	5424,
	5.5 emb; r5-6	182
	8.33 emb; r7-9; n3 f	8052
Microtus gregalis	5-13; r8-9; n2 locations	8052
	mode 7-9; r7-15	4645
	7.5 emb; r7-8; n2 f	8052
	11 emb; n1 f	10516
Microtus gud	6 emb; n1 f	8052
Microtus guentheri	mode 4-8; r1-21	1037
	5.5; r4.5-6.3; n~365 lt	1036
	5.6 emb; median 6; r4-8; n21 f	8115
	6.1; mode 7; r1-10	1965
	6.2; max 15	3610
	8 emb; max 13	6419
	8.82 emb; sd1.29; r6-11; n17 f	1962, 1966
	9.3 emb; r9-11; n3 f	7905
	10.73 ps; sd5.35; r6-19; n11	1966
Microtus longicaudus	2 emb; n1 f	2544
	3.5 emb; r3-4; n2 f	10598
	3.7 emb; r2-5; n9 f	12049
	3.9; r2-6	6632
	3.95 reared; r1-7; n37 lt	1995
	4-7 emb	502
	4-8	1474
	4 ps; n1 f	7809
	4.5 emb; r4-5; n2 f	11646a
	4.76 emb; r4-6; n17 f	10954
	4.8 emb; r4-6; n5 f	1119
	4.9 emb; r4-6; n10 f	4897
	5; r3-7	2088
	5.0; n26	10161
	5.2; mode 6; r3-6; n5 lt	6488
	5.5 yg; r5-6; n2 nests	10598
	5.6; r2-8; n39	4226
	6.0; r5-7; n3	10161
	7 emb; n1 f	6560
	7 emb; n1 f	4745
	7 ps; n1 f	12049
Microtus maximowiczii	5-9 emb	8052
Microtus mexicanus	1.8	9176
	r2-3 emb; n3 f	523a
	3.5 emb; r2-5; n2 f	502
	2.23 emb; ±0.32; mode 2; r1-4; n22 f	1365
	2.25 emb; ±0.22; mode 2; r1-4; n8 f	1365
	2.3 emb; mode 3; r1-4; n59 f	1813
	2.6 emb; r1-3; n7 f	4228
	2.6; r1-4; n15 lt	522
	3 emb	2341
	mode 3-4; r2-5	1474
Microtus middendorffi	mode 5-6; r3-8	8052
Microtus miurus	2.8 wean	7544
	3.9 birth	7544
	4-12 emb	1474
Microtus montanus	r4-7 emb; n10 f	2226
	4-8 emb	7809
	4-10 emb	502
	4.16 yg; r1-10; n254 lt, 31 f	7811

	4.3 cl; ±0.6	5672
	4.5 yg; r3-6; n4 lt	9967
	4.65 yg; r1-10; n60 lt, 12 f	7811
	4.7 emb; se0.5; 5.6 cl; se0.4; n12 f	2330
	r4.8-6.6 emb, ps; n17 seasons	8561
	5 emb; r4-6; n8 f	11646a
	5.1 cl; se0.93; n12	4014
	5.5 emb; r5-6; n2 f	6488
	5.8 emb; se0.25; r2-10; n46 f	11230
	5.98 reared; r3-6; n42 lt	1995
	6.02 ps; n75 f	4893
	6.47 emb; r2-10; n109 f	4893
	6.75 cl; r1-13; n194 f	4893
	7 emb; n1 f	10305
	7.0 emb; r6-8; n2 f	2544
	r7-8 ps	7809
Microtus montebelli	~3	3969
	3.5 emb	7392
	3.75 emb; se0.28; r2-5; n12 f	5511
	4.11 yg; ±1.49; mode 4; r1-8; n168	7762
	4.3; n88 lt	15
	4.5	9443
	4.9 emb; r2-8; n8 f	9919
	5.1 emb; se0.3; mode 5; r2-9; n40 f	5510, 5512
	6.00 emb; se0.29; r5-7; n9 f	5511
Microtus nivalis	2-4 emb	5984
	2.67; r1-4; n3 lt	3457
	3.0 emb; r2-4; n23 f	6313
	3.1 emb; mode 4; r2-4; n12 f	5486
	mode 4-8 emb; r1-9	6919
	4 emb; n1 f	6293
	5 emb; n1 f	6293
Microtus oaxacensis	1.2	9176
Microtus oeconomus	2-12	7433
	2.2-2.3 weaned	2563, 7544
	3-11	8845
	3.9-4.0 yg	2563, 7544
	mode 4-7; r2-12	3571
	4.04 emb; mode 7; r4-8; n25 f	3468
	4.41 emb; n80 f	3468
	4.6; r4-6	3655
	5.0-7.8	716
	5.19 emb; n31 f	6015
	5.3 emb; r4-6; n3 f	10516
	5.33 yg; n6 lt	9355
	5.48 emb; mode 5; r2-8; n25 f	11679
	5.67 emb; mode 6; r3-8; n18 f	11679
	5.7 emb; r4-7; n3 f	10516
	5.7 yg; mode 5; r3-9; n4 lt	659
	5.7; mode 5; r3-9; n25 lt	10340
	mode 6-9; r3-12	8052
	6.0 emb; r5-9; n8 f	10339
	6.1 emb; n16 f	12049
	6.4; r3-8; n10 lt	10344
	6.5 emb; r6-7; n2 f	8808
	6.74; se0.27; n41 f	5969
	6.94 emb; mode 7; r1-12; n373 f	5493
	7.15; mode 7; n119 f	10681
	7.7 emb; r6-9; n6 f	1913

Microtus oregoni	2.79 yg; mode 3; r1-5; n28 lt	2087
	3.11 emb; mode 3; r1-5; n26 lt	2087
	3.2 emb, ps (3.1 emb, 3.5 ps); mode 3; r1-8; n29 lt	3622
	3.3 emb; n8 f	1615
	3.41 cl; n34 lt	3622
	3.5; se0.21; n12 lt	1370
	3.8 reared; r1-6; n15 lt	1995
	4.0 emb; r2-6; n21 f	4970
Microtus pennsylvanicus	2-9	501
	2.6 weaned; ±1.6	2562, 7544
	3.7; se0.6; n12 lt	5673
	3.72 yg; sd1.88; r1-9; n416 lt	8642
	3.78 yg; se0.23; n37	7116
	4-8 emb	502
	4 yg; r1-9	6347, 6348
	4 emb; n1 f	495
	4 yg; n1 nest	9612
	4 yg; r3-5; n3 nests	2902
	4.0 yg; mode 4; r1-6; n23	5976
	4.1 yg; ±1.7	2562, 7544
	4.12 emb; r3-6; n8 f	2548
	4.18 yg; n51 lt	2059
	4.25 emb; r4-5; n4 f	2549
	4.37 emb; sd1.43; n51 f	1832
	4.46 emb; se0.13; mode 4; r1-9; 4.80 cl; n153 f	2059
	4.48 emb, ps; r1-8; n35 f	2376
	4.5	10648
	4.5 emb; r3-6; n2 f	261a
	4.54 emb; sd1.51; n152 f	5629
	4.72 yg; se0.26; n25	7116
	4.8 yg; mode 5; r2-7; n73	5976
	5 emb; n1 f	6796
	5 emb ea; n2 f	2902
	5.0; r4-6; n2 f	7809
	5 yg; r2-9; n17 lt, 1 f	501
	5.0 emb; mode 5; r3-8; n24	5976
	5.00 yg; ±0.17; r3-9; n41 lt	2082
	5.05; mode 5; r3-7; n39 lt	5149
	5.09 emb; n87 f	6015
	5.1 yg; 4.8 yg alive; r1-11; n106 lt, 27 f	10463
	5.24 cl; sd1.10; n151 f	5629
	5.25 emb; r4-7; n16 text; 5.20 emb; n15 table	10839
	5.3 emb; r5-6; n4 f	5798
	5.3 emb; r2-10; n67 f	12049
	5.32 uterine swellings; sd1.61; n25 f	1832
	5.49 yg; se0.27; n47	7116
	5.52 yg; se0.31; n23	7116
	5.52 reared; r2-8; n31 lt	1995
	5.5 emb; n4 f	11646a
	5.5 emb; mode 5; r1-11; n440 f	4278
	5.6-5.9 cl	1919
	5.67 ps; r5-6; n3 f	261a
	5.8 yg; r4-9; n11 lt	11956
	5.86; r2-10; n144 f	2660
	5.9 yg; mode 6; r4-8; n28	5976
	6 yg; n1 f	6769
	6 emb ea; n2 f	10218, 10219
	6 emb; r4-8; n2 f	10220
	6.0 emb; r5-7; n2 f	4745
	6.0 yg; n14 lt	11518
	6.00 cl; sd1.27; 4.25 emb; sd0.83; n16	4018
	6.09 cl; se0.115; r3-12; 5.59 emb; se0.110; r1-11; n251 f	721, 723
	6.2 emb; r4-9; n12 f	5405b$_1$
	6.2 emb; mode 7; r3-8; n13 f	5976
	6.3 emb; r5-8; n15 f	3703

	6.3 yg; n24 lt	4272
	6.6 emb; r2-9; n36 f	10954
	6.7 emb; r5-9; n10 f	6166
	6.9 emb; r4-12; n17 f	10101
	7 emb; n1 f	2545
	7 emb; n1 f	10652
	7 ps; n1 f	3703
	7.8 ps; r6-9; n5 f	10101
	8 emb; n1 f	10218
	8 yg; n1 lt	5210
	8.14 emb; r6-9; n7 f	11504
Microtus richardsoni	4-7 yg	503
	5 yg; r2-8	2088
	5.17 emb; se0.44; n19 f	6630
	5.42 ps; se0.58; n12 f	6630
	5.59 emb; se0.28; n37 f	6630
	5.60 emb; n16 f	6630
	5.6 emb; n20 f	6630
	5.62 emb; se0.18; mode 5; r2-9; n86 f	6631, 6632
	5.91 cl; se0.24; n46 f	6630
	6 emb; n1 f	4969
	6 emb; r5-9; n10 f	7809
	6.11 ps; se0.43; mode 8; r2-10; n26 f	6631, 6632
	6.35 ps; se0.32; n23 f	6630
	7.0 emb; n3 f	6630
	7.10 ps; se0.34; n7	6630
	7.33 emb; n3 f	6630
	7.85 emb; ±0.51; mode 8; r6-10; n26 f	1369
Microtus roberti	2.0 emb; r2-4; n11 f	10274
Microtus sachalinensis	7.9	11366
Microtus socialis	2.00-6.88	2995
	4 emb; n1 f	6293
	5.67 emb; r5-7; n3 f	5985
	mode 6; r4-11	8052
	max 9	4645
Microtus townsendii	2-10	2053
	4; r1-9	2088
	4-5; r1-9	1474
	4.92-5.42 emb; se1.00-1.19; n3 seasons	2053
	4.8 yg; r2-6; n13	6693
	5.4 yg; r2-7; n12	6693
	6.0 emb; r5-7; n20 f	6693
	7; r5-8	2053
Microtus transcaspicus	2.1; r1-4; n17 lt	8153
Microtus xanthognathus	8.0 yg; r7-10; n11 lt	12049
	8.0 emb; r6-10; n7 f	11878
	8.33 emb; r7-10; n3 f	6376
	8.8 emb; sd1.62; r6-13; n71 f	11878
Neofiber alleni	1.5 yg; n12 lt	11519
	2.2 emb, ps; mode 2; r1-4; n104 f	944
	2.7 cl; r1-5; mode 3; n46 f	944
	3 yg ea; n2 lt	4282
Ondatra zibethicus	1.5 emb; r1-2; n2 f	1290
	2.65 yg; r1-5; n17 nests	10580
	3-9; (7-9 start of season, 3-5 end of season)	7291
	3.3 yg; mode 3; r1-5; n108 lt	11762
	3.47 emb; r1-5; n103 f	8015
	3.62 emb; mode 4; r1-6; n66 f	10580
	3.7 emb; mode 4; r1-6; n206 f	11762
	3.9 yg; r1-8; n23 lt	4832a
	4.5; 95%ci 4.0-4.9; r1-10	2243
	4.8; 95%ci 4.3-5.3; r1-10; lt <9 d	2243
	5.0; 1st lt; 5.6; 2nd lt; r1-12	7093
	5.2 yg; n55 families	8762

	5.38 emb; mode 5; r3-7; n13 f	9529
	5.4 ps; n50 lt, 36 f	1915
	5.4 yg; r2-9; n62 nests	3621
	r5.5-8.5; n25 locations	5884
	5.8	11174
	5.8 yg; r4-8; n5 lt	6805
	6-8 yg	90
	6 emb; r3-9; n23 f	2616
	6.0 emb; r5-7; n2 f	4491
	6.0 ps; n44 sets, 38 f	3061
	6.1 yg; mode 6; r2-13; n15 lt	11934
	6.1; mode 6-7; r1-11; n418 lt	2697
	6.15 yg; n27 lt, 0-5 d	3061
	6.27; se0.18; r1-13; n132 lt	705
	6.28 yg; n60 lt	9503
	6.33 emb; n27 f	2685
	6.4 yg; n26 f	10112
	6.5 yg; r1-11; n158 lt <1 wk old	3081
	6.6 emb; se0.3; r5-8; n16 f	10410
	6.6 ps; mode 8; r2-11; n25 f	9003
	6.7 emb; mode 7; r1-16; n28330 f	4892
	6.78 yg; n188 lt	3082
	6.79; se0.16; r1-13; n115 lt	705
	6.79 emb; mode 7; r1-12; n1294 lt	4891
	6.87 ps; n510 sets, 197 f	9503
	6.9 ps; r3-15; n24 f; prob 2 lt/f	351
	6.91 emb; n65 f	703
	7 emb; n1 f	3558
	7 emb; n1 f	5333
	7.0 emb; r1-10	11467
	7.0-9.0; varies with parity	10066
	7.0 yg; mode 6; r2-9; n35 lt	9003
	7.03 emb; n39 f	2685
	7.1 yg; r5-11; n22 lt	0000
	7.1 emb; r2-10; n45 f	3621
	7.2 emb; r5-10; n11 f	6805
	7.3 yg; r5-10; n31 lt; 8.0; ≤1 wk	9003
	7.3 yg; modes 6,8; r4-10; n80 lt	8104
	7.4; n76 lt	719
	7.5 emb; n8 yg f	8237
	7.5 yg; n364 lt; r6.35-8.41; n22 yr	3082
	7.7 ps; r7-9; n3 yg f	8761
	7.7 emb; r4-11; n33 f	9690
	8.2 emb; r5-11; n6 f	3081
	8.4 emb; sd1.2; n9 f	8237
	8.7 ps; n7 f	3558
	9 emb; n1 f	8234
	9	6267
	9.1 emb; r2-10; n16 f	9690
	10.93 ps; n16, 2 sites	2685
	12.06 ps; n103	2685
	12.7 ps; n48 f	8762
	17 ps; r12-24	8236
	18.31 ps; ±1.01	10112
	19.8 ps; n36 f	8237
Phenacomys intermedius	mode 4-5; max 8	2088
	4.24 yg; se0.12; mode 4; r2-7; n59 lt	5150
	4.4 yg; n8 lt	4219
	4.8 yg; r2-8; juveniles: 3.8, adults: 5.9	555a
	4.8 emb; se0.75; r3-6; n4 f	11230
	4.8 emb, yg; r2-8; n43 lt	3429
	4.9 yg, emb, ps; r3-8; n37 f	10101
	5.3 ps; se0.52; r2-9; n11 f	11230
	7 emb; n1 f	10220

	7 emb ea; n2 f	279
Pitymys afghanus	3.5 emb; r3-4; n2 f	8052
Pitymys duodecimcostatus	2.0 emb; sd0.8; mode 2; r1-4; n37 f	8216
	2.2 emb; mode 2	11255
	2.4 yg; r1-3; n5	11820
	2.66; mode 3; r1-4; n32 lt	6361
	2.7 emb; r1-5	1843a
	3.0 emb; r1-4; n9 f	11820
Pitymys juldaschi	3.43; r2-6; n60 lt	1057
	3.63; r2-7; n70 lt	1057
Pitymys leucurus	2 emb; n1 f	7382
Pitymys lusitanicus	2.0 yg; r1-3; n56 lt	11820
	2.26 emb; r1-4; n57 f	11820
Pitymys majori	2.75 live emb; r1-4; n4 f	10274
	3-7 emb	8052
Pitymys multiplex	3	6370
Pitymys ochrogaster	1-8	2137
	2.3 yg; se0.5; n6	2529
	3.18; se0.24; median 3; r1-6; n65 lt	6881
	3.27 emb	5629
	3.27 emb; sd1.28; n160 f	5629
	3.3; se0.2; n22 lt	10333
	3.37 yg; ±0.075; n82 lt	3283
	3.38 emb; 5.16 cl; n20 f	4019
	3.40 emb, ps; r1-5; n35 f	2376
	3.4 emb; r1-7; n58 f	5241
	3.42 emb; 3.79 cl; n182 f	9273
	3.5 emb; r2-5; n11 f	6298
	3.6; r1-6; n28 lt	10954
	3.6 emb; mode 3; r1-6; n110 f	5405b$_1$
	3.7 emb; r1-6; n3 f	3703
	3.73 cl; sd1.06; n158 f	5629
	3.77; r1-8; n280 lt	9111
	3.87 yg; n54 lt	2059
	3.89 emb; se0.10; mode 4; r2-7; 4.35 cl; n134 f	2059
	3.93 reared; r1-7; n28 lt	1995
	4.0 emb; r3-5; n2 f	495
	4.19 emb; r1-9; 4.17 ps; n198 f	9235
	4.25; n8 lt	6051
	4.4 emb; r3-6; n9 f	261a
	4.5; se0.6; n13 lt	5673
	4.7 ps; r3-7; n7 f	6298
	5.0; se0.4	3648
	5.0 emb; r4-6; n2 f	1812a
	5.0 emb; r3-7; n2 f	2547
	5.8 emb; r5-7; n5 f	11957a
	6 ps ea; n2 f	261a
	7.2 ps; r6-10; n15 f	11957a
	10 ps; n1 f	4525
Pitymys pinetorum	1.9 emb	11059
	mode 2-4	783
	2 emb; n1 f	2547
	2 emb; n1 f	8988
	2 emb ea; n2 f	6475
	2 emb ea; n3 f	6298
	2.2 yg; se0.1; r1-3; n19 lt	6533
	2.25 emb; r1-5; n138 f	8268
	2.33; r2, 2 emb, 3 yg; n2 f	6487
	2.33 emb; r2-3; n3 f	5405b$_1$
	2.44 ps; r2-5; n9 lt	3852
	2.5 emb; r2-3; n2 f	3747
	2.5 emb; r2-3; n2 f	3803
	2.54 emb; r2-5; n13 f	3852
	2.59 wean; r1-6; n150 lt	9530

	2.75 wean; se0.09	9536
	2.79; r2-4; n14 lt	4273
	3-4; r2-7	1474
	3.0 ps; r2-4; n3 f	3747
	3 yg ea; n4 lt	3852
	3.2 yg; se0.1; n50 lt	2493
	3.31 yg; se0.09; mode 3; r1-6; n150 lt	9530, 9536
	3.4 yg; se0.3; n25 lt	2493
	4 ps; n1 f	9164
	8 emb; n1 f	9164
Pitymys quasiater	1 emb; n1 f	2341
	2.1 emb; r1-4; n12 lt	4228
Pitymys sikimensis	2 emb; n1 f	2256
	2.32 emb; r1-4; n22 f	7382
	3 emb; n1 f	182
Pitymys subterraneus	mode 2-5; r1-7	3571
	2.0; r1-4; n6 lt	7904
	2.17-2.45 ovulations	5281
	2.2	10379
	2.2 emb; r1-3; n5 f	4114
	2.25 emb; se0.077; r1-4; n71 f	6004
	2.4; r1-4	9656
	2.4 yg; se0.2; n28 primiparous	5282
	2.5 emb; r2-3; n2 f	5986
	2.5; r2-3; n22 lt	6199
	2.65 yg; 2.36 wean; se0.1; mode 3; r1-5; n317 lt	5282
	2.66 emb; se0.095; r1-4; n38 f	8335
	2.74 yg; mode 3; r1-5; 2.28 wean; n143 lt	1407
	2.875 ps; n8 lt	4114
	2.88; r2-4; n32 lt	7903
Pitymys tatricus	3 emb; n1 f	5986
	3.0 emb; se0.081; r2-4; n61 f	6004
Pitymys thomasi	4.45 emb; sd1.58; r1-8; n20 f	7905
Prometheomys schaposchnikowi	2.7 emb, ps; r1-4; n6 lt	10275
	3.5 emb; r3-4; subadult f	12007
	mode 4-5 emb; r1-5; adult f	12007
Synaptomys borealis	4 yg; r4-8	555a
	mode 4-5; r2-8	2088
	4-5; r2-8	1474
	4.4 emb; r3-6; n10 f	12049
Synaptomys cooperi	1 emb; n1 f	5405b$_1$
	1-7 yg	1469
	2-4 emb	10329
	2 emb; n1 f	2025
	2.5 yg; r2-3; n2 f	10329
	2.8 emb; r2-4; n5 f	6475
	mode 3 yg; r2-5	5210
	3 yg; r1-7	555a
	3 emb; n1 f	6298
	3 emb; n1 f	11956
	3.20 yg; ±0.19; r2-5; n9 lt	2082
	3.33 emb; r3-4; n3 f	4525
	3.7; n6 lt, 1 f	2026
	4.0 emb; r3-5; n4 f	5798
	4.5 emb; r4-5; n2 f	6486
	4.7 emb; r4-6; n6 f	5405b$_1$
SEXUAL MATURITY		
Arvicola terrestris	f: 1st conceive: 15 cm	11204
	f: 1st preg: 76-96 g	11250
	f: 1st birth: 1st summer	4645
	f: ~6 wk; 120 g	8392
	f: 35-40 d	10944
	f: 60-65 g	8052
	f: 116 g	8392

	m: spermatogenesis: 15 cm	11204
	m: testes mature: 6-10 wk; 65-95 g	11250
	m: testes mature: 85-90 g	11668
	m: 1st mate: 43-47 d	11668
	m: 110 g	8392
	f/m: mate: not yr of birth	10447
	f/m: 2-2.5 mo	3571
Clethrionomys gapperi	f: 1st birth: 1st summer	11230
	f: ≤4 mo	10569
	m: sperm: 1st summer	11230
	f/m: mate: 7 mo	7238
	f/m: mate: yr of birth	6796
	f/m: 2-4 mo	7238
Clethrionomys glareolus	f: vag perf: 26.9 d; se3.1; n14	1075
	f: vag perf: 29.3 d; se4.6; n17	1075
	f: mate: yr of birth	2077
	f: 1st cl: 16 g	12076
	f: 1st conceive: 27 d; n1	4172, 12081
	f: 1st preg: ~5 wk	12077
	f: 1st preg: 22 g	2447
	f: 1st birth: 60 d	4645
	f: 1st birth: 3.5 mo	11362
	f: 1 mo	10589
	f: 1-1.5 mo	1408
	f: est 13 g	516
	m: 2 mo	8425
	m: <3 mo	12077
	m: est 14 g	516
	m: testes mature: 18 g	12076
	m: 1st sire: ~67 d	1408
	f/m: 1st mate: ~1 yr	11679
	f/m: 1st mate: ~1 yr	1076
	f/m: mate: 1 mo	11670
	f/m: mate: yr of birth	1222, 9342
	f/m: mate: yr of birth	7988
	f/m: 1st repro: yr of birth	4341
	f/m: 8-9 wk	3571
	f/m: ~3 mo	5307
Clethrionomys rufocanus	f: 1st birth: ~110 d	6134
Clethrionomys rutilus	f: preg: 1st yr	6782
	f: yr of birth	3775
Dicrostonyx groenlandicus	f: vag perf: 25-27 d	4335
	f: 1st birth: 49 d; n1	4335
	f: 1st birth: 58-90 d	4469
	m: 1st sire: 85 d	4469
	m: ~1 yr	6013
Ellobius talpinus	f: 1st birth: ~5 mo	4645
	f/m: 1st repro: 90 d	12111
Eolagurus luteus	f/m: 3-4 wk	9951
Eothenomys smithi	f: 1st birth: 120 d	294b
	f/m: 1st repro: >25 g	12026
Lagurus curtatus	f: 1st mate: min 47 d; mode 60 d	5240
	f: 1st mate: 2 mo	5146
	f: preg f: 103 mm	6925
Lagurus lagurus	f: 1st mate: 35 d-1.5 mo	8052
	f: 1st birth: 49-54 d	3516
	f: 1st birth: 50-65 d	4645
	f: 30 d; r19-35	3652
	m: 1st sire: 51-56 d	3516
	m: 4-6 wk	8052
	m: 45-46 d	3652
	f/m: 1st mate: 40-45 d	8634

Lemmus lemmus	f: vag perf: 10 d	6806
	f: 1st mate: 20 d	6806
	f: 1st birth: 39 d; mode 2nd mo	3462
	f: 16-49 d	9762
	f/m: 1-1.5 mo	5967
	f/m: ~80 d, 45-60 g	8051
Lemmus schisticolor	f: vag perf: 22 d; 24 g	5135
	f: 1st mate: 3 wk	3466
	f: 1st birth: 2 mo; n15/73	5135
	f: 16 g; 82 mm	10016
	m: testes descend: 44 d; 28.2 g	5135
	m: 22 g; 88 mm	10016
	f/m: ≥20 g	5135
Lemmus sibiricus	f: ~2 mo	9071
	f: 110 d	716
	f: 3-4 wk	7595
	m: 3-4 wk	7595
	f/m: summer of birth	6013
Microtus agrestis	f: vag perf: 19.8 d; se0.2; n21 f with m present	1875
	f: vag perf: 20 d	8920
	f: vag perf: 22.0 d; se0.5; n21 f no male present	1875
	f: fecund: 3 wk	6397
	f: 1st conceive: 42 d	6397
	f: 1st birth: 7-8 wk	9011
	f: 2-3 wk	3571
	f: 3 wk	1872
	f: 6 wk	11294
	f: >12 g; 5-6 wk; n1	1218
	m: testes large: 6 wk	521, 6397, 8920
	m: fertile: ~20 g	1218
	m: 6 wk	1872
	f/m: 50-60 d	12070
Microtus arvalis	f: vag perf: 11 d	3460
	f: 1st mate: 21 d	6902
	f: 1st birth: 33-34 d	3460
	f: 1st birth: 40 d; mode 60-65	4645
	f: 1st birth: 40 d; mode 80-89	3459
	f: 1st birth: 41-57 d	9013
	f: 1st birth: 47-144 d	9269
	f: 1st birth: 50 d	3465
	f: 20-25 d	2465
	f: 47-90 d or 59-90 d	8052
	m: epididymal sperm: 35 d	6902
	m: 2nd mo	8052
	f/m: 1-2 mo to >4 mo	6903
	f/m: 9-12 wk; full grown 6-10 wk	3571
	f/m: 68 d	6237
Microtus brandti	f: vag perf: 3 wk	9012
	f: 1st birth: 8 wk	9012
Microtus californicus	f: 1st mate: 20 d	4490
	f: 1st mate: 25-30 g	4893
	f: 1st preg: 3 wk-9 mo	4053
	f: 10-15 g	4052
	m: sperm: 40 g	4053
	m: 1st mate: 6 wk	4490
	m: 1st mate: 35-40 g	4893
	f/m: 25-30 g	6434
Microtus canicaudus	f: 1st conceive: 18-22 d	4196a
	f: 1st conceive: 30-50 d; n1/20; 90-120 d; n4/8	8410
Microtus chrotorrhinus	f: preg f: 21 g	5796
	f: preg f: TL: 131 mm; n1	5796
	f: HB: >140 mm; 30 g	6896
	m: HB: >150 mm; 30 g	6896
Microtus fortis	f: 3.5-4 mo; 106 mm	8052

Microtus gregalis	m: ~2 mo; 20 g	8052
	f/m: repro: yr of birth	10516
Microtus guentheri	f: vag perf: 30 d	1965, 1966
	f: 1st mate: 50 d	3610
	f: 1st preg: 60 d	1965
	m: sperm: 30 d; n1	1966
	m: sperm: 85 d; mode >100 d	1965
	m: 1st mate: 62 d; mode 95 d	3610
	m: 1st mate: 11 mo	1966
	f/m: mate: most 9-12 mo; few season of birth	1962
Microtus longicaudus	f/m: repro: yr of birth	502
Microtus middendorffi	f/m: ~2 mo	8052
Microtus montanus	f: vag perf: 18-42 d; n42	9518
	f: 1st mate: 19-29 g	4893
	m: 35 g	4893
	f/m: 3-4 wk	7808
	f/m: summer of birth	11230
Microtus montebelli	f/m: 1st repro; 60 d; 32.2 g	7762
	f: 25 g	5510
Microtus nivalis	f/m: repro: yr of birth	6919
Microtus oeconomus	f: vag perf: 51.38 d	5313
	f: 1st birth: 6 wk	3571
	f: 3 wk	3571
	f: summer of birth	10681
	m: testes large: 48.4 d	5313
	f/m: mate: yr of birth	10516
Microtus oregoni	f: 1st mate: 22-24 d	2087, 2088
	f: 1st conceive: 27-36 d	2087
	m: 1st mate: 34-38 d	2087
	m: 42 d	2088
Microtus pennsylvanicus	f: 1st mate: 25 d	501
	f: 1st conceive: 4 wk; 24 g	4272, 4278
	f: 1st birth: 45 d	501
	f: 3-4 wk	2850
	f: median 18.8 g; 95%ci 17.7-20.1	7293
	f: yr of birth; 45-49 g	12007
	m: testes: 1 mo; 25-30 g	4278
	m: 1st mate: 45 d	501
	m: 5-6 wk	2850
	m: median 23.6 g; 95%ci 20.2-27.4	7293
	f/m: mate: 35 g	6347
	f/m: 30-40 d	2088
Microtus richardsoni	f: mate: season of birth	6632
Microtus socialis	f: 1st birth: 45-60 d	4645
Microtus townsendii	f: vag perf: 2.5-7.5 wk; n16/20; 15-19 g	6691
	m: >30 g	6693
Neofiber alleni	f/m: 90-100 d; 230-280 g	944
Ondatra zibethicus	f: vag perf: 4 mo	10066
	f: preg: 3-4 mo	4891
	f: preg: 1 yr; some yr of birth	702, 705
	f: 1st birth: 1st summer	4645
	m: 4-5 mo; mode 7-8 mo	10490
	m: 7-8 mo	11027
	m: ~1 yr	704
	f/m: 4-8 mo	3571
	f/m: 1+ yr	11467
	f/m: yr of birth; n3/594	8761
Phenacomys intermedius	f: 1st mate: 1st yr	1474
	f: 1st mate: 4-6 wk	3429
	f: 1st birth: 7 wk	555a
	f: 1st birth: yr of birth	5150
	f: 4-6 wk	1474
	f/m: 1st mate: 1st summer	11230

Pitymys lusitanicus	f: 1st preg: 5 wk; n1	11820
	m: 1st sperm: 33 d	11820
	m: 1st sire: 7 wk; n1	11820
Pitymys ochrogaster	f: 1st conceive: ~1 mo; mode 7-11 wk	3283
	f: 1st mate: 30 d	1474
	f: 1st mate: 33-34 d not fertile; >60 d fertile	9111
	m: 1st sire: 49 d	9111
	f/m: ≤30 d	1871a
Pitymys pinetorum	f: 1st cl: 10-12 wk	9536
	f: 1st conceive: 62 d; median 50; r32-115; n13	6388, 6389
	f: 1st conceive: 11-24 wk; n21	9530
	f: 1st conceive: 105 d	9536
	f: 1st preg: TL: 108 mm; n2	8268
	m: 1st sperm 6-8 wk or 60 d	9536
	m: 1st sire: 51-77 d; n12	9530
	m: 1st sire: 69 d; median 57; r44-99; n12	6388, 6389
Pitymys subterraneus	f/m: by 100 d	1407
	f: 16.3 g	8335
	f: 1st cl: 13.5 g	6004
	f: 1st conceive: 90 d	9656
	f: 1st preg: 13-14 g	10379
	m: 18.6 g	8335
	m: fertile: 13-17 g	10379
	m: testes large: yearlings	6005
Pitymys tatricus	f: 1st cl: 23 g; sometimes 18 g	6004
Prometheomys schaposchnikowi	f: 45-49 g; yr of birth	12007
CYCLE LENGTH		
Clethrionomys glareolus	anestrus: 0.5 d; r0-5; proestrus: 1.5 d; r0-5	10204
	estrus: 1.8 d; 0-7; postestrus 2.7 d; 0-7	10204
	mode 3 d; most 2-8; n300 cycles, 30 f	6824
	4 d; smears	1408
	4-8 d	8425
	6.5 d; r3-12; n29 f	10204
Dicrostonyx groenlandicus	9.66 d; sd2.2; n41 cycles, 18 f	9521
Lagurus lagurus	7 d; vaginal smears	11647
Microtus agrestis	3-4 d; during lactation	1806
	5.18 d; r2.5-11.5; n28 cycles, 7 f; diestrous intervals	7334
Microtus arvalis	mode 4 d; most 2-9; n70 cycles, 10 f	6824
Microtus guentheri	6-10 d	3610
Microtus oeconomus	8.5 d; r6-11; n6 cycles	5039
Microtus townsendii	estrus: 4.5 d; r1-19; diestrus: 3.2 d; r1-18; n12 f	6692
	8 d; mode 4; most 2-8 70%; 5 cycles >14; r2-28; n40 cycles, 12 f	6692
Neofiber alleni	≤15 d	7983
Ondatra zibethicus	estrus: 1 d	719
	6.1 d; mode 4; r2-22; n136 cycles, 10 f	7094
	28.7 d; r24-34; n11 cycles, 7 f	719, 720
Pitymys pinetorum	estrus: 1-22 d; diestrus: 1-9 d; n10 f, no cycles	9530
GESTATION		
Arborimus longicaudus	r27.0-28.5 d; n4 f	4264
	33-42 d; n2 f	4264
Arvicola terrestris	2.5-3 wk	8052
	21 d	7772
	21 d	7433
	mode 21 d; r21-26	981
	21-22 d	3571
	22-24	4660
Clethrionomys gapperi	mode 17 d; r17-19; n6 cases, 5 f	10569
Clethrionomys glareolus	16-35 d; lactation delay possible	8425
	17.5 d	7002
	17.5-18 d	8052
	18 d	10589
	18 d	5477
	18 d; r18-23; n25; primiparous	4173

	18.3 d; se0.07; n12; primiparous	4172
	18.32 d; r17.5-21.5; n7	11362
	19 d	8179
	19-20 d; 3 d longer with lactation	2077
	19.5 d; se0.01; n74	1875, 1881
	est 20 d; min ibi	1408
	mode 20-21 d; r19-34; pairing to birth	8021
	20 d	10383
	20 d; n1	11679
	21 d	7433
	21-21.5 d; with lactation	7002
	25 d; r20-30	4645
	max 37 d; with lactation	11362
Dicrostonyx groenlandicus	implant: 5 d	4468
	19-21 d	6781
	19.5-20.5 d	8958
	20 d	4335
	20.0 d; se0.1; r19.5-21.5; n38 no lactation	4468, 4469
	20.7 d; se0.2; r19.5-23.0; n25 with lactation	4468, 4469
	21.0 d; se0.6; r20-22; n3 no lactation	6759
	22.4 d; se0.7; r20-27; n9 f with lactation	6759
	28.0 d; se1.0; r24-31; n8 f with lactation	6759
Dicrostonyx hudsonius	19-21 d	1474
Dicrostonyx torquatus	15-21 d	8810
	18 d; r18-19	8051
Dicrostonyx unalascensis	mode 21 d	2560
Dinaromys bogdanovi	1 mo	7983
Ellobius talpinus	26 d	12111
Eothenomys smithi	19 d; min ibi	294b
Lagurus curtatus	25 d; r24-26	5240
Lagurus lagurus	implant: 6-7 d	142
	0.5 mo	8052
	19-20 d; min ibi	11647
	20 d; r15-23	4645
	20+ d; mode 20; r20-21	3516
	20-21 d; vaginal smears	3652
	20-22 d	8634
Lemmus lemmus	~16 d	8051
	20.2 d; r19.54-21.29; n67; pairing to birth, min ibi	6806
	20 d; r20-21; mating to birth, min ibi	3462
	3 wk	2386
	mode 3 wk	9762
Lemmus schisticolor	14 d	7705
	est 22-25 d	5135
	24 d	3466
Lemmus sibiricus	18 d	8051
	20-20.5 d; n12	7595
	21-21.5 d; n2	6013
	23 d	716
Microtus agrestis	implant: 3-5 d	1249
	18-21 d	3571
	18-22 d	1875
	20-23 d	1250
	20.5 d; nursing	12070
	20.7 d; n32	12070
	20.76 d; r20.5-21.5; n13	1806
	21 d	7433
	21 d	6397, 8920
Microtus arvalis	18 d	7292
	18 d	10911
	19 d; r16-23	4645
	mode 19-21 d; r16-24	3571
	20 d	2466
	20-28 d	6339

	20 d; r16-36; ibi	3459, 3460
	20.56 d; se0.08; mode 20; r18-24; n273	9013
	21 d	11225
	21 d	2465
	21 d; r18-32	3459, 3460
	21-23 d; r20-31; n21; pairing to birth	9269
	22.8 d; r20-27; ibi	3465
	< 22 d	2462
Microtus breweri	est 21 d	10648
Microtus californicus	21 d	4052
	~21 d	4490
	21.1 d; r20.5-22; n4	9747
Microtus canicaudus	≤23 d	4196a
Microtus guentheri	21 d	8115
	21 d; r21-25; min ibi	1965, 1966
Microtus miurus	21 d	12049
Microtus montanus	20.5-21 d	9967
	21 d; n314 lt, 43 f	7811
	mode 21 d; r21-26; ibi	4014
Microtus montebelli	29 d; ibi	3969
Microtus nivalis	20 d; estrus to birth; 22 d; ibi	3457
	21 d	7433
	21 d	5486
Microtus oeconomus	20-23 d; mating to birth, ibi	3468
	21 d	7433
Microtus oregoni	23.8 d; r23.33-24.5; n4 lt	2087
Microtus pennsylvanicus	20.0 d; sd0.0; n6 without lactation	9898
	21 d	2562
	21 d	10648
	21 d	501
	21 d	4278
	21 d; r19-22	6347, 6348
	21.0 d; se0.2; n12	5673
	21.5 d; sd1.4; n11 with lactation	9898
Microtus richardsoni	22 d; n2; pairing to birth	5252
	22 d; n3 min ibi	5252
	42 d; with lactation	1369
Microtus socialis	19-20 d	4645
Microtus townsendii	~21 d	1474
	21-24 d; n3	6693
	24 d; n1 nursing 5 babies	6693
Neofiber alleni	26 d; min ibi; 29 d; death of m to birth; n2	944
	1 mo; n1; removal of m to birth	4282
Ondatra zibethicus	21 d	1355
	21-23 d	8051
	25 d	6267
	28 d	11934
	4 wk	11174
	29 d; r28-30; n2	3061
	30 d	7291
	est ~30 d	11934
	~30 d; ibi	7093
	~1 mo	2243
	~1 mo; n76	3080, 3082
Phenacomys intermedius	19 d; capture to birth	5150
	21-24, 19-24 d; n2	3429
Pitymys lusitanicus	20 d; n2; min ibi	11820
Pitymys ochrogaster	~21 d	1474
	~21 d	3648
	21.5 d; r20.5-23; n18	9110, 9111
	22.8 d; se0.2; n13	5673
	24.8 d; se0.3; n22; pairing to birth	10333
Pitymys pinetorum	~21 d	1474
	24 d; mode 24; r24-25; n8	5809

	24 d; n several hundred	6822, 9530, 9531
Pitymys subterraneus	20 d; n2; min ibi	7904
	21 d	9656
	21 d; mode 21; r20-21; n4	6199
	22.9 d; se0.3; r21-25; n28 primiparous	5282
Synaptomys cooperi	23 d; nursing	2026
LACTATION		
Arborimus longicaudus	den emergence: 29 d	5026
	solid food: 30-40 d	4264
	wean: 21 d	4264
Arvicola terrestris	solid food: 12 d	4660
	wean: 11 d	4660
	wean: ~14 d	2045
Clethrionomys gapperi	solid food: 14 d; 8.1 g	7238
	solid food: at eye opening	10569
	wean: 12-17 d	7238
	wean: 14.7 d	5148
	lact: 17-21 d	10569
	independent: 1 mo	1871a
Clethrionomys glareolus	solid food: >14 d	3571
	solid food: 2 wk	8052
	wean: ~3 wk	3571
	leave nest: 3 wk	8425
	wean: 18 d	5477
	wean: 20-21 d	8052
Dicrostonyx groenlandicus	solid food: 14 d	4335
	wean: 14 d	6781
	wean: 16-18 d	4335
Dicrostonyx torquatus	wean: 1 mo	8051
Ellobius talpinus	wean: 60 d	4645
Eothenomys smithi	solid food: 15 d	294a
	nurse: 25-30 d	294a
Lagurus curtatus	leave nest: 12 d; r10-15; n7 lt	4905
	solid food: 14 d; r12-16	4905
	may nurse: 25+ d	4905
	wean: 21 d	5240
Lemmus lemmus	solid food: 10 d	6806
	solid food: 11 d	8051
	den emergence: 13-30 d	8051
	independent: 15 d	2186a
	den emergence: 2-3 wk	11679
Lemmus schisticolor	solid food: 11 d	5135
	leave nest: 12-14 d	5135
	lact: ~20 d	5135
Microtus agrestis	wean: 12-14 d	8920
	lact: 14 d; 13-16	8921
	independent: 14 d	1805
Microtus arvalis	solid food: 14 d	3571
	solid food: 14 d	8052
	independent: 14 d	9269
	wean: 15-20 d	2465
	wean: 16 d	7292
	wean: 16 d	10911
	wean: 17-20 d	3460
Microtus brandti	den emergence: 2-3 wk	9012
Microtus californicus	solid food: ~10 d	9747
	solid food: 2 wk	4490
	nurse: 14 d	4052, 4053
	wean: 16 d	6041
	nurse: min 23 d	9747
	nurse: 3 wk	4490
Microtus canicaudus	wean: ≤18 d	4196a
Microtus gregalis	solid food: 10 d	4645
	solid food: 10 d	8052

Microtus guentheri	solid food: 10 d	1965
	wean: 2 wk	3610
	wean: 15-21 d	1965, 1966
	wean: 18-20 d	12003
Microtus montanus	solid food: 11 d	9967
	wean: 15 d	9967
	abandon young: 13-16 d	5250
	independent: 2-3 wk	1871a
Microtus montebelli	wean: 20 d	7762
Microtus nivalis	wean: 3 wk	3571
Microtus oeconomus	leave nest: 11-12 d	10681
	solid food: 2 wk	3571
	wean: 20 d	3468
Microtus oregoni	wean: ~15 d	4970
Microtus pennsylvanicus	solid food: 9-12 d	4278
	solid food: 13-14 d	7066
	wean: 11-12 d	4272, 4275, 4278
	wean: 12 d	501
	nurse: 12-14 d	7066
	wean: 14 d	6347
	wean: 14.4 d	5148
	wean: 18-21 d	10463
Microtus socialis	solid food: 20 d	4645
	solid food: 20 d	8052
	den emergence: 15 d	4645
Microtus townsendii	wean: 15 d; n15 lt	6693
Microtus xanthognathus	wean: ~1 mo	12049
Neofiber alleni	wean: 3 wk	944
Ondatra zibethicus	den emergence: 3 wk	10364
	den emergence: 3 wk	3571
	wean: ~24 d	2695
	wean: 4th wk	3081
	independent: 4-6 wk	11934
	wean: 30 d	8051
	independent: 2 mo	1871a
Phenacomys intermedius	solid food: 14 d	3429
	solid food: <18 d	5150
	nurse: 19 d	3429
	independent: ~2 mo	1871a
Pitymys lusitanicus	independent: 20 d	11820
Pitymys ochrogaster	solid food: 10-14 d	9111
	solid food: 12-16 d	7066
	solid food: 17 d	3283
	nurse: 18-21 d	7066
	wean: 21 d	9111
Pitymys pinetorum	solid food: 15-16 d	6533
	solid food: 15-18 d	7066
	wean: 17 d	4273
	wean: 21 d	6533
	wean: 3rd wk	783
	nurse: 20-21 d	7066
Pitymys subterraneus	wean: ~20 d	9656
	wean: 27 d	6199
Prometheomys schaposchnikowi	wean: 3 wk	12007
Synaptomys cooperi	wean: ~3 wk	2026
INTERLITTER INTERVAL		
Arvicola terrestris	3-12 wk	10944
	mode 21-22 d; r21-25	11668
	22 d	4660
Clethrionomys gapperi	17 d min	7238
Clethrionomys glareolus	mode 19 d; n large; no lactation	4172, 4173
	20.16 d; most 16-35; r20-100; n212	1408
	20-21 d	10383
	21 d	157

	mode 21; r17-24, 78%; r37-42, 7.5%; n1164 lt	1875, 1881
	mode 22-23 d; n large; with lactation	4172, 4173
	max 5.5 wk; with lactation	11362
	45-55 d	8052
	50-60 d	4645
Clethrionomys rufocanus	20.2 d; sd1.4; n12 intervals	5593
	40-50 d	6134
Clethrionomys rutilus	17.5 d; r14-22; n36 intervals	7544
	min 20.5 d	2564
	r28-25 d; n12 intervals	7544
Dicrostonyx groenlandicus	19-37 d	6781
	mode 21 d; r19-45	4469
Dicrostonyx torquatus	min 23 d	8051
Dicrostonyx unalascensis	20 d; r16-27	7544
Ellobius talpinus	34-36 d	12111
Eothenomys smithi	mode 22 d	294a
Lagurus lagurus	76% 19-23 d; r18-25	3652
	19-91 d	11647
	25-30 d	4645
Lemmus lemmus	20 d; r18-22	7544
	21 d	6806
	21 d	3462
	40 d	8051
Lemmus schisticolor	mode 25 d; 60% 23-28; r22-40; n170 intervals	5135
Lemmus sibiricus	21 d; r18-27	7544
Microtus abbreviatus	18-26 d	7544
Microtus agrestis	20.26 d; r19-24; n42	1806
Microtus arvalis	20 d; r16-36	3459
	mode 20 d; r16-36; n417	9013
	18-132 d	9269
	22.8 d; r19-49	3460, 3465
	25-30 d	4645
Microtus fortis	40-45 d	8052
Microtus guentheri	25 d	1966
	33.2 d	3610
Microtus miurus	21 d; r18-30	7544
Microtus montanus	min 21-26 d; n10	4014
	21 d; n314 lt, 43 f	7811
Microtus montebelli	min 18 d; mode 20-21	7762
Microtus nivalis	33; r22-44; n2	3457
Microtus oeconomus	20 d; r16-25	2563, 7544
	~3 wk	10681
	25.3 d; n17 lt, 1 f	3468
	7 wk	659
Microtus oregoni	23.5-54 d	2087
Microtus pennsylvanicus	mode 20 d; 70% 18-24; r18-48; n71 lt, 19 f	10463
	21 d; r19-23	7544
	95% ~21 d	7116
	21 d	501
	21 d	4278
	21-22 d	6347, 6348
Microtus socialis	30-35 d	4645
	30-35 d	8052
Ondatra zibethicus	27.9 d; r18-35; n10 f	8104
	30 d	7093
	33 d; r29-35; n11 lt, 5f	2695
Pitymys juldaschi	25.6 d; r20-41	1057
	27.6 d; r27-66	1057
Pitymys lusitanicus	23.4 d; mode 24; r20-27; n19 intervals	11820
Pitymys ochrogaster	mode 21 d; r19-63; n431 intervals	9110, 9111
	r23-27 d; n5 intervals, 5 f	3283
	25-50 d	1871a
	~27 d	9273

Pitymys pinetorum	24-25 d	9530
	27.6 d; se1.8	2493
Pitymys subterraneus	66.1%; 18-25 d; 13.5%; 26-36 d; 20.4%; 37-88 d	5282
	22-29 d	1407
	23.8 d; mode 24; r20-26; n78	7904
	24.58 d	9656
LITTERS/YEAR		
Alticola roylei	10	8646
Alticola strelzowi	1-2	3088
	2-3	8646
	3 summer	8052
Arvicola terrestris	1.5-2.3 lt	10447
	2-4/season; south, more north	8052
	3-4	7433
	3-4; r2-4	3571
	4; mode 2 54%; r1-4	10944
	4-5	11668
	13-14	4660
Clethrionomys centralis	10	8646
Clethrionomys gapperi	>1	11230
	2	2088
	2-3	10839
	2.2-3.3	5151
	2.4; r1-6	6632
Clethrionomys glareolus	mode 1; r1-3	1844
	mode 2; r1-3	12077
	2; r1-3	7271
	2-4	4645
	3-4	3571
	3-4	7433
	3-4	8052
	max 4	11362
	4-5; wild; 4-7; lab	10589
	est 4-5	1222
Clethrionomys rufocanus	1-3	5140
	2-3	8646
	3-4	5501
	max 7	6134
Clethrionomys rutilus	3.8; ±3.3	2564
Dicrostonyx groenlandicus	0-17	4469
Dicrostonyx hudsonius	1-2	1474
Dicrostonyx torquatus	2; occasionally 3 in summer	8051
Ellobius talpinus	6-7	12111
	4	8052
	2; adult f	4645
Eolagurus luteus	min 3-4	9951
Hyperacrius fertilis	2-3	9175
Lagurus lagurus	mode 4-5; r4-7	4645
	6	8052
	10-12	8634
	11/8 mo; n1 f	3516
Lemmus lemmus	2/winter; r1-3	5967
	3; favorable yr	8051
	4?	6806
	10-12	3462
Lemmus schisticolor	2	8051
	max 2	10016
Lemmus sibiricus	min 2	1474
	3-4	8051
	min 12	2088
Microtus agrestis	2-5	1872
	3-4	3571
	3-4	7433
	4	12070

	4	8052
Microtus arvalis	1-11	9269
	2; r1-5	9013
	3-10; r3-12	3571
	5-7	8052
	16	11225
Microtus brandti	min 2-3	8052
	3; r1-5; n5	9012
Microtus californicus	4.4-5.6	4053
	9.30-11.22; over 3 yr	4893
Microtus chrotorrhinus	min 2	6632
Microtus fortis	6; good yr	8052
Microtus gregalis	2	8052
	3-4	10516
Microtus gud	2-3	8052
Microtus guentheri	mode 1-3; r1-5	1965
	6-7; 7 is theoretical	1962, 1966
	10	3610
Microtus longicaudus	1.9; r1-4	6632
Microtus montanus	4.38-5.15; over 3 yr	4893
Microtus nivalis	1-2; r1-3	3571
	2	7433
	2	6313
	2-3	6919
Microtus oeconomus	2.9	2563
	min 3	8052
	3-4	10681
	3-4	4645
	3-4	7433
	3-4	3571
	3.8	2563
Microtus oregoni	4.8	3622
	max 5	2088
Microtus pennsylvanicus	1.6-2.3	5151
	2-3	10839
	4.0; ±3.9	2562
	max 5 lt/season	7116
	5-10	4278
	est 5-6; theoretical max 17	4278
	max 17	501
Microtus richardsoni	2	1369
	max 2/season	6632
Microtus sachalinensis	0-2	11366
Microtus socialis	3-5	4645
	4-8	8052
Microtus xanthognathus	2 lt/season	11878
Neofiber alleni	4-5	944
Ondatra zibethicus	>1	9003
	2	3558
	1-2	11934
	1-2; north; 3; south	6266
	mode 1; r1-3	2243
	1-4	8104
	mode 5-6; r1-8	8015
	1; r1-2; 2 rare, 4/133 f	7093
	2	719
	2	10410
	2	6267
	mode 2	8762
	2	10410
	2; 3 rare	11467
	2	7015
	2	8762
	mode 2	3080

	est 2-3 lt/season	12112
	2-3	1474
	2-3	720
	2-3	90
	2-3; north; 3-4 south	4645
	2.0; mode 2; r1-4; n76 f	3082
	2.3	9529
	2.36	8237
	2.7; r1-4; 2.5; 2 yr	9503
	2.8	10112
	3	11027
	3-4; r3-5	3571
	3-4	9164
	3-5	6214
	3.28	2697
	4-5	7433
	mode 4	10421
	mode 4; max 6	7291
	3-4	8051
Phenacomys intermedius	min 2	1474
	max 3	5150
Pitymys ochrogaster	3-4/season	1474
	min 5 lt/summer	2137
Pitymys pinetorum	1-4	3852
	3-4	1474
Pitymys subterraneus	min 2	8052
	2-3; r2-6	3571
	5-6	7433
	8.7/9 mo	9656
	max 10	7904
Prometheomys schaposchnikowi	2	12007
	2?	8051
Synaptomys cooperi	2-3	1474
SEASONALITY		
Alticola strelzowi	repro: Mar-Oct, peak Apr-June; Kazakhstan	8020a
	repro: end Apr-mid July; USSR	3088
Arborimus longicaudus	mate: yr rd; CA	782
	mate: yr rd; n Canada, nw US	5026
	birth: yr rd; OR, CA	1474
Arvicola sapidus	preg f: Feb-Oct; Spain	11248
Arvicola terrestris	fertile m: Feb-Nov; n Austria	11668
	testes: large Feb-Aug; Netherlands	11204
	testes: large Mar-May, regressed Dec; s France	2471
	mate: Mar-Apr; GB	2045
	mate: end Mar-Sept; lab	8392
	mate: end Mar-early Oct; Germany	3571
	mate: Apr-Sept; USSR	10944
	mate: Apr-Oct; Europe	7433
	mate: peak July-Aug; lab	7772
	preg f: Jan-Mar; Switzerland	7261
	preg f: Mar-Sept; s France	2470
	preg f: Mar-Nov; Spain	11250, 11251
	preg f: Apr-Oct; Czechoslovakia	6007
	preg f: Apr-mid Oct; n Austria	11668
	birth: Feb-Oct; n Austria	11668
	birth: Mar-Oct; Netherlands	11204
	birth: begins Apr-May; USSR	8052
	birth: Apr-Sept; s France	2470
	birth: Apr-Oct; Europe	7433
	birth: Apr-Oct; Germany	3571
	birth: May-Aug; Scotland, UK	10447
	lact f: Sept; n1; Iran	6293
Clethrionomys californicus	preg f: Aug; n1; OR	6708a

Clethrionomys centralis	testes large: June-mid Aug; USSR	8052
	birth: June-mid Aug; USSR	8052
	birth: yr rd; lab	634
Clethrionomys gapperi	mate: end Mar-Nov; PA	9164
	mate: May-Sept; MI	6796
	mate: May-Nov; CO	11230
	preg f: end Apr-early Sept; s QUE, Canada	11956
	preg f: May, July; n3; NS, Canada	10144
	preg f: mid June-early Sept; MAN, Canada	10101
	preg f: July; n4; ND	3703
	preg f: Sept; n1; NM	502
	birth: Mar-Oct; boreal N America	1474
	birth: Apr-Aug; MAN, Canada	8383
	birth: June-Sept; BC, Canada	2088
	repro: Mar-Oct; WI	5210
Clethrionomys glareolus	fertile: yr rd, peak summer-fall; lab	8021
	sperm: Feb-Oct, peaks Apr-May, low Aug; England, UK	9342
	spermatogenesis: Mar-Oct?; Czechoslovakia	5041
	spermatogenesis: peak June, low Jan; UK	516
	spermatogenesis: peak July-Aug; UK	2077
	testes: large mid May-mid Sept; Norway	11679
	testes: large June-Aug, small Aug-Sept; AK	716
	epididymal sperm: Apr-Aug; Wales, UK	5307
	ovulation: early Apr; UK	2077
	ovulation: Apr-Aug; s Sweden	7988
	mate: Jan-mid Apr; lab	6052
	mate: Mar-Sept; Alps, Europe	3571
	mate: Mar-Oct; Poland	6053
	mate: Apr-Aug; USSR	12095
	mate: Apr-Sept; Europe, UK	7433
	mate: Apr-Sept, some Dec; France	943
	mate: Apr-Sept, some winter; Czechoslovakia	12076
	mate: Apr-Oct, peak May, some winter; England, UK	7873
	mate: mid Apr-early Oct; Germany	7271
	mate: May-Aug; USSR	8052
	mate: May-summer, peak June; Channel I, UK	955
	mate: peak May, low Oct-Jan; Czechoslovakia	12078
	mate: June-Aug; AK	716
	mate: June-Aug; Smolensk, Russia	8052
	mate: ends Aug-Sept; n Sweden; 2 mo earlier s Sweden	7988
	mate: Dec; England, UK	10175
	mate: Dec-Mar; s Sweden	4342
	mate: Dec-Mar; s Sweden	7251
	mate: low winter; s Sweden	3073
	preg or vag perf: Mar-Oct; England, UK	3342
	preg f: Feb-July, peak May; Czechoslovakia	12077
	preg f: Feb-Oct; Germany	6548
	preg f: Mar-Nov, peak July; Pyrenees, Spain	11255
	preg f: Apr-?; Germany	10340
	preg f: Apr-June; Germany	10344
	preg f: Apr-Sept, peak 76.9% Aug; Germany	7271
	preg f: Apr-mid Sept; Germany	10339
	preg f: Apr-Oct, peak June; England, UK	1222
	preg f: peak Apr-Sept, low Nov-Mar; Czechoslovakia	12079
	preg f: May-Aug; Wales, UK	5307
	preg f: May-Oct; s Sweden	7988
	preg f: mid May-Sept; Czechoslovakia	5103
	preg f: mid May-Sept; Norway	11679
	preg f: peak June-Aug; UK	2077
	preg f: 6% Oct; USSR	12095
	preg/lact f: May-Sept; Switzerland	1844
	birth: Mar-Sept, peak June-July; lab	11362
	birth: Mar-Oct, peak July-Sept; France	2470
	birth: Apr-Oct; Alps, Europe	3571

	birth: Apr-Oct; Europe, UK	7433
	birth: begins May; ne Turkey	10274
	birth: May-Aug; Poland	6055
	birth: June-Sept; Norway	11679
	birth: Nov, Jan; Germany	6091
	birth: peak May-Aug; lab	10383
	repro: end Mar-Sept; s Sweden	810
	repro: Apr-Sept; Denmark	5302
	repro: Apr-Sept; Germany	10339
	repro: yr rd, low winter; Germany	6091
	repro: yr rd, low Jan-May; se France	6370
Clethrionomys rufocanus	testes: large Apr-Aug; USSR	8052
	mate: Apr-early Oct; Mongolia	10516
	mate: mid May-mid July; n Norway	5140
	mate: June-Aug; n Sweden	11603
	preg f: ?-Aug; few data; Manchuria	12105
	preg f: May-Sept; Norway	11679
	preg f: May-Sept; Finland	5501
	preg f: mid May; USSR	8052
	birth: May-early June; Finland	5493
	birth: synchronized early June; Japan	5593
	lact f: ?-Sept; few data; Manchuria	12105
Clethrionomys rutilus	spermatogenesis: peak mid May; USSR	8052
	mate: ?-Aug; n Norway	5039
	mate: Apr-Aug; Altai, Mongolia	8052
	mate: mid June-?; w Siberia	8052
	mate: May-Sept; sw YUK, Canada	3774
	mate: June-?; n Canada	6782
	mate: Jan; n3; n Norway	6645
	preg f: ?-mid Aug; n Canada	6782
	preg f: low Mar; Finland	5493
	preg f: July; Manchuria	12105
	birth: May-early June; Finland	5493
	birth: May-Sept; nw Canada	1474
	birth: June-Aug; USSR	8052
	repro: f: May-Aug; m: end Apr-mid Sept; YUK, Canada	6015
Dicrostonyx groenlandicus	mate: June-July; NWT, Canada	6013
	preg f: June-July; MAN, Canada	10101
	preg f: end June-early Aug; NWT, Canada	6710
	repro f: July-Sept; NWT, Canada	9211
Dicrostonyx rubricatus	mate: ?-mid Aug; Arctic	177
	birth: end June-early July; Arctic	177
Dicrostonyx torquatus	mate: Mar-July, Oct-Nov; Russia	1779
	preg f: peak end June; USSR	8051
	birth: end winter-early July; USSR	8051
Dinaromys bogdanovi	repro: Mar, June; Yugoslavia, nw Greece	7983
Ellobius lutescens	spermatogenesis: yr rd; captive	10323
Ellobius talpinus	mate: yr rd; USSR	12111
	birth: Apr-Oct; USSR	8052
	repro: Apr-Oct; Jarantaj, Mongolia	10516
Eolagurus luteus	repro: summer; USSR; yr rd; lab	9951
Eothenomys andersoni	yg appear: July-Oct; Japan	7393
Eothenomys melanogaster	mate: most of yr; China	182
	conceive: Oct; China	182
Eothenomys smithi	preg f: yr rd, peaks end spring, Oct-Nov, low July-Aug; Japan	10665
	repro: m: Nov-Mar, f: Nov-Apr; Japan	12026
Hyperacrius fertilis	birth: spring, end summer; Pakistan	9175
Lagurus curtatus	sterile: July-Nov; WA	5240
	mate: yr rd; OR	6925
	mate: yr rd; OR	7491
	mate: yr rd, peak summer; ID	7606
	mate: yr rd; w US	1474
	birth: yr rd, low summer; WY	1871a
	lact f: June; n1; ND	3703

	repro: est yr rd; CA	2807
Lagurus lagurus	mate: Apr-Oct; Balkash, Kazakhstan	9950
	mate: Apr-Oct; USSR	8052
	birth: end Apr-early Oct; USSR	8052
	birth: peak warm season, low winter; USSR	4645
	repro: yr rd, peak fall; USSR	9903
	repro: yr rd; lab	3652
Lemmus lemmus	testes: large May-Oct; Norway	11679
	testes: large July; Lapland	811
	testes: small Oct; Jantland, n Sweden	811
	vaginal plugs: mid Aug-?; Norway	5301
	mate: May-Aug; Sweden	6806
	mate: spring-fall; USSR	5967
	preg f: May-Sept; Norway	11679
	preg f: July; Lapland, USSR	811
	birth: May-early June; Finland	5493
	birth: June-Aug; Lapland, USSR	8051
	birth: July; n1; Telemarkin, Norway	10210
	repro: est yr rd; lab	2386
Lemmus schisticolor	preg f: May-Sept, peak July; Finland	10016
	preg f: July-Sept, maybe earlier; USSR	8051
	birth: June-Aug; USSR	8051
	juveniles: winter; Norway	7706
	repro: Mar-Dec; lab	3466
Lemmus sibiricus	mate: June-Aug; NWT, Canada	6013
	preg f: Mar-Apr, July 57%, Dec-Feb; Yamaf Peninsula, n Siberia	653
	preg f: end June-early Aug; NWT, Canada	6710
	preg f: peak July; AK	716
	preg f: peak June-Aug; AK	7595
	repro: May-Sept; USSR	9071
	birth: peak mid June-July; USSR	8051
	repro f: mid July-Aug; n11; NWT, Canada	9211
Microtus agrestis	anestrus f: winter; UK	1218
	testes: peak Apr-Oct; England, UK	11932
	testes large: June-mid Oct; Czechoslovakia	6008
	mate: Jan-Mar; Sweden	4342
	mate: Feb-?; UK	521
	mate: Mar-Oct; Germany	3571
	mate: mid-Mar-Oct; UK	1218
	mate: end June-?; s Sweden	11603
	mate: spring-fall; Europe, UK	7433
	mate: winter if acorn crop good; England, UK	10175
	preg f: ?-Dec; UK	521
	preg f: Mar-Dec; France	2470
	preg f: Mar-Oct; Finland	8645
	preg f: Apr; England, UK	1804
	preg f: May-Sept; Brandenburg, Germany	10340
	preg f: end May-Aug; s Ural, USSR	8052
	preg f: June-Oct; Czechoslovakia	6008
	preg f: Nov-Dec; Berlin, Germany	6091
	preg/lact f: June, none winter; England, UK	1878
	birth: Mar-Sept; captive	519
	birth: Mar-Nov; Germany	3571
	birth: Mar-Dec; France	2470
	birth: May-?; USSR	8052
	birth: May-early June; Finalnd	5493
	birth: Apr-Dec; captive	1872
	birth: June; England, UK	1878
	birth: spring-fall; Europe, UK	7433
	lact f: mid Mar; Sweden	6223
Microtus arvalis	scrotal testes: Jan; Belgium	819
	testes: large Mar-Oct; Poland	6337
	testes: large Apr-July, small winter; France	2460

	testes: regress Oct; Poland	5091
	testes: regress Oct; Poland	6336
	mate: mid Jan-Oct; France	6902
	mate: Feb-Sept; Channel I, UK	955
	mate: Feb-Oct, low winter; w Europe	3571
	mate: Apr-Sept; Czechoslovakia	741
	mate: Apr-Oct; USSR	2650
	preg f: Jan; Belgium	819
	preg f: Jan-Feb; Germany	11593
	preg f: Jan-May; Germany	10344
	preg f: Feb-Oct; Mecklenburg, Germany	9013
	preg f: Feb-Oct; Poland	10343
	preg f: Mar, June, Aug, Oct; n8; Channel I, UK	955
	preg f: Mar-Sept; France	2466
	preg f: Mar-Sept; France	2465
	birth: Feb-Oct; Poland	10343
	birth: Mar-Sept, peak July-Sept; France	2470
	birth: Mar-early Oct; Germany	10343
	birth: peak Mar-Oct; w Europe	3571
	birth: May-Oct; captive	9013
	birth: yr rd; captive, Russia	8153
	lact f: Oct; n2; Channel I, UK	955
	lact f: Nov; Belgium	819
	repro: mid Jan-Nov; France	6902
	repro: low Dec; Germany	3463
	repro: yr rd, low Mar-May; Alps, France	6370
Microtus brandti	birth: mid Mar-?; lab	9012
	birth: May, Aug/Sept; n2; Mongolia	182
	birth: mid May-?; USSR	8052
	repro: mid Feb-Aug; lab	9012
Microtus breweri	testes: quiescent Oct-Dec; MA	64
	preg f: Apr 90%-Oct 30%; MA	9351, 10648
	lact f: Mar-Oct; MA	10647
Microtus californicus	mate: yr rd, peak spring; CA	4052, 4053
	mate: yr rd; CA, OR	1474
	preg f: yr rd, peaks spring, fall; CA	4893
	preg f: low July-Oct, Jan; CA	4053
	birth: yr rd, peak Jan-May; CA	6040, 6041
Microtus chrotorrhinus	mate: end Mar-mid Oct; Canada	6896
	preg f: July n6/12, Oct n1/13; no data other mo; WV	5794
Microtus fortis	mate: Apr-Oct; USSR	8052
	conceive: Mar; China	182
	repro: end Mar-mid Nov; Korea	5424_1
Microtus gregalis	mate: summer; USSR	8052
	birth: May-Sept; Mongolia	10516
	birth: warm season; Europe, USSR	4645
	birth: est yr rd as yg found Nov-Dec; Sibeńa, Russia	7919
Microtus gud	birth: July; USSR	8052
Microtus guentheri	mate: Feb-Oct; lab	1965
	preg f: Dec-Feb; n21; no data other mo; Greece	8115
	birth: peak Mar; lab	1965
	birth: yr rd; Lebanon	6419
	birth: yr rd; captive	1036
	birth: yr rd; lab	3610
	repro: Oct-Apr; captive, Israel	1966
	repro: Nov-Apr; Israel	1962
Microtus longicaudus	testes: large May-June; CO	10255
	mate: Apr-Sept; BC, Canada	2088
	preg f: May, Sept; n2; ID	11646a
	preg f: end May-June; no data other mo; NV	6488
	preg f: June-July; no data other mo; SD	10954
	preg f: June-Aug; n9; YUK, Canada	12049
	preg f: Aug; n10; AZ	4897
	birth: May-Sept; CA	1474

Microtus mexicanus	mate: yr rd?; NM	502
	preg f: Jan, July-Aug; Coahuila, Mexico	522
	preg f: July; n3; Durango, Mexico	523a
	preg f: Aug, Oct; NM	502
	preg f: yr rd; Mexico	9176
	preg f: est yr rd; Mexico	1813
Microtus middendorffi	birth: early Apr-mid Oct; USSR	8052
Microtus miurus	mate: June-Aug; AK, nw Canada	1474
Microtus montanus	mate: Mar-mid Nov, none mid Nov-early Mar; WA	9724
	mate: Apr-Aug; no winter data; CO	11230
	preg f: Jan, Mar, Apr; n8; ID	11646a
	preg f: Feb-Sept; UT	7808
	preg f: Apr; n1; UT	10305
	preg f: Apr-Oct; CA	4893
	preg f: yr rd, low winter; WY	5251
	birth: yr rd; WY	1871a
Microtus montebelli	testes: regress Sept; Japan	7392
	mate: Apr-Dec, peaks early summer, end fall; Japan	15
	mate: May-June, Sept-Oct; low winter; Japan	5510
	mate: yr rd, peaks spring, fall; Japan	9443
	preg f: Apr 70%, Aug 60%; Japan	9443
	preg f: May-Nov, peak June; Japan	7392
	preg f: none July; Japan	5510
Microtus nivalis	testes: large June-Oct, small Nov-Jan, no data Jan-June; Alps, Germany	5486
	estrus: Sept; Germany	6128
	estrus: Apr; captive	6128
	mate: Apr; n1; captive	3457
	mate: May-Aug; France	6313
	mate: June-Aug; USSR	7433
	mate: June-early Aug; Germany	3571
	preg f: July-Sept; no data Jan-June; Bavarian Alps, Germany	5486
	preg f: Aug; n1; Iran	6293
	birth: Apr; n1; captive	3457
	birth: May; captive	6128
	birth: June-July; Germany	3571
	birth: June-Aug; USSR	7433
	repro: mid Mar-early Sept; USSR	6919
Microtus oaxacensis	preg f: Apr-June, Nov; n11; Oaxaca, Mexico	9176
Microtus oeconomus	testes: large May-Sept; Norway	11679
	testes: regress Aug; n1; n Norway	5039
	testes: small Sept; Germany	10340
	mate: ?-Sept; Mongolia	10516
	mate: Apr-Sept; Scandinavia, Germany	7433
	mate: mid Apr-mid Sept, peak May-June; USSR	8052
	mate: prob mid Apr-Oct; Finland	5969
	mate: May-Sept; Finland	10681
	mate: end July-early Sept; s Norway	5301
	mate: winter; s Norway	4342
	preg f: Apr; n1; USSR	8052
	preg f: Apr-early Aug; Germany	10339, 10340
	preg f: May-Sept; Norway	11679
	birth: Feb-Apr, Aug-Nov; lab, Berlin, Germany	3468
	birth: yr rd; lab, Oldenburg, Germany	3468
	birth: Apr-Sept; Scandinavia, Germany	7433
	birth: May-early June, low winter; Finland	5493
	birth: yr rd, peak Mar-Oct; Germany	3571
	repro: f; mid May-Aug; m; May-Aug; YUK, Canada	6015
Microtus oregoni	sperm: Feb-Sept; OR	3622
	mate: mid Mar-mid Sept; BC, Canada	2087, 2088
	mate: May-Aug+; w US, s BC, Canada	1474
	preg/lact f: Mar-Sept; OR	3622
	yg trapped: yr rd, peak June-July; OR	4970

Microtus pennsylvanicus	mate: end Mar-Nov; MAN, Canada	7293
	mate: Apr-June, Oct-Nov; MI	3741, 3742
	mate: end Apr-Sept; s QUE, Canada	11956
	mate: yr rd, low winter; NY	4278
	mate: none winter; MN	10839
	preg f: Jan, Mar-Apr, Oct-Nov; NC	1290
	preg f: Feb-Nov; MN	495
	preg f: Mar-Nov; NY	4278
	preg f: Mar-Dec; MI	3878
	preg f: May-June; n2; SD	261a
	preg f: Apr-May, July; n12; NE	5405b$_1$
	preg f: peak Apr-Sept; OH	2376
	preg f: Apr-Oct; ONT, Canada	2082
	preg f: est May-Aug; MAN, Canada	10101
	preg f: June; ID	11646a
	preg f: June-Aug; n15 f; ND	3703
	preg f: mid June-Aug; no data other mo; SD	10954
	preg f: July; n10; se MT	6166
	preg f: peak mid July-mid Sept; YUK, Canada	12049
	preg f: Nov-Feb, prob also other mo; MA	10648
	preg f: yr rd; MN	721
	birth: yr rd; WY	1871a
	birth: yr rd; IL	10463
	lact f: Mar-Oct; MA	10647
	lact f: June; n1; SD	261a
	lact f: July; n4; se MT	6166
	sex active f: few Oct-Mar; OH	2376
	repro f: mid May-Aug; m: May-mid Sept; YUK, Canada	6015
	repro: yr rd, peak Mar-June; IN	2059
	repro: yr rd, most Mar-Nov; WI	5210
Microtus richardsoni	conceive: end May-early Sept; ALB, Canada	6631, 6632
	preg f: July-Aug; WY	1871a
	preg/lact f: July; n4; OR	4969
	birth: end June-Sept; BC, Canada	2088
Microtus roberti	yg trapped: early June-?; Turkey	10274
Microtus sachalinensis	mate: May-Sept; ne Sachalin, Russia	11366
Microtus socialis	mate: mid Jan-Apr, end Aug-mid Sept; USSR	8052
	preg f: Nov; Iran	6293
	birth: spring-fall; Europe, USSR	4645
	juveniles: Aug-Sept; Iran	6293
Microtus townsendii	scrotal testes: yr rd, peak 100% Apr-July, low 50% Nov-Dec; BC, Canada	3792
	mate: Mar-Sept; BC, Canada	2088
	mate: Apr-Aug, no data after Aug; BC, Canada	3792
	birth: Mar-Sept; w US, s BC, Canada	1474
Microtus transcaspicus	birth: yr rd; captive, Turkmenia	8153
Microtus xanthognathus	mate: est May-Oct; NWT, Canada	2704
	preg f: May-mid July; AK	11876, 11878
Neofiber alleni	mate: est Jan-fall; few data; GA	4386
	preg f: yr rd, not Mar; FL	944
	birth: mid June; few data; FL	8267
Ondatra zibethicus	gonads: increase Mar-July, regress Aug-Nov; WI	719
	sperm: Dec-Oct; MD	3392
	sperm: yr rd; TN	9529
	spermatogenesis: Feb-Aug; Germany	7291
	spermatogenesis: Mar-July; WI	719
	testes: enlarge Dec, large Mar-Aug; Germany	704
	testes: enlarge Mar; Canada	3558
	testes: large Feb-Sept; Netherlands	11174
	testes: large Apr-May; Kazakhstan	10490
	testes: max May, prob sperm yr rd; MD	12112
	testes: large May-Aug, small Sept-Dec; Germany	7291
	testes: large June-Sept; USSR	9690
	m fertile: Mar-Sept; Netherlands	2697

cl present: Jan-Feb; MD	3392
cl present: Jan-Aug; OH	2685
uterus: large Feb-June; Berlin, Germany	705
mate: Jan-Oct; Germany	3571
mate: Feb-Mar; Berlin, Germany	705
mate: mid Feb-mid Aug, peak Apr-June; WI	720
mate: end Feb-Mar; UK	11467
mate: end Feb-mid Nov; IA	3082
mate: Mar-?; Kazakhstan	10066
mate: Mar-Apr; Netherlands	11174
mate: Mar-Aug; N Am	6214
mate: mid Mar-mid Apr; NY	3061
mate: Apr-May; WY	1871a
mate: Apr-Aug; n N America	1474
mate: Apr-early Oct; Mongolia	2362
mate: Apr-Oct; Europe, UK	7433
mate: end Apr-early May; Canada	3558
mate: May-Sept; s BC, Canada	2088
mate: July; MAN, Canada	7093
mate: June-Aug; e Siberia	5946
mate: yr rd, peaks Nov, Mar, low July-Aug; LA	8015
mate: yr rd; CA	2616
ovulation: Mar-June; WI	719
preg f: Feb-Aug; OH	2685
preg f: Mar, July; n2; NC	1290
preg f: Mar-early Aug; s Bavaria	705
preg f: Mar-Sept, peak June; Netherlands	2697
preg f: Mar-Sept; Germany	10421
preg f: Apr-Aug; TN	9529
preg f: Apr-Sept; Netherlands	11174
preg f: end May-Aug, sometimes Sept; ne Sweden	6805
preg f: June-Dec; East Germany	4891
preg f: low Oct-Mar; Germany	7291
preg f: Nov; n1; NB, Canada	8234
preg f: yr rd, peak Nov-Apr; LA	10580
birth: Feb-Nov; Germany	3571
birth: peak mid Feb-Oct; CA	2616
birth: Mar-early Sept; WI	2695
birth: Mar-Oct, peak Apr-July, rare Oct; IA	3082
birth: Apr-?, peak end June; USSR	9690
birth: Apr, June-Sept; N Am	6214
birth: Apr-July; ONT, Canada	11934
birth: Apr-Aug, peak Apr-June; NE	9503
birth: Apr-Aug; e Canada	8237
birth: Apr-Sept; N America	6214
birth: Apr-Oct; Europe, UK	7433
birth: end Apr-early July, peak after mid May; WI	719
birth: end Apr-July; s ONT, Canada	11934
birth: end Apr-early Oct; UK	11467
birth: end May-July; Canada	3558
birth: end May-June; MAN, Canada	7093
birth: end May-mid Aug	9003
birth: May-Aug; NB, Canada	8236
birth: May-mid Aug; USSR	8051
birth: May-Sept; Sweden	2243
birth: May-Sept; captive	6267
birth: May-Sept, peaks monthly; WI	2695
birth: mid May-Aug; MAN, Canada	8104
birth: mid May-Aug, peak early June; captive	7094
birth: peak May-June; QUE, Canada	10410
birth: peak May-July; IA	3080
birth: June-July; Netherlands	11174
birth: yr rd, few Nov-Feb; w US	5146
lact f: July; n1; ND	3703

	repro: mid Feb-mid Aug; WI	5210
	repro: Mar-Aug; Germany	702, 704
	repro: Apr-Sept; Germany	11027
	repro: ends Sept; Dnepr, Ukraine	787
Phenacomys intermedius	preg f: end May-early Sept; ALB, Canada	5150
	preg f: June-Sept; CO	11230
	preg f: mid June-early Sept; MAN, Canada	3429, 10101
	birth: May-Sept; ALB, Canada	5150
	birth: June-Sept; N America	1474
Pitymys duodecimcostatus	preg f: peak Oct-June; s Spain	8216
	preg f: Dec-Apr; Pyrenees, Spain	11255
Pitymys majori	mate: peak June; USSR	8052
Pitymys ochrogaster	m fertile: yr rd; KS	5241
	spermatogenesis: yr rd; KS	5241
	mate: yr rd; captive	9111
	mate: yr rd, low winter, July; KS	9273
	preg f: peak Mar-June, none Jan; KS	5241
	preg f: Mar-Sept; SD	10954
	preg f: Mar, Oct; n2; KS	1812a
	preg f: Apr; n2; MN	495
	preg f: Apr-Nov; n110; NE	$5405b_1$
	preg f: peak Apr-Sept, few Oct-Mar; OH	2376
	preg f: May, July-Aug; n5; MAN, Canada	11957a
	preg f: May-June; n9; SD	261a
	preg f: June-July; n3; ND	3703
	preg f: Dec-Jan; n1; KS	5241
	preg f: yr rd, low Dec-Mar; TX	3284
	preg f: yr rd, low Dec-Jan; KS	3283
	birth: peak Mar-Sept; central, nw US	1474
	birth: mid May-Sept; MAN, Canada	2137
	lact f: May; n2; SD	261a
	repro: f: yr rd, peak Apr-July; IL	682
	repro: yr rd, peak Mar-June; IN	2059
	repro: yr rd, peak July-Sept, low Dec-Feb; WI	5210
Pitymys pinetorum	scrotal m: yr rd; OK	3852
	mate: Jan-Oct; n US	1474
	mate: Jan-Oct; NY	783
	mate: mid Feb-mid Nov; CN	3747
	mate: Apr-Sept; captive; NY	4273
	preg f: Feb-Nov, peak July-Aug, low Feb-Mar; no data Dec-Jan; PA	9987
	preg f: Apr, Aug, Dec; n3; NE	$5405b_1$
	preg f: Apr-Oct; VA	11059
	preg f: yr rd, low Dec-Jan; NC	8268
	repro: yr rd, most Mar-Sept; WI	5210
	repro: yr rd, peak Oct-May; OK	3852
Pitymys sikimensis	preg f: Mar, Aug-Nov; n22; Nepal	7382
Pitymys subterraneus	spermatogenesis: Mar-Nov; France	2470
	testes large: June-Oct; Germany	4114
	m fertile: Mar-Nov; no data Nov-Feb; Germany	6199
	m fertile: Apr-Sept; no data before Apr; Czechoslovakia	6005
	mate: ?-Oct; Europe	7433
	mate: Feb, Apr-early Oct; Germany	3571
	mate: Apr-?; Tatry Mt, Czeckoslovakia	10379
	preg f: mate: mid Feb-Sept, peak May; Czechoslovakia	8335
	preg f: Apr-Sept, peak June; Czechoslovakia	6004
	preg f: Aug-Oct; n5; Germany	4114
	preg f: yr rd; Germany	6199
	birth: ?-Sept; Tatry Mt, Czechoslovakia	10379
	birth: ?-Nov; Europe	7433
	birth: Apr-Sept; Europe, USSR	4645
	birth: May-Nov; Germany	3571
	birth: June-Nov; France	2470
	repro: peak Mar-Sept; lab	9656

Pitymys tatricus	m fertile: Apr-Sept; no data before Apr; Czechoslovakia	6005
	preg f: Apr-Aug; Czechoslovakia	6004
Prometheomys schaposchnikowi	mate: Apr; USSR	12007
	birth: May; USSR	12007
	birth: end May-mid July?; USSR	8051
	repro: spring-fall; Turkey	10275
Synaptomys borealis	birth: May-Aug; n N Am	1474
	birth: May-Aug; BC, Canada	2088
Synaptomys cooperi	mate: Apr-Nov, peak May; no data Nov-Apr; NJ	2026
	mate: yr rd; mid e N America	1474
	preg f: May, Oct; n7; NE	5405b$_1$
	preg f: yr rd; KS	1469
	birth: May-Sept; e central N America	555a
	repro: Mar-Nov; WI	5210
	repro f: yr rd; IL	682

Order Rodentia, Family Muridae

Rats and mice (*Acomys* 11-90 g; *Aethomys* 25-150 g; *Anisomys* 500-600 g; *Anonymomys* ~125 mm; *Apodemus* 13-64 g; *Apomys* 19-50 g; *Arvicanthis* 35-183 g; *Bandicota* 100-1500 g; *Batomys* ~205 mm; *Berylmys* 143-400 g; *Bullimus* 240-275 mm; *Bunomys* 60-150 g; *Carpomys* ~200 mm; *Celaenomys* ~195 mm; *Chiromyscus* >47 g; *Chiropodomys* 15-43 g; *Chiruromys* 115-175 mm; *Chrotomys* 115-160 g; *Colomys* 50-75 g; *Conilurus* 100-200 g; *Crateromys* ~255 mm; *Cremnomys* 30-60 g; *Crossomys* ~205 mm; *Crunomys* 55 g; *Dacnomys* 230-290 mm; *Dasymys* 47-164 g; *Diomys* 40-70 g; *Diplothrix* ~230 mm; *Echiothrix* 200-250 mm; *Eropeplus* 195-240 mm; *Gatamiya*, *Golunda* 50-80 g; *Grammomys* 24-80 g; *Hadromys* 66 g; *Haeromys* 55-75 mm; *Hapalomys* 120-165 mm; *Hybomys* 30-70 g; *Hydromys* 400-1300 g; *Hylomyscus* 8-42 g; *Hyomys* 1 kg; *Kadarsanomys* 176-230 g; *Komodomys* 125-200 mm; *Leggadina* 15-25 g; *Lemniscomys* 18-72 g; *Lenomys* 235-290 mm; *Lenothrix* 81-273 g; *Leopoldamys* 200-495 g; *Leporillus* 140-200 mm; *Leptomys* 145-160 mm; *Limnomys* ~125 mm; *Lophuromys* 23-111 g; *Lorentzimys* 12-13 g; *Macruromys* 150-250 mm; *Malacomys* 48-145 g; *Mallomys* 980 g-2 kg; *Margaretamys* 115-150 mm; *Mastacomys* 100-200 g; *Maxomys* 29-284 g; *Mayermys* 17-21 g; *Melasmothrix* 40-58 g; *Melomys* 30-200 g; *Mesembriomys* <1 kg; *Microhydromys* ~80 mm; *Micromys* 3-12 g; *Millardia* 47-100 g; *Muriculus* 70-95 mm; *Mus* 2-50 g; *Mylomys* 46-190 g; *Neohydromys* ~92 mm; *Nesokia* 112-185 g; *Niviventer* 26-240 g; *Notomys* 20-60 g; *Oenomys* 15-121 g; *Papagomys* 130-175 mm; *Parahydromys* ~240 mm; *Paraleptomys* 120-140 mm; *Paruromys* 350-500 g; *Pelomys* 46-170 g; *Phloeomys* 1.5-2 kg; *Pithecheir* 58-146 g; *Pogonomelomys* 51-65 g; *Pogonomys* 35-75 g; *Praomys* 8-100 g; *Pseudohydromys* 20 g; *Pseudomys* 12-105 g; *Rattus* 30-550 g; *Rhabdomys* 25-75 g; *Thynchomys*; *Solomys* 230-330 mm; *Srilankamys* ~145 mm; *Stenocephalemys* 120-195 mm; *Stochomys* 27-104 g; *Taeromys* 185-250 mm; *Tarsomys* ~135 mm; *Tateomys* 35-98 g; *Thallomys* 37-128 g; *Thamnomys* 30-100 g; *Tokudaia* 125-175 mm; *Tryphomys* 130-175 mm; *Uranomys* 31-53 g; *Uromys* 600-1000 g; *Vandeleuria* 10 g; *Vernaya* ~90 mm; *Xenuromys* ~1000 g; *Xeromys* 40-60 g; *Zelotomys* 38-85 g; *Zyzomys* 35-143 g) are primarily terrestrial herbivores from Europe, Asia, Africa, and Australia.

Males are larger than females in *Acomys* (2532, 4329), *Aethomys* (4329; cf 9893), *Apodemus* (2193), *Bandicota* (8500; cf 728), *Golunda* (8500, 8501), *Millardia* (730), *Nesokia* (728), *Oenomys* (5760), *Praomys* (4329; cf 4350, 4925), *Pseudomys* (10117), *Rattus* (899, 2841, 4841, 10501, 10696, 11475, 11832, 11833; cf 2954, 4925, 8500), and *Uranomys* (4350), while females are larger than males in *Notomys* (1258, 1261, 10117; cf 2532). In *Grammomys* either males or females may be larger (4329, 4350). A copulatory plug is present in *Acomys* (2532), *Hydromys* (8103), *Notomys* (1258, 1266; cf 2532; *Pseudomys* 1263), *Praomys* (1215, 5366), *Pseudomys* (5643), and *Rattus* (6162, 6237, 8393). Sex chromosome abnormalities occur (3492; *Acomys* 6974).

The 50 g (*Rattus* 6366), hemochorial, hemoendothelial, or epitheliochorial, labrinthine, discoid or diffuse placenta has a mesometrial or antimesometrial, interstitial implantation into a bicornuate uterus (231, 432, 1215, 1256, 3011, 5735, 7563, 8326, 8794, 9356) and is eaten after birth (4992, 9175, 9213, 10696). Estrus occurs postpartum (427, 845, 947, 1215, 1255, 1335, 1814, 1815, 2045, 2131, 2132, 2443, 2445, 2446, 2450, 2518, 2530, 2566, 2568, 2836, 2911, 3133, 3461, 3756, 4353, 4355, 5170, 5366, 5643, 5760, 6237, 7688, 7791, 7792, 8034, 8103, 8393, 8904, 8965, 9429, 10117, 10158, 10481, 10696, 10698, 10704, 10707, 11434, 11614, 11711; cf 1771, 8103). Lactation may delay gestation (1256, 3047, 3798).

Acomys are relatively precocious at birth often with fur and open eyes (1035, 2532, 2566, 2567, 5051, 8686, 10481, 11397; cf 187, 299, 1035, 1080, 2568, 4329, 5760). Other murids have hair at birth (*Mastacomys* 1542, 10698; *Pithecheir* 6448) and open their eyes at 4-7 d (*Lophuromys* 299, 1771, 2445, 5760; cf 5051), 7-8 d (*Lemniscomys* 299, 5051, 5760, 8429; cf 11614), 7-10 d (*Rhabdomys* 1335, 1814, 10158; cf 5051), 1-2 wk (*Aethomys* 331, 1333, 1771, 1815, 4329, 5051, 7168), 9-21 d (*Pseudomys* 670, 3259, 3438, 5642, 10117; cf 4042, 4355), 10-12 d (*Grammomys* 299, 334, 5051, 5760, 8218, 10158), 11 d (*Mesembriomys* 2131), 11-15 d (*Thallomys* 7168), or 17-28 d (2132, 2532, 4355, 10117). *Praomys* neonates are either naked or furred and open their eyes after 10-19 d (247, 299, 514, 1815, 2445, 4353, 5051, 5760, 7166, 7168, 8034, 10158). Many murids are born naked and blind (187, 334, 2091, 2452, 4329, 4410, 5760, 6317, 8500, 8501, 8861, 9175, 9265, 10158, 10709, 11710). Eyes open at 5-22 d (*Rattus* 1330, 2954, 3133, 3260, 3437, 3571, 7433, 9175, 10696; cf 7165, 11833), 7 d (*Arvicanthis* 5051; cf 2450), 7-10 d (*Micromys* 2045, 2565, 3461, 3571, 5553, 7433), 10 d (*Melomys* 7209), 10-16 d (*Hydromys* 8103), 11-16 d (*Apodemus* 2045, 2569, 3571, 7433), 12 d (*Millardia* 12034), 12-15 d (*Mus* 245, 329, 670, 845, 5685, 7433, 8501, 9175, 10158, 11711), 13-17 d (*Bandicota* 108, 9175, 9429, 10268), 15-21 d (*Hylomyscus* 2445), 16-17 d (*Zelotomys* 947, 10158), or 17-18 d (*Niviventer* 12105).

Females have 2 mammae (*Phloeomys* 9562), 2-4 mammae (*Melomys* 7209), 4 mammae (*Colomys* 4495; *Conilurus* 7209, 10698; *Chrotomys* 7983; *Hydromys* 7108, 8103, 10698; *Hyomys* 7209; *Leporillus* 7983; *Macruromys* 7209; *Mallomys* 7209; *Pithecheir* 7983; *Pseudohydromys* 7983; *Pseudomys* 5619, 10704; *Tateomys* 7675; *Thamnomys* 4495; *Uromys* 7209; *Xeromys* 10698), 4-6 mammae (*Acomys* 331, 5760; *Aethomys* 331, 4495, 9945, 10158; *Grammomys* 331, 988, 4495, 8218; *Pogonomelomys* 7209), 4-8 mammae (*Oenomys* 4495, 5760), 6 mammae (*Anisomys* 7209; *Chiropodomys* 4408; *Chiruromys* 7209; *Dasymys* 331, 4495, 5760, 9945, 10158; *Lophuromys* 331; *Lorentzimys* 7209; *Malacomys* 331, 4495; *Melasmothrix* 7675; *Pogonomys* 7209; *Thallomys* 9945), 6-8 mammae (*Apodemus* 182, 3571, 7433; *Hylomyscus* 5760; *Maxomys* 573, 11186; *Millardia* 9175), 6-14 mammae (*Rattus* 573, 610, 1254, 2321, 2555, 2911, 3260, 3571, 4041, 5111, 5277, 7052, 7209, 7433, 9175, 10696, 10917), 6-24 mammae (*Praomys* 184, 186, 331, 1215, 4495, 5760, 9468, 9945, 10158), 8 mammae (*Kadarsanomys* 7672; *Lemniscomys* 331, 4495, 9945; *Leopoldamys* 182; *Micromys* 182, 3571, 7433; *Mylomys* 4495; *Nesokia* 728; *Pelomys* 331, 9945; *Rhabdomys* 9945), 10 mammae (*Komodomys* 7983; *Lenothrix* 7674; *Tryphomys* 573; *Zelotomys* 5760, 10158), 10-12 mammae (*Mus* 331, 573, 4495, 5111, 6202, 7209, 7433, 9175, 9945, 11711), 10-22 mammae (*Bandicota* 108, 728, 7907, 9175, 11436). Commonly, males have a baculum, penial spines, seminal vesicles, and coagulating glands (1257, 1259, 1264, 1265, 1276, 4683, 7546a, 7791, 8034, 8983, 11883). *Mus* exhibits no photoperiodic regulation (1314).

Details of anatomy (988, 1258, 1263, 1266a, 1275, 1326, 3068, 5513, 8979, 8980, 9431, 9432, 9878, 10565), endocrinology (*Acomys* 714, 715; *Notomys* 1256, 1261, 1275; *Praomys* 7737; *Rattus* 5163, 5695), spermatogenesis (1260, 1262, 1274a, 1276, 1277, 3292, 3293, 7205, 8324, 10927; *Acomys* 8327; *Apodemus* 5093, 5094, 5479; *Rattus* 5695), copulation (1266, 2525, 3097, 3133), birth (*Apodemus* 9213), reproductive energetics (669, 670), and light (*Bandicota* 9433, 9434; *Notomys* 1252, 1253) are available.

NEONATAL MASS

Acomys cahirinus	4-5 g; term emb	7795
	f: 4.9 g; n67	7545
	5-6 g; r3.0-8.5	1035
	5.49 g; sd0.64; n20 lt	11397
	m: 5.6 g; n81	7545
	5.78 g; r4-8; n74	2566, 2567
	6.37 g	2568
Acomys wilsoni	3-4 g; term emb	7795
Aethomys chrysophilus	4.1 g	1333
	4.8 g	7797a
Aethomys kaiseri	6.1 g; r5.2-7.4; n42	1771
Aethomys namaquensis	2.5 g	7797a
Apodemus agrarius	1.8 g	3571
Apodemus flavicollis	2.3 g	1032
Apodemus mystacinus	3 g	2569
Apodemus peninsulae	2.0 g; r1.6-2.5; n38	12105
Apodemus sylvaticus	1-2 g	2045
Arvicanthis niloticus	2.5-3.5 g	7793
	~3 g	2446a

	4-6 g	8811
	4.25 g; n6	2450
	4.65 g; ±0.18; n19	2450
Bandicota bengalensis	3-5 g	9175
	4.85 g; r4.0-5.7; n26	10268
	5.8 g; n25, 3 lt, youngest litters seen	4409
Bandicota indica	f: 9.43 g; se0.14; n65	10370
	m: 10.54 g; se0.15; n77	10370
Grammomys dolichurus	4.2 g; n8	8218
Hybomys univittatus	4-4.8 g	3692a
Hydromys chrysogaster	f: 22.23 g; sd2.72; n17	8103
	m: 22.89 g; sd4.38; n19	8103
	f: 25.49 g; sd0.27; n161	8103
	m: 26.93 g; sd0.20; n209	8103
Hylomyscus alleni	1.2-1.8 g	2446a
Hylomyscus stella	1.47 g; r1.3-1.7; n5	2445
Lemniscomys striatus	1.6-1.8 g	7792
	~3 g	2446a
	~3 g	8429
Lophuromys flavopunctatus	4-5 g; near term	2452
	5-8 g	5760
Lophuromys sikapusi	6.5-7.5 g	2572a
	6.6-9 g	3692a
	7.89 g; r6.6-8.9; n8	1771
Lophuromys woosnami	9-10 g; 2 d	2572a
Mastacomys fuscus	7.25 g; n1	1542
Mesembriomys gouldii	34.70 g; r33.00-37.50; n6	2131
Micromys minutus	0.7 g; r0.65-0.8	2045
	0.8-0.9 g	3571
	1.1 g; n41 ?	5553
Mus caroli	1.2 g	10370
Mus cervicolor	1-2 g	10370
Mus domesticus	1.70 g; se0.05	6864
Mus minutoides	0.8 g	245
	0.8 g	11711
	~1.3 g	2446a
Mus musculus	0.8 g; n6, 1 lt	4409
	~1 g	845
	1.14 g	7704
Mus triton	1.3 g; n1	4329
Niviventer niniventer	3-4.5 g	12105
Notomys alexis	2.5 g	10117
Praomys daltoni	1.5 g	247
Praomys morio	2.435 g; n8	2954
	2.97 g; r2.57-3.09; n1 lt, 4 yg, 2 wk	2445
Praomys natalensis	1.8 g; late term	7791
	2.2 g; r1.9-3.0; n137, 15 lt, 3 f	7166, 7168
Praomys tullbergi	2.7 g; r2.0-3.0	4353
Pseudomys australis	~4 g; from graph	10117
Pseudomys gracilicaudatus	f: 4.2 g; 95%ci 3.6-4.8; n23	3438
	m: 4.5 g; 95%ci 3.6-5.4; n28	3438
Pseudomys higginsi	5 g; from graph	4042
Pseudomys novaehollandiae	1.0-2.4 g	5642
	2 g ea; n2	8801
Rattus exulans	1.6-3.5 g	1330
	2.0 g; n11 lt	4409
	2.2 g	10370
	f: 2.2 g; ±0.4; n44	1330
	m: 2.4 g; ±0.4; n60	1330
	f: 2.8 g; n7	11833
	m: 3.1 g; n3	11833
	3.4 g; r3.2-3.6; n3 f	10578
Rattus fuscipes	4.5 g; r2.5-6.0; n156	10696

Rattus lutreolus	f: 4.14 g; n1	4041
	f: 5.0 g; 95%ci 4.9-5.2; n69 (m&f)	3437
	m: 5.12 g; r4.89-5.34; n2	4041
	m: 5.2 g; 95%ci 5.1-5.2; n69 (m&f)	3437
Rattus norvegicus	5.27 g; r2.800-6.380; n204 yg, 18 lt	7708
	5.6 g; n11, 2 lt	4409
	5.72-5.99 g	409
	5.8 g; n333, 32 lt	5503
	7.35 g	1843
Rattus rattus	4.0 g; n21, 5 lt	4409
	4.15 g; r2.6-5.1; n16	10268
	f: 4.6 g; ±0.1; d2; n92 m&f	2091
	m: 4.9 g; ±0.1; d2; n92 m&f	2091
	5.1 g; median 5.4 g; sd1.46; n136	8861
Rattus sordidus	~3 g; from graph	7053
Rattus tiomanicus	4.1 g	10370
Rhabdomys pumilio	~2 g	1814
	2.5 g; se0.05	1335
	2.5-3.0 g	4640
Thallomys paedulcus	2.5-2.8 g	7168
Uromys caudimaculatus	20 g	10698
NEONATAL SIZE		
Acomys cahirinus	HB: 40-46 mm	5051
	51 mm; n1, 12 h	1080
	52 mm; r44-57	2566, 2568
Acomys wilsoni	HB: 40 mm; n2	5051
Aethomys chrysophilus	m: 42 mm	1333
Aethomys kaiseri	42-45 mm; n6	5051
Bandicota bengalensis	35 mm	9175
	HB: 47 mm; n25, 3 lt, youngest litters seen	4409
Bandicota indica	f: 52.0 mm; se0.4; n65	10370
	m: 55.0 mm; se0.3; n77	10370
Grammomys dolichurus	HB: 35 mm; n5	299
	HB: 40.0 mm	8218
Hydromys chrysogaster	CR: 52.4 mm; n3, 33 d emb	8103
	f: HB: 83.06 mm; sd1.39; n17	8103
	m: HB: 83.84 mm; sd1.83; n19	8103
	f: HB: 87.14 mm; sd0.34; n161	8103
	m: HB: 89.22 mm; sd0.28; n209	8103
Lemniscomys griselda	<80 mm; n1	9179
Lophuromys flavopunctatus	HB: 48 mm; n3	299
Mastacomys fuscus	63 mm; n1	1542
Mesembriomys gouldii	f: 92 mm	2131
Mus minutoides	HB: 25 mm	11711
	HB: 26 mm	11711
	HB: 26 mm	245
Mus musculus	HB: 22 mm; n6, 1 lt	4409
Mus triton	26 mm; n1	4329
Praomys natalensis	HB: 25-33 mm; n12	299
	HB: ~30-35 mm	7166
Pseudomys gracilicaudatus	f: 43.6 mm; 95%ci 40.0-47.2; n25	3438
	m: 44.8 mm; 95%ci 40.9-48.6; n29	3438
Pseudomys higginsi	48 m	4042
Rattus exulans	HB: 37 mm; n11 lt	4409
	f: HB: 42.4 mm; n7	11833
	m: HB: 42.6 mm; n3	11833
	56 mm; r55-57; n3	10578
Rattus fuscipes	46 mm; r38.9-51.4; n110	10696
Rattus lutreolus	f: 43.8 mm; 95%ci 42.9-44.6; n69 m & f	3437
	m: 45.6 mm; 95%ci 44.6-46.6; n69 m & f	3437
	f: 50 mm; n1	4041
	m: 50 mm; r49-51; n2	4041
Rattus norvegicus	HB: 52 mm; n11, 2 lt	4409

Rattus rattus	HB: 40 mm	9265
	HB: 45 mm; n21, 5 lt	4409
Rhabdomys pumilio	HB: 35 mm; se0.0	1335
	HB: 45 mm; n4	299
WEANING MASS		
Acomys cahirinus	~10 g; 1st trapped	7795
	15.16 g; n14	11397
Acomys wilsoni	~10 g; 1st trapped	7795
Aethomys chrysophilus	30-32 g; 1st trapped	7797a
Aethomys kaiseri	f: 46.0 g	1771
	47.8 g; n33	1771
	m: 48.9 g	1771
Aethomys namaquensis	10-15 g; 1st trapped	7797a
Arvicanthis niloticus	15-25 g	7793
	~25 g	2450
Bandicota bengalensis	22.9 g; r13.1-41.6; n12, 30 d	10268
Bandicota indica	41.0 g; r31.4-46.4; n9, 30 d	10268
Dasymys incomtus	30 g	2570
Hydromys chrysogaster	~150 g; 30 d; from graph	8103
Lemniscomys striatus	10-15 g	7792
	20 g	2570
Lophuromys flavopunctatus	6.5-9.5 g; from graph	2445
	30 g	2570
Lophuromys sikapusi	13.61 g; n8 yg, n2 lt	1771
Melomys burtoni	11 g; lightest trapped	10109
	50 g; juvenile	735
Melomys cervinipes	17-18 g; lightest trapped	10109
Mesembriomys gouldii	481 g; n1, 42 d; from graph	2131
Mus minutoides	3.18 g; 24 d	245
	~3.5 g; 3 wk; from graph	11711
Mus musculus	6.07 g; n36 yg, 3 lt of 12, 21 d	10095
	~10 g	845
	10.33 g; n15 yg, 5 lt of 3, 21 d	10095
	f: 10.7 g; se0.3	2723
	f: 12.8 g; se0.1	2723
Mus triton	4-5 g	2570
Notomys alexis	10-15 g; 25 d	10117
Oenomys hypoxanthus	30 g	2570
Praomys daltoni	~8 g	247
Praomys erythroleucus	12 g; lightest trapped; est 20-25 g at weaning	1215
Praomys jacksoni	20 g	2570
Praomys natalensis	10-15 g; lightest trapped	7791
	10-20 g	2570
	11.7 g; n125, 14 lt, 4 f, 21 d	7166
	f: 13 g; ±0.72; r10-16.5 d; n12, 21 d	5366
Praomys tullbergi	8-13 g; 28 d	4353
Pseudomys apodemoides	est 9 g; sd0.8; n6	1934
	13 g; 6 wk	3259
Pseudomys australis	10-15 g; 25 d; from graph	10117
Pseudomys gracilicaudatus	~24 g; from graph	3438
Rattus exulans	18-23 g; solid food	11833
	m: 26.6 g; sd2.8; n9	4821
	f: 27.2 g; sd1.8; n9	4821
Rattus fuscipes	m: 12.3 g; r11-13; n3, 5 wk	3260
	f: 13 g; n1, 5 wk	3260
	17.6 g; 3 wk	10696
	29.9 g; 28 d	10696
	40.9 g; r36.9-42.1; n2	10696
Rattus lutreolus	25 g	4041
	< 40 g; young juveniles	614
Rattus norvegicus	f: 61.2 g; sd8.7; n10	4821
	m: 66.3 g; sd9.9; n10	4821
Rattus rattus	28.6 g; r25.3-40.1; n7, 30 d	10268
	30-50 g; est 6-7 wk	8861

	m: 42.4 g; sd7.7; n10	4821
	f: 47.4 g; sd11.7; n10	4821
Rhabdomys pumilio	8.0 g; n9	4640
	8.9 g; 14 d	1335
Tateomys rhinogradoides	82.8 g; sd12.7; n4	7675
Zyzomys argurus	14 g m; lightest trapped	733
WEANING SIZE		
Mus minutoides	HB: ~50 mm; 3 wk; from graph	11711
Praomys tullbergi	f: HB: 70-80 mm, 28 d	4353
LITTER SIZE		
Acomys spp	1.88 emb; r1-4; n25 f	299
Acomys cahirinus	mode 1-2; r1-4	2449, 2452, 4344
	1-2 emb; n9+ f	4925
	1-4	10481
	1.7 wean	7544
	1.94 emb; ±0.05; 1.85 live emb; ±0.05; r1-5; n179 f	7795
	2.0 yg	7544
	2 emb; n1 f	331
	2.0; n26	2532
	2.08; sd0.85; r1-4; n87 lt	11397
	2.13 emb; se0.09; mode 2; r1-4; n55 f	11475a
	2.37; mode 2; most 1-4; r1-7; n408 lt	12038
	2.38 yg; n75 lt	2568
	2.44; mode 2; r1-5; n123 lt	2566, 2568
	2.5 emb	11475a
	2.68 emb; mode 3; r1-4; n25 f	2567
	2.75; n4 lt	12128
	2.76 yg; r1-5; n84 lt	1035
	2.8 ps; n21	4329
	3; r2-4	1080
	3 emb, ps; r1-6; n13 f	8134
	3.2 emb; n17 f	4329
	4.5 yg; r2-11; n17 lt	12128
	4.5 yg; 5.51 emb; r3-7; n49 lt	109
Acomys russatus	2.5; r2 yg, 3 emb; n2 lt	8134
Acomys spinosissimus	2.5 emb, ps; sd1.3; mode 2; r1-5; n6 f	3821
	2.6 emb; r2-3; n5 f	9893
	3.0 emb; r2-5; n12 f	10154, 10158
	3.3 emb; r2-6; n9 f	8965
Acomys wilsoni	2.23 emb; 2.21 live emb; ±0.09; r1-5; n79 f	2443, 2452, 7795
	2.5 emb; r1-4; n2 f	4925
Aethomys chrysophilus	2-4 emb	9945
	2 emb; n1 f	337
	2.5 emb; r2-3; n2 f	9893
	2.6 emb; r1-4; n9 f	299
	2.8 emb; n5 f	331
	3 emb; n1 f	2096
	3.1; r1-5; n37 lt	1333
	3.2 emb; r2-5; n15 f	4329
	3.3 ps; r1-7; n12 f	4329
	3.3 emb; r1-5; n41 f	8965
	3.9 emb; r3-6; n10 f	10154, 10158
	4.1 yg; ±0.27; n14 lt	7797a
Aethomys hindei	2.05; r1-4; n18 lt	8083
Aethomys kaiseri	2.6; r1-4	1771
	3 emb; n1 f	4495
	3 yg; n1 lt	187
	3.0 emb ea; n4 f	2443, 2446a, 2452
	3.4 emb, yg; r1-5; n5 f	299
	3.5 emb; r3-4; n2 f	4925
Aethomys namaquensis	2-5	11864
	2-5	9945
	2.6 ps/emb; sd1.1; mode 3; r1-4; n15 f	3821
	3.1 emb; r2-7; n42 f	10154, 10158

	3.3 emb; r1-5; n43 f	8965
	3.4 yg; ±0.16; n53 lt	7797a
	3.6; r1-5; n11	10158
	4; n3 lt	7168
	5 yg; n1 f	7165
	7.0 ps; r3-11; n2 f	4329
Aethomys nyikae	3.67 emb; n3 lt	331
Aethomys walambae	3 emb; n1 f	337
Apodemus spp	6.8 emb; r6-7; n5 f	2442
Apodemus agrarius	2-8	7433
	mode 4-9; r2-12	3571
	5-6	4645
	mode 5-6 yg; r2-9	5424_1
	5.0; r4-6; n4 lt	11710
	5.8 emb; mode 6; r2-8; n15 f	10340
	6.64; r4-9; n79 lt	8330, 8331, 8332
Apodemus argenteus	4.0 emb; n34 f	3546
	4.1; n14 lt	7925
	4.13; mode 5; r3-6; n23 lt	12025
Apodemus flavicollis	3-8	7433
	mode 3-8; r3-12	3571
	3.9; n104 lt	5170
	4.42 emb; mode 5; n91 f	11255
	4.76 emb; r3-7; n13 f	10340
	5.04 emb	8331
	5.12 emb; sd1.28; mode 5; r1-8; n31 lt, n159 emb	10358
	6	6370
	6; n1 lt	9213
	6; r1-12	4645
	6.8 emb; mode 7; 66% 5-7; r3-11; n61 f	5461
Apodemus microps	4.83 emb; sd0.76; mode 5; r4-6; n6 f	10358
	5.89 emb; r3-8; n73 lt	10480
	6.40; se0.159; mode 6; r3-10; n80 lt	8329, 8331
Apodemus mystacinus	1.67 yg; r1-2; n3 lt, 1 f	2569
	4 emb; r2-9	8317
	5 emb; n1 f	6419
Apodemus peninsulae	4; r1-7; n20 lt	12105
	2-8 emb, yg	5424_1
Apodemus speciosus	4.3 emb; r2-7; n18 f	12024
	4.6 emb; 3.2 spring, 5.6 fall	10632
	4.7 emb	7394
Apodemus sylvaticus	mode 2-8; r2-9	3571
	2-9	7433
	4 emb; n1 f	4403
	4 emb; n1 f	182
	4 emb; n1 f	6419
	4 emb; n1 f	2448
	4 emb ea; n2 f	7185
	4.1 emb; se0.3; n9 f	1877
	4.1; mode 4; r2-9; n10 lt	9438
	4.23; se0.10; n100 lt	3067
	4.3 emb; r2-6; n3 f	10340
	4.3 yg; se0.2; n51 primiparous	1874
	4.5-5.7; across breeding season	2990
	4.8 emb; r3-7; n25 f	9477
	4.85 emb; r2-6; n33 f	741
	mode 5-6; r1-12	4645
	5 emb; r3-7	516
	5.05 yg; se0.2; n405 lt	1874
	5.2 cl; ±0.3; n9 f	1877
	5.25; r4-7; n8 lt	955
	5.37 emb; sd1.31; r4-7; n16 f	10358
	5.4 emb	5307
	5.4 emb; r2-10; n6 f	6293

	5.4 emb; ±0.2; n18 f	3154
	5.5 emb; r5-6; n2 f	7382
	5.57 emb	8331
	5.6 emb; r5-7; n5 f	846
	5.67 emb; r2-9; n3 f	1590
	5.84; se0.147; mode 5-6; r3-10; n143 lt	8329
	6.5 emb; mode 7; 78% 5-7; r4-11; n69 f	5461
	6.6 emb; r5-7; n6 f	7779
	6.70; r4-9; n10 lt	955
Arvicanthis niloticus	2-10	4344
	mode 3; r3-10	2449
	3.0; r2-4; n2 lt	12128
	3.60 emb; se0.27; n15 f	7793
	3.7 yg; ±1.0; r1-6; n12 lt	2450
	3.7; r1-7; den emergence	8193
	3.93 emb; se0.29; n46 f	7793
	4-6 yg	5760
	4 emb ea; n2 f	2443, 2452
	4.5 emb; r3-10; n44 lt	7790
	4.64 yg; mode 6; r2-7; n14 lt	8431
	4.66 emb; mode 4; r3-10; n41 f	2446a
	4.875 emb; r3-6; n8 f	4925
	5-6	4484
	5-10	8811
	5 emb; n1 f	10799
	5 emb, yg; r3-7; n3 f	299
	5.38 emb; se0.23; n53 f	7793
	5.4; mode 4-8; max 12	3756
	6 emb; n1 f	186
	6.00 emb; r2-12; n130 f	10712
	7.125 emb; r5-9; n8 f	2454
	8 ps; r7-9; n2 f	10799
	max 8 yg	4347
Bandicota bengalensis	2 yg	11507b
	2-14	8724
	2-13 emb; n33 f	4409
	3.87-4.56 cl	9432
	4.0 emb; n3 f	8905
	mode 5-10; most of yr; mode 14-18; Sept, Nov	9175
	5.4; r1-10; n29 lt	9429
	6.10 emb; n345 f	10267
	6.2 emb; se0.67; n14 f	728
	7.4 emb; sd2.4; mode 7; r1-14; n227 f	11436
	8.88 emb (or ps?); modes 7-11; r5-13; 2 yr	5586
	8.89; n137 lt	3554
	mode 10-12	8500, 8501
	10.3; se0.67; n65 lt	5718
	14-18; Oct-Nov: 5-10; rest of yr	108
Bandicota indica	5.3 emb; r4-7; n16 f	8794
	5.49 yg; n43 lt	10821
	6.84; se0.52; r2-13; n25 lt	10370
	10 yg; n1 lt	10268
Berylmys bowersi	4.0 emb; r2-5; n3 lt	4406
Chiropodomys gliroides	1.25; n4 lt, 1 f	7159
	2.2 yg; mode 2; r1-3; n18 lt	4406
Chiruromys forbesi	3 emb; n1 f	7209
Chiruromys vates	1 yg; n1 f	7209
Colomys goslingi	2; n1 lt	4495
	2 emb; max 3; n6 f	8855
	2.0 yg; n2 lt	2574
Conilurus penicillatus	1-3	10698
	2 emb ea; 3 lt	10705
Cremnomys blanfordi	2.5; r2-3; n4 lt	1339

Cremnomys cutchicus	4.03 emb; 64.6% 3-4; r2-8	8723, 8729
	7.2 cl; mode 6-9; r3-11	8723, 8729
Dasymys incomtus	1; n1 lt	12128
	1.5 emb; r1-2; n2 f	7362
	mode 2-3; r1-4	5760
	2-4	9945
	2; n1 lt	187
	2.3 emb; r2-3; n6 f	10799
	2.5 emb; r2-3; n2 f	4329
	2.5 emb; r2-3; n2 f	4495
	2.5 emb; r2-3; n6 f	2452
	2.75 emb; max 4; n32 f	8855
	3 yg; n1 f	186
	3 emb or yg; n2 f	4925
	4 emb; n1 f	334
	4 emb; n1 f	9893
	4.5 emb; n2 f	331
	5.2 ps; r3-9; n5 lt	10799
	5.3 emb; r2-9; n4 f	10154, 10158
Golunda ellioti	mode 3	8500
	mode 3-4 yg	8501
	4 yg; n1 lt	9175
	6 emb; n1 f	8501
	6.6 emb; r5-10	8723, 8724
Grammomys cometes	2.9 emb; r2-5; n6 f	4329
	2.9 ps; r1-4; n6 f	4329
	3 emb ea; n3 f	4925
Grammomys dolichurus	70% 2-4; r1-7; n634 lt	988
	2.25 emb; max 4; n20 f	8855
	2.7 yg; r1-4; n3 lt	8218
	3 emb; n1 f	4925
	3 emb ea; n2 f	4925
	3 emb ea; n3 f	2452
	3 nestlings; n1 lt	6272
	3.6 emb, yg; r3-5; n5 lt	299
	3.8 ps; r1-8	4329
	4.0 emb; n3 f	331
	4.6 emb, yg; r4-6; n5 lt	4329
	4.67 emb; r4-6; n3 f	334
Grammomys rutilans	1 yg; n1 lt	2574
	1.3 emb; mode 1; r1-3; n12 f	2754
	2.0 emb; r1-3; n3 f	2446a
	2.1 emb; mode 2; r2-3; n7 f	4350
Grammomys surdaster	2 yg; n1 f	186
Graomys spp	6 yg; r3-10	4682
Hybomys fumosus?	4 emb; n1 f	2754
Hybomys trivirgatus	2 emb; n1 f	10799
	2 emb ea; n3 f	7362
	2.11 emb; r1-3; n9 f	2154
	2.8 emb; mode 3; r2-3; n4 f	4350
Hybomys univittatus	2 emb; max 4; n50 f	8855
	2.2 emb; r1-3; n5 f	8856
	2.24 emb [2.35]; mode 2; r1-5; n126 f, 282 emb	2573, 2574
	2.5 emb; mode 2; r1-5; n38 f	2754
	2.7 emb; r2-3; n3 f	2443
	2.7 emb; r2-3; n3 f	8854
	3 yg	3692a
	3 emb ea; n2 f	2452
Hydromys chrysogaster	3; n1 lt	2214
	3.29; ±1.26; median 3; r1-7; n131 lt, 40 f	8103
	4 yg	7209
	mode 4-5 yg; r1-6	7108
Hylomyscus aeta	4.3 emb; r3-6; n3 f	2955

Hylomyscus alleni	2.5 emb; max 3; n6 f	8855
	mode 3; r1-5	2449
	3 emb; n1 f	4495
	3 emb ea; n3 f	7362
	3.0 emb; mode 2; r2-6; n11 f	4406
	3.21 emb; mode 3; r2-4; n19 f	2446a
	3.38 emb; mode 3; r2-5; n26 f	2573, 2574
Hylomyscus denniae	4.75 emb; max 5; n6 f	8855
Hylomyscus stella	2.9 emb; mode 3; r1-4; n14 f	4350
	3 emb; n1 f	10799
	3.2 ps; n19 lt	2445
	3.4; mode 4; r1-6; n35 lt	2754
	3.57 ps; n5 lt	2445
Hyomys goliath	1 yg	7209
Kadarsanomys sodyi	max 4 yg	7672
Lemniscomys barbarus	4; mode 4; r3-6	12128
	4.5 emb; mode 3; r4-5; n4 f	10799
	5 emb; n1 f	2443, 2452
	5.5 ps; r4-7; n2 f	10799
Lemniscomys griselda	2-5	10158
	3 yg; n1 lt, 12 d	331
	3; n1 lt	10154
	4.0 emb, yg; r2-6; n3 lt	299
	5.7 emb; r1-3; n9 f	8965
Lemniscomys striatus	2.7-7.0 emb; varies with season	3645
	3.0 emb; r2-4; n5 f	4925
	3.91 yg; r2-7; n11 lt	8429
	4; mode 5; max 12	5760
	4.25 emb; max 5; n43 f	8855
	4.4 emb; r3-7; n9 f	10799
	4.7 emb, yg; n3 f	187
	4.7 emb; r1-5; n3 f	8856
	4.78 live emb; se0.17; r2-8 1st; n58 f	7790, 7792
	4.8 yg; n5 lt	2574
	mode 5; r1-8	2449
	5 emb; n1 f	2443
	5.14 emb; r3-6; n7 f	8429
	6 ps ea; n2 f	10799
Lenothrix canus	3.0 emb; mode 2; r2-6; n11 f	4406
Leopoldamys edwardsi	2 emb; n1 f	182
	14 ps; n1 lt	10823
Leopoldamys sabanus	3.1 emb; mode 3; r1-7; n54 f	4406
	5 emb; n1 f	5403
	5.0 ps; r4-6; n2 lt	10823
Leporillus conditor	3.5; r3-4; n2	6317
Lophuromys flavopunctatus	r1-16 ps; n43 f	4328
	1.67 emb; r1-2; n3 f	4925
	1.83 emb; mode 2; r1-2; n70 f	2572a, 2573, 2574
	1.99 emb; mode 2; r1-4; n107 f	2572a
	2.0 emb ea; n6 f	2443
	2.0 emb; r1-3; n18 f	8856
	2 emb; max 4; n105 f	8855
	2.16 emb; mode 2; r1-4; n38 f	299
	2.16 emb; mode 2; r1-4; n275 f	2572a
	2.17 emb; mode 2; r1-3; n53 f	2445
	2.17 emb; n276 lt; r2.00-2.47; by month	2570
	2.20 yg; mode 2; r2-3; n5 lt	2445
	2.3; r2-3; n6 lt	2452
	2.33; r2-3; n3 f	337
	2.4 emb; mode 3 51%; r1-4; n43 f	4328
	2.5 emb; r2-3; n2 f	186
	2.5 emb; r2-3; n2 f	8854
	2.5 emb; r1-3; n4 f	10231
	3 emb; n1 f	187

	3.0 emb; n2 f	331
	3.0 yg ea; n2 lt	299
	3.7 emb; n6 f	4495
	4 emb; n1 f	187
	7.0 emb; n2 f	10231
Lophuromys luteogaster	1 emb; max 2; n5 f	8855
	2.0 yg; n10 lt	2572a, 2574
Lophuromys nudicaudus	1.6 emb; r1-2; n5 f	2572a
Lophuromys rahmi	2.0 emb; n2 f	2572a
Lophuromys sikapusi	2 emb; n1 f	2443, 2452
	2 emb; n1 f	2953
	2.13 emb; mode 2; r2-3; n8 f	7362
	2.5 emb; r2-3; n6 f	2154
	3 yg	3692a
	3 emb ea; n2 lt	10799
	3.0 emb; mode 2; r2-5; n13 f	4350
	3.24 emb; mode 3; r1-5; n46 f	2446a, 7790
	4.0; r3-5; n2 lt	1771
	4.7 ps; r4-5; n3 lt	10799
Lophuromys woosnami	2 emb ea; n3 f	2443, 2452
	2 emb; n20 f	8855
Lorentzimys nouhuysi	2 emb; n1 f	7209
Malacomys edwardsi	2.0 emb ea; n4 f	7362
	2.5 emb; r2-4; n4 f	2154
	3 emb; n1 f	10799
Malacomys longipes	2.5 emb; mode 2-3; r1-4; n12 f	2754
	3 emb; n1 f	331
	3 emb; n1 f	2452
	3 emb; r1-5; n27 f	8855, 8856
	3.00 emb; mode 2; r1-5; n41 f	2573, 2574
Mallomys rothschildi	1 emb ea; n2 f	3294a
Mastacomys fuscus	1 yg; n1	9194
	2	9716
	2.0; r1-3; n3 lt, 1 f	1542
	2.0 ea; n7 lt	4042
Maxomys inas	3 emb; n1 f	7161
Maxomys rajah	3.3 emb; mode 3; r2-5; n19 f	4406
Maxomys whiteheadi	3.0 emb; mode 3; r1-6; n57 f	4406
Melomys burtoni	2	10917
	2.5; max 4; n105 lt	3606
	2.85 emb; r1-4; n20 f	7053
Melomys cervinipes	2.7 yg; n3 lt	1255
	2.8; r2-3; n11 lt	2309
Melomys leucogaster	2 yg	7209
Melomys levipes	1-2 yg	7209
	1.45 emb; 1.09 live emb; mode 1; r1-2; n11 f	2836
Melomys lorentzi	1	7209
Melomys moncktoni	2 yg; n1 f	7209
Melomys platyops	2 emb; n1 f	6436a
Melomys rubex	mode 2 yg	7209
	2.00 emb; 1.60 live emb; mode 2; n5 f	2836
Melomys rufescens	mode 2 yg	7209
	2.14 emb; r2-3; n7 f	6436a
	2.27 emb; 1.91 live emb; mode 2; r1-4; n11 f	2836
Mesembriomys gouldii	1.75; r1-3; n4 lt	2131
	2	2214
	2	10917
	2 yg; n1	9194
Mesembriomys macrurus	2	2214
Micromys minutus	2.68; n16 lt	5553
	mode 3-6; r3-12	3571
	3.77 yg; r2-6; n35 lt	3461
	4; n1 lt	2565
	5-9 yg	5424,

	5.40; ±0.16; n62 lt	4396
	6.0 emb; r5-7; n2 f	5968
	max 12	7433
Millardia gleadowi	2.3; r2-3	8723, 8724
Millardia meltada	3.44 yg; r1-8; n27 lt	940
	4.1; r1-10; n21 lt	12034
	5.03; r3-7	8904
	5.53 emb; r3-9; n32 f	730
	5.62; n101 lt	3554
	5.9 yg; r4-10	8723, 8724
	6-8 yg	8500, 8501
	6 ps; n1 f	7382
	6; n1 lt	9175
	6 yg; r5-8; n6 lt	940
	6.15; r4-9	8904
	6.78 ps; r3-11; n23 f	730
	8	107
Mus spp	2-4.5 emb	3645
Mus booduga	4 emb; n1 f	7382
	4-8	9175
	5.58 yg; mode 3-4; r2-11; n96 f	1688
	6-10	8501
	6.1 emb; mode 6; r2-12; n199 f	1688
Mus bufo/minitoides/triton	4 emb; max 7; n40 f	8855
Mus cervicolor	4.4; r2-6	8723
Mus domesticus	1-8	6774
	4.4; ±0.09; n180 lt	4031
	5.25 emb; r3-6; n4 f	10616
	5.33 ps; r4-6; n3 lt	10616
	10.9 yg; se0.7	6864
Mus gratus	3 emb; n1 f	185
	4; n1 lt	187
	4 emb; n1 f	4495
	5 yg; n1 nest	4495
Mus indutus	4.9 emb; r2-8; n17 f	10158
Mus mayori	3 emb; n1 f	8500, 8501
	3 yg; n1 nest	8501
Mus minutoides	2.0 emb; r1-3; n2 f	7362
	3 yg; n1 f	334
	3 yg; n1 f	186
	3.0; r2-4; n4 lt	5685
	3.00 yg; mode 3; r1-5; n18 lt, 7 f	245
	3.38 emb; mode 2; r2-6; n13 f	245
	3.5 emb; r3-5; n4 f	4350
	3.6 yg; n5 lt	2574
	3.85 ps; mode 3; r2-6; n33 f	245
	4 yg; n1 lt	329
	4.0 emb; r2-7; n6 f	9893
	4.0 yg; r1-7; n27 lt, 12 f	11711
	4.3 emb; r4-5; n3 f	2443, 2452
	4.3 ps; r3-6; n12 f	10799
	4.5 emb; n2 f	331
	4.5 emb; mode 4; r2-10; n13 f	10799
	4.67 emb, ps; n3 lt	4329
	4.9; r2-8; n17 lt	10154
	5 emb; n1 f	184
	5.67 emb; r4-8; n3 f	10158
	6.0 yg; r3-8; n4 lt	187
	6.75 emb; r5-10; n4 f	9892
	7 emb; n1 f	187
Mus musculus	1.63 ps; n63 f	1246
	2.7 emb; 1.7 healthy emb; n3 f	6432
	3-8	4344
	3.9 emb	6855

	4.29 yg; n55 lt, 21 f	2725
	4.3 emb; mode 5; r1-7; n82 f	4406
	4.5 emb; sd1.3; n150 f	11437
	4.7-6.3; n244 lt; monthly	844, 845
	4.8 emb; se0.09; n292 f	10151
	4.9 ps; n24 f	6432
	5 emb; n1 f	8907
	5 emb; n1 f	2341
	5 yg; n1 f	11956
	5 emb ea; n2 f	4409
	mode 5-6; r4-8	8500, 8501
	5.0-10.7 ps; n58 f; monthly	844, 845
	5.2 yg; n13 lt, 6 f	8102
	5.3 yg; 95%ci 5.1-5.5; 5.2 wean; 95%ci 5.0-5.4; n205 lt	2459
	5.37 emb; se0.10	9323
	5.4; ±0.04; n1400 lt	4031
	5.7 emb; n4 f	1547
	5.7 emb; n~550 f	10233
	5.76 emb; n582 f	6254
	5.9 emb; mode 6; r1-10; n48 f	1702
	mode 6 emb; r3-9; n10 f	1702
	6.0	10632
	6.0 emb; r5-7; n2 f	6298
	6.1 emb; ±1.5; r4-9	8804
	6.12 yg; sd2.32; mode 6-7; r1-12; n431 lt	11490
	6.17; r5-9; n6 f	2547
	6.3 ps; r5-8; n6 f	6298
	6.5; n41 lt	8768
	6.9 cl; n8 f	6432
	7.5; r5-9; n4 lt	1785
	7.5 emb; n8 f	847
	max 8 yg	4347
	9.1; se0.7; r1-9; n16 lt	1315
	9.78	7704
	r12.3-14.3; n2624 lt; across months	2723
Mus platythrix	3-10 emb	8723, 8724
	3.8 yg; r2-7; n15 lt	8924
	4.56 emb; mode 4,6; r2-8; n22 f	8924
	7.6 yg; mode 7; r2-12; n87 f	1688
	8.3 emb; mode 6,8,12; r4-14; n83 f	1688
Mus setulosus	3.25 emb; mode 3; r2-5; n8 f	2754
	5 emb ea; n3 f	2955
Mus spretus	5.3 emb; sd1.37; mode 5; r2-10; n193 f	11222a
Mus tenellus	2 emb; 1 f	4344, 4345
Mus triton	4.5 emb; mode 5; r2-6; n14 f	7790
	6; r5-7	4329
Mylomys dybowskii	4.0 emb; r3-5; n3 f	2443, 2444
	4.32 emb; r2-6; n28 f	2446a, 7790
Nesokia indica	r2-8 emb; n8 f	6293
	4.0; r1-7; n2 lt	12128
	4.07 emb; se0.54; n13 f	728
	4.18; n101 lt	3554
	7 emb; n1 f	10616
Niviventer cremoriventer	3.7 emb; mode 3; r2-5; n10 f	4406
Niviventer eha	5 emb; r3-7; n2 f	7382
Niviventer fulvescens	2.5 emb; r2-3; n2 f	182
	5 emb; r3-6; n3 f	7382
Niviventer niviventer	mode 1-7	12105
	5.0 recent ps; r4-6; n2 lt	10824
	6 emb; n1 f	7382
Notomys alexis	mode 3 emb; r2-5	3258
	3.0; sd1.0; mode 3; n23 lt	4355
	4.0; sd0.96; mode 4-5; n14 lt	4355
	4.0; mode 4; r1-9; n176 lt	10117

	4.04 yg; mode 4; r2-8; n3 f	1255
	4.4; n8	2532
	4.6 emb; se0.3; n150 f	1255
Notomys cervinus	1-4 yg; 50% of 3; n8 lt	2132
	2.5	670
	3.5; sd0.84; mode 4; r2-4; n6 lt	4355
Notomys fuscus	2.7; mode 3; r1-5; n74 lt	427
Notomys mitchellii	2-3	12128
	3	670
	3.6; sd0.85; mode 4; r2-5; 9 lt	4355
	4	10917
	4; r4-6	10917
Oenomys hypoxanthus	2.25 emb; max 4; n54 f	8855
	2.3 emb; r1.85-2.71; n130 f; across months	2570
	2.5 emb; mode 3; r1-5; n33 f	2754
	2.61 emb; mode 2; r1-5; n13 f	2573
	2.71 emb; mode 3; n14 f	4495
	3-4; max 6	5760
	3; n1 lt	187
	3.17 yg; mode 2-3; r1-5; n18 lt	2573, 2574
	3.4 emb; r3-4; n9 f	2452
	5-8	9468
Pachyuromys dupresi	5; n1 lt	8614
Pelomys fallax	2-3 yg	5760
	3 yg; n1 f	5051
	4 emb; max 8; n21 f	8855
	5 emb; n1 f	334
	7.9 emb; r6-9; n12 f	9893
	8.0 emb; n2 f	331
Pelomys minor	2-3 yg	5760
Phloeomys cumingi	1	9562
Pithecheir parvus	mode 2 yg	6448
Pogonomelomys sevia	mode 1 yg	7209
	1.00 emb, 0.75 live emb; mode 1; r1; n4 f	2836
Pogonomys macrourus	2.45 emb, 2.36 live emb; mode 2; r2-3; n11 f	2836
	4.00 emb/ps; sd2.00; n3 f	7114
Pogonomys mollipilosus	2.45 emb, 2.36 live emb; mode 2-3; r2-3; n11 f	2836
Pogonomys sylvestris	2.64 emb, 2.45 live emb; mode 3; r2-3; n11 f	2836
Praomys daltoni	3.3-6.5 emb; varies with season	3645
	3.93 yg; 4 mode; r1-7	246
	5.55 emb; mode 5; r3-10; n18 f	245, 246, 247
	6.0 emb; r5-7; n4 f	10799
	7 ps ea; n2 f	10799
Praomys delectorum	3.67 emb; r2-6; n12 f	5051
	4 emb ea; n3 f	337
	6 yg; n1 lt	5051
Praomys derooi	3.75; r2-5	4354a
Praomys erythroleucus	4.0; n2 lt	12128
	7-15	5058
	11.8 emb; r7-17; n17 f	1215
	12.1 cl; r5-19; n51 f	1215
Praomys fumatus	3.0 emb; r2-4; n2 f	4925
Praomys huberti/erythroleucus	14.7 emb; max 28; n21 f	3646
Praomys jacksoni	78% 2-4 ps; r1-9; n18 f	4329
	2-6	5760
	2 yg; n1 lt	187
	mode 3; r1-8	2449
	3.0 emb; r2-4; n6 f	8856
	3.19 emb; mode 3; r1-5; n99 f	2573, 2574
	3.75 emb; max 8; n65 f	8855
	3.75 emb; n102 f	2570
	3.8 emb; mode 4; r2-5; n11 f	2754
	mode 4 emb	4329
	4 emb; n1 f	334

	4.0 emb; n3 f	4495
	5 emb; n1 f	331
Praomys morio	mode 3-4 yg; r2-6	2443, 2444, 2452
	3 emb; n1 f	2953
	3.0 yg; mode 4; r1-4; n9 lt	2445
	3.25 emb; mode 3; r1-5; n44 f	2754
	3.33 emb; mode 3; r2-5; n67 f	2445
	4.02; mode 4-5; r2-6; n61 lt	2954
	4.4 emb; r3-5; n5 f	2443, 2444, 2452
Praomys natalensis	2-4	9945
	4.25; r1-8; n4 lt	12128
	4.7 emb; max 12; n33 f	8855
	5 emb; n1 f	10231
	5.1; r1-13; n28 lt	4588
	5.4; sd2.9; n5 lt	8385
	mode 6-12; max 22	10158
	6.5 yg; r6-7; n2 nests	334
	7 ps; n1 f	10799
	7.32	8091
	7.5 ps, emb; sd0.9; mode 7; r7-9; n6 f	3821
	8 emb; n1 f	334
	8; max 16	8034
	8.0 emb; r6-11; n4 f	6612
	8.1 emb; r2-16; n30 f	299
	8.2 emb; r7-9; n5 f	2454
	8.4 weaned; r2-14; n16 lt	7166
	8.53 yg; mode 7; r1-17; n19 lt, 4 f	7166
	9.0 emb ea; n2 f	2154
	9.2 emb; r3-16; n5 f	4495
	9.4 emb; r7-13; small f	4329
	9.5; ±0.57; r6-15; n19 f	9892
	9.5 emb	1954
	9.97; r7-22; n34 lt	10154
	10.0 emb; r9-11; n2 f	7362
	10.3 yg; r4-21; n57 lt	4579
	10.4 emb; r7-13; n5 f	4925
	11.0 emb; se0.46; r2-20; n72 f	9893
	11.2 emb; r3-16	1704
	11.27 yg; sd2.28; r8-17; n11 lt	514
	11.63 emb; r5-17; n41 f	8965
	11.99 emb; r5-24; n76 f	10712
	mode 12; r1-19	2449, 2452
	12.0 emb; r9-14; n3 f	10799
	12.1 emb; se0.26; r6-19; n41 f	7790, 7791
	12.4 emb; r1-20; n95 f	1789
	12.8 emb; r8-21; n43 f	1788
	13.4 emb; r7-19; n34 f	9014
	13.6 emb; r11-17; large f	4329
	14.0 emb; r13-15; n2 f	184
	15 emb; n1 f	187
Praomys tullbergi	2.71 ps; r1-4; n7 f	10799
	3.0 yg; mode 3; r1-6; n103 lt	4353
	3.4 emb; mode 4; r2-5; n16 f	10799
	3.45 emb; r2-4; n15 f	2154
	3.46 emb; mode 3; r3-4; n11 f	7362
	3.6 emb; r2-6; n36 f	4350, 4353
	3.67 emb; r3-4; n3 f	4925
	4 yg; n1 f	186
Pseudohydromys murinus	1 emb; n1 f	6436a
Pseudomys albocinereus	3.8; sd0.96; mode 4; r2-5; n21 lt	4355
	4 yg; n1	9194
Pseudomys apodemoides	3-6 yg	3259
	4-7 emb	3259

Pseudomys australis	2.5 yg; r2-3; n2 f	11495
	3.0; sd0.52; mode 3; r2-4; n10 lt	4355
	3.6; mode 4; r1-7; n140 lt	10117
Pseudomys delicatulus	3 yg; n1	9194
	3.25 emb; r3-4; n8 f	10704
Pseudomys desertor	3.0; sd0.66; mode 3; r1-4; n24 lt	4355
Pseudomys gracilicaudatus	2-4 cl	10708
Pseudomys hermannsburgensis	2 yg; n1 f	11495
Pseudomys higginsi	2.3; r2-3; n3 lt	4042
Pseudomys nanus	3 yg; n1	9194
Pseudomys novaehollandiae	2.5; r2-3; n2 lt, 1 f	4355
	4.0; sd1.03; r1-6; n95 lt	5643
	4.56 emb; r2-6	5645
	5 emb; n1 f	9194
Pseudomys praeconis	3.5; r3-4; n2 lt	9189
Pseudomys shortridgei	3 yg ea; n6 lt	4355
Punomys lemminus	2 emb; n1 f	7983
Rattus argentiventer	6.0 emb; r5-7; n9 f	4404
	6.94 yg; se0.19; mode 7; r1-18; n395 lt	6162
	10.81 ps; se0.29; 10.63 emb; se0.29; r1-18; n118 f	6162
	11.06 ps; se0.31; 10.87 emb; se0.31; r2-18; n103 f	6162
	11.50 ps; se0.25; 11.30 emb; se0.25; r4-18; n129 f	6162
	12.70 ps; se0.55; 12.50 emb; se0.56; r4-18; n40 f	6162
Rattus exulans	2.5 live emb; n201 f	5216
	2.5-4.5 yg	5217
	3; r1-5	7209
	3.0 emb; r1-5; n9 f	7886
	3.1; r1-6; n14 f	6436a
	3.12 emb; sd0.70; 2.88 live emb; sd1.05; n17 f	2839
	3.2 ps; r3-4; n5	10702
	3.35 emb; 3.09 live emb; mode 3; r1-5; n75 f	2836
	3.43 emb; sd0.65; 3.14 live emb; sd0.94; n51 f	2839
	3.6 yg; sd0.21; r2-6; n30 lt	1330, 11434
	3.66 emb; sd1.06; 3.50 live emb; sd1.30; n50 f	2839
	3.69 live emb; sd1.11; n23 f	7114
	3.8 emb	10578
	3.8 yg; mode 4; r1-10; n221 lt, 37 f	2911
	3.81 emb; mode 3-4; r1-8; n87 f	11832
	3.9; r3-6; n7	4821
	3.96 emb; sd1.24; mode 4; r1-9; n152 f	11434
	3.97, 4.07; n59 lt, 2 groups	11833
	3.97 yg; mode 4; r2-7; n212 f	11832
	3.98 emb; ±1.05; mode 4; r2-6; n62 f	4408, 4409
	4.0 emb; 3.7 live emb; n7 f	10650
	4.0 emb; r1-7; n25 f	10702
	4.07 emb; r2-19	10545
	4.07 ps/emb; mode 3-4; r1-11; n59 lt	11832
	4.3 emb; mode 4; r1-8; n98 f	4406
	4.48 ps; mode 3-5; r2-11; n125 f	11832
	4.5 emb; mode 4; r1-7; n38 f	4404
	4.61 emb, ps; sd1.37; n13 f	7114
	4.7 emb; mode 5; r2-6; n10 f	11477
	4.7 emb; n~140 f	10955
Rattus fuscipes	3 yg; n1	9194
	3.81 yg; mode 4; r1-7; n79 lt	10696
	4 yg; n1 lt	3260
	4.2; r3-7	9188
	4.21 yg; n24 lt	10696
	4.7 emb, yg; r2-7; n45 f	4988, 4990, 10707
	5.0 emb, ps; r3-7; n11 f	10709
	6 emb ea; n2 f	3260
	6.0 cl; r2-13; n16	10709
	6.1 new cl; n16	1254
	6.2 cl; r1-13; n68	4990, 10707

	7 cl; r3-11	10696
Rattus hoffmanni	4-5 emb	7675a
Rattus hoxaensis	5 ps; n1 lt	10824
Rattus leucopus	2-5	7209
	3 cl; n1	10709
	3.35 emb; r2-5; n22 f	10702
	3.5 new cl; se0.4; n5	1254
	5 emb; n1 f	9194
Rattus lutreolus	3.85; r3-5	4041
	3.9 emb; sd1.0; r1-6; n16 f	1208
	4.1 emb; sd0.9; r3-6; n11 f	4041
	5.0 emb; sd2.0; r1-11; n35 f	1208
	6 yg; n1	9194
	7.0 new cl; se1.0; n5	1254
	12.0 ps; sd4.2; r7-24; n21	1208
	13.4 ps; sd8.2; r2-31; n68	1208
Rattus mindoriensis	4-9 emb	3025
Rattus muelleri	3 emb; n1 f	2313
	3 emb; n1 f	2313
	3.8 emb; mode 3; r1-9; n28 f	4406
Rattus niobe	2; r1-3	7209
	2.0 emb; r1-3; n9 f	10702
	2.00 emb; 1.91 live emb; mode 2; r1-3; n11 f	2836
	2.7 ps; r2-4; n3	10702
Rattus norvegicus	2 emb; n1 f	4483
	3-9 emb; n9 f	4409
	5-14	2555
	5-20	7433
	5.2; r3-8; n6	4821
	5.8	2455
	6.4; se1.0; n20 lt	1849
	6.84 emb; n270 medium f	10456
	7 emb; n1 f	1590
	7-8	8393
	mode 7-8; r5-12	4645
	7.0-9.1; r of x	7033
	7.9; n83	2317b
	8.09 yg; sd2.81; n11 f	956
	8.2 emb; se0.309; n59 f	2317a
	8.4 emb; n402 f	2317b
	8.7 emb; sd2.1; r5-12; n21 f	11435
	8.83 emb; ±2.62; r5-17; n18 f	6242
	8.9 emb; n288 f	2317b
	mode 9-10	9175
	9.0; n216 lt	4485
	9.0; r8-12; n7	8802
	9.00 emb; n2768 large f	10456
	9.036; r1-17; n775 lt	6399
	9.28 emb; se0.28; n105 f	2321
	9.3 emb; r3-16; n22 f	7886
	9.35 emb, ps; 61% 7-9; r2-19; 10.39 cl; r7-17; n17 f	6242
	9.5 emb; r7-12; n4 f	2547
	9.6 emb; r6-13; n95 f	4234
	9.69 emb; sd2.66; n29 f	956
	9.9 emb; 9.0 late gest emb; n212 f	2996
	9.9 emb; n345	2317b
	10 emb; n1 f	5111
	10.0 ps; r2-19; n12 lt	6242
	10.1 emb; se0.188; n213 lt	2317a
	10.4 yg; n32 f	5503
	10.5; n8 lt	6238
	11.2 cl; n95 f	4234
	max 12	7209

Rattus praetor	3 emb; r2-5	6436a
	3 emb; r1-6	7209
	3.10 emb; 2.60 live emb; mode 1-4; r1-6; n10 f	2836
	4.5 emb; r2-7; n4 f	10702
	5 yg; n1 lt	10702
Rattus rattus	1-9	8723
	1-9	8723
	2.65; mode 3-4; r1-5; n46 lt	8861
	3-4; r3-8	8501
	3-8 emb; n15 f	4409
	3.6 live emb; n97 f	5216
	4 emb; n1 f	1590
	mode 4-9; r3-15	3571
	4.17 yg; r1-9; n6 lt	9265
	4.49 emb; se0.10; r1-8; n176 f	899
	4.83 emb; r2-9; n6 f	9265
	4.9; r3-7; n8	4821
	5 emb; n1 f	5403
	5-10	2555
	5 emb; r3-6; n6 f	7886
	5.2 emb; n19 f	10650
	5.25 emb; r1-11; n357 f	4404
	5.4	2455
	5.6 emb; n118 medium f	10456
	5.67 emb; r4-9; n3 f	10616
	5.8 emb	8423
	5.88 yg; se0.59; n99 lt	1765
	5.89 emb (6-8 cl); se0.31; n194 f	1765
	5.89 emb; n3015 f	11632
	5.9; n21 lt	858
	6	7209
	mode 6-7; max 17	5760
	6-8 emb	10823
	6 emb; n54 f	2452
	6.05 emb; se0.23; mode 6; r1-11; n74 f	11475a
	6.1; ±0.233; r1-15	11475
	6.4; r4-9; n9 lt	5315
	6.44 wean; r1-9	10545
	6.60; se0.18; mode 5-7; r1-13	8902
	6.7 emb; r4-8; n6 f	10702
	6.77 emb; r3-11; n13 f	2590
	6.8 emb; n99 f	2316
	6.9; ±0.147; r1-16	11475
	7 emb; n1 f	182
	r7-8 emb; n3 f	10824
	7.2 yg; ±0.5; r4-10; n13 lt, 9 f	2091
	7.3 emb; n978 large f	10456
	7.5 emb; ±0.181	11475
	8-20	7433
	mode 8; r3-10; n13 lt	3133
	8 ps; n1 f	10616
	9 emb; n1 f	185
	9 emb; n1 f	7498a
	9.23 emb; r2-18	10545
	9.5 emb; r9-10; n2 f	1590
	10.5 yg; r9-12; n8 lt	6237
Rattus sordidus	1.5 emb; r1-2 n4 f	7209
	4.5 ps; r4-5; n2	10702
	5.8 emb; r5-9; n14 f	10702
	6 yg; n1 f	11495
	6 emb; max 13	7053
	7.4 new cl; n19	1254
Rattus steini	2.7 emb; r1-4; n7 f	10702
	3.4 ps; r2-5; n8	10702
	4 yg; n1 lt	10702

	4.67 emb, ps; sd1.37; n6 f	7114
Rattus taerae	4.5 emb; r4-5; n2 f	4925
Rattus tiomanicus	3.0 yg; n8	11885
	3.7 yg; n24 nests	11885
	3.98 emb; r1-10; n82 f	4404
	4.39 emb; mode 4; r2-7; n442 f	11885
Rattus tunneyi	5.5 new cl; se0.6; n5	1254
	7.5; r4-11; n263 lt	3606
Rattus turkestanicus	3.8 emb; r2-6; n5 f	7382
	4-6 emb	9175
Rattus verecundus	2.13 emb; mode 2; r1-3; n15 f	2836
	2.3 emb; r2-3; n3 f	10702
Rhabdomys pumilio	r2-31 cl; n20 f	2530
	3-10	1814
	4-12	5760
	4.0 emb; r3-5; n2 f	4925
	4.3 emb; r3-5; n3 f	2452
	4.36 emb; ±0.24; n42 primiparous f	2298
	4.5; ps: 5.0; r2-7	4329
	4.75 emb; r3-6; n4 f	299
	4.90 healthy emb	2298
	5.0; r3-9; n11 lt	10154
	5.00 emb; ±0.12; n85 parous f	2298
	5.43 emb; r2-10; n80 f	10712
	5.75 emb; r4-7; n4 f	337
	5.9; r2-9	10158
	6-7	9945
	6 emb; n1 f	334
	6.19 yg; n20 lt	8384
	6.5 yg; r3-10; n42 lt	1335
	7 emb; n1 f	7822, 7823
Stochomys defua	2.75; mode 3; r2-3; n4 lt	7362
Stochomys longicaudatus	2 emb; r1-4; n18 f	8855
	2.4 emb; mode 2; r1-4; n30 f	2573, 2574
	2.5 emb; r1-4; mode 2; n42 f	2754
	3 yg ea; n2 lt	4354a
Thallomys paedulcus	2 yg; n1 f	4329
	2.5 emb; r1-4; n2 f	9945
	2.7 yg	7822
	2.7; n23 lt	7168
	3 yg; n1 f	9945
	3.6; r2-5; n5 lt	10154, 10158
	4 emb; n1 f	4925
	4 emb; n1 f	4329
	4 yg; n1 f	1836
	7 emb; n1 f	9893
Thamnomys venustus	1.42 emb; r1-2; n12 f	2446a
Uranomys ruddi	2.6-5.7 yg	7983
	2.9; r1-4	4350
	3.96 emb; n112 f	752a
	4 ea; n3 lt	4350
	5 emb; r4-6; n4 f	10799
	8 ps; n1	4329
Uromys caudimaculatus	1 yg; n1	9194
	1-3	7209
	1-3	10698
Vandeleuria nolthenii	4 yg ea; n2 nests	8501
Vandeleuria oleracea	3; 4 occasional	8501
	mode 3-4	8500
Xeromys myoides	2	10698
Zelotomys hildegardeae	5 emb; n1 f	2444
	5 emb; n1 f	2452
	7 emb; n1 f	4925
Zelotomys woosnami	4.7; r4-5; n3 lt	947

	5.0	10154
	7.0 emb; r5-11; n3 f	10158
Zyzomys argurus	2.7 emb; sd1.2; r1-6; n17 f	1192
	4.3 ps; sd2.8; r1-11; n24	1192
SEXUAL MATURITY		
Acomys cahirinus	f: vag perf: 45 d; ±2.7; n26 f; 36.2 g; ±1.66	8326
	f: 1st mate: 39-50 d	2568
	f: 1st birth: 103 d; ±4.04	8326
	f: 80-84 mm	4329
	m: sperm present: 56 d; n1; 90-94 mm	4329
	m: 46 d	1080
	m: 2-3 mo	2568
	f/m: 2-3 mo	5760
Aethomys chrysophilus	f: vag perf: 56-70 d	1333
	f: 1st birth: 138 d; r108-187	1333
	f: 110-119 mm	4329
	m: 120-129 mm	4329
Aethomys hindei	f: vag perf: 70 d; ±3; r54-84; r101.9-108.2 g	8083
	m: scrotal testes: 44-51 d; n5; r54.9-109 g	8083
Apodemus agrarius	f: 1st cl: 15 g; 50% 18.16	8330, 8333
	m: testes large: 17 g; 50% 17.79	8330, 8333
	m: sire yg: 30% 16.1-18.0 g; 100% 22.1	8332
	f/m: 1st mate: 2.5 mo	4645
	f/m: > 22 g	8343
Apodemus argenteus	f/m: 2+ mo	3546
Apodemus flavicollis	f: 1st cl: 10 g	54
	f: 1st cl: 50% 23.81 g	8333
	f: ~1 mo; 14-16 g	10358
	m: testes large: 50% 26.96 g	8333
	m: testes large: ~1 mo	10358
	m: sperm present: 15-20 g	54
	m: 1 yr	6548
	f/m: 7-8 wk	3571
Apodemus microps	f: 1st cl: 50% 12.24 g	8333
	f: 1st preg: 16 g; rarely 10.1-14	10480
	f: 18 g; active	10358
	m: testes large: 50% 14.73 g	8333
Apodemus speciosus	f/m: season of birth	12024
Apodemus sylvaticus	f: 1st cl: 50% 16.05 g	8333
	f: 1st birth: 80-90 d	4645
	f: 16 g	10358
	m: est 14 g	516
	m: testes large: 60-65 d	3070
	m: testes large: 21% 13.0 g; 91% 21	8333
	m: 1 yr	6548
Arvicanthis niloticus	f: 1st repro: 40-50 d	8811
	f: 1st birth: 60 d	2450
	f: 3 mo	4484
	f: 20-35 g	7793
	f: 45 g	10712
	f: 55-65 g	7793
	m: scrotal testes: 40-50 d	2450
	m: testes large: ~6 mo	7600
	m: 33-40 g	7793
	m: 60-80 g	7793
Bandicota bengalensis	f: 1st birth: 3 mo	9175
	f: 60 d	9429
	f: vag perf: 50% 40-79 g; 100% > 90	3554
	f: vag perf: 90-100 d	10268
	m: scrotal testes: 51 d	9429
	m: testes descend: 58-70 d; n3	10268
	m: visible epididymal tubules: 70-159 g	3554

Bandicota indica	f: ovarian mass: 350 g	8794
	f: vag perf: 190-200 d, ~300 g	10268
	f: 1st birth: 130-415 d	10370
	m: epididymal sperm: 50% 150 d	10370
	m: testes descend: 160 d; n1	10268
Berylmys bowersi	f: 1st preg: 290 g	4406
	m: fertile: 50% 166.5 g; 95% 338.1	4406
Chiropodomys gliroides	f: 1st preg: 11 g	4406
	m: fertile: 50% 7.2 g; 95% 165	4406
Dasymys incomtus	f: fecund: 127 mm	4329
	f/m: 60-80 d	2570
Grammomys cometes	f: 1st preg f: 105-109 mm	4329
	f: 1st multiparous: 115-119 mm	4329
Grammomys dolichurus	f: 1st estrus: 58-69 d	988
	f: 1st mate: 75-94 d	988
	f: ps present: min 95-99 mm; mode 105	4329
Grammomys rutilans	f: 1st preg: 40 g	4350
	m: scrotal testes: 40-45 g	4350
Hybomys trivirgatus	f: 1st preg: 40 g	2154
Hybomys univittatus	f: 1st mate: 36-90 d; most 4.5 mo	2446a
	f/m: 45 g	2573
Hydromys chrysogaster	f: vag perf: ~95 d	8103
	f: 1st estrus: 124-130 d; 507.00 g; se46.59; n8	8103
	f: 1st birth: 158 d	8103
	f: 425 g	7108
	m: scrotal testes: 90-120 d; 518.67 g; se48.05; n12	8103
	m: sperm present: 130-140 d	8103
	m: 400-600 g	7108
Hylomyscus alleni	f: 1st preg: 110 g	4406
	m: fertile: 50% 67.4 g; 95% 145.4	4406
	m: scrotal testes: 12-14 g	2573
Hylomyscus stella	f: 1st mate: 38 d; n1	2445
	f: 1st preg: 15 g	4350
	m: scrotal testes: 15 g	4350
Leggadina forresti	f/m: 90 d	11493
Leggadina lakedownensis	f/m: 90 d	11493
Lemniscomys striatus	f: fertile: 3 mo	11614
	f: vag perf: 7 wk	7477, 8429
	f: 1st mate: ~1 yr	5760
	f: 1st preg: 5-9 mo	11614
	f: 1st birth: 32 g	4350
	m: fertile: 2 mo+	11614
	m: scrotal testes: 30 g	4350
	f/m: 70-90 d	2570
Leopoldamys sabanus	f: 1st preg: 190 g	4406
Leporillus conditor	f/m: 240 d	11493
Lophuromys flavopunctatus	f: 1st preg: 45 g	2572a
	m: scrotal testes: 45 g	2573
	m: testes large: 50 g	2572a
	f/m: 50-70 d	2570
Lophuromys sikapusi	f: 1st preg: 40-50 g	4350
Lophuromys woosnami	f: 1st preg: 39 g	2572a
	m: testes large: 40-44 g	2572a
Malacomys edwardsi	f: 1st preg: 61 g	2154
	m: testes large: ~60 g	2154
Malacomys longipes	m: scrotal testes: 90 g	2573
Maxomys burtoni?	f: 1st repro: 50% 52-60 g	5647
	m: spermatogenesis: 50% 49-81 g	5647
Maxomys rajah	f: 1st preg: 130 g	4406
	m: fertile: 50% 67.5 g; 95% 145.1	4406
Maxomys whiteheadi	f: 1st preg: 30 g	4406
	m: fertile: 50% 35.6 g; 95% 44.7	4406
Melomys cervinipes	f: vag perf: 80 d; se5	1255
	f: 1st mate: 7 mo	2309

Micromys minutus	f: 1st birth: 52 d; n1	3461
	f/m: 4.5-6 wk	3571
Millardia meltada	f: vag perf: 50 d; r35-66	12034
	f: 1st birth: 3-4.5 mo; r108-335 d; n6 f	940
	f: 30-49 g	3554
	f: 40 g	730
	m: epididymal sperm: 45 g	8904
	m: testes large: 50 g	730
	m: 30-59 g	3554
Mus domesticus	f/m: 4-8 wk	11217
	f: 1st birth: 177 d; ±6.2; n57	4031
Mus minutoides	f: 1st birth: 62 d; n1	11711
	f/m: 1st mate: ~42 d	10158
	f/m: 70-84 d	11711
Mus musculus	f: vag perf: 42.2 d; se1.4; n18 with males	2725
	f: vag perf: 57.2 d; se1.7; n20	2725
	f: vag perf: 68.4 d; se2.2; n17 with females	2725
	f: vag perf: 8-12 g; n258/546	9323
	f: 1st cl: 8-12 g; n323/546	9323
	f: 1st estrus: 27-47 d	2726
	f: 1st estrus: 38.1-39.9 d; n98, 5 samples	6935
	f: 1st estrus: < 8 wk	8501
	f: 1st preg: 5 g	4406
	f: 1st birth: 102 d; ±1.1; n317	4031
	f: <2 mo	7209
	f: 2.5 mo	10151
	f: 12 wk	9175
	m: scrotal testes: 8-12 g; n87/151	9323
	m: fertile: 50% 9.0 g; 95% 12.1	4406
	f/m: 1st mate: 2 mo	4645
Mus spretus	f: 1st cl 1/18 3-5 wk; 100% 5 wk)	2814
	f: 1st preg: 28-35 d	2814
	f: 6-7 wk	11222a
	m: sperm present: 3-5 wk	2814
	m: 8 wk	11222a
Mus triton	f: 70-74 mm	4329
	m: > 74 mm	4329
	f/m: 50 d	2570
Nesokia indica	f: 63-89 g	3554
	m: 110 g	3554
Niviventer cremoriventer	f: 1st preg: 40 g	4406
	m: fertile: 50% 19.0 g	4406
Notomys alexis	f: vag perf: 40-55 d	1255, 10792
	f: 1st estrus: 53.0-82.3 d; n42, 5 gps	10792
	f: 1st cl: min 44 d; mode >54 d	1255
	m: sperm present: 60 d; 22.5 g min	1255
	f/m: 1st repro: 12 wk	10117
	f/m: 2-3 mo	4355
	f/m: 80 d	11493
Notomys cervinus	f/m: 180 d	11493
Notomys fuscus	f: 1st conceive: 70 d	427
	f: 1st repro: 105 d	427
	f/m: 80 d	11493
Notomys mitchellii	f/m: 90 d	11493
Oenomys hypoxanthus	f/m: 100-130 d	2570
Praomys daltoni	f/m: 1st birth: 4.5-5.5 mo	247
Praomys erythroleucus	f: 40 g	1215
	m: 45 g	1215
	f/m: 1st breed: 4-5 mo	5057, 5058
Praomys jacksoni	f: 1st cl or ps: 90-94 mm	4329
	m: 18% 80-89 mm; 100% >109 mm	4329
	m: testes large: 30 g	2573
	f/m: 50-90 d	2570
	f/m: ~2.5 mo	5760

Praomys morio	f: 1st mate: 2.5 mo	2445
	f: 1st birth: 72-211 d	2954
Praomys natalensis	f: vag perf: 38-47 d	8034
	f: vag perf: 76 d; ±5.88; r64-145 d; n12	5366
	f: 1st birth: 77 d; n1	7166
	f: 1st birth: 99.2 d; n231	8034
	f: 35 g	7791
	m: epididymal sperm: > 40 d	514
	m: 35 g	7791
	f/m: 104 d; ~39 g	8091
Praomys tullbergi	f: 1st conceive: 62 d; 27 g	4350, 4353
	f: 1st preg: 34 g	2154
	f: 1st birth: 108 d; r86-139; n13	4353
	f: 1st birth: 4 mo	4354
	m: scrotal testes: 25 g, sometimes 15 g	4350
	m: scrotal testes: 40-60 d	4353
	m: testes large: 40 g	2154
	m: ~100 d	4353
Pseudomys albocinereus	f/m: 2-3 mo	4355
Pseudomys australis	f/m: 180 d	11493
Pseudomys desertor	f/m: 2-3 mo	4355
	f/m: 80 d	11493
Pseudomys hermannsburgensis	f/m: 90 d	11493
Pseudomys higginsi	f/m: 270 d	11493
Pseudomys nanus	f/m: 100 d	11493
Pseudomys novaehollandiae	f: vag perf: 13.1 wk; sd4.59; r7-23; n30	5643, 5644
	f: 10.5 g; 7 wk; lab; yr of birth; wild	5645
	f: 14.6 g; sd2.06; r10.5-19.0; n29	5643
	m: 1st mate: 20.2 wk; sd9.03; r11-36; n11	5643, 5644
	m: 14.275 g; r13.6-14.8; n4	5643
	f/m: 120 d	11493
Pseudomys shortridgei	f/m: 10-11 mo	4355
Rattus argentiventer	f: vag perf: 48.6 d; ±0.7; r43-53	6162
	f: vag perf: 32.9 d; ±0.4; 50% 31; r26-46; n130	6162
	f: 1st mate: r43-53 d	6162
	f: 1st preg: ~85 g	4404
	m: scrotal testes: 55% by 30 d; r26-82; n98	6162
	m: sperm present: 89.6 d; ±2.5; r85-105	6162
Rattus exulans	f: vag perf: min 48 d; mode 8 wk	11833
	f: vag perf: 40 g	11833
	f: 1st preg: 22.9 g; se2.1	4404
	f: 1st birth: mode 30+ wk; min 17	11833
	f: 1st birth: 137.1 d; r72-239	2911
	f: median 39.2 g; 95%ci 33.2-46.2	10650
	m: scrotal testes: 6-7 wk	11833
	m: fertile: 12 wk	11833
	m: 1st repro: 63 d	11833
	m: median 36.6 g; 95%ci 30.2-44.3	10650
Rattus fuscipes	f: vag perf: 21-56 d	10706, 10707
	f: vag perf: 46 d; r35-57	10696
	f: 1st ovulation: 56-66 d	10706
	f: 1st conceive: 16-17 wk	10696
	f: 1st birth: 4.5 mo	10696
	m: mature sperm: 8-9 wk	10706
	m: epididymal sperm: 10-11 wk	10706
	m: 1st sire: 2.5 mo	10696
	m: 1st sire: 17 wk	11494
Rattus leucopus	f: 1st birth: 23 wk	11494
	m: 1st sire: 16 wk	11494
Rattus lutreolus	f: vag perf: 34 d; r26-45	3437
	f: 1st conceive: 73 d	3437
	f: 1st birth: 16 wk	11494
	f/m: >100 g; immature <75 g	614
Rattus mindoriensis	f: 120 g	3025
Rattus muelleri	f: 1st preg: 220 g	4406

	m: fertile: 50% 172 g; 95% 387.7 g	4406
Rattus norvegicus	f: vag perf: 42.5 d; ±5.7; 81.1 g; ±9.2	1849
	f: 1st estrus: 45.7 d; ±6.9; 110.7 g; ±16.5	1849
	f: 1st conceive: 55.7 d; ±22.6; 103.2 g; ±23.3	1849
	f: 1st preg: 105 g	8393
	f: 1st birth: 3 mo	4645
	m: 95-104 g	6399
	m: sperm present: 42.1 d; ±1.9; 111.5 g; ±17.3	1849
	m: 1st mate: 64.6 d; ±20.5; 185.5 g; ±62.8	1849
	f/m: 2 mo	2555
Rattus rattus	f: vag pref: 50-70 d; n3	10268
	f: 1st mate: 12 wk	9175
	f: 1st conceive: 50% 80 g	3250
	f: 1st preg: 62-115 g	4404
	f: 1st birth: 95-143 d	3133
	f: 1st birth: 5-6 mo	8500, 8501
	f: 3-4 mo	11475a
	f: 1.5-6 mo	10545
	f: 7 mo, 165 g	5315
	f: median 76.7 g; 95%ci 70.9-82.9	10650
	f: 78-97 g; n3 locations	11475
	m: testes descend: 40-50 d; n3	10268
	m: sperm present: 24% 80-100 g; 74% 120-140	3250
	m: median 107.9 g; 95%ci 101.2-115.0	10650
	m: 80 g	8902
	m: 6 mo; 165 g	5315
	f/m: 3 mo	2555
	f/m: 3-4 mo	5760
	f/m: 3-6 mo	3571
Rattus sordidus	f: vag perf: 37 d; 35 g	7053
	f: 1st estrus: 63-70 d; 50-55 g	7053
	f: 1st birth: 11 wk	11494
	m: 6 wk	11494
	m: 9-10 wk	7053
Rattus tiomanicus	f: vag perf: 21-30 g	11885
	m: scrotal testes: 50% 46-59 g; r30-100	11885
Rattus tunneyi	f: 1st birth: 8 wk	11494
	f/m: 1st mate: ≤ 2 mo	3606
Rhabdomys pumilio	f: 1st mate: 5-6 wk	2298
	f: 1st conceive: 34 d	1335
	f: 1st birth: 3 mo	5760
	f: 1st birth: 90 d; r57-107; n11	1335
	f: 1st birth: 100% >119 mm; min 100-104	4329
	f: > 20 g	10712
	m: sperm: 11 wk	2298
	m: 52-58 d	1814
	m: 85 mm	4329
	f/m: 2 mo	10158
Stochomys longicaudatus	m: testes large: 75 g	2573
Thallomys paedulcus	f: 1st birth: min 107 d	7168
Uranomys ruddi	f: HB: 90 mm	752a
	m: HB: 95 mm	752a
Uromyx caudimaculatus	f/m: 180 d	11493
Zyzomys argurus	m: scrotal testes: 5-6 mo	733
	f/m: > 29 g	733
Zyzomys woodwardi	m: scrotal testes: 5-6 mo	733
	f/m: > 70 g	733
CYCLE LENGTH		
Acomys cahirinus	8-14 d	2568
	11.1 d; sd1.9; mode 11; r6-18; n116 cycles, 88 f	8325, 8326
Arvicanthis niloticus	6 d; n1 f	1997
Bandicota bengalensis	proestrus: 9.5 h; se0.75; estrus: 43.52 h; se4.33	9436
	metestrus: 24.02 h; se1.5; diestrus: 47.46 h; se5.7	9436
	4.19 d; se0.30; n24 f	9435
	5.19 d; n20 cycles	9436

Chiropodomys gliroides	10 d; sd3.7; r7-13; n6 cycles, 1 f	7159
Dasymys incomtus	7 d; n1 f	7477
Grammomys dolichurus	4-5 d	988
Grammomys rutilans	7 d; n1 f	1997
Grammomys surdaster	5 d; n1 f	1997
Hybomys univittatus	4-5 d; n4 f	1997
Hydromys chrysogaster	proestrus 1 d, estrus 2, metestrus 2, anestrus 5	8103
	10.79 d; se1.85; median 10; r7-17; n169 cycles, 38 f	8103
Leggadina forresti	7 d	7983
Lemniscomys striatus	6 d; r5-7	7477, 8429
Lophuromys sikapusi	11 d; n1 f	1997
Malacomys longipes	4-6 d	1997
Mesembriomys gouldii	26.4 d; ± 3.5; mode 27; r21-35; n21 cycles, 7 f	2131
Mus minutoides	4 d	7477
Notomys alexis	6.2 d; ±2.94; r4-8; mode 6; n42 cycles, 5 f	10117
	6.5 d; r6-7; n6 cycles, 1 f	4355
	6.8 d; se0.2; n29 cycles, 10 f	10732
	8.0; mode 7; r5-10; n30 f	2132
	8.1 d; se0.9; n29 cycles, 6 f	1253, 1275
	8.3 d; se0.2; n21 cycles, 9 f	10732
Notomys cervinus	median 20 d; r8-38; n86 cycles, 18 f	2132
Notomys fuscus	7.4 d; mode 7; r5-10; n68 cycles, 13 f	427
	8.5 d; modes 7,9; r5-13; n18 cycles, 4 f	2132
Notomys mitchellii	8.3 d; mode 7; r6-17; n69 cycles, 8 f	2132
Pelomys campanae	6.5 d; n1 f	7477
Praomys jacksoni	4-5 d	1997
Praomys morio	6 d	1997
Praomys natalensis	7-8 d	8034
	8.8 d; ±0.40; mode 6; r4-20; n83 cycles	5366
Praomys tullbergi	6-7 d	4353
Pseudomys albocinereus	prob 7-10 d; modes 6-9, 14; r6-19; n19 cycles, 6 f	4355
Pseudomys australis	8.5 d; ±1.12; mode 7; r5-21; n52 cycles, 9 f	10117
Pseudomys desertor	7.75 d; r7-9; n4 cycles, 4 f	4355
Pseudomys novaehollandiae	6.01 d; sd1.26; r4-10; n73 cycles, 25 f	5643
Rattus argentiventer	4.3 d; ±0.1; r3-6	6162
Rattus fuscipes	4.5 d; mode 4-5 99%; r3-6; n1603 cycles, 37 f	10696
	5.3; se0.2; n34 cycles, 5 f	1254
Rattus leucopus	4.8; se0.5; n33 cycles, 6 f	1254
Rattus lutreolus	5.2 d; se0.2; n40 cycles, 6 f	1254
Rattus norvegicus	4.5; se1.0; n11 cycles, 11 f; 1st cycle	1849
Rattus rattus	4.5; se0.1; mode 4; median 4.1; n1077 cycles, 68 f	2527
	10 d	5760
Rattus sordidus	4.3 d; r3-6; n132 f	7053
	5.5 d; n88 cycles, 14 f	1254
Rattus tunneyi	4.8 d; se0.1; n35 cycles, 6 f	1254
Rhabdomys pumilio	proestrus 1.1 d; estrus 4.6 d; nonestrus 6.2 d	2530
	11.0 d; mode 7; median 7.4; sd8.9; se0.4; n475 cycles, 37 f	2530
Stochomys longicaudatus	5-6 d; n2 f	1997
Uromys caudimaculatus	7 d	10698
GESTATION		
Acomys cahirinus	11 d 14 h; n1	1080
	35.875 d; mode 36; r35-37; n32	2568
	36-40 d	2449
	37-45 d; n27	1035
	37.6 d; mode 38; r36-40; n87; ibi	2566
	38 d	8326
	38 d	9356
	38.1 d; n26	2532
	39 d	12038
	39.9 d; sd1.19; r38-44; n46 lt	11397
	44.2 d; r42-46; n5	109

Aethomys chrysophilus	21-23 d	5760
	~26 d; min ibi	1333
	31 d; n1 ibi	4329
Aethomys hindei	r24-25 d; n18	8083
Aethomys kaiseri	27 d; ±1	1771
Aethomys namaquensis	> 22 d	7165
Apodemus agrarius	18 d; r18-21; min ibi	11710
	mode 21-23 d; r20-23	3571
	21-23 d	7433
Apodemus flavicollis	23-26 d	3571
	23-26 d	7433
	24-25 d; mode 28; r21-35; min ibi	5170
Apodemus mystacinus	23 d; n3	2569
Apodemus sylvaticus	21 d; n6; 2-3 d delay if nursing	3067
	mode 21-23 d; r21-25	3571
	25-26 d	2045
Arvicanthis niloticus	~18 d; via 1205 Davis 1963, ill 24 se 91	2446, 2449
	20 d	4484
	prob 21-23 d	3756
	22-24 d; min ibi	2450
	24-26 d	8811
Bandicota bengalensis	17 d	3554
	21 d	9429
Bandicota indica	20-21 d; pairing to birth	10370
Chiropodomys gliroides	19-21 d; n1	7159
Conilurus penicillatus	35 d	10698
	36 d	11493
Grammomys dolichurus	24 d	988
Grammomys rutilans	25 d	2446a, 2449
	~25 d	5760
Hybomys univittatus	29-31 d	2446a, 2449
Hydromys chrysogaster	34.87 d; se1.82; mode 34; r33-41; n38	8103
	35 d	11493
	36.64 d; se2.20; r35-41; n11 f with lactation	8103
Hylomyscus alleni	25-33 d	2449
Hylomyscus stella	29.5 d; r25-33; n6	2445
Leggadina lakedownensis	33 d	11493
Lemniscomys striatus	21 d	11614
	22.6 d; ±1.4; n29	3644
	28 d	2446a
	28 d; min ibi	8429
Leporillus conditor	44 d	11493
Lophuromys flavopunctatus	30.5 d; r30-31; n2	2445
Lophuromys woosnami	≥32 d	2572a
Mastacomys fuscus	5 wk	9716
	38-40 d	11493
Melomys cervinipes	38-40 d	11493
Mesembriomys gouldii	43.75 d; mode 44; r43-44; n4	2131
Micromys minutus	18-21 d	3571
	21 d	7433
	21 d; ibi	3461
Millardia meltada	16 d	3554
	20 d	12034
	20 d; min ibi	940
Mus caroli	18.1 d; lactating 23.1 d	10370
Mus cervicolor	17.6 d; lactating 21.0 d	10370
Mus domesticus	20.0 d; se0.5	6864
Mus minutoides	est 18-19 d	11711
	23.2 d; r22-24; n6	245
Mus musculus	18-20 d	7433
	18-21 d	9175
	19 d	8501
	19-21 d	845
	19.6 d	7704

	~20 d	4645
	31 d; n1 f nursing 8 yg	6238
Nesokia indica	17 d	3554
Notomys alexis	implant: 7-8 d	1255
	implant: 11-13 d with 3-4 yg suckling	1255
	implant: 14-17 d with 5 yg suckling	1255
	32 d	2132
	32-47 d	11493
	32-47 d; mode 35, 41; n13 sperm in urine to birth	10117
	32.5 d; se1.5; n4 f; postpartum estrus to birth	1255, 1258
	32.6 d; n8	2532
	32.67 d; r32-34; n3	4355
	34 d with 1-2 yg; 39.8 d with 4-6 yg	1255
Notomys cervinus	40 d; mode 38; r38-43; n11 f	2132
	47 d; r39-51; n8 f with lactation	2132
Notomys fuscus	32-35 d	11493
	34.3 d; r32-38; n6; vaginal sperm-birth usually > 61 d	427
Notomys mitchellii	35.5 d; r34-37; n4	2132
	34-42 d	11493
Praomys daltoni	29 d; r21-37; sd1.4; n40	247
Praomys erythroleucus	21 d	5058
Praomys jacksoni	34-37 d	2445, 2446, 2449
Praomys morio	26.5 d; r26-27; n2	2954
	36.0 d; r34-37; n4	2445
Praomys natalensis	23 d	2446a
	23 d	8034
	~23 d	5366
Praomys tullbergi	23-25 d; min ibi	4353
Pseudomys albocinereus	>23 d	9189
	38.5 d; r38-39; n2; ibi	4355
Pseudomys apodemoides	34-55 d	11493
Pseudomys australis	30-31 d; mode 31; n5; sperm in urine to birth	10117
Pseudomys delicatulus	29-30 d	11493
Pseudomys desertor	27-34 d	11493
	27.5 d; n3	4355
Pseudomys gracilicaudatus	27 d	3438
Pseudomys hermannsburgensis	30-33 d	11493
Pseudomys higginsi	31-32 d	11493
Pseudomys nanus	23-25 d	11493
Pseudomys novaehollandiae	30 d	11493
	31.5 d; sd0.96; r29-33; n25	5643
	33.2 d; sd1.75; r32-37; n8 with lactation	5643
Pseudomys praeconis	30 d	11493
Rattus argentiventer	21.4 d; ±0.2 d; r20-23; n18	6162
Rattus exulans	19 d	10545
	19-21 d	11833
	≥ 21 d	10578
	21.7 d; lactating 23.6 d	10370
	23 d; pairing to birth; 3-7 d longer with lactation	2911
Rattus fuscipes	22.8 d; mode 23 60%; r22-24; n20 cases, 15 f	10696, 10707
	28-32 d; with lactation	10696, 10707
Rattus norvegicus	20-24 d	7433
	21-22 d	2555
	22 d; n8	6238
Rattus rattus	15-20 d	8500, 8501
	mode 20-24 d; r20-26	3571
	21 d	10545
	21 d	2452
	21-22 d	3133
	21-30 d	5760
	22 d	5216
	22 d ea; n7	6237
	22 d	2555
	25 d	8861

	26 d; n2	5315
Rattus sordidus	r21-22 d; n3	7053
Rattus tiomanicus	22.0 d; lactating rats 24.3 d	10370
Rhabdomys pumilio	22 d; ibi	1814
	25.4 d; min 23	10158
	25.5 d; n6	2298
	26.4 d; r26-28; n7	2530
Uromys caudimaculatus	36 d	7983
	41 d	10698
Zelotomys woosnami	~31 d; n2 cases, 1 f; min ibi	947
Zyzomys argurus	25 d	11493
LACTATION		
Acomys cahirinus	solid food: 1 d	10481
	solid food: 5 d	8686
	den emergence: 6-7 d	2566
	independent: 14 d	5051
	wean: 2 wk	5760
	wean: 28 d; can wean: 17 d	12038
	wean: ~1 mo	8686
Aethomys chrysophilus	wean: 24 d	4329
	wean: 31-33 d	7797a
Aethomys kaiseri	teat attachment: 12 d	1771
	solid food: 12 d	1771
	wean: 26 d	1771
Aethomys namaquensis	teat attachment: 18 d	7168
	teat attachment: 3 wk	1815
	wean: 18 d	7165
	wean: 21-26 d	7797a
	wean: 27-28 d	7168
Apodemus sylvaticus	wean: ~18 d	2045
Arvicanthis niloticus	wean: 3 wk	2450
Bandicota bengalensis	solid food: ~20 d	10268
	wean: 28 d	9175
	no milk: ~30 d	10268
	leave nest: ~1 mo	108
Bandicota indica	wean: 25 d	10821
	no milk: ~30 d	10268
Conilurus penicillatus	independent: 20 d	10698
	wean: 30 d	11493
Cremnomys blanfordi	teat attachment: 1 mo	1339
Dasymys incomtus	wean: 30 d	2570
Grammomys cometes	wean: 19-26 d	10158
Grammomys dolichurus	teat attachment: 1st few d	8218
	wean: 19 d	8218
	wean: 30 d; n3	988
Hydromys chrysogaster	solid food: 3 wk	8103
	wean: 29.45; sd2.68; median 29; r17-35; n85 lt	8103
	wean: 30 d	11493
	independent: 35 d	10698
Leggadina forresti	wean: 30 d	11493
Leggadina lakedownensis	wean: 30 d	11493
Lemniscomys striatus	solid food: 14 d	11614
	wean: 19-21 d	11614
	wean: 30-50 d	2570
Leporillus conditor	wean: 40 d	11493
Lophuromys flavopunctatus	wean: ~1 mo	2445
	wean: 50-70 d	2570
Lophuromys sikapusi	solid food: 8 d	1771
	wean: 12 d	1771
Mastacomys fuscus	teat attachment: 3 wk	10698
	wean: 40 d	11493
Melomys cervinipes	wean: 37 d; se3	1255
	wean: 40 d	11493

Mesembriomys gouldii	teat attachment: 4 wk	2131
	solid food: 43 d	2131
	wean: 40 d	11493
	wean: 6-8 wk	2131
Micromys minutus	den emergence: 11-12 d	5553
	den emergence: 12-13 d	3461
	solid food: 12-13 d	3461
	solid food: 12-13 d	3571
	wean: 15 d	5553
	wean: 16 d; r15-16	3571
Mus minutoides	wean: 17 d	11711
	wean: 24 d	245
Mus musculus	wean: 21 d	9175
	wean: 3 wk	545
	wean: 3 wk	10095
	wean: 21-22 d	5501
	wean: 26 d	7704
Mus triton	wean: 25-30 d	2570
Notomys alexis	wean: 24-30 d	10117, 11493
	wean: 30 d	1252, 1255
	wean: 1 mo	2132
	wean: 33 d	4355
Notomys cervinus	wean: 1 mo	2132
	wean: 35 d	11493
Notomys fuscus	wean: 30 d	427, 11493
	wean: 1 mo	2132
Notomys mitchellii	wean: 30 d	11493
	wean: 1 mo	2132
Oenomys hypoxanthus	wean: 30 d	2570
Praomys jacksoni	wean: 30 d	2570
Praomys morio	solid food: 17-19 d	2954
	wean: 27 d	2954
Praomys natalensis	solid food: 17 d	514
	independent: ~19 d	11241
	wean: 19-21 d	7168, 7169
	wean: 21 d	2445a
	wean: 21 d	8084
Praomys tullbergi	wean: 24-25 d	4353
Pseudomys albocinereus	wean: 25 d	4355
	wean: 35 d	11493
Pseudomys apodemoides	wean: 35 d	11493
Pseudomys australis	wean: 22-30 d	10117, 11493
Pseudomys delicatulus	wean: 30 d	11493
Pseudomys desertor	wean: 20 d	4355
	wean: 30 d	11493
Pseudomys gracilicaudatus	solid food: 12-14 d	3438
	wean: 25 d; r20-35	3438
Pseudomys hermannsburgensis	wean: 30 d	11493
Pseudomys higginsi	wean: 30 d	11493
Pseudomys nanus	wean: 25 d	11493
Pseudomys novaehollandiae	solid food: 4th wk	5642
	wean: 25.2 d; r23-28; n14 lt	5642, 5643, 5644
	wean: 31.5 d; r29-33	5645
Pseudomys praeconis	wean: 30 d	11493
Rattus exulans	solid food: end 3rd wk	11693
	wean: 28 d	11693
	wean: 4 wk	4821
Rattus fuscipes	wean: 21-28 d	10696
Rattus lutreolus	solid food: 19-21 d	3437
	wean: 25 d; r23-30	3437
	wean: 25 d	4041
Rattus norvegicus	wean: est 21-25 d	2321
	wean: 4 wk	4821

Rattus rattus	leave nest: 17-23 d	3133
	solid food: 15-18 d	3133
	wean: ~25 d	9265
	wean: 28 d	2091
	wean: 4 wk	4821
	wean: mode 6-7 wk; r4-10	8861
Rhabdomys pumilio	leave nest: 14 d	1335
	wean: 14 d	7168
	wean: 14 d	4640
	wean: 16 d	1335
Thallomys paedulcus	teat attachment: 15 d	7168
	wean: 28-31 d	7168
Uromys caudimaculatus	wean: 40 d	11493
Zelotomys woosnami	leave nest: 16 d	947
Zyzomys argurus	wean: 35 d	11493
Zyzomys woodwardi	wean: 35 d	11493
INTERLITTER INTERVAL		
Acomys cahirinus	34 d	7544
Apodemus agrarius	18-21	11710
Apodemus flavicollis	mode 20-35 d; r20-126; n90	5170
	40-60 d	4645
Apodemus sylvaticus	27.56 d; mode 23-24; r18-54; n86	3067
	50-60 d	4645
Arvicanthis niloticus	20.4 d; wet season; 49 d; dry season; Rojewero Plains	7793
	22-24 d	2450
	30 d; wet season; 70 d; dry season; Crater Track	7793
	42.4 d; wet season; 68.4 d; dry season; Mweya Pen	7793
Bandicota bengalensis	61.9 d	11436
Bandicota indica	71.25 d; r40-105; n12	10370
Grammomys dolichurus	mode 28 d; r20-80; n500	988
	48 d	8218
Lemniscomys striatus	21-23 d	11614
Micromys minutus	min 21 d	3461
Mus minutoides	r22-58 d; n5	245
	22.4 d; r19-26; n11	11711
Notomys alexis	32.75 d; r32-34; n4	4355
Notomys fuscus	min 41-42 d; mode 80-89	427
Notomys mitchellii	36.5 d; r36-37; n2	4355
Praomys natalensis	25 d	8034
	26 d	8091
	4 wk	4588
	min 29 d	514
	33.1 d; ±2.5; r24-58; n15 intervals, 4 f	7166
Praomys tullbergi	23-25 d; 29 lt; modes 24,29 d	4353
Pseudomys albocinereus	38.5 d; r38-39; n2	4355
Pseudomys desertor	29.75 d; r28-34; n4	4355
Pseudomys novaehollandiae	33.5 d; sd1.29; r32-39; n18	5643
Rattus argentiventer	mode 20-25 d; r20-60+; n196	6162
Rattus exulans	74 d	5216
Rattus sordidus	22-35 d	7053
Rattus rattus	< 1 mo	3133
	31 d; r22-57; n7	6237
	31.5 d; ±0.1; r27-38; n10	2091
	50 d	5216
Rhabdomys pumilio	min 23 d	1335
	26.8-43.5 d; n39 intervals, 5 pr	2530
Thallomys paedulcus	26 d min	7168
Zelotomys woosnami	31 d min	947
Zyzomys argurus	7.2 mo; sd3.0; r3-13	733
Zyzomys woodwardi	7.6 mo; sd3.1; r2-14; prob overestimate	733

LITTERS/YEAR		
Acomys cahirinus	potential 7.6; mode 3-4	7795
Acomys wilsoni	potential 7.9; mode 3-4	7795
Aethomys namaquensis	2	11864
Apodemus agrarius	1-3	10340
	3-4	7433
	mode 3-4; r3-5	3571
	mode 3-4; r3-5	4645
Apodemus argenteus	max 4	7925
Apodemus flavicollis	2-3	7433
	2-4	4645
	mode 2-4; r2-5	3571
	max 11	5170
Apodemus mystacinus	min 2	6419
Apodemus sylvaticus	2-4	4645
	3	9438
	3; r2-4	3571
	4-5	7433
	4-5	9175
	max 6	107
Arvicanthis niloticus	3-4	4484
Bandicota bengalensis	>1	8501
	5.9	11436
	max 9	9429
Grammomys cometes	max 3	4329
Hydromys chrysogaster	1	7108
	2-3	7209
	2.55/season; ±0.97; r1-5	8103
	2.64; ±1.10; r1-5	8103
Lemniscomys striatus	1 each rainy season	2446a
	2; mode 2; r1-3	7792
Melomys cervinipes	1.3/season; ±0.77; r1-2; n18	10109
Mesembriomys gouldii	4	2131
Micromys minutus	2-3	7433
	2-3; r2-7	3571
	3	2045
Millardia meltada	9.28	8904
Mus minutoides	2-3 each rainy season	2446a
	3-8	5760
Mus musculus	3-5	8500, 8501
	4-5	4645
	4-6	7433
Mus triton	3?	10158
	max 3	4329
Praomys natalensis	2	7791
Pseudomys apodemoides	max 3	1934
Pseudomys shortridgei	1	4355
Rattus exulans	1-2; 2 rare	11832, 11833
	3.9	5216
	5.2; r1-13	2911
	max 17	10578
Rattus fuscipes	3.6; n34 f	10696
Rattus norvegicus	1	8802
	2-5	4645
	2-7	2555
	2-7	7433
	2.65	6242
	max 3	2455
Rattus rattus	2+	7433
	2-5	2555
	2-6	3571
	max 3	2455
	3	5315
	3.2	5216

	4	10544
	4+	7185
	4-5	8501
	4.2; calculated	858
	6-7	9175
Rhabdomys pumilio	4	5760
	max 10; 1 record 21 lt/27 mo	1814
Zyzomys woodwardi	86% 1 lt/2+yr; 14% 2 lt/2+yr	733
SEASONALITY		
Acomys cahirinus	fertility: Feb-July, minor peak Nov; sw Saudi Arabia	109
	fertility: 80+% Dec-Feb, Apr-May, July, 40% Oct; Kenya	7797
	conception: peak Dec-May, low June-Sept; Malawi	4329
	mate: Feb-Apr; Negev desert, Israel	715
	mate: 1-3 mo after rains; Sudan	4344
	mate: yr rd; Kenya	7795, 7796
	mate: yr rd; captive	715
	preg f: Jan, Apr, July, Oct-Sept; Kenya	4925
	preg f: Feb-Oct; few data; Egypt	8134
	preg f: peak Apr-Sept; Cyprus	11475a
	preg f: May, July-Aug; few data; Tanzania	299
	preg f: Nov-Dec; Sudan	4344
	birth: low Nov, high Apr-Oct; lab	2568
	birth: yr rd, peak Apr-Sept, n58/85; captive	1035
	birth: yr rd, low Oct-Jan; captive	12038
	birth: yr rd; n17; zoo	12128
	placental scars: Sept-Jan; Sudan	4344
Acomys spinosissimus	breed: Nov-Mar; Zimbabwe	1815
	preg f: Dec-Jan, Mar-Apr; n12; Botswana	10154
	birth: Oct-Apr; Transvaal, S Africa	8965
	birth: Nov-Apr; S Africa	10158
	repro: Oct-Apr; Mozambique	3821
Acomys wilsoni	fertility: low Mar; Kenya	7797
	mate: yr rd; Kenya	7795, 7796
	preg f: July; n2; Kenya	4925
	preg f: Nov-Feb; Tanzania	299
Aethomys chrysophilus	spermatogenesis: peak Feb-May; Malawi	4329
	mate: yr rd; lab	1333
	preg f: peaks Feb-May, Oct-Nov, none Aug-Sept; Malawi	4329
	preg f: Oct-Nov; Namibia	9945
	birth: Oct-Apr; n10; Botswana	10154
	birth: yr rd, peak summer; Zimbabwe	1815
	birth: yr rd; Transvaal S Africa	8965
	juveniles: yr rd; Zambia	331
	repro: Oct-Apr; Mozambique	3821
	repro: yr rd; S Africa	10158
Aethomys hindei	preg f: peaks Oct-Dec, Mar-May; s Uganda	8083
	birth: peaks Dec-Feb, June-Aug; s Uganda	8083
Aethomys kaiseri	mate: Oct-Jan; Tanzania	299
	preg f: Jan, Aug; n2; e Africa	4925
Aethomys namaquensis	preg f: Jan; n2; Malawi, Nyasaland	4329
	preg f: Jan-May, Oct, peak Mar-Apr; Botswana	10154
	preg f: Feb; Namib Desert, Namibia	11863
	preg f: Aug-Sept; Namibia	9945
	birth: Mar; Namibia	9945
	birth: yr rd, peak summer; Zimbabwe	1815
	lact f: Nov; n1; Malawi	4329
	repro: Oct-Apr; Mozambique	3821
Aethomys nyikae	juveniles: most mo; Zambia	331
Apodemus agrarius	m sexually active: Mar-Dec; Poland, Czechoslovakia	8330
	testes: peak Mar-Sept; Poland	5094
	testes: regress Sept; Poland	5093
	mate: Mar-Sept; Germany	3571
	mate: peak Mar-Apr; Korea	12048
	mate: spring-early fall; Finland	7433

	preg f: Mar-Nov; east Hessen, Germany	8343
	preg f: Apr-Sept; Poland, Czechoslovakia	8330, 8331, 8332
	preg f: end Apr-Aug; Germany	10340
	preg/lact f: May-Oct; Poland	5093
	birth: Apr-Oct; Germany	3571
	birth: spring-early fall; Finland	7433
	repro: end Mar-mid Oct; Korea	5424_1
Apodemus argenteus	testes: large Mar-Sept; Japan	3546
	mate: Feb-Apr, Sept-Nov; Japan	7925
	mate: Oct-Apr; Japan	12025
	birth: May-Aug, Oct-Nov; Japan	7761
	yg appear: June-Sept; Japan	7393
Apodemus flavicollis	f without cl: Feb, May-Nov; s Sweden	3072
	fertile m: Apr-Oct; Austria	8848
	fertile m: yr rd; Denmark	5302
	spermatogenesis: Feb-Oct; s Sweden	3071
	testes: increase Jan, May, regress Nov; Germany	5461
	testes: large Feb-Aug, small Sept-Dec; Austria	10358
	testes: large Apr, small Aug-Jan; s Sweden	3071
	mate: Feb-Mar; s Sweden	4342
	mate: Feb-Sept; Europe	7433
	mate: Feb-Sept; Germany	6548
	mate: Feb-Oct; Germany	3571
	mate: early Apr-Oct; s Sweden	810
	mate: low winter, 37% f, 42% m fertile; s Sweden	3073
	preg f: Feb-June, Aug-Oct; s Sweden	3072
	preg f: Feb-Sept; Austria	10358
	preg f: Feb-Oct, 293 d; s Moravia, Czechoslovakia	8331
	preg f: Feb-Oct; Austria	10358
	preg f: mid Feb-mid Oct; Brandenburg, Germany	10340
	preg f: Mar-Sept, none July; n Germany	5461
	preg f: June-July, Oct; Pyrenees, Spain	11255
	birth: Mar-Oct; Germany	3571
	birth: yr rd; lab	5170
	lact f: Jan-Feb; Berlin	6091
	yg appear: Apr-Aug or Aug-Dec; 2 generations; Poland	54
	repro f: peak Mar-Oct, rare Dec-Jan; Denmark	5302
	repro: Mar-Sept	5461
	repro: end Mar-Oct; Sweden	810
Apodemus microps	sperm: Feb; Bulgaria	10480
	testes: large Feb-Sept, small Oct-Dec; Austria	10358
	preg f: Feb-Mar, Aug-Sept; Austria	10358
	preg f: Mar-Oct; Czechoslovakia	8331
	preg f: Mar-Oct, peak June; Bulgaria	10480
	preg/lact f: Mar-Oct; May 62%, none Nov-Feb; s Moravia, Czechoslovakia	8329
Apodemus mystacinus	testes; large Apr-May, July-Sept; Lebanon	6419
	birth: Feb, Apr; captive	2569
	birth: warm mo; Lebanon	6419
	repro: Mar-Oct; Bulgaria	8317
Apodemus peninsulae	preg/lact f: Apr-Oct; Korea	5424_1
Apodemus speciosus	testes: peaks Feb-Mar, Aug-Oct; Japan	7620
	testes: increase Feb, peak Mar-Oct, small Nov; Japan	7394
	testes: large Sept-Mar, small Apr; Japan	12024
	estrus f: Oct-Feb; Japan	12024
	mate: mid Apr-early May, end Aug-mid Sept; Japan	5951
	preg f: Mar-May, Sept-Dec; Japan	10632
	preg f: Apr-Sept, peak Aug; Japan	7394
	preg f: May, Sept-Oct, not June-Aug; Japan	5951
	preg f: Oct-Apr; Japan	12024
	birth: May, mid Sept-mid Oct; Japan	5951
	lact f: Oct-May; Japan	12024
	repro f: Mar-Apr, Oct; Japan	7620

Apodemus sylvaticus	f without cl: Jan-June, Aug-Dec; s Sweden	3072
	testes: large Feb-early Sept, small Sept-Dec; Austria	10358
	testes: large Feb-Oct; Germany	5461
	testes: enlarge Mar; sw France	8982
	testes: large Apr, small June; Sweden	3071
	testes: large Sept-Feb; Algeria	5987
	testes: large summer-Oct, regress: Dec-Jan; s France	8985
	testes: large summer, small winter, increase Mar; UK	516
	testes: small Oct-mid Jan; France	9438
	fertile m: 42-100% Feb-Apr, June-Sept, none May, Oct-Jan; Austria	10358
	fertile m: Mar-Oct; s France	5247
	fertile m: Apr-Sept, none Oct-Nov, Jan; few data, Mellum I, Germany	7779
	fertile m: few end fall-early winter; UK	516
	conceives: end Apr; St Kilda, Scotland, UK	5307
	mate: Feb; Germany	11530
	mate: Feb-Sept; Germany	6548
	mate: Feb-mid Oct; Scillly, Cornwall, England, UK	9254
	mate: end Feb-Sept, peak June-Aug; Fair Isle, Scotland, UK	2448
	mate: Mar-Sept; Germany	3571
	mate: Mar-Nov, peak June-July; England, UK	7332
	mate: Apr-Sept; nw Bohemia, Czechoslovakia	741
	mate: Apr-mid Oct; s Sweden	810
	mate: May-fall; Europe	7433
	mate: June-Aug; Ireland, UK	3154
	mate: Dec; England, UK	10175
	mate: Dec-Mar; Sweden	4342
	mate: summer-fall; Channel I, UK	955
	mate: low winter, 26% f, 21% m fertile: s Sweden	3073
	mate: winter if acorn crop good; England, UK	10175
	mate: yr rd; France	943
	preg f: Jan-Nov; 296 d; s Moravia, Czechoslovakia	8331
	preg f: Feb-Apr, Aug-Oct; s Sweden	3072
	preg f: Feb-Aug; Austria	10358
	preg f: Mar-Sept; Germany	5461
	preg f: Mar-Sept; France	9438
	preg f: Mar-Oct; England, UK	3342
	preg f: Apr-Aug; St Kilda, Scotland, UK	5307
	preg f: July; few data; Mellum I, Germany	7779
	preg f: peak July, low Oct-Mar; UK	516
	preg f: Aug; n1; China	182
	preg f: Oct; n1; Lebanon	6419
	preg f: Oct-Dec; Algeria	5987
	preg/lact f: 72.6% May, none Dec-Jan; no data Nov-Jan; s Moravia, Czechoslovakia	8329
	birth: Mar; Scilly, Cornwall, UK	9254
	birth: Apr-Oct; Germany	3571
	birth: May-Aug; St Kilda, Scotland, UK	5307
	birth: May-fall; Europe	7433
	birth: May-Oct, winter; France	10897
	birth: yr rd; lab	3067
	lact f: July-Aug; n3; Fair Isle, Scotland, UK	2448
	lact f: Aug-early Nov; Iran	6293
	lact f: Dec-Feb; Algeria	5987
	repro: Feb-Oct; s Moravia, Czechoslovakia	8329
	repro: Mar-Sept; Germany	5461
	repro: Apr-; Germany	10340
	repro: end Mar-early Oct; s Sweden	810
	repro: summer-Oct, none winter; sw France	8982, 8983
	repro: yr rd; lab	8985
Arvicanthis niloticus	fertile f/m: yr rd; Uganda	7793
	testes: large Aug-Dec; Ethiopia	7600
	testes: regress Feb; Kenya	10712

	mate: June-Nov; Egypt	4484
	mate: 1st half dry season; Ethiopia	7600
	mate: rainy-early dry season, peak end rains-early dry; Kenya	10712
	mate: wet season; Kenya	2454
	mate: most mo; Khartoum, Sudan	4347
	mate: yr rd	3756
	preg f: Jan, May-June; few data; e Africa	4925
	preg f: June; n1; Tanzania, Uganda	186
	preg f: July, Sept, Feb; few data; Sudan	4344
	preg f: peak Sept-Oct; Ethiopia	7600
	preg f: Nov-Feb, May-Sept; Uganda	7793
	preg f: yr rd, low dry season; n179; Uganda	2446a, 2453
	birth: Mar-May; lab	8193
	birth: shortly after rain period; Ethiopia	7600
	juveniles: yr rd; Sudan	4344
Bandicota bengalensis	mate: mid Feb-mid Oct, peak Apr, Aug-Sept; Punjab, India	5586
	mate: low winter; n42; Ludhiana, India	6775
	mate: yr rd; Pakistan	108
	preg f: yr rd; Calcutta, India	10267
	birth: peak Oct-Nov; Pakistan	108
Bandicota indica	preg f: 50% Mar, none Feb, 10-30% other mo; n199 f; India	8794
	birth: yr rd; Sri Lanka	8500, 8501
Chiropodomys gliroides	preg f: 18% Jan-Mar, 6% Apr-Sept; Malaysia	4406
Colomys goslingi	fertile m: Mar; w Uganda, e Zaire	5760
	fertile f: July; few data; Zaire	8855
	birth: June-July; n4; w Uganda, e Zaire	5760
	preg f: Apr; n1; w Uganda, e Zaire	5760
	lact f: June-July; n1; w Uganda, e Zaire	5760
Conilurus penicillatus	repro: through winter; Australia	10698
Cremnomys cutchicus	epididymal sperm: yr rd, low Oct-Dec; India	8729
	mate: Mar-Oct; w Rajahstan, India	8723
	birth: Mar-Oct; India	8729
Dasymys incomtus	preg f: June; Malawi	4329
	preg f: Aug; n1; Kenya	4925
	preg/lact f: 60-80% Jan-June, 45% July, 0-30% Aug-Nov; Zaire	2570
	lact f: Jan; n1; Kenya	4925
	lact f: Apr; n1; Tanzania	186
	lact f: Aug; few data; Malawi	4329
	birth: June-Oct; Namibia	9945
	birth: Aug-Jan; S Africa	10158
Golunda ellioti	testes: regressed Dec-Jan, no mating until Apr; Pakistan	9175
	mate: Mar-Sept; n33; Ludhiana, India	6775
	preg f: Mar-Aug; Rajahstan, India	8724
	birth: early part of the yr; Sri Lanka	8501
	birth: May; n1; Pakistan	9175
Grammomys cometes	preg f: Aug-Sept; n2; Kenya	4925
	preg f: peak wet season; Malawi	4329
Grammomys dolichurus	testes: peak Feb-Mar n8; Malawi	4329
	mate: extended season; S Africa	10158
	preg f: Feb; n1; e Africa	4925
	preg f: Nov; n2; Kenya	4925
	birth: Dec-June, peak Feb-May; Malawi	4329
	yg found: almost yr rd; Zambia	331
	repro: peak Feb-May; S Africa	8218
Grammomys surdaster	lact f: Feb; n1; Tanzania	186
Graomys spp	preg f: Dec-Mar; neotropics, S America	4682
Hybomys univittatus	repro f: yr rd, peak Dec-Jan, low Apr-Aug; Zaire	2573
	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	preg f: Jan; n3; Zaire	8854
	preg f: Apr-Aug, Oct, Dec-Jan; Zaire	8855
	yg found: Feb-Mar, June; Zaire	8856
Hybomys trivirgatus	preg f: Mar, July, Sept; Nigeria	4354a

Hydromys chrysogaster	sperm: yr rd, some regression fall-winter; captive	8103
	preg f: July-Feb; VIC, Australia	7108
	birth: yr rd, peak Sept-Mar 88%, low June; captive	8103
Hylomyscus aeta	preg/lact f: Nov-Dec, Feb; Cameroun	2955
Hylomyscus alleni	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	preg f: Jan, July; Zaire	8855
	repro f: peak Dec-Mar, low June-Aug, Oct; Zaire	2573
Hylomyscus stella	mate: peak May-June; Uganda	2445
	mate: peak June, Dec; w Uganda	5760
	preg f: Jan-Feb, Nov; Nigeria	4352
	preg f: Mar-Oct; Nigeria	4350
	lact f: Feb-Apr; Uganda	2445
	juveniles: Oct; July-Aug, Dec; e Africa	5760
Lemniscomys griselda	birth: Dec; Malawi	4329
	birth: perhaps summer; S Africa	10158
	birth: summer; Transvaal S Africa	8965
Lemniscomys striatus	fertile: Mar-May, Aug-Dec; Uganda	7797
	preg f: Jan-Feb, June-July; n5; e Africa	4925
	preg f: Jan, Nov; Zaire	8855
	preg f: Mar-May, Sept-Dec; Uganda	7792
	preg f: Apr-June, Sept-Dec, peaks May, Oct-Nov; n213; Uganda	2446a, 2453
	preg f: Oct-Nov; Nigeria	5190a
	preg f: Oct-Dec; Cameroun	9468
	preg f: Dec-Feb; n4; Zaire	8856
	preg/lact f: 50-70% Dec-Mar, June-July; 10-30% Apr-May, Aug-Nov; Zaire	2570
	birth: Apr-July, Sept-Nov; Ivory Coast	3645
Leopoldamys sabanus	m without sperm: 24% Jan-Mar, 4-5% Apr-Sept, 49% Oct-Dec; Malaysia	4406
	preg f: 8% Jan-Mar, 11% Apr-June, 22% July-Sept, 14% Oct-Dec; Malaysia	4406
	preg f: yr rd; Malaysia	4405
Lophuromys flavopunctatus	sexually active: yr rd; Zaire	2573
	testes non-spermatic: July-Sept 3/7, Nov-Jan 5/31; Malawi	4329
	mate: peak Sept-Nov, Mar-June; Uganda	2445
	mate: peak with rainfall; s Tanzania	5760
	preg f: Jan; n2; Tanzania	186
	preg f: Jan-Feb; Zaire	8856
	preg f: Jan-May, Oct-Dec, none June-Sept; n171; Tanzania	299
	preg f: Jan, Sept; n2n; Zaire	8854
	preg f: peak Mar, low May-Sept, increase Oct-Dec; Zaire	2570
	preg f: May-Aug; few data; Ethiopia	7600
	preg f: Sept-Oct; n2; e Africa	4925
	preg f: peak Sept-Dec; Zaire	2572a
	preg f: Oct-May; Nyasaland	4328
	preg f: yr rd, not Jan, low Dec, July-Aug; Uganda	2445
	preg/lact f: 80% Feb-Mar, 10% July-Aug, 60% Dec; Zaire	2570
	yg present: few May-June, abundant July-Aug; no data other mo; Uganda, Kenya	10231
	yg: peak end rainy-early dry season; Zaire	8706
	repro: peak Nov; Malawi, Nyasaland	4329
Lophuromys sikapusi	preg f: Mar-July, Sept-Oct; Nigeria	4350
	preg f: May, Aug, Oct-Nov; Nigeria	5190a
	preg f: yr rd, peak wet season; Uganda	2446a, 2453
	preg f: end dry season, few data probably; Nimba, Liberia	7362
	fertile: 100% Mar-June, Aug-Dec; < 20% July; Uganda	7797
Lophuromys woosnami	preg f: Sept-May; Zaire	2572a
Malacomys edwardsi	preg f: Nov-July; Nigeria	4352
Malacomys longipes	sexual activity f: peak Dec-Feb; none May, Aug; Zaire	2573
	mate: Mar-June; British Cameroun	9468
	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	preg f: Mar, May, July, Sept, Dec-Jan; Zaire	8855

Mastacomys fuscus	preg f: Oct-Feb; TAS, Australia	4042
	repro: spring-summer; Australia	10698
Maxomys rajah	m without sperm: 35% Jan-Mar, 4% Apr-June, 14% July-Sept, 21% Oct-Dec; Malaysia	4406
Maxomys whiteheadi	m without sperm: 30% Jan-Mar, 6% Apr-June, 4% July-Sept, 26% Oct-Dec; Malaysia	4406
	preg f: yr rd; Malaysia	4405, 4406
Melomys burtoni	mate: none dry periods; QLD, Australia	3606
	mate: rainy season; Australia	10917
	mate: yr rd, higher wet season; NT, Australia	735
	preg f: 1/32 June, 1/9 Aug; none July, Sept-Nov; QLD, Australia	7053
	preg/lact f: yr rd; NT, Australia	735
	birth: Mar-June, Aug; QLD, Australia	7053
	repro: May-Aug; 26o14'S 153o02'E	10109
Melomys cervinipes	preg f: Sept; n1; QLD, Australia	2841
	repro: May-July; 26o14'S 153o02'E	10109
Melomys levipes	mate: low June-Sept; New Guinea	2836
	mate: end yr; New Guinea	7209
Melomys lorentzi	preg f: Jan, May; New Guinea	7209
Melomys rubex	mate: peak end yr; New Guinea	7209
Melomys rufescens	mate: low July-Sept; New Guinea	2836
	repro: yr rd; New Guinea	7209
Mesembriomys macrurus	mate: est yr rd, n Australia	2214
Micromys minutus	mate: peak Apr-Aug; Germany	3571
	mate: May-Sept; Europe	7433
	mate: end May-Dec; Great Britain	2045
	birth: Apr-Sept; Germany	3571
	birth: May-Sept; Europe	7433
	birth: May-Dec, 74% Aug-Sept; Britain	4396
	repro: peak Sept-Oct; Korea	5424,
Millardia gleadowi	mate: Aug-Oct; India	8723
Millardia meltada	scent gland: large Oct-Nov, Mar-June; India	6097
	scrotal m: Feb-Oct; large sample; Ludhiana, India	6775
	testes: large Oct-Dec, quiescent Jan-May; India	2428
	fertile m: yr rd; Pakistan	730
	preg f: Jan-Oct; large sample; Ludhiana, India	6775
	preg f: none winter; Pakistan	730
	preg f: yr rd, peak June-Aug; India	8904
	preg f: yr rd, peak Oct; w Rajahstan, India	8723
	preg/lact f: peaks Mar-May, Aug-Oct; lab, India	940
Mus spp	repro: Apr-May, Oct; Ivory Coast	3645
Mus booduga	scrotal m: Mar-Oct; Ludhiana, India	6774
	preg f: Apr; n1; Nepal	7382
	preg f: July-Mar; India	1688
	birth: yr rd, peak Mar-Apr; Sri Lanka	8501
Mus bufo/minitoides/triton	preg f: Aug, Nov; Zaire	8855
Mus cervicolor	preg f: Dec, July; India	8723
Mus domesticus	sub adult: yr rd; Ludhiana, India	6774
Mus flavipectus?	birth: Apr, June, Oct, Dec; Canton, China	7185
Mus gratus	preg f: Jan; n1; e Africa	185
	juveniles: yr rd; Katanga, Zaire	8565
Mus indutus	repro: yr rd; S Africa	10158
Mus mayori	preg f: Feb; n1; Sri Lanka	8501
	lact f: Dec; n1; Sri Lanka	8501
Mus minutoides	mate: yr rd; e Africa	5760
	preg f: May, Dec; Nigeria	5190a
	preg/lact f: Feb, Apr, July; Malawi	4329
	birth: Dec, Feb, June; n3; Zambia	331
	birth: wet season; S Africa	10158
	birth: yr rd; n17; Botswana	10154
	lact f: May; n1; Tanzania	186
	lact f: July; n1; Kalahari Gemsbok Nat Park, S Africa	7823
	juveniles: peak July-Aug, low Nov-May; Katanga, Zaire	8565

	repro: Apr-June, Sept-Dec; Uganda	2446a
	repro: Aug-May; sw Nigeria	245
Mus minutoides/triton	% preg/lact f: 80% Jan, 20% Apr-May; 50% June, 30% July, 0% Aug-Sept; 14-31% Oct-Dec; Zaire	2570
	preg f: low Mar-Dec; Zaire	2570
Mus musculus	repro active f: end Apr-July; CA	6432
	m with sperm: yr rd; CA	1246
	mate: Oct-May, est yr rd; Japan	10632
	mate: spring-fall; Europe	7433
	mate: est yr rd; Sudan?	4344
	preg f: Mar-Sept; UK	844
	preg f: Apr-Sept; PA	9164
	preg f: mid Apr-mid Dec; CA	1246
	preg f: May-June, Aug, Oct-Nov; n41; Nepal	7382
	preg f: May-Oct; MD	10413
	preg f: June-Oct, Dec; NC	1290
	preg f: yr rd, low mid June-mid July, peak mid July-Aug; England, UK	9323
	birth: spring-fall; Europe	7433
	birth: yr rd, peak spring-summer; lab	2723
	birth: yr rd; Sri Lanka	8501
	repro: yr rd, peaks Apr-May, Aug-Sept; WI	5210
	repro: est yr rd; Sudan	4344
	repro: est yr rd; Khartoum, Sudan	4347
	repro: yr rd; MS	10151
	repro: yr rd; New Guinea	7209
Mus platythrix	preg f: Aug, Oct; few data; India	8723
	preg f: Aug-Mar; India	1688
	preg f: Sept-May, peak Oct-Jan; India	8924
	lact f: Jan-Oct; Ludhiana, India	6774
Mus setulosus	preg/lact f: Jan; n5; Cameroun	2955
Mus spretus	scrotal testes: 50% Mar-Oct; s Spain	11222
Mus triton	repro m: peak Apr-May; Malawi	4329
	preg f: Apr-July, none Aug-Nov; Malawi	4329
	preg f: peak May-Dec; Uganda	2453
	repro: low Aug-Nov; S Africa	10158
	repro: Mar-May, Aug-Jan; Uganda	7797
Mylomys dybowskii	fertile: Mar-May, Aug-Dec; no data Jan-Feb; Uganda	7797
	preg f: July; n2; Uganda	2444
	preg f: yr rd, none July-Aug; n47; Uganda	2453
Nesokia indica	mate: yr rd; lab	6293
Niviventer eha	preg f: Sept; n2; Nepal	7382
	lact f: Oct; n5; Nepal	7382
Niviventer fulvescens	preg f: May, Aug; n3; Nepal	7382
	preg f: July-Aug; n2; China	182
	lact f: May, Nov; n7; Nepal	7382
Niviventer niviventer	lact f: June, Aug-Sept; Nepal	7382
Notomys alexis	mate: yr rd, peak after heavy rain; Australia	4355
	preg f: Dec-Jan; n5; NT, Australia	1266a
Notomys cervinus	mate: winter; ne SA, sw QLD, Australia	11495
	mate: yr rd; captive	4355
Notomys fuscus	birth: yr rd; captive	427
Notomys mitchellii	repro: yr rd; lab	4355
Oenomys spp	juv found: yr rd, peak Nov-Dec, Feb; Kantanga, Zaire	8565
Oenomys hypoxanthus	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	preg f: Feb, Apr, June, Aug, Oct; Zaire	8855
	preg/lact f: 50-60% Nov-Mar, 10% July-Aug; Zaire	2570
	birth: dry season; British Cameroun	9468
Pelomys fallax	preg f: Mar, June, Aug, Nov; Zambia	9893
	preg f: May-June; n2; Malawi	4329
	birth: est Aug-Apr; S Africa	10158
	juveniles: yr rd; Zambia	331
	sub adults: June-July; Malawi	4329
Pithecheir parvus	mate: est yr rd; Malaysia	6448

Pogonomelomys sevia	mate: end yr; New Guinea	7209
Pogonomys macrourus	mate: low July-Sept; New Guinea	2836
Pogonomys sylvestris	mate: low July-Sept; New Guinea	2836
Praomys daltoni	mate: yr rd, peaks start end rainy season; Nigeria	246, 247
	repro: Feb-May, Nov-Dec; Ivory Coast	3645
	repro: yr rd, peaks Feb-Apr, Oct-Nov; Nigeria	4354a
Praomys erythroleucus	mate: yr rd, peak end wet-early dry season; Sierra Leone	1215
	preg f: Sept-Oct; Senegal	5057
	preg f: peak Oct-Nov; Sierra Leone	1215
Praomys fumatus	preg f: Sept, Nov; n2; Kenya	4925
Praomys jacksoni	fecundity: 100% Dec-Jan, 16% June-July; Malawi	4329
	sexually active: yr rd, peak Oct-Apr, low June, Aug; Zaire	2573
	testes regress: dry season; Malawi	4329
	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	preg f: Feb-June, Aug-Sept, Dec; Zaire	8855
	preg f: Nov-Dec; n6; Zaire	8856
	preg f: Sept-Nov, Apr; Malawi	4329
	preg/lact f: 50-70% Nov-July; 20-30% Aug-Oct; Zaire	2570
	yg: increase Feb, abundant Apr-June, rare Sept; Zaire	8564
Praomys morio	mate: peak Apr-May; Kenya, Uganda	10231
	preg f: Aug-Sept; n3/17; Uganda	2444
	birth: yr rd; s Uganda	8080
	lact f: May-Aug; n6; Uganda, Kenya	10231
	repro f: yr rd, low 24% Aug, peak 77% June; Uganda	2445
Praomys natalensis	fertile: Apr-May, Sept-Dec; Uganda	7797
	fertile m: Oct-June peak Nov-Apr; Zambia	1789
	fertile m: Nov; Sudan	4345
	scrotal testes: Sept; Zimbabwe	10591
	testes: large Mar-Apr, Oct-Dec; Uganda	7791
	testes: large Nov-June, peak Jan-Mar; Zambia	1788
	testes: small June-Sept; Zambia	9893
	preg f: May-mid June, Oct-Dec; Uganda	7791
	mate: none Mar-May; Kenya	10712
	mate: Aug-May; Transvaal, S Africa	1954
	mate: yr rd, low Oct-Apr; Zambia	331
	mate: yr rd; Botswana	9892
	preg f: Jan-Mar, July, Sept, Nov; n30/225; Tanzania	299
	preg f: Jan, May; n5; e Africa	4925
	preg f: Feb-May; Zambia	1788
	preg f: Feb, Apr-May, Oct; Nigeria	5190a
	preg f: Apr-May, Oct-Dec; Uganda	7791
	preg f: May-June, Oct-Dec, peak May, Oct-Nov; n119; Uganda	2446a, 2453
	preg f: none June-Jan; Malawi	4329
	preg f: Sept; n2; Liberia	184
	preg f: Sept-Feb; Natal, S Africa	10592
	preg f: Oct-Dec; British Cameroun	9468
	preg f: Dec-June; Zambia	1789
	preg f: yr rd, peak Apr; Botswana	10154
	preg/lact f: June, Sept; no data Dec; Zimbabwe	10591
	birth: peak Feb-Apr; Transvaal, S Africa	8965
	birth: peak Feb-May; Malawi	4329
	birth: peak end rainy season; Zambia	331
	birth: yr rd, low winter; Zimbabwe	1815
	lact f: mid May-mid Aug; n2; Uganda	10231
	juveniles: peak spring, autumn; Transvaal, S Africa	7200
	yg: abundant June-Oct, Jan; Zaire	8564
	repro: Oct-Apr; Mozambique	3821
	repro: yr rd, low June-July; S Africa	10158
	repro: yr rd; Namibia	9945
Praomys tullbergi	scrotal testes: yr rd; Nigeria	4353
	mate: yr rd, peak Dec-Apr; Sierre Leone	2154
	mate: yr rd, peak Dec-Apr; Nigeria	4354
	preg f: Jan-Feb; n3; e Africa	4925

	preg f: end dry period; Mt Nimba, Liberia	7362
	preg f: yr rd; Nigeria	4350
	preg f: yr rd; W Africa	4353
	lact f: Feb; n1; Uganda, Tanzania	186
	repro: peaks July-Aug, Nov-Mar; Nigeria	4352
Pseudomys albocinereus	mate: yr rd; lab	4355
Pseudomys apodemoides	testes: enlarge Mar-Apr; SA, Australia	3259
	mate: fall-winter; SA, Australia	3259
	lact f: peak Aug-Feb; n2 pop; VIC, Australia	1934
Pseudomys australis	fertile: May; QLD, Australia	10708
	mate: yr rd; captive	4355
Pseudomys desertor	mate: yr rd; lab	4355
Pseudomys gracilicaudatus	mate: June; NT, Australia	10708
Pseudomys higginsi	mate: summer; TAS, Australia	4042
	birth: mid Nov-mid Mar; TAS, Australia	4042
Pseudomys novaehollandiae	preg f: Jan-Mar; TAS, Australia	8801
	preg f: Aug-Mar, peak Aug-Dec; NSW, Australia	5645
Pseudomys shortridgei	preg f: Oct-Dec; VIC, Australia	4355
Punomys lemminus	birth: Nov-Apr; Peru	7983
Rattus argentiventer	bimodal, peaks harvesting rice; Malaysia	6162
Rattus exulans	testes: large yr rd; Ponape I, Micronesia	5216
	mate: July-Sept; Ponape I, Micronesia	5216
	mate: Dec-Aug, peak Dec-Mar; New Guinea	2839
	mate: yr rd, low Aug-Sept; Marshall I, n Pacific Ocean	2911
	mate: yr rd, low June-Oct; New Guinea	2836
	preg f: est yr rd, peak Feb-July; New Guinea	10702
	preg f: yr rd; Burma	11434
	preg f: yr rd; Malaysia	4405, 4406
	preg f: yr rd, peak Feb-Apr, Sept-Nov; Java, Indonesia	10955
	birth: Jan-Sept, peak Mar-Aug; HI	11833
	birth: peak Mar-Aug; HI	11832
	repro: peaks Nov, Apr; New Calidonia	7886
	repro: yr rd; New Guinea	7209
Rattus fuscipes	m fertile: yr rd; NSW, Australia	10696
	scrotal m: July-Feb; VIC, Australia	3260
	mate: Nov-Dec; Bass Strait, Australia	4841
	mate: peak Dec-Jan; s Australia	6344
	mate: spring-autumn; QLD, Australia	2841
	birth: Jan-Mar; VIC, Australia	9188
	birth: yr rd; NSW, Australia	10696
	birth/lact: Sept-Feb; VIC, Australia	3260
Rattus hoffmanni	lact f: Mar-May, Aug-Dec; Sulawesi, Indonesia	7675a
Rattus leucopus	preg f: Jan-Feb, Apr, Oct, Dec; New Guinea	10702
Rattus lutreolus	mate: none Apr-July; SA, Australia	614
	mate: Oct-Feb; TAS, Australia	4041
Rattus muelleri	m without sperm: Oct-Dec 27%, Jan-Mar 22%, Apr-June 0%, July-Sept 10%; Malaysia	4406
	preg f: 3% Jan-Mar, 13% Apr-June, 26% July-Sept, 19% Oct-Dec; Malaysia	4406
Rattus niobe	mate: Sept-June; New Guinea	2836
	preg f: est yr rd; New Guinea	10702
	repro: yr rd; New Guinea	7209
Rattus norvegicus	m fertile: yr rd; MD	6242
	testes: large Dec, small July; s GA	8802
	testes: large yr rd; MD	2320
	mate: peak Mar-June; Wales, UK	8393
	mate: Dec-Feb; s GA	8802
	mate: spring-fall; Europe	7433
	mate: yr rd, low Jan, peak Mar; n620 f; VA	4485
	preg f: Feb-Oct, peak June-July; MD	6242
	preg f: yr rd; Wales, UK	8393
	preg f: yr rd; India	5141
	birth: Jan-Feb; s GA	8802
	birth: spring-fall; Europe	7433

	repro: low Jan; New Caledonia	7886
	repro: yr rd; New Guinea	7209
Rattus praetor	preg f: May-June, Oct; n5; New Guinea	10702
	lact f: July; n1; New Guinea	10702
	repro: yr rd, low June-Oct; New Guinea	2836
Rattus rattus	testes: large yr rd; Ponape I, n Pacific Ocean	5216
	testes: peak Jan-Feb, June-July; India	8902
	mate: peaks Jan-Feb, Sept-Oct; Kenya	10607
	mate: peaks Mar-Oct, Nov-Feb; Germany	3571
	mate: Apr-Sept; w Rajahstan, India	8723
	mate: peaks Apr, Sept, low Jan-Feb; Cyprus	11475
	mate: spring-fall; Europe	7433
	mate: yr rd, peak Jan-Mar; Ponape I, n Pacific Ocean	5216
	mate: yr rd, peak Feb-Mar, May-June, low July-Aug; TX	2316
	mate: 7 mo; Philippines	10544
	preg f: peaks Jan-Mar, Sept, or Feb-Mar, Nov; n2 populations; New Zealand	858
	preg f: Jan, Mar, Sept-Oct; n8; New Guinea	10702
	preg f: Jan 0/98; Feb 4.1% n103; Mar 7% n67; Apr-Dec 11-35.3%; India	1765
	preg f: 38% end Jan-early Mar; Vietnam	10823
	preg f: Feb-Dec, peak Mar-Apr, Sept; Cyprus	11475a
	preg f: peak Feb-May, low Nov-Jan; n2041; India	315
	preg f: peaks Apr-July, Oct-Nov, low Mar; n11,096; Sri Lanka	4826
	preg f: Apr-May 9-10%, Oct-Feb 20%; Perth, Australia	1897
	preg f: peak June, Dec; Jodhpur, India	8902
	preg f: Aug 25%, Nov-Jan 4%; HI	3090
	preg f: Oct; n1; FL	7498a
	preg f: peak summer, low Nov-Jan; n1877 f; Egypt	8423
	preg f: yr rd, 22% Apr, 80% July; few data; Czechoslovakia	3250
	preg f: yr rd, low May; India	899
	preg f: yr rd; Nigeria	1513
	preg f: yr rd; Malaysia	4405, 4406
	preg f: yr rd; India	8902
	birth: May-June; Rajasthan, India	8720
	birth: spring-fall; Europe	7433
	repro: low Mar, Aug; no data other mo; Zaire	2570
	repro: low May-June; New Calidonia	7886
	repro: mid Nov-mid Mar; England, UK	11475
	repro: yr rd; Bohemia, Czechoslovakia	5315
	repro: yr rd; New Guinea	7209
Rattus richardsoni	birth: May-June; few data; New Guinea	10702
Rattus steini	lact f: est Apr-Dec; New Guinea	10702
Rattus sordidus	mate: yr rd; QLD, Australia	7053
Rattus taerae	preg f: Nov; n2; Kenya	4925
Rattus tiomanicus	preg f: yr rd, peak Dec-Jan, low Apr-June; Malaysia	11885
Rattus tunneyi	birth: none dry period; QLD, Australia	3606
Rattus turkestanicus	preg f: May, Aug, Oct; n5; Nepal	7382
Rattus verecundus	mate: end Oct-May; New Guinea	2836
	preg f: Mar; n3; New Guinea	10702
	lact f: Sept; n1; New Guinea	10702
	repro: Nov-June; New Guinea	7209
Rhabdomys pumilio	sperm: 50% m Aug-Feb; dry summer; S Africa	2298
	testes: large Aug, declines Dec; Kenya	10712
	mate: Oct-Apr; S Africa	1335
	mate: peak wet season; Malawi	4329
	preg f: Jan; n1; Kalahari Gemsbok Nat Park, S Africa	7823
	preg f: Jan-Apr, July, Nov-Dec; n45; Transvaal, S Africa	8965
	preg f: Sept; Kenya	5760
	preg f: Sept; n3; Kenya	4925
	preg f: Sept-Apr; S Africa	2298
	birth: Mar, May, Nov; Malawi	4329
	birth: few May-July; captive	1814
	birth: Sept-Mar; Cape Flats, S Africa	4640

	juveniles: Oct-May; Cape Flats, S Africa	2298
	juveniles: peak fall, spring; Transvaal, S Africa	7200
Stochomys longicaudatus	sexually active f: Feb-Apr 100%, Aug-Nov 80%; Zaire	2573
	mate: peak Oct-Mar, low June-Aug; e Zaire	2574
	mate: est yr rd; Nigeria	5760
	preg f: Jan, Apr-Sept; Zaire	8855
	preg f: Mar, June; Nigeria	4354a
Thallomys paedulcus	preg f: Apr; n1; Kenya	4925
	preg f: Oct; n2; Namibia	9945
	preg/lact f: Jan, Apr, May; Malawi	4329
	birth: Aug-Apr; S Africa	10158
	lact f: Sept; n1; Namibia	9945
	juveniles: July-Aug; e Africa	5760
Uranomys ruddi	conceive: est Mar-July; Malawi	4329
	preg f: May-July, Nov; Ivory Coast	3645
	preg f: yr rd, peak Oct-Dec; Ivory Coast	752a
Uromys caudimaculatus	birth: Nov-Dec; New Guinea	7209
Zelotomys hildegardeae	preg f: July; n1; w Uganda	5760
	preg f: Nov; n1; s Kenya	5760
	preg f: Nov; n1; Kenya	4925
Zelotomys woosnami	mate: Dec-Mar; Botswana	947
	preg f: Feb; n3; Botswana	10154
	birth: Dec-Apr; S Africa	10158
	repro: Dec-Mar; S Africa	947
Zyzomys argurus	mate: yr rd; NT, Australia	733
	preg f: yr rd, peak Apr; nw Australia	1192
	preg/lact: yr rd; nw Australia	733
	birth: peak Mar-May; nw Australia	733
Zyzomys woodwardi	mate: yr rd, low Nov-Dec; NT, Australia	733
	preg/lact: yr rd; nw Australia	733
	birth: peak Mar-May; nw Australia	733
	repro: May-Sept; nw Australia	733

Order Rodentia, Family Gliridae

Dormice (*Dryomys* 17-44 g; *Eliomys* 45-170 g; *Glirulus* 16-17 g; *Graphiurus* 15-85 g; *Muscardinus* 15-40 g; *Myomimus* 60-110 mm; *Myoxus* 70-180 g) are primarily granivorous, terrestrial inhabitants of Africa south of the Sahara (*Graphiurus*) or the Palaearctic. *Eliomys* has a vaginal plug after mating (6237). Neonates are blind and naked (4180, 8050). Their eyes open at 16-26 d, (*Muscardinus* 6237, 8050, 9672, 11376), 18-21 d (*Eliomys* 3571, 7433), 21 d (*Dryomys* 3571, 7433), or 20-22 d (*Myoxus* 8050). Females have 6-12 mammae (331, 3571, 4495, 5760, 7433, 8050). Males have seminal vesicles and a baculum (4105, 5040, 8050). Details of anatomy (3570), endocrinology (5236, 5236a, 5237), and spermatogenesis (5317) are available.

NEONATAL MASS		
Dryomys nitedula	1.3 g	7862
	1.9 g	8050
Graphiurus murinus	3.5 g; n2	5760
Muscardinus avellanarius	0.80 g	9672
NEONATAL SIZE		
Graphiurus murinus	40 mm; n2	5760
WEANING MASS		
Graphiurus oculatis	25.6 g; lightest trapped	1695
LITTER SIZE		
Dryomys laniger	4.0 emb; r3-5; n5 f	10273
Dryomys nitedula	2 ps ea; n2 f	3578
	mode 2-4; r1-7	3571
	2-4	7433
	2.9 yg; r1-5; n7 lt	3578
	3	7862
	mode 3-4; r2-5	8050
	mode 3-4; r2-6	4645
	4 emb; n1 f	3578

Eliomys quercinus	2-8	7433
	2.86; r2-5; n7 lt	5484
	mode 3-6; r2-8	3571
	3-6	4645
	3-8	5490
	4.12; r1-7; n64 lt	5488
	5 emb; n1 f	7900
	5 emb; n1 f	5489
	5.1 emb; mode 5; r3-7; n9 f	5487
	5.75 yg; r5-7; n4 lt	5489
	6.6 emb; n9 f	6314
	7 yg; 1 nest	8050
Glirulus japonicus	4 yg; r3-7	7983
Graphiurus crassicaudatus	2 yg; n1 lt	184
Graphiurus hueti	2 emb; n1 f	7362
Graphiurus murinus	1 emb; n1 f	7362
	r2-3 nestlings	4495
	mode 3 yg	4354a
	3; n1 lt	4180
	3 emb; n1 f	4495
	3 emb; n1 f	331
	3 yg; n1 f	334
	3.0 emb; r2-4; n3 f	4925
	4 emb; n1 f	9892
	4 yg; n1 nest	4925
	max 5	5760
Graphiurus ocularis	4-6 yg	1695
Graphiurus parvus	4 emb ea; n2 f	4925
Graphiurus platyops	5.5 emb; r5-6; n2 f	331
Muscardinus avellanarius	2-4	11376
	3-5	4645
	3-5	8050
	mode 3-7; r1-9	3571
	3-9	7433
	3.65; r3-5; n17 lt	5491
	3.87; mode 3,4; r2-8; n78 lt	9672
	4.3; r4-5; n3 lt	641
	4.5 yg; r4-5; n4 lt	3578
	4.9 ps; r3-6; n7 f	3578
Myoxus glis	2.0-6.4; r2-9	11359
	mode 2-9; r2-11	3571
	2.7 yg; r2-3; n3 lt	3578
	3-7	7433
	3-10 emb	4645, 8050
	4.9 ps; r3-8; n10 f	3578
	5.5 emb; r5-6; n2 f	3578
	5.7; r1-9; n3 lt, 1 f	11360
	10 emb ea; n2 f	6293
SEXUAL MATURITY		
Eliomys quercinus	f/m: 1 yr	3571
	f: 1st mate: mode 2nd yr	5488
Graphiurus spp	m: testes scrotal: 14-15 g	4350
	f: 1st preg: 15 g	4350
Graphiurus murinus	m: spermatogenesis: 14-15 g	4350
	f: 1st mate: 15 g	4350
Muscardinus avellanarius	f: 1st birth: 3 yr; n1	641
	f/m: 10-12 mo	3571
Myoxus glis	f: 1st birth: ~10-11 mo	4645
	f: 2nd yr	8050
	m: 10-11 mo	8050
	m: sperm: 1 yr	5040
	f/m: 10-12 mo	3571
CYCLE LENGTH		
Eliomys quercinus	mode 10 d; r6-10; interestrous interval	5488

GESTATION		
Dryomys nitedula	21-28 d	7433
	mode 23-25 d; r21-28	3571
	min 1 mo	8050
Eliomys quercinus	21-22, 23 d	5488
	22 d; n1; vaginal plug to birth	6237
	23 d	7433
	23 d; r21-28	3571
Glirulus japonicus	~1 mo	7983
Graphiurus murinus	~24 d	5760
Muscardinus avellanarius	20 (?)-28 d	3571
	20 d	7433
	~26 d	5491
	30 d; r20-43; pairing to birth	9672
Myoxus glis	20-25 d	4645, 8050
	est ~4 wk	11359
	28-30 d	3578
	30-32 d	3571, 7433
LACTATION		
Eliomys quercinus	wean: 4 wk	3571
Muscardinus avellanarius	solid food: 19 d	6237
	wean: 1 mo	8050
	independent: 5-6 wk	4645, 8050
	independent: 40 d	11376
	wean: ~1 mo	3571, 7433
Myoxus glis	solid food: 24-25 d	8050
	wean: ~3 wk	4645
	wean: 25-30 d	8050
	wean: >5 wk; r4-5 wk	3571
INTERLITTER INTERVAL		
Graphiurus ocularis	8 wk	1695
Muscardinus avellanarius	50 d; n1	3578
LITTERS/YEAR		
Dryomys laniger	1	10273
Dryomys nitedula	1	3578
	1	4645
	1; perhaps 2	8050
	1; r1-2 (?)	3571
	1-2	7433
Eliomys quercinus	1	3570
	1	5488
	1; r1-2	3571
	1; r1-2	4645
	1-2	7433
	2?	8050
	est 2	5487
Glirulus japonicus	1-2	7983
Graphiurus ocularis	2	1695
Muscardinus avellanarius	1	11376
	1+; max 2	3578
	1; r1-2 (rare)	9672
	1.5; r1, 2	641
	1-2 (3)	3571
	2	4645
	2	7433
	2/summer	8050
Myoxus glis	1	3578
	1	4645
	1	11360
	1	8050
	1; r1-2 (?)	3571
	1-2	7433

SEASONALITY

Dryomys laniger	testes: large June, small Aug; s Turkey	10273
	preg f: June; few data; s Turkey	10273
	lact f: end July-early Aug; few data; s Turkey	10273
Dryomys nitedula	testes: active June-July; no May data; n Moravia, Czechoslovakia	3578
	testes: decrease mid June-Aug; USSR	8050
	mate: Apr-May; central Europe	3571
	mate: Apr-May; Eurasia	7433
	mate: May-June; n Moravia, Czechoslovakia	3578
	ovulate: May-June; n Moravia, Czechoslovakia	3578
	CL: present May-June, absent July-Aug; n Moravia, Czechoslovakia	3578
	birth: peaks Mar-Apr, July, Sept-Oct; Israel	7862
	birth: May-June; Eurasia	7433
	birth: May-Aug, peak May-June; central Europe	3571
	birth: June; USSR	8050
	birth: June-July; n Moravia, Czechoslovakia	3578
Eliomys quercinus	spermatogenesis: Feb-Aug; peak May-June; lab	3570
	testes: large Feb-July; Germany	5488
	testes: large Mar; n15; Germany	5490
	testes: large Apr; Formentera I, Spain	5487
	m active: Apr-Aug; Alps, Europe	6314
	estrus: Apr-May; Germany	7433
	mate: Feb-?; few data; Tunisia	5490
	mate: Feb-July, peak May-June; Germany	5488
	mate: Apr-May, Aug; Germany	3571
	mate: mid June-mid Aug; captive outside	1140
	ovulate: May; lab	3570
	preg f: Apr, Aug; no data other mo; Formentera I, Spain	5487
	preg f: Apr-June, Aug, Oct; Germany	5488
	preg f: Oct, few data; Mallorca, Baleares, Spain	5489
	preg/lact f: May-Aug, peak end June-early July; Alps, Europe	6314
	birth: Mar; Germany	5488
	birth: May-June, Sept; Germany	3571
	birth: May-June; Germany	7433
	birth: mid June-July; France	6314
	nest: June; n1; USSR	8050
Graphiurus murinus	preg f: Jan; n1; Zambia	331
	preg f: Feb-Mar, May, Aug; Nigeria	4354a
	preg f: June, Nov; central Ivory Coast	3645
	preg f: Nov; n3; Kenya	4925
	lact f: Jan; e Africa	185
	lact f: May-June, Dec; central Ivory Coast	3645
	juveniles: Nov; Zambia	331
	yg found: Feb-Mar; S Africa	10158
Graphiurus ocularis	birth: spring-summer; S Africa	1695
Graphiurus parvus	preg f: Nov; n2; Kenya	4925
Graphiurus platyops	birth: Feb; n1; Zambia	331
	juveniles: Nov-Dec; Zambia	331
Muscardinus avellanarius	testes: large May-Sept; n Moravia, Czechoslovakia	3578
	estrus: May-July; Europe	7433
	mate: May-Aug; central, s Germany	3571
	mate: mid May-early July; n Moravia, Czechoslovakia	3578
	mate: spring; Europe, USSR	4645
	conception: ~end May; near Munich, Germany	11376
	cl present: end Apr-Aug; no Sept data; n Moravia, Czechoslovakia	3578
	birth: May-mid Sept; s Harz Mtns, Germany	9672
	birth: June-Aug; Europe	7433
	birth: June-Sept; central, s Germany	3571
	birth: June-mid Oct; USSR	8050
	birth: mid June-July; n Moravia, Czechoslovakia	3578
	birth: peak end June-early July; near Munich, Germany	11376

	birth: July-Aug; German & Austrian Alps	5491
	birth: spring-early summer; Europe, USSR	4645
Myoxus glis	sperm: July, none May-June, Aug-Sept; s Slovakia, Czechoslovakia	5040
	testes: active June-Aug; Moravia, Czechoslovakia	3578
	testes: large end May-mid July; captive, Germany	11360
	estrus: May-July; Europe	7433
	mate: end May/early June-mid Aug; Germany	3571
	mate: end June; ~3 wk after emergence; USSR	8050
	mate: end June-early Aug; Germany	11359
	mate: July; n Moravia, Czechoslovakia	3578
	mate: 3 wk after hibernation; Europe, USSR	4645
	preg f: June-Aug; n Moravia, Czechoslovakia	3578
	preg f: Aug-Sept; n2; Iran	6293
	birth: June-Aug; Europe	7433
	birth: June-Sept; Germany	3571
	birth: July, perhaps Aug; USSR	8050
	birth: Aug-early Sept; n Moravia, Czechoslovakia	3578
	birth: Aug-Sept; lower Saxony, Germany	11360
	birth: spring; Europe, USSR	4645

Order Rodentia, Family Seleviniidae, Genus *Selevinia*

Selevinia betpakdalaensis is a 20-25 g inhabitant of the Bet-Pak-Dala Desert in Kazakhstan. This crepuscular rodent preys on terrestrial invertebrates. *Selevinia*'s molt includes epidermal tissues as well as fur.

LITTER SIZE		
Selevinia betpakdalaensis	6-8 emb	7983
SEASONALITY		
Selevinia betpakdalaensis	mate: May, July; Kazakhstan	7983

Order Rodentia, Family Zapodidae

Jumping mice (*Eozapus*, *Napaeozapus*, *Sicista*, *Zapus*) are 5-35 g, nocturnal, terrestrial omnivores from North America and northern Eurasia. We found no reproductive information on *Eozapus*. Female *Napaeozapus* and *Zapus* are larger than males (5034, 11624, 11957). Neonates are naked and open their eyes at 18-21 d (*Sicista* 3571, 7433), 22-25 d (*Zapus* 2112, 4271, 8812, 10582, 11624), or 26-31 d (*Napaeozapus* 4271, 6304, 9883, 9884, 10183). Females have 6-8 mammae (502, 1474, 2548, 3571, 6304, 7433, 9612, 11957). Males have penial spines, a baculum, and seminal vesicles (1473, 5983, 8051, 11957).

NEONATAL MASS		
Napaeozapus insignis	0.87 g; n1	6304
	1 g	9883, 9884
	1.0 g; n9 yg, 2 lt	4271
Zapus hudsonius	0.78 g; r0.65-0.9; n6	8812
	0.8 g; r0.7-1.0; n14	11624
Zapus trinotatus	0.82 g; r0.7-0.9; n6	10582
NEONATAL SIZE		
Napaeozapus insignis	HB: 24.2 mm; n1	6304
	HB: 29.5-31 mm; n3	9883
	HB: 30 mm; n9	4271
Zapus hudsonius	TL: 34.4 mm; r30-39; n19	11625
	TL: 38.3 mm; r37-39; n7 stillborn	10954
	52 mm; n2	4271
WEANING MASS		
Napaeozapus insignis	8.5-9.3 g	6304

LITTER SIZE		
Napaeozapus insignis	3-5; r1-6	1474
	3.75 emb; r2-6; n4 f	9883
	r4-6 emb; n14 f	8738
	4.0 yg; r2-5; n3 f	9883
	4.2 emb; r4-5; n10 f	4271
	4.3 ps; r4-5; n6 f	4271
	4.3 ps; r2-7; n45 f	11957
	4.6 emb; ±0.11; r2-7; n80 f	11957
	mode 5 yr; r2-6(8?)	5210
	5 yg; n1 nest	10183
	5.3 emb; r4-7; n6 f	5798
	7 ps; n1 f	10839
Sicista betulina	2-6	3571
	2-6	7433
Sicista napaea	1-6	9954
Sicista tianschanica	3-6 emb	8051
Zapus hudsonius	3.5; r1-6; n2 f	9883
	3.5 yg; r2-5; n2 lt	4271
	3.5 emb; r2-4; n4 f	4271
	4 ps ea; n2 f	4271
	4.5; r3-7; n17 lt	11625
	4.5 emb; r2-8; n62 f	11625
	4.6 ps; r2-7; n12 f	10954
	5 emb; n1 f	11504
	5.3 yg; n8 lt	8812
	5.4 emb; r2-8; n62 f	5405b,
	5.53 emb; r2-9; n78 f	11624, 11625
	5.8 emb; r4-7; n14 f	8812
	5.9 emb; r4-8; n25 f	10954
	6 emb; n1 f	495
	6 emb; n1 f	2025
	6 ps; 6 emb; n2	10101
	6 yg; r5-7; n4 lt	11956
	6.0 emb; n6 f	6166
	7 emb; n1 f	261a
	7 emb; n1 f	1290
	7 emb; n1 f	9164
	7.8 emb; r6-9; n5 f	3703
	8 yg; n1 f	1290
Zapus princeps	2-7	1474
	4 emb; n1 lt	6488
	mode 4-6	502
	4.46 emb; se0.23; n22 f	3159
	4.7 emb, yg; r2-7; n139 lt	2112
	5 ps; n1 f	11957a
	5.4 emb; r4-7; n5 f	1119
	5.5 emb; r5-6; n2 f	10597
	5.9 emb; r4-8; n18 f	11957a
	6.0 emb; r5-7; n3 f	7809
Zapus trinotatus	4-6 yg	5146
	6 yg; n1 f	10582
SEXUAL MATURITY		
Napaeozapus insignis	f: not 1st yr	8152
Zapus hudsonius	f: 1st birth: 2 mo	8812
	f: median 14.5 g; 95%ci 12.1-17.4	68
	m: median 12.5 g; 95%ci 6.4-24.6	68
	f/m: 2 mo	7882
Zapus trinotatus	f/m: yr after birth	3600
GESTATION		
Napaeozapus insignis	19, 22, 23, ~29+ d; n4; capture to birth	9883
	20-23 d	5210
	23 d; n1; pairing to birth	9884
	25 d; n1; capture to birth	9884

Sicista betulina	28-35 d	3571, 7433
Sicista napaea	~20 d	9954
Zapus hudsonius	17-22 d	1871a
	17.5, 18, 20/21 d; n3 f with lactation	8812
	18-21 d	1474
Zapus princeps	18 d	1871a
LACTATION		
Napaeozapus insignis	solid food: 34 d	6304
	wean: ~5 wk	6304
Sicista betulina	wean: >5 wk	3571
Zapus hudsonius	solid food: at eye opening	8812
	nurse: 3-4 wk	8812
	wean: 4 wk	1871a
Zapus princeps	den emergence: ~30 d	2112
	wean; >28 d	2112
	nurse: 1 mo	1871a
	nurse: 33 d; ±2.5	2112
	independent: 2-3 mo	1871a
Zapus trinotatus	independent: 1 mo	3600
LITTERS/YEAR		
Napaeozapus insignis	1	11956
	1	5186
	mode 1	4271
	1-2	8152
	2/season	11957
	possibly 2	1474
Sicista betulina	1	7433
	1; r1-2	3571
Sicista napaea	est 1	8051
Sicista subtilis	1	4645
Sicista tianschanica	1	8051
Zapus hudsonius	1	2088
	1; possibly 2	11956
	max 2	10839
	2-3	11624
	2-3	8812
	2-3/season	1474
	2-3/season	7882
Zapus princeps	1	1474
SEASONALITY		
Napaeozapus insignis	epididymal sperm: end May-early June; MAN, Canada	5186
	mate: May-Sept; ne US, se Canada	11957
	preg f: end May-Aug; NH	8738
	preg f: June; s QUE, Canada	11956
	preg f: peak June; NS, Canada	8152
	birth: early June-mid July; NY	4271
	birth: June-Sept; ne Canada, ne US	1474
Sicista betulina	estrus: May-June; Scandinavia, s Norway, Germany, Russia	7433
	birth: June-Aug; Scandinavia, s Norway, Germany, Russia	7433
	birth: June-Aug; Germany	3571
	mate: mid May-June; Germany	3571
Sicista napaea	repro: end June-mid Aug; USSR	9954
Sicista tianschanica	testes: large June, decrease end summer-fall; USSR	8051
	preg f: mid June; USSR	8051
	lact f: July; n1; USSR	8051
Zapus hudsonius	testes: scrotal May-Sept; MI	7882
	mate: May-Oct; NY	11624
	mate: June-Aug; n N America	1474
	preg f: May; n1; MN	495
	preg f: June; n1; SD	261a
	preg f: June; few data; SD, WY	10954
	preg f: June-Aug; n5; ND	3703
	preg f: June-early Sept; ONT, Canada	5034
	preg f: July; n6; se MT	6166

	preg f: Sept; n1; NC	1290
	preg/lact f: June-Aug; MI	7882
	preg/lact f; June-Sept; MI	7585
	birth: peak end May-early June; WI	5210
	birth: June; NY	4271
	birth: June; BC, Canada	2088
	birth: mid June-Aug; MN	8812
	birth: peak spring; s QUE, Canada	11956
	lact f: June; n4; ND	3703
	lact f: June; n1; NC	1290
	lact f: Aug-Sept; n2; NE	5405b$_1$
Zapus princeps	testes: large 12 d after emergence; UT	2112
	mate: end May-mid July; UT	2112
	conceive: peak mid June-early July; ALB, Canada	3159
	preg f: June-July; BC, Canada	10597
	preg f: June, Aug; MAN, Canada	11957a
	birth: June-July; w Canada, w US	1474
	birth: mid July-?; BC, Canada	10597
Zapus trinotatus	birth: July-Aug; OR	3600

Order Rodentia, Family Dipodidae

Jerboas (*Alactagulus* 45-60 g; *Allactaga* 45-420 g; *Cardiocranius* 50-75 mm; *Dipus* 70-95 g; *Euchoreutes* 70-90 mm; *Jaculus* 50-135 g; *Paradipus* 110-155 mm; *Pygeretmus* 75-120 mm; *Salpingotus* 40-55 mm; *Salpingotulus* 35-45 mm; *Stylodipus* 100-130 mm) are nocturnal, saltatorial, often semifossorial herbivores from arid regions from northern Africa through central Asia. Male *Jaculus* are slightly larger than females (8134). Corpora lutea are found only in pregnant females suggesting that ovulation is induced (*Jaculus* 3755). A postpartum estrus has been observed (8051). Neonates are naked and blind (4346, 4349, 4351, 7778, 8051). *Jaculus* neonates open their eyes at 38 d (4349, 4351). Females have 8 mammae (8051). Males have seminal vesicles and penial spines, but usually no baculum (4105, 8051, 11822).

NEONATAL MASS		
Jaculus jaculus	2 g; n6	4346, 4349
NEONATAL SIZE		
Jaculus jaculus	25 mm; n6	4346, 4349
Salpingotulus michaelis	HB: 17 mm	9175
LITTER SIZE		
Alactagulus pumilio	1-6 emb	10516
	2 yg	10516
	3 emb ea; n2 f	11307
	3.0 emb; ±0.08; mode 3; r1-4; n71 f	7778
	3.3-4.2 emb, ps; r1-6 emb	9888
	3.3 emb; ±0.06; mode 3; r1-6; n231 f	7778
	3.4 emb; ±0.17; mode 3; r2-6; n43 f	7778
	3.4 emb; ±0.11; mode 3; r1-6; n104 f	7778
	3.9 emb; ±0.22; mode 4; r2-7; n28 f	7778
	4; r3-6	8051
	5.0 uterine blood clots; r3-7; n4 f	6293
Allactaga bobrinskii	5.3 emb; ±0.22; r3-7; n32 f	7778
Allactaga bullata	2.5 emb; r2-3; n2 f	8051
	2.7 emb; n3 f	10516
Allactaga elater	2-8 emb	10516
	4 emb ea; n2 f	6293
	4.1; r2-6	7005
	4.1 emb; ±0.21; mode 5; r1-6; n34 f	7778
	4.4 emb; ±0.14; mode 4; r2-8; n88 f	7778
	4.5 emb; 1st lt; 3.7 emb; 2nd lt	742
	4.5 emb; ±0.09; mode 4; r2-8; n147 f	7778
	4.7 emb; ±0.10; mode 5; r2-8; n148 f	7778
	4.8 emb; ±0.21; mode 5; r2-8; n33 f	7778
	4.9 emb; ±0.20; mode 4,6; r2-8; n48 f	7778
	5; r2-6	8051
Allactaga euphratica	2-8	7778
Allactaga jaculus	1-6	8051

	3.4; r2-5	7005
Allactaga major	3.0 emb; ±0.16; r2-4; n8 f	7778
	4.1 emb; ±0.49; r4-6; n9 f	7778
	4.6 emb; ±0.38; r2-7; n3 f	7778
Allactaga severtzovi	3.5 emb; r1-8	7778
	4.0 emb; r3-6	7778
	4.1 emb; ±0.14; r2-7; n>48 f	7778
	5.0 emb; ±0.38; r2-7; n14 f	7778
Allactaga sibirica	2.8 emb; ±0.19; mode 3; r1-4; n18 f	7778
	~3; 5-6 rare	8051
	3.4 emb; ±0.29; mode 3; r2-5; n9 f	7778
	3.7; r2-6	7005
	3.7 emb; ±0.26; mode 3; r1-7; n40 f	7778
	3.75 emb; r1-7; n4 f	10516
Dipus sagitta	3; r2-5	8051
	2.9 emb; mode 3; r2-5; n80 f	3213
	3.0 emb; ±0.08; mode 3; r2-5; n80 f	7778
	3.01 emb; mode 3; r2-5; n85 f	10516
	3.5 yg; r3-4; n2 nests	11307
	3.6 emb; ±0.08; mode 4; r1-5; n20 f	7778
	4 emb; n1 f	11307
	4.2 emb; ±0.09; mode 4; r1-8; n201 f	7778
	4.4 emb; ±0.08; mode 4; r2-8; n127 f	7778
Jaculus blanfordi	3 emb; n1 f	6293
Jaculus jaculus	1.8, 3.8, 4.8 emb; r1-5; n3 samples	3755
	mode 3 emb; r2-5	4344, 4346, 4347, 4349, 4351
	3-7 cl; r2-7	5480
Jaculus lichtensteini	mode 4-6 emb; r2-8	7778
	4.7, 6.5 emb; n2 samples	7778
	5 emb; n1 f	8051
Jaculus orientalis	2 emb; n2 f	8134
	2 ps; n1 lt	8134
	mode 3 yg; r2-4	4960
	3 ea; n2 lt	12128
Jaculus turcmenicus	4.0 emb; ±0.27; r3-6; n13	7778
Paradipus ctenodactylus	3 emb; n1 f	8051
	3.5 emb; r2-6; n8 f	7778
Pygeretmus platyurus	4 ps; n1 f	7778
	5 emb; n1 f	7778
	6 yg; n1 lt	7778
Salpingotus crassicauda	2.3-2.7; mode 2-3	9952
	2.7 yg; ±0.07	7778
Salpingotulus michaelis	2-4	9175
Stylodipus telum	2 yg	10516
	3-4	4645, 8051
	3-4 emb	10516
	4.1 emb	7778
	5.2 emb; ±0.19; mode 5; r2-8; n164 f	7778
SEXUAL MATURITY		
Allactaga elater	f/m: 3-3.5 mo	742
Dipus sagitta	f/m: yr following birth	8051
Jaculus jaculus	f: 1st birth: 8-12 mo	4346
	f: 12-14 mo	8051
	m: sperm: 4-5 mo; 42-49 g	3755
GESTATION		
Alactagulus pumilio	25-30 d	10516
Dipus sagitta	est 25-30 d	3213
	25-30 d	10516
Jaculus jaculus	44-46 d	3755

LACTATION		
Jaculus jaculus	solid food: 38 d	4349
	independent: ~2 mo	8051
LITTERS/YEAR		
Alactagulus pumilio	2	8051
Allactaga elater	1	8051
	2 lt/season	742
	2	7005
	est 2	9175
Allactaga sibirica	1	7821
	1	8051
Dipus sagitta	2	10516
	2	8051
Jaculus jaculus	1-2	8051
Jaculus lichtensteini	1	7778
Paradipus ctenodactylus	several, few data	9774
Salpingotus crassicauda	2	9952
Stylodipus telum	1	8051
SEASONALITY		
Alactagulus pumilio	testes: large end July, small Aug; USSR	8051
	mate: spring, fall; no data summer, winter; USSR	9888
	preg f: est Oct; few data; ne Iran	6293
Allactaga bobrinskii	mate: Apr-June, Sept-Oct; w Kizil-Kum, n Turkmenia	7778
Allactaga elater	testes: large end Mar; USSR	8051
	testosterone: high Feb, Aug, low June; Kyzylum, Ustyurt, Uzbekistan	742
	mate: Apr-?; USSR	8051
	mate: est spring-fall; sw Pakistan	9175
	preg f: peak Apr-Aug; Kazakhstan	7005
	preg f: Oct; n2; Iran	6293
	birth: May-?; USSR	8051
	lact f: Oct; n1; Iran	6293
	repro: Mar-Aug; n Kyzylum, Ustyurt, Uzbekistan	742
Allactaga euphratica	testes: large Feb-Apr; Iraq	5481
	preg f: May-Oct; Armenia	7778
Allactaga major	preg f: Apr-June; Aral Sea, Kara-Kum, Turkmenia	7778
Allactaga sibirica	testes: peak May-June, decrease July; Transbaikal, Russia	8051
	preg f: May-June; Kazakhstan	7005
	birth: end June-July; USSR	8051
	repro: ?-mid July; se Transbaikal, Russia	7821
Cardiocranius paradoxus	repro: est spring; USSR	7778
Dipus sagitta	mate: mid Mar; USSR	8051
	preg f: Apr-Aug; w Kazakhstan	3213
	preg f: May; n1; Turkestan	11307
	lact f: June; n2; Turkestan	11307
	repro: Feb-mid June; Karakumy desert, Turkmenia	9499
	repro: Mar-Aug, peak June-July; USSR	8051
	repro: peak spring-summer; nw Kyzylkumy, Uzbekistan	9409
Jaculus blanfordi	preg f: Jan; n1; Iran	6293
Jaculus jaculus	spermatogenesis: Feb-Mar, May, Aug, Oct-Nov; n Iraq	5480
	spermatogenesis: yr rd; Khartoum, Sudan	4346
	testes: large Feb, May, Oct; n Iraq	5480
	testes: large yr rd; Sudan	3755
	mate: June-July, Oct-Dec; Khartoum, Sudan	4346, 4347
	preg f: Feb, June, Nov; n Iraq	5480
	preg f: Apr-Jan; Kazakhstan	7005
	preg f: peak Oct-Nov; USSR	4351
	preg f: peak after rain ~80%, none June; Sudan	3755
	preg f/birth: peak Oct-Nov; Sudan	4344
	birth: end Apr-June; USSR	8051
	birth: Oct-Feb; n Sudan	4344

Jaculus lichtensteini	repro: Feb-mid June; Karakumy desert, Turkmenia	9499
Paradipus ctenodactylus	repro: Feb-mid June; Karakumy desert, Turkmenia	9499
Salpingotus crassicauda	birth: May-June; Zaisan Hollow, Kazakhstan	9952
Stylodipus telum	birth: May-June; Europe, USSR	4645
	birth: est May-mid July; USSR	8051
	repro: est end Mar/early Apr-mid June; USSR	8051

Order Rodentia, Family Hystricidae

Old world porcupines (*Atherurus* 1.5-4 kg; *Hystrix* 8-30 kg; *Thecurus* 4-5.5 kg; *Trichys* 1.5-2.5 kg) are terrestrial, nocturnal herbivores from Africa and southern Asia. Usually, males and females are of equal size, however, female *Aethurus* are larger than males, while male *Hystrix* have been reported to be larger than females (8500, 8501, 8853, 10158, 11202, 11203). The placenta is hemochorial with a distinct subplacenta and interstitial implantation (6628). A postpartum estrus is recorded at 10 d (10831; cf 11076, 11078). The well-developed and mobile neonates have quills and open eyes at birth (2384, 5760, 8501, 8853, 9175, 10158). Females have 4-6 teats (331, 4495, 5760, 9945, 10158). Males have seminal vesicles, a baculum, and a sigmoid penis (8246, 11081). Details of anatomy (9161, 11075, 11080, 11565), endocrinology (11079, 11081a, 11208a), and copulation (7532) are available.

NEONATAL MASS		
Atherurus africanus	146.25 g; r100-175; n12	8853
Hystrix africaeaustralis	r107-294 g; n1 lt; term emb	10158
	~300 g	2384
	311.3 g; r300.0-330.0	10158
	351 g; sd47.4; r300-400; n19	11076, 11077
Hystrix brachyura	350-540 g	10831
NEONATAL SIZE		
Atherurus africanus	HB: 180 mm	8853
Hystrix indica	200-225 mm	8501
LITTER SIZE		
Atherurus africanus	1	8853
	1	12128
	1-4	5760
	mode 1; r1-3	2449
	mode 1-2; r1-4	8798
	1 emb ea; n2 f	4495
Atherurus macrourus	1 ea; n2 lt	12128
	1-3 yg	10829
Hystrix spp	mode 1-2; r1-4	8798
Hystrix africaeaustralis	1-3	2384
	1-4	331
	1-4	2439
	1.5 yg; sd0.66; mode 1 59%; r1-3; n165 lt	11076, 11078
	1.51; r1-4; n151	10158
	1.81; mode 2; r1-3; n77 lt	1230
	2	11253
	2	4180
Hystrix cristata	mode 1; r1-2	12128
	2-4	5277
	2-4 yg	4354a
Hystrix indica	1	8500, 8501
	1-2	12128
	2-4	9175
	2; n1	10616
	2; r2-4	106
SEXUAL MATURITY		
Hystrix africaeaustralis	f: 1st conceive: 619 d; ±172	11076
	m: testosterone: 413 d; ±118; r273-532	11076
	m: 8-18 mo	11081

CYCLE LENGTH		
Hystrix africaeaustralis	follicular phase: 1.9 d	11075
	follicular phase: 6.0 d; r4-8; n2	7532
	luteal phase: 19.5 d; sd7.2; r11-26; n4	7532
	luteal phase: 29.3 d; sd4.7	11075
	28.4 d; sd3.86; r25-35; n5 cycles	7532
	31.2 d; se6.5; most 28-36; r17-42; n43 cycles, 12 f	11076, 11078
	33 d; sd11.6; r18-51; n16 cycles	11081a
	34 d; sd6.6; r23-42; n12	11075
	36.9 d; sd11.5; r16-62; n34 cycles	11081a
Hystrix cristata	35 d	11551
GESTATION		
Atherurus africanus	100-110 d	8853
Hystrix spp	6-8 wk	8798
Hystrix africaeaustralis	42-56 d	2384
	est~8 wk	5760
	93.5 d; r93-94; se0.6; n4	10036, 11076, 11078, 11080
	112 d	2439
Hystrix brachyura	~4 mo	10831
Hystrix cristata	est ~8 wk	5760
Hystrix indica	~112 d	9175
LACTATION		
Hystrix africaeaustralis	den emergence: 1+ wk	5760
	wean: start 4 wk	10158
	wean: 120 d; r90-148	10036
	wean: ~7 wk	10158
Hystrix cristata	den emergence: 1+ wk	5760
INTERLITTER INTERVAL		
Hystrix africaeaustralis	345 d; sd66; r200-500; n22 young live	11078
	385 d; se60.4; r296-500; n8	11076
	405.8 d; se86.2; n5	10036
Hystrix indica	109 d; n1	8724
LITTERS/YEAR		
Atherurus africanus	2-3	8798
	max 3	5760
Hystrix africaeaustralis	1	10036, 11078
	2	11253
	2	4180
Hystrix cristata	2 possible	5760
SEASONALITY		
Atherurus africanus	birth: yr rd, peak end spring-early fall; location not given	8798
	repro: not seasonal; w Africa, w,s Uganda, s Kenya	5760
Atherurus macrourus	preg f: Apr, July-Aug; n3; n Vietnam	10829
Hystrix africaeaustralis	estrus: yr rd; S Africa	11076
	mate: May-Dec; Orange R, S Africa	10036
	mate: summer; s Africa	5760
	mate: yr rd; zoo	1230
	birth: July-Dec; S Africa	2384
	birth: Aug-Mar; S Africa	10158
	birth: Aug-Mar; Orange R, S Africa	10036, 11076
	birth: rainy season; Zaire	11253
	birth: yr rd; S Africa	11078
Hystrix cristata	birth: yr rd; n32; zoo	12128
Hystrix indica	birth: peak June-July, none Oct, Jan, Mar; w Rajahstan, India	8723
	birth: peak Oct-Apr; Sri Lanka	8501

Order Rodentia, Family Erethizontidae

The New World porcupines (*Coendou* 1-5 kg; *Echinoprocta* TL 500 mm; *Erethizon* 3-18 kg; *Sphiggurus*) from North, Central, and South America are nocturnal herbivores. *Erethizon* males are larger than females (2665). Ovulation is usually from the right ovary and may be induced (*Erethizon* 2665). A copulatory plug is present after mating (*Erethizon* 11906). The discoidal, labyrinthine, hemochorial placenta has a partially interstitial, mesometrial attachment usually into the right horn of the uterus which is transitional between duplex and bicornuate (*Erethizon* 2665, 7571, 8391). A postpartum estrus occurs after the birth of precocious, furred or quilled, open-eyed neonates (502, 2665, 5146, 9166, 9167, 9823, 10258, 10508, 10923, 11906). Females have 4-12 teats (219, 502, 1474, 5111, 11906). Males have a baculum (1473). Details of anatomy (2665, 7571) are available.

NEONATAL MASS		
Coendou prehensilis	390 g	5858
	f: 413.0 g; r398-432; n4	9167
	m: 413.5 g; r410-422; n4	9167
	415.3 g; sd26.1; r364-447; n4	9166
Erethizon dorsatum	340-640 g	11906
	~450 g	2665
	491.2 g; r402.7-604.5; n7	9818, 9819, 9823
	510 g; n1	10508
	588 g; n1	10923
LITTER SIZE		
Coendou mexicanus	1 emb; n1 f	2228a
	1 emb; n1 f	4228
	1 emb; n1 f	5423b
	1 yg ea; n3 lt	219
Coendou prehensilis	1.0	5858
	1 emb; n1 f	8562a
	1 ea; n23 lt, 4 f	9166, 9167
Erethizon dorsatum	mode 1	7571
	1	2665
	1	502
	1; 2 possible	5146
	1 emb; n1 f	10839
	1 emb; n1 f	10954
	1 yg; n1 lt	10923
	1 emb ea; n2 f	9834
	1 ps ea; n2 f	10954
	1 ea; n4 f, 4 lt	3171
	1 yg ea; n7 lt	9818, 9819
	1 emb ea; n20 f	6016
	1.0196; r1-2; n52 f	10508
Sphiggurus villosus	1; n1 lt	12128
SEXUAL MATURITY		
Coendou prehensilis	f: 19 mo; n1	9166, 9167
Erethizon dorsatum	f: 1st ovulate: 1.5 yr	2665
	f: 18 mo	11906
	m: 1st sire: 16 mo; n1	9820
	f/m: 2 yr	1871a
	f/m: 2.5 yr	8798
	f/m: 3 yr	1474
CYCLE LENGTH		
Erethizon dorsatum	estrus: 8-12 hr	11906
	25-30 d	2665
GESTATION		
Coendou prehensilis	192 d; min ibi	9167
	est 195-210 d	9166
Erethizon dorsatum	16 wk; n1	10508
	6.5-7 mo	2665
	29-31 wk	1871a
	7 mo; n2 f	5146

	~7 mo	1474
	210.25 d; r205-217; n4	9818, 9819
	213 d	11564
LACTATION		
Coendou prehensilis	solid food: 4 wk	9166, 9167
	independence: 10 wk	9167
	independence: 15 wk	9166
Erethizon dorsatum	solid food: a few hours after birth	1474
	nurse: 2-3 wk	1871a
	wean: min 2 mo	11906
INTERLITTER INTERVAL		
Coendou prehensilis	192-674 d	9167
SEASONALITY		
Coendou mexicanus	preg f: Jan; n1; Veracruz, Mexico	4228
	preg f: Feb; n1; Campeche, Mexico	5423b
	preg f: May; n1; San Luis Potosí, Mexico	2228a
	lact f: Apr; n2; Yucatan, Mexico	5423b
Erethizon dorsatum	testes: scrotal June-Jan; MA	2665
	mate: Aug/Sept-Nov; MA	2665
	mate: Sept-Oct; US	1474
	mate: Nov; BC, Canada	2088
	mate: Nov-early Dec; NH	10508
	mate: fall; UT	6560
	preg f: Mar; n1; SD	10954
	birth: Mar; BC, Canada	2088
	birth: Apr-May; US	1474
	birth: end Apr-early May; MA	2665
	birth: ~early May; IL, WI	2060
	birth: May-June; w US	5146
	birth: spring; WI	7571
	lact f: July; n1; ND	3703

Order Rodentia, Family Caviidae

Cavies and guinea pigs (*Cavia* 200-500 g; *Dolichotis* 1.5-16 kg; *Galea* 300-700 g; *Kerodon* 450-1000 g; *Microcavia* 200-500 g) are herbivores from South America. *Galea* males may be larger than females (5953). *Cavia* sperm form rouleaux in the epididymis (3519). Coitus triggers ovulation in *Galea* (9264, 11559, 11561, 11565) and a copulatory plug follows mating in *Cavia* (6237, 11553). Four-cell embryos and morulas are found in the uterus of *Galea* at 4.5 d (9161). The 9.5 g (n1, *Galea*) labyrinthine, hemoendothelial, discoidal placenta with subplacenta has a mesometrial, interstitial implantation into a bicornuate uterus (3011, 6237, 7563, 9161, 9162, 9484) and is eaten after birth (2759, 9256). A postpartum estrus follows (2759, 5953, 6146, 6147, 6237, 9161, 9168, 9256, 9257, 9264, 10494, 10680, 11553) the birth of open-eyed, furry, precocial neonates (2759, 6147, 6601, 8798, 9168, 9256, 9257, 10493, 10680, 11553). Females have 2 (*Cavia*), 4 (*Galea*, *Dolichotis*, *Microcavia)*, or 8 mammae (*Dolichotis*) (5111, 6237, 9257, 11563). Males have penial spines and a baculum (6237). Details of anatomy (9161, 9162, 11565), endocrinology (4528, 6601, 10645), spermatogenesis (4936), copulation (2759, 10811a), zygote transport (7968), birth (2759), and milk composition (7211) are available.

NEONATAL MASS		
Cavia aperea	m: 56.6 g; se4.7; r39-86; n9	9257
	f: 58.7 g; se3.1; r40-91; n17	9257
	m: 61.2 g; se1.5; n27	11553
	f: 61.8 g; se1.7; n38	11553
Cavia porcellus	f: 75.3 g; se4.7; r59-108; n9	9257
	m: 78.45 g; r65-114; n50 1 d	8643
	f: 80.34 g; r61-125; n50 1 d	8643
	m: 91.6 g; se5.2; r55-120; n11	9257
Dolichotis patagonum	524 g; se55; n5	992, 6601
	639.7 g; r481-733; n6	9291
Dolichotis salinicola	199.0 g	5858
Galea musteloides	f: 32.0 g; sd50; n9	5953
	m: 36.4 g; se1.0; r24-50; n42	9257

	f: 37.6 g; se1.2; r23-54; n46	9257
	f: 39.4 g; se0.9; n81	11553
	m: 40.0 g; sd8.6; n17	5953
	m: 40.5 g; se0.7; n85	11553
Kerodon rupestris	76.0 g; sd11.99; n10	6147
	80.0 g	5858
	85 g; singleton lt heavier	9168
Microcavia australis	m: 29.9 g; se1.9; r17-44; n17	9257
	f: 30.9 g; se1.2; r24-39; n16	9257
NEONATAL SIZE		
Cavia aperea	m: TL: 122.0 mm; se3.3; r108-136; n9	9257
	f: TL: 123.9 mm; se1.9; n17	9257
Cavia porcellus	f: TL: 128.0 mm; se2.7; r112-138; n9	9257
	m: TL: 138.0 mm; se3.1; r121-156; n11	9257
Dolichotis patagonum	CR: 249 mm; se9; n5	992, 6601
Galea musteloides	m: TL: 99.8 mm; se1.0; r88-116; n42	9257
	f: TL: 99.9 mm; se1.0; r87-112; n46	9257
Kerodon rupestris	145.2 mm; sd10.50; n10	6147
Microcavia australis	f: TL: 97.1 mm; se1.2; r89-104; n16	9257
	m: TL: 99.4 mm; se1.8; r81-104; n17	9257
WEANING MASS		
Cavia aperea	f: 169.6 g; se8.0; n37	11553
	m: 172.5 g; se8.5; n22	11553
Cavia porcellus	228 g	6610
Galea musteloides	f: 127.5 g; se5.5; n58	11553
	m: 146.0 g; se5.7; n52	11553
LITTER SIZE		
Cavia aperea	1-2 emb	596
	2.1 yg; se0.2; r1-5; n28	9257, 9264, 11553
	2.5 yg; r1-5; n31 lt	9264
Cavia porcellus	2.1 yg; r1-4; n25 lt	9257, 9264
	2.68 yg; r1-8; n7285 lt, 39 lineages	11958
	3.7 wean; n297 lt	6610
	4.3-4.5 yg; n16 yr (thousands of litters)	5033
	4.4 yg; n297 lt	6610
	7; n1 lt	6237
Dolichotis patagonum	mode 1; r1-3	12128
	1.33 yg; r1-2; n6 lt	9291
	1.54; r1-3; n39 lt	2109
	1.67; r1-2; n3 f	6601
	1.75 yg; n8 lt	6709
	1.79 yg; mode 2 70%; r1-3; n63 lt	2759
	2; r1-3	7425
	2.03; r1-3; n73 lt	1230
Dolichotis salinicola	1.5; r1-3	5858
	2 emb; n1 f	2944
	2; r1-3	7432
Galea musteloides	mode 2-3; r1-5	9264, 11553
	2 emb; r1-4; n12 f	8302
	2.1; sd1.2; r1-5; n10 lt	5953
	2.7; se0.1; n69 lt	9257
	4.47 cl	7968
Galea spixii	1-5	10493
	2.2 yg; sd0.87; n10 lt	6147
	3.0; sd1.0; r1-5; n24 lt	10494
Kerodon rupestris	1.38 or 1.41 yg; sd0.5; r1-2; n17 lt	6146, 6147
	1.53; mode 1; r1-3; n75	9168
	1.58; se0.73; r1-3; n26 lt	10680
	1.6; r1-2	5858
Microcavia australis	2.21; mode 2 64%; r1-3; n14 lt	9256
	2.8 yg; se0.3; r1-5; n16 lt	9257, 9264

SEXUAL MATURITY

Cavia aperea	f: 1st conceive: 30-89 d	9257, 9264
	f: vag perf: 58.6 d; se3.0	11553
	f: 1st conceive: 84.4 d; se9.9; min 27	11553
	f/m: 10-12 wk	8798
Cavia porcellus	f/m: 1st mate: >2.5 mo	6237
	f/m: 2 mo	8798
Dolichotis patagonum	f: 1st birth: 8 mo	2759
	m: 6 mo	2759
Galea musteloides	f: vag perf: 48.5 d; se1.2	11553
	f: 1st conceive: 50 d	11553
	f: 1st conceive: 58 d; r28-93; n9 spring-born	9257, 9264
	f: 1st conceive: 60.6 d; r17-152; n342	9264
	f: 1st conceive: 167 d; r113-255; n6 fall-born	9257, 9264
	m: 1st spermatids: 21-25 d	4936
	m: 1st sperm: 31-35 d	4936
	m: epididymal sperm: 60 d	4936
Galea spixii	f: vag perf: 80 d	6147
	m: testes descend: 135 d; n2	6147
Kerodon rupestris	f: vag perf: 60 d	6147
	f: 1st conceive: 55 d prob unusual	9168
	f: 1st conceive: 82 d; n1	6146
	f: 1st birth: 156 d	6147
	f: 133 d	9168
	m: testes descend: 115 d; ±2.7; n4	6147
Microcavia australis	f: 1st mate: 45 d; r40-50; n2	9256, 9257
	f: 1st conceive: 82-85 d	9256, 9257, 9264

CYCLE LENGTH

Cavia aperea	20.6 d; se0.8; n67 cycles, 21 f	9264, 11553
Cavia porcellus	estrus: ~24 h	6237
	estrus: ~24 h	10440
	16 d; r13-21; 2o sources 2 strains	2715
	50 d; n1 cycle, 1 f	6237
Dolichotis patagonum	31.4 d; se1.3; r24-39; n6 f	7669
	35.3 d; se5.0; r28-42; n7 f	991
Galea musteloides	17.3 d; r17-18; n3 cycles, 3 f	9257, 9264
	19.0 d; se1.2; n4 f with vasectomized m	11553
	19.1 d; r12-36; n21 cycles with vasectomized m	9264
	19.5 d; r12-27; n2 cycles, 2 f	11559
Microcavia australis	15 d; n1 cycle, 1 f; post partum	9257

GESTATION

Cavia aperea	60.9 d; se0.3 or 0.4; mode 60; r59-63; n9	9257, 9264, 11553, 11564
	63.9 d; se0.9; mode 60; r59-74; n26	9264
Cavia porcellus	>57 d	6237
	65-68 d	11551
	67.0 d; se0.8; n6	9257
	68 d; r58-75	2715
Dolichotis patagonum	~58 d	8798
	3 mo; min ibi	2759
	96 d; se0.7; n6	991, 992, 6601
Dolichotis salinicola	~77 d; n4	5858, 11563
Galea musteloides	implant: 5 d	7968
	implant: 6 d	9161
	mode 54 d; r48-61; n90; ppe to birth?	9264
	52 d	4528
	52-53 d	11553, 11564
	53 d; n1	10646
	59 d; sd5; r65-71	5953
Galea spixii	49-52 d	6147
	~7-8 wk	10494
Kerodon rupestris	75.0 d; sd1.42; r72-77	6146
	75.87 d; sd1.57; n24 lt	9168
	75.9 d; se1.14; r74-78; n17	10680
	77 d	5858
Microcavia australis	54 d	11564

LACTATION		
Cavia aperea	wean: 1 wk; most 2-3 wk	11553
	wean: 3 wk	8798
Cavia porcellus	solid food: 2-3 d	8798
	wean: 2 wk	8798
Dolichotis patagonum	solid food: 2 d	6709
	wean: 6-11 wk	2759
Galea musteloides	wean: 1 wk	11553
	wean: 3 mo	5953
	nurse: ~50 d	9257
Galea spixii	wean: 42.25 d; ±5.7; n4	6147
Kerodon rupestris	wean: 35.17 d; ±10.1; n6	6147
INTERLITTER INTERVAL		
Cavia porcellus	96.3 d	6610
Dolichotis patagonum	3-3.6 mo	2759
Galea musteloides	53.2 d; se0.2; r50-58; n41	9257
Microcavia australis	54.2 d; sd0.4; r53-55; n6	9256, 9257, 9264
LITTERS/YEAR		
Cavia aperea	max 5	9257, 9264
Dolichotis patagonum	3-4	2759
Galea musteloides	max 7	9257
Galea spixii	est max 6-7	10494
Microcavia australis	max 4	9257
	5 possible; 4 observed	9256
SEASONALITY		
Cavia aperea	mate: Sept-May; few data; Uruguay	596
	birth: yr rd, peak Nov; captive	9257, 9264, 11553
Dolichotis patagonum	mate: yr rd; zoo	1230
	birth: Feb-Nov; captive	2109
	birth: Mar-Oct; n19; zoo	12128
	birth: yr rd, peak May-Sept; captive	8798
	repro: yr rd; free ranging	2759
Dolichotis salinicola	preg f: Aug; n1; Bolivian Chaco	2944
Galea musteloides	mate: yr rd; captive	11553
	preg f: Dec; n12; Lake Titicaca, Peru	8302
	birth: yr rd; Argentina	9257
	birth: yr rd; captive	9257, 9264
Galea spixii	preg/lact f: yr rd; Caatinga, ne Brazil	10493, 10494
	repro: yr rd; Brazil	6808
Kerodon rupestris	birth: yr rd, peak July-Mar; captive	9168
	birth: yr rd; captive	6146
Microcavia australis	preg/lact f: Sept-Apr; no Aug data; Argentina	9256, 9257
	preg f: Oct-Feb; captive	9257, 9264

Order Rodentia, Family Hydrochaeridae, Genus *Hydrochaeris*

At 40-60 kg, capybaras (*Hydrochaeris hydrochaeris*) are the largest extant rodents. These social, semi-aquatic, crepuscular and nocturnal herbivores inhabit marshy areas of South America. They mate in the water (10892) and subsequently give birth to well-furred and open-eyed neonates that can walk within 15 minutes (8798, 10892, 12066). Females have 14 teats (5111). Details of anatomy (11565) and copulation (8075) are available.

NEONATAL MASS		
Hydrochaeris hydrochaeris	1.4 kg	8076
	1.5 kg	7466
	1.75 kg; r136-226	12066
LITTER SIZE		
Hydrochaeris hydrochaeris	2; n1 lt	12128
	2 yg ea; n2 f	5111
	3 yg; n1 f	10885
	3.3 yg; r2-5; n13 lt	12066
	3.5; r1-6	8076

	4; r1-8	10892
	4 emb; n1 f	10885
	4.8 emb; r1-8; n68 f	7466
	7	4558
SEXUAL MATURITY		
Hydrochaeris hydrochaeris	f/m: 15 mo	12066
	f/m: 15-18 mo	10892
	f/m: 1.5 yr, 30-40 kg	7466
GESTATION		
Hydrochaeris hydrochaeris	104-111 d; n1	10885
	119-126 d	8798
	120 d	8076
	148 d; r144-154; n3	4558
	152.5 d; r149-156; n2	12066
LACTATION		
Hydrochaeris hydrochaeris	solid food: 2-3 d	12066
	nurse: 3-4 mo	7466
	wean: 4 mo	12066
LITTERS/YEAR		
Hydrochaeris hydrochaeris	1	8798
	1.5	10892
SEASONALITY		
Hydrochaeris hydrochaeris	mate: peak Apr-May; Venezuela	7466
	mate: Oct-Nov; Mato Grosso, Brazil	7466
	birth: peak Sept-Nov; Venezuela	7466
	birth: peak Sept-Dec; e Argentina to Panama	10892

Order Rodentia, Family Dinomyidae, Genus *Dinomys*

The herbivorous pacaranas (*Dinomys branickii* 10-15 kg) inhabit the Andes Mountains of western South America. Term embryos have fur and open eyes (1988). Females have 8 teats (11563). Details of mating (1988) are available.

NEONATAL MASS		
Dinomys branickii	622.7 g; r570-660; n3	7228
	950.75 g; r825-1076.5; n2 term emb, dead with placenta	1988
NEONATAL SIZE		
Dinomys branickii	HB: 275 mm; r265-285; n2 term emb, dead	1988
LITTER SIZE		
Dinomys branickii	1-2 yg	11563
	2 emb; n1 f	1988
	2 emb; n1 f	3847
	2 yg; anecdote	10682
	2.4 yg; r1-4; n5 lt	7228
GESTATION		
Dinomys branickii	223-283 d; n1	1988
	< 252 d; n1	7228
LACTATION		
Dinomys branickii	solid food: 2 d	7228

Order Rodentia, Family Agoutidae, Genus *Agouti*

The 4-12 kg paca (*Agouti paca*) from lowland habitats of Central and northern South America is a solitary, nocturnal herbivore. Pacas have a hemichorial, discoidal placenta which implants into a bicornuate uterus (4728). A postpartum estrus occurs (1981). Neonates are furred with open eyes and can walk as well as swim shortly after birth (4728). Females have 2 (5111) or 4 (4728) teats. Males are ascrotal with seminal vesicles and coagulating glands (4728). Details of anatomy (1981, 4728) are available.

NEONATAL MASS		
Agouti paca	500-800 g	1981
	650 g; r600-800	4728
NEONATAL SIZE		
Agouti paca	24-30 cm	4728
LITTER SIZE		
Agouti paca	1 emb; n1 f	3209
	1 emb; n1 f	8562a
	1 ea; n2 lt	12128
	1 emb ea; n2 f	2228a
	1 ea; n55 lt	1981
	1.03; r1-2; n31 lt	4728
	2	3633
	max 18 cl	4728
SEXUAL MATURITY		
Agouti paca	f: 1 yr	1981
	m: testes size, spermatogenesis: 1 yr	1981
	m: 3 mo	4728
GESTATION		
Agouti paca	96.75 d; sd3.44; r93-101; n3	4728
	est 116 d	1981
LACTATION		
Agouti paca	wean: 82 d	4728
INTERLITTER INTERVAL		
Agouti paca	191 d; assuming 116 d gestation	1981
SEASONALITY		
Agouti paca	mate: early winter; Yucatan, Mexico	3633
	preg f: Jan; n1; El Salvador	3209
	preg f: June; n2; San Luis Potosí, Mexico	2228a
	preg f: yr rd; n50; ne Colombia	1981
	birth: Sept-Mar; lab, Costa Rica	4728
	birth: winter-early spring; Yucatan, Mexico	3633

Order Rodentia, Family Dasyproctidae

Agoutis (*Dasyprocta* 2-4 kg) and acouchis (*Myoprocta* 1 kg) are herbivores in tropical regions of Central and South America. A copulatory plug is formed (11551). The hemochorial, labyrinthine, zonary placenta, associated with a mesoplacenta, implants into a bicornuate uterus (692, 693, 3023, 7563, 9707, 10476), and is eaten after birth (3023, 5846). A postpartum estrus may occur midway through lactation, earlier if the young die, or not at all (5846, 5847, 11555). Neonates are furred with open eyes (3023, 3024, 5846, 10176) and suckle from 8 teats (3024, 5111, 11563). Details of anatomy (11555, 11556, 11565), endocrinology (9349, 10646), and copulation (10176) are available.

NEONATAL MASS		
Myoprocta acouchy	f: 74.3 g; se1.9; n24	5846
	m: 79.3 g; se2.5; n23	5846
WEANING MASS		
Myoprocta acouchy	300-400 g	5846
LITTER SIZE		
Dasyprocta cristata	2; n4, n2 lt	12128
Dasyprocta leporina	mode 1; r1-2	12128
	1.4 yg; r1-3; n53 lt	1230
	2-4 yg	5111
Dasyprocta prymnolopha	1-2	12128
Dasyprocta punctata	1; n1 lt	12128
	1 emb; n1 f	1475
	1 ea; n2 lt	12128
	1.25 emb; r2-4; n4 f	3023
	1.25; r1-2; n12 lt	10176
	1.9 yg; mode 2; r1-3; n20 lt	7227
	2	3317
	mode 2; max 4	3024

	2 emb; n1 f	1475
	2 yg; n1 f	3023
	2 yg ea; n7 lt	2109
Myoprocta acouchy	1.5; r1-2; n6 lt	5847
	mode 2	5846
	2-5 cl	9349
	2 emb; n6 f	9349
SEXUAL MATURITY		
Myoprocta acouchy	f: vag perf: 147 d; ±14	5846
	f: 1st conceive: 249 d; ±24	5846
CYCLE LENGTH		
Dasyprocta leporina	vag perf: 1-10 d	11556
	34.2 d; se2.1; n29 cycles, 3 f; fall to spring	11556
	87.2 d; ±8.9; n6; summer	11556
Myoprocta acouchy	vag perf: 10-22 d; r3-22; n62	5846
	40 d; se2; r16-68; n22 cycles	5846
	42.7 d; se2.5; n21 cycles	11555
GESTATION		
Dasyprocta leporina	104 d	1355
Dasyprocta punctata	35 d; n1	3024
	~3 mo	10176
	<127 d; min ibi	7227
Myoprocta acouchy	99 d; n5	11555
	99 d; se1; r94-115; n21; vag perf or mating to birth	5846, 9349
LACTATION		
Dasyprocta punctata	solid food: d of birth	3024
Myoprocta acouchy	solid food: middle 1st wk	5846
	wean: 6-8 wk	5846
INTERLITTER INTERVAL		
Dasyprocta punctata	min 127 d	7227
SEASONALITY		
Dasyprocta leporina	birth: yr rd; n27; zoo	12128
	birth: yr rd; zoo	1230
Dasyprocta prymnolopha	birth: yr rd; n17: zoo	12128
Dasyprocta punctata	preg f: July-Aug, Nov-Dec; few data; Panama	3024
	preg f: Dec; n2; El Salvador	1475
	birth: yr rd; zoo	7227
	birth: yr rd, peak Mar-July; Panama	10176
	young found: Jan, Apr, Aug, Dec; El Salvador	1475
	young found: yr rd; Panama	3317
Myoprocta acouchy	estrus: fall, spring, long cycles summer; zoo	11555
	estrus: Dec-May; zoo	5847
	mate: yr rd; zoo	5846
	birth: peak Aug, none Oct; zoo	5846

Order Rodentia, Family Chinchillidae

Chinchillas (*Chinchilla* 350-800 g) and viscachas (*Lagidium* 1-3 kg; *Lagostomus* 2-8 kg) are terrestrial to semifossorial, colonial herbivores in southern South America. *Chinchilla* and *Lagostomus* are primarily nocturnal, while *Lagidium* is diurnal (8301, 8798, 9345). *Chinchilla* females are larger than males (9266), while the reverse is true for *Lagostomus* (11557, 11562). *Lagidium* exhibits reproductive asymmetry with the right ovary and uterus used preferentially (8300, 8301, 11565). *Lagostomus* ovulates about 800 oocytes (9161, 9163, 11553, 11558, 11565), but only 2 embryos implant (3891, 6511, 6815, 11553, 11557). A copulatory plug is present after mating (8301, 11551, 11553, 11557).

The 9-11 g, hemochorial, discoidal, labyrinthine placenta with a subplacenta has a mesometrial, interstitial implantation into a duplex or bicornuate uterus (4800, 5738, 8301, 9161, 9162, 9163, 9266, 9707, 10819, 11558). Estrus may occur postpartum, but is more common after lactation (1499, 3891, 5263, 8301, 9161, 11551, 11553, 11557). The precocious neonates are furred with open eyes and can walk (2109, 3891, 6095, 8300, 11551, 11553, 11557, 11564). Females have 4 (*Lagidium*, *Lagostomus)*, or 6 (*Chinchilla*) teats (3891, 8301, 11557, 11563). Males have seminal vesicles, coagulating glands, and a baculum (4800, 8301, 9266).

Details of anatomy (4799, 4800, 5263, 8301, 9162, 9266, 11550, 11554, 11558, 11565), endocrinology (1559, 4095, 4096, 4528, 10646), copulation (936), and milk composition (3891, 11324) are available.

NEONATAL MASS		
Chinchilla lanigera	r37.4-43.8 g; n4	6095
	f: 38.7 g; n10	9266
	39.7 g; se1.39; n19	9266
	m: 41.0 g; n9	9266
Lagidium peruanum	~180 g	8300, 8301
Lagidium viscacia	f: 260 g; n1	11554
Lagostomus maximus	f: 185.0 g; se4.6; n49	11553, 11557
	m: 207.0 g; se5.2; n38 or se6.1; n28	11553, 11557
WEANING MASS		
Lagostomus maximus	m: 645.5 g; se35.7; n29	11553, 11557
	f: 659.0 g; se18.3; n42	11553, 11557
LITTER SIZE		
Chinchilla brevicaudata	1.45	10809
Chinchilla lanigera	1.0; n68275 f	4800
	mode 1-2; r1-6	11550, 11551
	1.05 yg; n44454 f	4800
	1.45; mode 1; r1-3; n20 lt	2109
	1.87 yg; n648 lt	4800
	1.90 yg; ±0.76; mode 2 46%; r1-4; n123 lt	6117
	2.3; r2-3; n3 lt; n7 live fet, 8 total	10643
Lagidium peruanum	1 yg	8300, 8301
Lagidium viscacia	1 ps; n1 f	8556
	2	2944
	mode 2; r1-2	12128
Lagostomus maximus	1.89 emb; mode 2; r1-3; n18 f	6511
	1.94 yg; mode 2 90%; r1-3; n62 lt	3891, 11553, 11557
	2.2 yg; r1-5; n5 lt, 3 f	6815
	43-255 oocytes; max 544	9161
	200-800 oocytes	9163, 11553, 11565
	max 845	11558
SEXUAL MATURITY		
Chinchilla lanigera	f: vag perf: 173.2 d; ±57.6; r<71-308; n45 f	6117
	f/m: 8+ mo	11551
Lagidium peruanum	f: 1st mate: ~6 mo	8301
	f: ~900 g	8300
	m: sperm: ~7 mo	8300, 8301
Lagostomus maximus	f: vag perf: 153.7 d; se3.4; n31	11557
	f: 1st conceive: 253.8 d; se20.2; min 83; n15	11557
CYCLE LENGTH		
Chinchilla lanigera	estrus: 2 d	1499
	estrus: 2-3 d	11551
	vag perf: 7.4 d; r3-18	5263
	vag perf: 8 d	1499
	vag perf: 8.0 d; ±1.11; n25 f	1327
	33 d; r11-49; n14 cycles	1327
	35.7 d; ±7.9; r15-62; n100 cycles, 24 f	6117
	35 d	11564
	40 d	1499
	40.91 d; r21-54; n118 cycles, 23 f	5263
	mode 41; most 30-50; n70 cycles	11550, 11551
	41.05 d; r37.4-43.4; n154 cycles, 23 f	5263
	42 d	11552
	68.17 d; r36-130; n36 cycles, 23 f	5263
Lagostomus maximus	vag perf: 3-15 d	11553
	vag perf: 7.3 d; se0.2; n213	11557
	41 d; r25-55 75%; r15-75	11553
	45.0 d; se1.2; 30-35 68%; n152 cycles, 46 f	11557

GESTATION		
Chinchilla brevicaudata	128 d	10809
Chinchilla lanigera	implant: 5.25 d	9161
	110.4 d; r108-112; n5 gest, 5 f	6117
	111 d; r105-115	11550, 11551, 11564
	111 d	10646
Lagidium spp	est 3 mo	8798
Lagidium peruanum	~100 d	8301
	104 d	9161
Lagidium viscacia	4-4.5 mo	11554
Lagostomus maximus	implant: 18.5 d	9161, 9163
	153.7 d; se0.5; r145-166; n23	3891, 9163, 11557, 11564
	155 d; r149-162	11553
LACTATION		
Chinchilla lanigera	nurse: min 3 wk	11564
	wean: 6-8 wk	11551
	wean: 60 d	2109
Lagidium peruanum	solid food: day of birth	8300
	wean: >1 mo	8300, 8301
Lagostomus maximus	wean: 29-70 d; m earlier	11557
	wean: 54 d	11553
LITTERS/YEAR		
Chinchilla brevicaudata	1-2	10809
Chinchilla lanigera	max 2	10809
	2	4800
Lagidium peruanum	est 2; r1-3; field data	8301
SEASONALITY		
Chinchilla brevicaudata	mate: biannual; n Chile	10809
Chinchilla lanigera	m fertile: yr rd; lab	11551
	mate: Nov-May; lab	11552
	mate: Dec-May; captive	1499
	birth: Mar-Aug 81%; captive	6117
	birth: Apr-May, Aug-Oct, Dec; n19; zoo	12128
	birth: spring; coastal Cordillera, Chile	10809
	birth: yr rd, peak Apr-May; captive	8798
	repro: Nov-May; lab	11551
Lagidium peruanum	sperm: est yr rd; Peru	8301
	anestrus: Sept-early Oct; Peru	8301
	mate: yr rd, peak mid Oct-Nov; Peru	8301
	birth: mid Jan-Feb; Peru	8301
Lagidium viscacia	mate: end May-early June; n1; Argentina	11554
	birth: yr rd; n18; zoo	12128
Lagostomus maximus	mate: peak est Mar-Apr; Argentina	11557
	birth: yr rd, peak Apr-July; location not given	8798
	birth: yr rd; lab	11553, 11557

Order Rodentia, Family Capromyidae

Hutias (*Capromys*, *Geocapromys*, *Isolobodon* (possibly extinct), *Plagiodontia*) are 700 g to 7 kg, crepuscular or nocturnal herbivores, primarily distributed in the West Indies. *Capromys pilorides* is usually solitary, but is sometimes found in pairs (8798). *Capromys melanurus* enters estrus 28 d postpartum (1421). Neonates are furred, have open eyes and prehensile tails, and can walk (*Capromys* 1421, 8798; *Geocapromys* 5022, 8092). *Capromys* have 4 teats, while *Plagiodontia* have 6 (7430, 8798). Males have inguinal testes (*Capromys* 311).

NEONATAL MASS		
Capromys pilorides	199 g; r150-244	5354
Geocapromys ingrahami	70 g; n1	5354
	84 g; n1; 70-90 g	5022
Plagiodontia aedium	101.7 g; n1	5354
	110.0 g	5858

WEANING MASS		
Geocapromys ingrahami	200-300 g; 1-2.5 mo	1914
LITTER SIZE		
Capromys melanurus	1.75 yg; r1-2; n4 lt	1421
Capromys pilorides	1-3 yg	9139
	1.8; r1-3	5858
	2; r1-4	5354
	2 yg; n2 lt	1421
	2.0; r1-3; n4 lt	12128
	2.8 yg; 90% 1-4; r1-6; n38	1570a
Capromys prehensilis	2 yg; n1 lt	1421
Geocapromys brownii	1-2 yg	7509
	1-3	5354
	1.49 yg; 6% triplets; r1-3; n47 lt	282
	2.3; r2-3; n3 lt, 3 f	8092
Geocapromys ingrahami	1 yg *ea*; n8 lt	1914, 5022
Plagiodontia aedium	1	10809
	1 ea; n4 lt	5354
	1.3; r1-2	5858
SEXUAL MATURITY		
Capromys pilorides	f: 10 mo	5354
	f: HB: 416 mm	1570a
	m: 7-10 mo	5354
	m: HB: 440 mm	1570a
CYCLE LENGTH		
Capromys pilorides	16.3 d; r13-19	5354
	~1 mo	9139
Geocapromys ingrahami	10.08 d; r9-11	5354
Plagiodontia aedium	10 d	5354
GESTATION		
Capromys melanurus	199 d; r195-203; n2	1421
Capromys pilorides	17-18 wk	9139
	120-130 d	8798
	123 d; ±2	5354
	123 d	1421
	125 d	5858
	4.7 mo	1570a
Geocapromys brownii	est 6 wk	5354
	123 d; n1	8092, 8094
	~123 d	282
Geocapromys ingrahami	88 d; n1	5022
	108 d; n1; mating to death carrying a near term emb)	5022
	126-156 d; n1	5354
Plagiodontia aedium	≤119 d; n1	5354
	125-150 d	10809
LACTATION		
Capromys melanurus	leave nest: 2 d	1421
	solid food: 10 d	1421
Capromys pilorides	solid food: 10 d	8798
	solid food: 16 d	1421
	wean: 6 wk	8798
	nurse: 153 d	5354
Geocapromys brownii	solid food: ~30 h	282
	solid food: 34 h	8092
Geocapromys ingrahami	solid food: ≤ 3 d	5022
INTERLITTER INTERVAL		
Capromys melanurus	222 d; r221-223; n2	1421
Geocapromys brownii	168 d; n27	282
Geocapromys ingrahami	min 325 d	5022
SEASONALITY		
Capromys pilorides	birth: yr rd, peak June; zoo	5354
	birth: Oct; n1; captive	1421
Geocapromys brownii	birth: Aug-Oct; zoo	8092
Geocapromys ingrahami	birth: yr rd; Bahamas	1914

Order Rodentia, Family Myocastoridae, Genus *Myocastor*

Nutrias (*Myocastor coypus* 3-14 kg) are semiaquatic herbivores from temperate wetlands of central and southern South America. Males are larger than females (2045, 3571, 11743). The 12.2-14.5 g (n2), labyrinthine, hemochorial, discoidal placenta has a prominent subplacenta and implants mesometrially into a bicorunate or duplex uterus (4797, 4798, 9161, 9162). A postpartum estrus occurs (2336, 3957, 7875) after the birth of furred, open-eyed neonates that can swim within a day of birth (443, 2045, 2336, 2680, 3960, 6253, 7433, 8798). Ovulation may be induced (3957). Females have 6-10 thoracic, lateral teats (2336, 3571, 3957, 3960, 7433). Males have seminal vesicles (4798, 6776). Details of anatomy (3825, 4798, 8518, 9161, 9162, 10049, 11565), endocrinology (4528, 5280, 9346, 9552), milk composition (2918), and reproductive energetics (3960) are available.

NEONATAL MASS		
Myocastor coypus	100-160 g	2919
	150-250 g	2336
	150-250 g	159
	f: 201 g; sd46; n59	3960
	m: 213 g; sd53; n58	3960
	225 g; r175-332	7875
	234 g; n15	443
WEANING MASS		
Myocastor coypus	1.5-2 kg; 13 wk	3960
LITTER SIZE		
Myocastor coypus	1-9 yg; 12-14 rare	2919
	3.02; rofx 3.6-4.8; across months	11742
	4.1; mode 4; r1-12; n3279 lt	158
	4.23 emb; r1-9; n26 f	443
	~4.5 yg; r1-9	3115
	5; max 11	2336
	5.03 emb; mode 5; r1-11; n233 f	66
	5.06 live emb; se0.13; r1-9; n136 f; 1st lt	7875
	5.5 born; 4.64 born alive; n123 lt	9838
	5.63 emb; r2-11; n35 f	443
	5.63 live emb; se0.11; r1-13; n287 f; 2+ lt	7875
	5.75 emb; ±1.51; mode 5; r3-12; n148 f	1368
	6 emb; n582 f	3962
	6.05	5724
	7.3 emb; r5-9; n3 f	6253
	mode 16-18 emb; r12-20	2919
SEXUAL MATURITY		
Myocastor coypus	f: 1st cl: 10-12 mo	5952
	f: 1st mate: 4-8 mo	2336
	f: 1st preg: 2-4 mo; n1	5724
	f: 1st birth: 6-15 mo	11907a
	f: min 4 mo	2919
	f: < 6 mo	11742
	f: 8-9 mo	6253
	m: epididymal sperm: 5.5-6.5 mo; n1	5724
	m: testes large: 6 mo	11742
	m: spermatogenesis: 4-5 mo	8518
	m: 12-15 mo	6253
	f/m: 1.5-2.5 kg	7875
CYCLE LENGTH		
Myocastor coypus	estrus: 1 d; nulliparous; 2-4 d; multiparous	10049
	estrus: 1-4 d	3115
	estrus: 2-8 d; metestrus: 2 d; proestrus: 1 d	7295
	estrus: ~4 d	2336
	r5-11 d; n30 f	7295
	r5-28 d; n10 cycles	11760
	17 d; nulliparous; 19 d; multiparous; vaginal smears	10049

	24-26 d	3115
	26 d; r1-6 wk	7875
	28 d	2336
GESTATION		
Myocastor coypus	implant: 10 d	3957
	120-130 d	6253
	128 d	2336
	mode 128-130 d; r120-132	2919
	mode 128-135 d; r110-150	3571
	130 d	1046
	~130-134 d	443
	131 d; 128-133 72%; r126-141; n446	158
	134 d; r129-139; n16	10049
	135 d	4528
	135-150 d	7433
	19 wk	3961
LACTATION		
Myocastor coypus	solid food: at once	7875
	solid food: 24 hr	6253
	solid food: 1 d	443
	wean: 5 wk	3115
	wean: 6-8 wk	8798
	wean: 6-10 wk	7875
	nurse: 7.7 wk; 95%ci 1.0	3957, 3960
	nurse: 10.9 wk; n5	3957, 3960
INTERLITTER INTERVAL		
Myocastor coypus	128 d	2336
	~132 d	3957
LITTERS/YEAR		
Myocastor coypus	2	5724
	2-2.5	2336
	2.76	11742
	3/13-14 mo	1046
SEASONALITY		
Myocastor coypus	mate: yr rd; e Anglia, UK	3962
	preg f: yr rd, except Apr; MD	11742
	birth: yr rd, peak Feb-May; captive	8798
	birth: yr rd; Hessen, Germany	3571
	birth: yr rd, England, UK	7875
	birth: yr rd; N America	1474

Order Rodentia, Family Octodontidae

Southern South America hosts the 60-300 g, herbivorous, octodontid rodents (*Aconaemys*, *Octodon*, *Octodontomys*, *Octomys*, *Spalacopus*). While all genera burrow, only *Spalacopus* is considered fossorial (9016). Only the diurnal *Octodon degus* has received significant study. Morulae enter the bicornuate uterus at 3.5 d. At 5.5 d the mesometrial, interstitial attachment of a 200-300 g, discoidal, hemochorial placenta and prominent subplacenta begins (9161, 9162). A postpartum estrus occurs, but few captive females mate (9083, 11553). The eyes of the sparsely furred neonates open after 1 wk, although neonates are able to stand and right themselves at birth (9083, 11553, 11564). Females have 6-8 teats (11563). Males have a baculum, penial spines, seminal vesicles, and abdominal testes (2027a). Details of anatomy (9161, 9162, 11565) are available.

NEONATAL MASS		
Octodon degus	13.52 g; sd1.72; n89 yg, 16 lt	9283
	14.0 g; se0.1; n114	11553
	14.4 g	5858
	14.6 g; se0.06; n35	9083
		5858
Octodontomys gliroides	20 g	
WEANING MASS		
Octodon degus	68.0 g table, 69.8 g text; 28 d	9083
	74.4 g; r40-122; n106	11553

LITTER SIZE		
Aconaemys fuscus	3 emb; n1 f	8305a
	5 emb; n1 f	4054
Octodon degus	4.9; r1-8	5858
	5 yg; r1-8	11553, 11564
	5 yg; r1-10	11563
	5; n1	9161
	5.0 emb; r3-7; n2 f	8556
	5.3; mode 3; r3-8; n9	3552
	5.7 emb; sd1.3; n9 f	7241
	5.75 yg; se0.48; n4 lt	9083
	6.8 yg	11907
Octodontomys gliroides	2 yg; r1-3	11563
	2.1; r1-4	5858
Spalacopus cyanus	3 emb ea; n2 f	9016
SEXUAL MATURITY		
Octodon degus	f/m: 6 mo	9083
	f: vag perf: 127.5 g	7241
CYCLE LENGTH		
Octodon degus	vag perf: 3-21 d	11553
GESTATION		
Octodon degus	implant: 6.5-7 d	9230
	implant: 6.5-7 d	9161
	84-90 d	9230
	90 d	9083
	90 d	5858
	90 d; r87-93	11553
	>90 d	11564
Octodontomys gliroides	104 d	5858
LACTATION		
Octodon degus	solid food: 6-7 d	9083
	wean: 4-6 wk	9083
	wean: 4-6 wk	11553
SEASONALITY		
Aconaemys fuscus	preg f: Oct; n1; Neuquen, Argentina	8305a
	preg f: Nov; n1; Malleco, Chile	4054
Octodon degus	mate: June-July; lab, Chile	9230
	preg f: June-July; Chile	7241
	preg f: July-Sept; Chile	7241
	birth: peak July-Aug, Dec; captive, VT	11907
	repro: peak Sept; Chile	3552

Order Rodentia, Family Ctenomyidae, Genus *Ctenomys*

The 33 species of tuco-tucos (*Ctenomys* 100-700 g) are fossorial herbivores distributed from Peru to Patagonia. Males may be larger than females and solitary, while females are colonial (596, 8304). The 100-200 g, hemochorial, discoidal placenta has a prominent subplacenta (9161) and implants in either horn mesometrially (8302, 8304, 9161). Neonates are naked and blind (11564), but their eyes open at 2-3 d (1449c). Females have 6 teats (8304, 11563). Males have a baculum and penial spines (6401). Details of anatomy (11565), endocrinology (4528), and sperm (3196) are available.

NEONATAL MASS		
Ctenomys peruanus	34.7; n3 term emb	8304
Ctenomys talarum	8 g	1449c
LITTER SIZE		
Ctenomys opimus	1.8 emb; r1-3; n11 f	8304
	2-3 cl	8302
	2 emb; n1 f	8556
	2 emb ea; n2 f	8302
	3 ps; n1 f	8556

Ctenomys peruanus	3.1 viable emb; r1-5; n19 f	8304
Ctenomys talarum	1-6	11564
	5 yg; r1-7	11563
Ctenomys torquatus	2-3	596
SEXUAL MATURITY		
Ctenomys opimus	f/m: 1st mate: <1 yr	8304
Ctenomys peruanus	f/m: 1st mate: <1 yr	8304
GESTATION		
Ctenomys opimus	est ~2 mo; indirect est	8304
Ctenomys peruanus	est 4 mo	8304
Ctenomys talarum	>100 d	11564
	102 d; r93-120	1449c
Ctenomys torquatus	107 d	10639
LACTATION		
Ctenomys talarum	35 d	1449c
LITTERS/YEAR		
Ctenomys opimus	1	8304
Ctenomys peruanus	1?	8304
SEASONALITY		
Ctenomys opimus	mate: est Aug-Feb; Peru	8304
	preg f: Sept-Apr; Peru	8304
	birth: Oct-Mar; Peru	8304
Ctenomys peruanus	mate: end Nov-Dec; s Peru	8304
	birth: Mar-Apr; s Peru	8304
Ctenomys torquatus	mate?: Jan-Oct; Uruguay	10639
	preg f: Sept-Jan; Uruguay	596

Order Rodentia, Family Abrocomidae, Genus *Abrocoma*

South American chinchilla rats, *Abrocoma bennetti* and *A. cinerea*, range from 150 to 300 g. Females have 4 mammae.

NEONATAL MASS		
Abrocoma cinerea	8.0 g	5858
LITTER SIZE		
Abrocoma bennetti	5 emb; r4-6; n3 f	3552
Abrocoma cinerea	2.2; r1-3	5858
GESTATION		
Abrocoma cinerea	102 d	5858

Order Rodentia, Family Echimyidae

Spiny rats (*Chaetomys* 430-455 mm; *Carterodon* 155-200 mm; *Clyomys* 21-39 g; *Dactylomys* 600-700 g; *Diplomys* 250-480 mm; *Echimys* ~640 g; *Euryzygomatomys* 170-270 mm; *Hoplomys* ~450 g; *Isothrix* 220-410 mm; *Kannabateomys* ~250 mm; *Lonchothrix* ~200 mm; *Makalata* 150-215 g; *Mesomys* 150-200 mm; *Proechimys* 300-500 g; *Thrichomys* 300-450 g; *Thrinacodus* 180-240 mm) are a diverse assemblage of herbivores from Central and South American forests. Ovulation in *Proechimys* is induced (6650) and occurs 24-48 h before birth (11565). The morula reaches the uterus at 3.5 d (*Proechimys* 9161). *Proechimys* has a hemochorial, discoid, 8.5 g placenta with prominent subplacenta that begins as an antimesometrial attachment 5.5 d postcoitus, but becomes mesometrially attached (9161, 9162). Females have a bicornuate uterus with an extremely muscular cervix (*Proechimys* 9161). Dystocia is common in captive *Proechimys* (5398). A postpartum estrus occurs (*Proechimys* 9161, 11560). Neonates have open eyes at birth (*Hoplomys* 10756) or after 1-2 d *Proechimys* 11560). They are furred (*Diplomys* 10757; *Hoplomys* 10756; *Proechimys* 11560) and ambulatory (*Hoplomys* 10756). *Makalata* has 2-6 teats, one pair of which may be nonfunctional (5111, 7983). *Proechimys* (5111, 11560) has 4-6 mammae. *Euryzygomatomys* has 6 mammae (7983), while *Echimys* has 8 mammae (5111). Details of anatomy (6650, 9162, 11565) and endocrinology (4528) are available.

NEONATAL MASS		
Hoplomys gymnurus	24.3 g; r20-28.5; n6	10756
Proechimys guairae	20.9 g; se0.3; n190	11560
	26 g; n1 emb	5398
Proechimys guyannensis	24.7 g; from growth curve	3122
Trichomys apereoides	21.1 g; sd4.7; n127	9174a
LITTER SIZE		
Diplomys labilis	1.2 emb; r1-2; n13 f	10757
Echimys chrysurus	2 emb; n1 f	5111
Euryzygomatomys spinosus	1 emb ea; n2 f	7983
Hoplomys gymnurus	2.1 yg, emb; r1-3; n8 f	10756
Isothrix bistriatus	1 emb; n1 f	7983
Kannabateomys amblyonyx	mode 1 yg	7983
Makalata armata	1 yg; n1 f	5111
Mesomys hispidus	1 emb; n1 f	7983
Proechimys guairae	r1-3 emb; n8 f	5398
	2.7; r1-6; n123 lt	11560
	2.8; r2-5; n10	7995, 7998
	4 emb; n1 f	8562a
Proechimys guyannensis	2 emb; n1 f	5111
	2.4 yg; max 4	3122
Proechimys semispinosus	2.4 yg; se0.02; r1-2; n51 lt	3319
Trichomys apereoides	3.2 yg; sd1.43; mode 3; r1-7; n174 lt	9174a
SEXUAL MATURITY		
Proechimys guairae	f: vag perf: 55.9 d; sd12.8; n15 f	6650
	f: vag perf: 65.6 d; se2.4; n78 f	11560
	f: 1st conceive: 62.5 d; se3.6; n29	11560
	m: 1st mate: 3 mo	11560
	f/m: 5 mo; n10	7995, 7998
Proechimys guyannensis	f: 1st birth: 208 d	3122
Proechimys semispinosus	f: 1st preg: 9.2 mo; r6-12	3319
Trichomys apereoides	f: 1st conceive: ~73 d	9174a
	f: 1st preg: 170 d; sd13.5; n5	9174a
CYCLE LENGTH		
Proechimys guairae	14.9 d; r2-51; n54 cycles, 6 f	6650
	20.7 d; se3.4; n41 cycles, 27 f	11560
GESTATION		
Proechimys guairae	implant: 6-6.5 d	9161
	62.6 d; se0.2; n106	11560
Proechimys guyannensis	62-64 d; r55-71	3122
Trichomys apereoides	97.0 d; sd1.2; r95-98	9174a
LACTATION		
Proechimys guairae	wean: min 5 d	11560
Proechimys guyannensis	wean: 21-35 d	3122
Trichomys apereoides	wean: 6 wk	9174a
INTERLITTER INTERVAL		
Proechimys guyannensis	66 d; n1	3122
LITTERS/YEAR		
Proechimys guairae	4.4	7995, 7998
Proechimys semispinosus	4.68	3319
SEASONALITY		
Diplomys labilis	preg f: est yr rd; Panama	10757
Proechimys guairae	estrus: yr rd; lab	11560
Proechimys semispinosus	preg f: yr rd, peak Jan-June; Panama	3319
Trichomys apereoides	preg f: yr rd; captive	9174a
	repro: yr rd, low Dec-Jan; Brazil	7983

Order Rodentia, Family Thryonomyidae, Genus *Thryonomys*

The 2-9 kg cane rats (*Thryonomys gregorianus*, *T. swinderianus*) are semi-fossorial or semi-aquatic, nocturnal herbivores of marshes and savannas in Africa south of the Sahara. The hemochorial placenta with a well-developed subplacenta has an interstitial, mesometrial implantation (8029, 8029a) and is eaten after birth (421, 10158). The precocial neonates are furred with open eyes and can walk or run (331, 2439, 5760, 9945, 10158). Females have 2-3 pairs of lateral mammae placed high up on the sides of the abdomen (331, 421, 3132, 5760, 9945, 10158). The testes are semi-internal (9945).

NEONATAL MASS		
Thryonomys swinderianus	70-84 g; n4	3132
	110.0 g; sd34.0; n66	422
	131.0 g; sd28.0; r72-190; n46	421, 422
LITTER SIZE		
Thryonomys gregorianus	2-6	5760
	2.5 emb; r2-3; n2 f	10158
	3 emb ea; n2 f	4925
Thryonomys swinderianus	2-3 emb	8027
	3.6; sd1.2; n66	422
	3.9; sd1.3; 2-6 96%; r1-8; n46	421, 422
	4 emb; n1 f	331
	4-6	3132
	r4-11 cl; n4 lt	8027
SEXUAL MATURITY		
Thryonomys swinderianus	f: vag perf: 5 mo	421
	f: 1st birth: 12-18 mo	421
	f/m: 1st mate: ~1 yr	5760
CYCLE LENGTH		
Thryonomys swinderianus	6.62 d; sd3.1; n35 cycles	9652
GESTATION		
Thryonomys gregorianus	~3 mo	5760
Thryonomys swinderianus	est 10 wk	3132
	~3 mo	5760
	155 d; sd9; r137-172; fetal growth data	421
LACTATION		
Thryonomys swinderianus	solid food: 1 d?	11253
	wean: ~1 mo	10158
LITTERS/YEAR		
Thryonomys gregorianus	2 possible	5760
Thryonomys swinderianus	2	3132
	2	5760
	3	11253
SEASONALITY		
Thryonomys gregorianus	preg f: Nov-Dec; n2; Kenya	4925
	birth: Feb-Mar; n Uganda	5760
Thryonomys swinderianus	mate: Mar-Nov; Africa	5760
	preg f: 26% Mar, 95% June; n Ghana	421
	preg f: 80%/mo except 64% Feb, Aug; s Ghana	421
	birth: Mar; n1; Zambia	331
	birth: June-Aug; sw Africa	9945
	birth: June-Dec; Zimbabwe, Zambia, Malawi	10153
	birth: Aug-Dec; S Africa	10158
	birth: once dry, twice rainy season; Zaire	11253
	young found: mid-Oct-Jan; w Africa	5760

Order Rodentia, Family Petromyidae, Genus *Petromus*

The 100-300 g, terrestrial *Petromus typicus* from arid regions of southwest Africa are solitary or found in pairs. These diurnal herbivores are furred at birth (2380, 9945). Females have 6 lateral teats at the scapular level (2380, 7983, 9945, 10158), while males have semi-internal, inconspicuous testes and a short baculum (2380, 9945).

LITTER SIZE		
Petromus typicus	1-2	2380
	1-2	9945
	1-2 table or 2-4 text	11864
LITTERS/YEAR		
Petromus typicus	1	11864
SEASONALITY		
Petromus typicus	mate: Nov-Mar; Namibia	9945
	preg f: Jan; Namib desert, Namibia	11863
	preg f: Sept, Nov; S Africa	2380

Order Rodentia, Family Bathyergidae

African mole-rats (*Bathyergus* 350-1500 g; *Cryptomys* 50-200+ g; *Georychus* 125-360 g; *Heliophobius* 140-170 g; *Heterocephalus* 25-80 g) are fossorial herbivores from Africa south of the Sahara. Colonial life is highly developed (*Cryptomys* 780; *Heterocephalus* 5270), usually only one female and one or two males are reproductively active in a colony (1449c, 3176, 6144a). Males have internal testes (*Cryptomys* 9945). The hemochorial placenta has a subplacenta (*Bathyergus* 6628) and implants into a bipartite uterus (4794). Neonates are naked and open their eyes by 9 d (*Georychus* 779), 13-50 d (*Cryptomys* 780, 1449b), 3-4 wk (*Heterocephalus* 5267, 5269). Females have 6 teats (331, 2379, 3692, 4495, 5760, 9945, 10158). Males have seminal vesicles, penial spines, and abdominal testes (4794, 11127). Details of endocrinology (779, 3175a, 3176, 3177, 8116a), copulation (779, 780, 4727a), and birth (779) are available.

NEONATAL MASS		
Bathyergus janetta	15.4 g	5270b
Bathyergus suillus	34 g	5270b
Cryptomys hottentotus	7.9 g; sd0.5; r7.2-8.8; n18	1449b
	8-10 g	5270b
Georychus capensis	5.37 g; sd0.28; n6	779
	7.55 g; sd1.74; n4	779
	12.27 g; sd0.76; n10	779
Heliophobius argenteocinereus	7 g	5270b
Heterocephalus glaber	1.86 g; r1-2	5270a
	1.9 g	5267, 5269
NEONATAL SIZE		
Cryptomys hottentotus	2-2.5 cm	778a
	TL: 59.9 mm; sd2.4	1449b
Georychus capensis	3-4 cm	779
WEANING MASS		
Cryptomys hottentotus	18 g	1449c
	34.1 g; sd0.3; r33.8-34.5	1449b
Georychus capensis	36 g	1449c
Heterocephalus glaber	11 g	1449c
SEXUAL MATURITY		
Cryptomys hottentotus	f: 1st conceive: 340 d; n1 [aborted]	1449b
	f: 1st conceive: 73 wk	1449c
LITTER SIZE		
Bathyergus janetta	3.5; r1-7	5270b
Bathyergus suillus	2.4; r1-4; n34 lt	5267
	3-4 yg	10158
Cryptomys hottentotus	1-5	5760
	1.75 emb; r1-2; n8 f	8965
	2-4	11253

	2 emb; r1-3; n4 f	331
	2.2 yg; sd0.7; r1-3; n9+ lt	1449b, 1449c
	2.8; r2-4; n10 lt	780
	3 yg; n1 lt	646
	4.0 emb; n2 f	10154
	5 emb ea; n2 f	2379
	5 emb ea; n2 f	9945
Cryptomys ochraceocinereus	2-4	11253
Georychus capensis	5.9; r3-10; n19	779
	6.0 yg; mode 6; r1-10; n9 lt	10718
Heliophobius argenteocinereus	1-4	10158
	2; n1 lt	5267
	2-4	11253
	2.5 emb; r2-3; n2 f	4925
Heterocephalus glaber	3-11 yg	5267, 5269
	7.7; se2.3; r4-13; n26 lt	6144a
	9.7 weaned; se2.81	1283a
	max 12	5270
	12.3; se5.7; r1-27; n84 lt	5270a
CYCLE LENGTH		
Heterocephalus glaber	estrus: ≥24 h	6144a
GESTATION		
Bathyergus suillus	est 2-2.5 mo	5267
Cryptomys hottentotus	55-66 d; n2	778a
	78-92 d	780
	85-87 d; progesterone levels	780
	98 d; sd9.2; r84-112; n9	1449b, 1449c
Georychus capensis	44 d	779
Heliophobius argenteocinereus	87 d; n1	5267
Heterocephalus glaber	44 d	9899
	66-76 d; n121 lt, 16 f	5270a
	72-77 d	6144a
LACTATION		
Bathyergus janetta	wean: 28 d	5270b
Bathyergus suillus	wean: 21 d	5270b
Cryptomys hottentotus	independent: 18 d	780
	solid food: 19 d	1449b
	wean: 3-4 wk	778a
	wean: 82.5 d; sd15.4; r72-105	1449b, 1449c
Georychus capensis	independent: 17 d	779
Heterocephalus glaber	wean: 3-4 wk	5270a
	wean: 4-5 wk	6144a
	wean: ~1-2 mo	5270
LITTERS/YEAR		
Bathyergus janetta	1-2	5270b
Bathyergus suillus	1-2	5270b
Cryptomys hottentotus	1-2	780
	min 2	11253
Cryptomys ochraceocinereus	2	11253
Georychus capensis	2/season	779, 10718
Heliophobius argenteocinereus	1	10158
	2	11253
Heterocephalus glaber	1-4	5270
	4-5	5270a
	max 5	6144a
SEASONALITY		
Bathyergus janetta	preg f: Aug-Dec; S Africa	5270b
Bathyergus suillus	testes: large June-Sept; S Africa	11127
	mate: Apr-early Oct; S Africa	11127
	preg f: Aug-Sept; S Africa	10158
	repro: mid June-mid Oct; S Africa	5267, 5270b
Cryptomys hottentotus	preg f: Feb-Mar, Aug; Transvaal, S Africa	8965
	preg f: Apr; n2; Namibia	2379
	birth: end rainy season; Zaire	11253

	lact f: Jan, Apr; Transvaal, S Africa	8965
	juveniles/subadult found: yr rd; Zambia	331
	repro: yr rd; S Africa	5270b
Cryptomys hottentotus	preg f: Dec; n1; Karamoja, Uganda	5760
	preg f: Apr; n2; Namibia	9945
	est yr rd; S Africa	10158
Cryptomys ochraceocinereus	mate: Apr-June; Zaire	11253
	birth: Apr-June; Zaire	11253
Georychus capensis	inguinal testes: May-Sept; S Africa	779
	preg f: Sept-Dec; S Africa	779
Heliophobius argenteocinereus	preg f: May; n1; Kenya	4925
Heterocephalus glaber	yg found: Feb-Apr; Kenya	5267
	repro: yr rd; S Africa	1283a
	repro: yr rd; captive	5270a, 5270b

Order Rodentia, Family Ctenodactylidae

Gundis (*Ctenodactylus*, *Felovia*, *Massoutiera*, *Pectinator*) are 150-350 g, diurnal, terrestrial herbivores from the Sahara. A copulatory plug is formed and placentophagia occurs as does a prepartum estrus (3971). Neonates are furred, homeothermic, and have their eyes open (3727, 3970, 6237). Females have vaginal closure membranes, a bicornuate uterus, and 2-4 mammae (3727, 8404). Males have a baculum and seminal vesicles (8404). Details of placentation (2392, 2399) are available.

NEONATAL MASS		
Ctenodactylus gundi	29.9 g; se11.9; r18-40; n15	3971
Ctenodactylus vali	17 g; n1	3972
	f: 18 g; n1	3727
	19.9 g; n14	3727
Massoutiera mzabi	20.5 g; r20-21; n2	3727
LITTER SIZE		
Ctenodactylus gundi	1.7; mode 2; r1-3; n17 lt	3971
	2	3727
	~5 emb; 2.45; mode 2; r1-5; n148 f	2398
Ctenodactylus vali	2; r2-3	3727
Massoutiera mzabi	2	3727
Pectinator spekei	1	3727
	2-3	8404
SEXUAL MATURITY		
Ctenodactylus gundi	f: vag perf: 75 d	3971
	9-12 mo	3727
Ctenodactylus vali	f: vag perf: 10-12 mo; n2	3727
Massoutiera mzabi	f: vag perf: 10-12 mo; n1	3727
	f: vag perf: 6 mo; n1; wild-caught	3727
Pectinator spekei	f: vag perf: 10-12 mo; n1	3727
CYCLE LENGTH		
Ctenodactylus gundi	vagina open: 4.7 d; closed 23.6 d	3727
	vagina open: 7.4 d	3971
	28.7 d; mode 31; r21-34; n15 cycles, 2 f	3727
Ctenodactylus vali	vagina open: 3.8 d; closed 19.6 d	3727
	23.4 d; mode 22; r14-33; n34 cycles, 5 f	3727
Massoutiera mzabi	vagina open: 4.9 d; closed 19.9 d	3727
	24.9 d; mode 26; r14-36; n57 cycles, 7 f	3727
Pectinator spekei	vagina open: 4.9 d; closed 17.6 d	3727
	22.7 d; mode 25; r14-29; n31 cycles	3727
GESTATION		
Ctenodactylus gundi	est >57 d	6237
	73.2 d; se6.6; r70-79; n5: vaginal plug-birth	3971
Ctenodactylus vali	est 56-67 d	3727
	68 d; n1 capture to birth	3972
INTERLITTER INTERVAL		
Ctenodactylus gundi	69.7 d; se0.9; r69-70; n3	3971
Ctenodactylus vali	64 d; n1 after stillborn	3972

SEASONALITY

Ctenodactylus gundi	vagina open: yr rd; lab	3971
	diestrus: May-Dec; Tunis	2398
	estrus: Jan-Mar; Tunis	2398
	estrus: Feb-Aug; Algeria	3727
	preg f: Feb-Mar; Tunis	2398
	birth: est Jan-Mar; Algeria	3727
	birth: Feb-Mar, May, Nov; lab	3971
	yg found: Feb-June; Algeria	3971
Ctenodactylus vali	birth: Mar, July, Nov; n3; captive	3727
	birth: Mar-Apr; semi-captive, Algeria	3727
	repro: Feb-June, peaks Mar-Apr, end June; Algeria	3727
Massoutiera mzabi	estrus: Apr-Aug; Algeria	3727
	yg found: Mar-June; Algeria	3727
Pectinator spekei	preg f: Sept-Oct; few data; ne Africa	8404